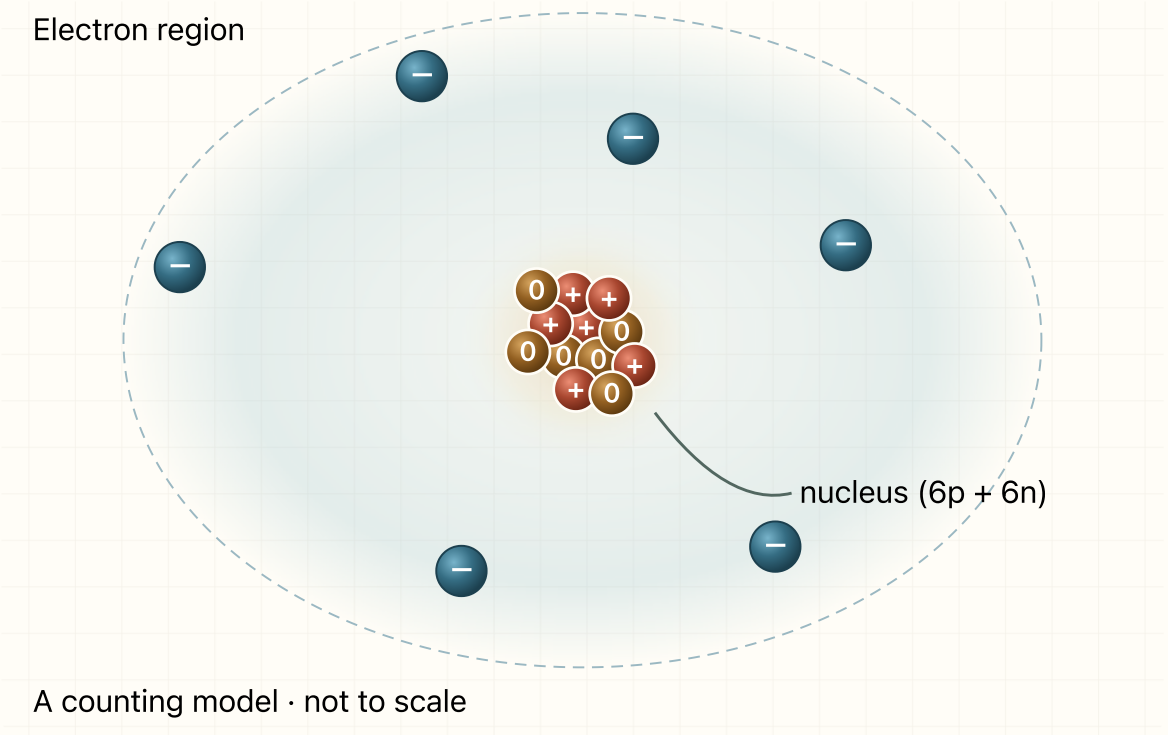
KYMAIA ARCHIVAL LABORATORY PRESS · COMPLETE THREE-VOLUME EDITION

Kymaia Chemistry: From Zero to Hero

A Living Textbook of Subatomic Particles, Experimental Evidence, Electrostatic Forces, and Quantitative Chemical Reasoning

Frontispiece: Carbon-12 (6p, 6n, 6e⁻)



Charge Tug-of-War Balance (+q vs. −q) Net charge 0 (Balanced ✓)

6 protons (+6 e)

6 electrons (−6 e)

Charge balanced ✓: +6 nuclear proton charge cancels −6 electron charge (net q = 0).

Tokens count particles. The shading is not a calculated electron cloud, and the tokens are not measured positions.

126C

3 Volumes · 12 Chapters · 56 Interactive Investigations · 19 Mathematical Model Engines · Exported 2026-10-28T18:47:04.149Z

AUTHOR'S FRONTISPIECE & LETTER TO THE INVESTIGATOR

To Kymaia, Upon Opening the Invisible World

On soft grey pencils, star-forged nuclei, honest maps, and what I hope you carry from this laboratory

> *"The ordinary world looks quiet and continuous—until you learn to count the tiny, restless particles holding it together."*
>
> *— From the Laboratory Archival Frontispiece*

My dearest Kymaia,

Look at the dark grey tip of a sketching pencil resting beside your notebook, and then picture a cut diamond catching the afternoon sun. One is soft enough to leave a whispered trail of grey across warm rag paper whenever you draw; the other is hard enough to scratch glass. Yet if you could shrink yourself ten billion times—until a single atom stood as tall as a cathedral—you would discover one of nature's most marvelous jokes. Both the humble pencil mark and the brilliant diamond are built from the exact same kind of atom: carbon, an element whose identity is fixed forever by the six positively charged protons huddled in its nucleus ($Z = 6$). Nothing magical was sprinkled into the diamond. Only the three-dimensional geometry of how those carbon atoms hold hands changed—sliding hexagonal sheets in graphite versus a rigid four-way tetrahedral scaffold in diamond.

Every proton in a raindrop on your window was born in the first minutes of the universe, and every carbon, oxygen, and iron atom in your sketching hand was forged inside the heart of an ancient star long before the Earth had a name. Chemistry is not a dusty shelf of rules invented to quiz students on Tuesday mornings. It is the secret grammar of ordinary things: why a copper kettle sings when heated, why table salt vanishes into warm water yet returns as tiny cubic crystals on a saucer, why lemon juice stings a cut while baking soda calms it, and how a handful of carbon chains fold into the scent of vanilla, the indigo of denim, and the living machinery of your own cells.

As you turn the pages and turn the dials of this living book, I want to let you in on a secret that many textbooks hide until university: our diagrams are honest maps, never photographs or magic spells. In schoolrooms, people sometimes say things like "a sodium atom wants to lose an electron so it can be happy with a full shell." Whenever you hear talk like that, raise a skeptical eyebrow! Atoms do not have wishes, diaries, or

ambitions. An electron stays or leaves because of real physical interactions—the electrostatic tug-of-war of Coulomb's Law ($V(r) \propto q_1 q_2 / r$), the ceaseless thermal jiggling of particles at temperature T, and the strict conservation of every nucleus and charge.

Even our sharpest screen diagrams are simplified counting models. If a real carbon nucleus were the size of a glass marble on your desk, the atom's outer electron region would stretch across a football stadium! Because a screen drawn to true scale would look like an empty sheet of paper with an invisible speck in the middle, we draw schematic plates and interactive 3D models so you can count particles, rotate geometries, and test cause and effect.

Four Habits of Mind I Hope You Take From This Book

I — Macro ↔ Particle ↔ Symbol

Hold three lenses in your mind at once (Johnstone's Triangle)

Whenever you see a macroscopic clue on the laboratory bench—a colour shift, a rising manometer, or a fizzing beaker—trace it down to the submicroscopic dance of particles and forces, and only then translate it into symbolic notation (^_Z X^q, balanced equations, and mole ratios).

II — Four First Principles Spine

Replace "atoms want" with forces, energy, and counting

Ask which particle count changed, how positive and negative charges pull or push across distance, whether thermal motion (k_B T) can surmount an activation barrier (E_a), and where every single atom went on both sides of the arrow.

III

Evidence vs. Model Output

Interrogate every model until you find where it breaks

Every scientific model leaves something out on purpose so you can think clearly. Ask not only “What does this model explain?” but also “Where does this model hit its boundary, and what experimental evidence forced scientists to upgrade it?”

IV

Predict → Explore → Explain

Cherish the moment a prediction surprises you

A prediction that turns out to be wrong is not a mistake to erase—it is the exact compass needle pointing to a new discovery. That is why this notebook has no grades, no timers, and no penalties: your first prediction and your later explanation belong side by side.

Take your time. Nudge one particle at a time in the interactive plates below, spin the 3D crystal lattices with your finger or keyboard, sketch what you think is happening in the margins, and never stop asking “How do we know?” This laboratory was built for your curiosity.

With wonder, a magnifying loupe, and a pocketful of protons,

The Author · Written for Kymaia’s Chemistry Laboratory

P.S. Try the interactive Curiosity Cabinet and the live Particle & Force Sandbox right beside this letter—even the frontispiece of this book is an instrument you can experiment with!

SYNOPSIS OF THE THREE VOLUMES

Illustrated Table of Contents

Every chapter combines macroscopic laboratory evidence, submicroscopic particle mechanics, and symbolic notation across 12 chapters and 56 investigations.

FROM ZERO TO PARTICLE THINKER (CHAPTERS 0–3)

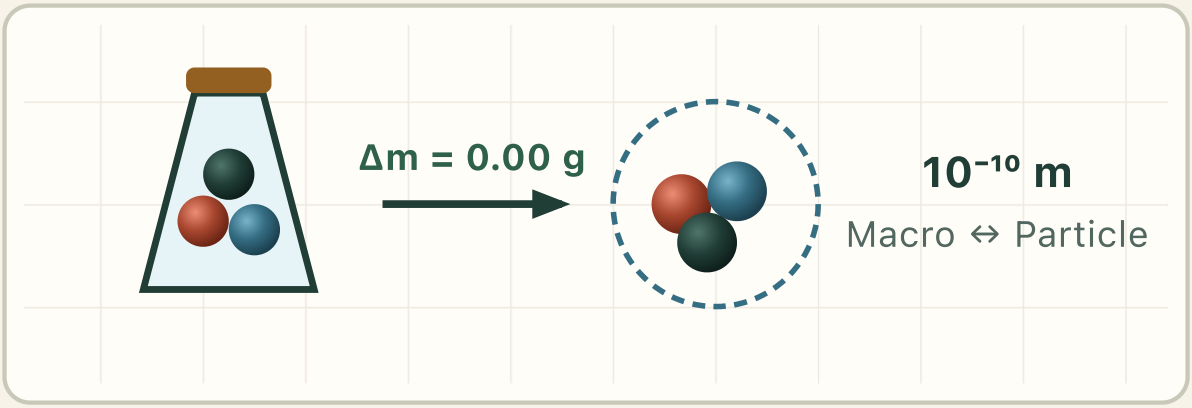
Volume I · Particles, Identity & Structure

Build the chemist's submicroscopic lens from scratch: observation versus inference, subatomic particle counting, electrostatic bonding, 3D molecular geometry, and intermolecular forces across states of matter.

CHAPTER 0 · CHAPTER 0 PRELUDE · HOW DO WE KNOW ATOMS EXIST IF WE CANNOT SEE THEM?

Chapter 0: The chemist's lens: scale, models and conservation

Big Question: A single drop of water weighs about 0.05 grams and looks completely smooth and continuous to your eye. Yet chemists claim that drop holds more than 1.6×10^{21} discrete, jostling H_2O molecules! Before we count protons or balance equations, how do scientists connect what we measure on a laboratory bench to an invisible particle world ten billion times smaller?



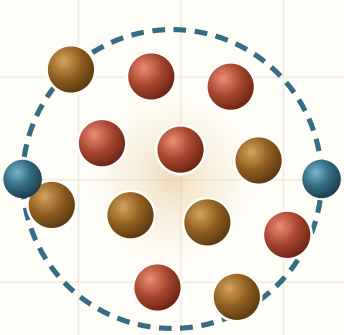
0.1 Observation versus model inference ..... Principle 1: Conservation of Count & Charge

0.2 Powers of ten: from flask to atom ..... Principle 4: Dynamic & Statistical Ratios

CHAPTER 1 · CHAPTER 1 PRELUDE · WHY DOESN'T LOSING AN ELECTRON TURN SODIUM INTO NEON?

Chapter 1: What makes an atom?

Big Question: Gold and mercury sit right next to each other on the periodic table (atomic numbers 79 and 80), yet one is a precious yellow solid and the other is a dense silvery liquid. Meanwhile, table salt contains sodium ions (Na^+) that have the exact same number of electrons (10 e^-) as neon gas—so why doesn't Na^+ turn into neon? To answer, we must open the atom and track protons, neutrons, and electrons separately.



Z = 6 Protons (+)
N = 6 Neutrons (0)
e^- = 6 Electrons (–)

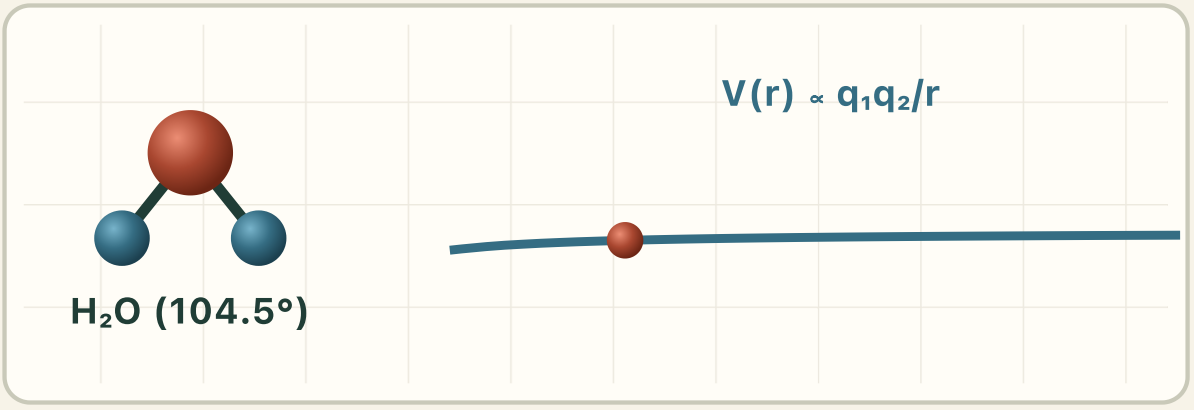
$^{12}_{6}C$

1.1 Inside an atom ..... Principle 1: Conservation of Count & Charge
1.2 What identifies an element? ..... Principle 1: Conservation of Count & Charge
1.3 Same element, different isotope ..... Principle 4: Dynamic & Statistical Ratios
1.4 Where does charge come from? ..... Principle 2: Electrostatic Balance
1.5 Reading an atom's identity ..... Principle 1: Conservation of Count & Charge
1.6 What might explain patterns? ..... Principle 2: Electrostatic Balance

CHAPTER 2 · CHAPTER 2 PRELUDE · HOW DO TOXIC SODIUM AND CHLORINE GAS BECOME EDIBLE SALT?

Chapter 2: How atoms form substances

Big Question: Pure carbon can be soft grey pencil lead (graphite) or the hardest gemstone on Earth (diamond). Meanwhile, water (H_2O) is a liquid made of discrete 3-atom molecules, whereas table salt (NaCl) forms hard cubic crystals with no individual 'NaCl molecules' at all. How do electrostatic forces and 3D geometry turn the same set of elements into radically different substances?

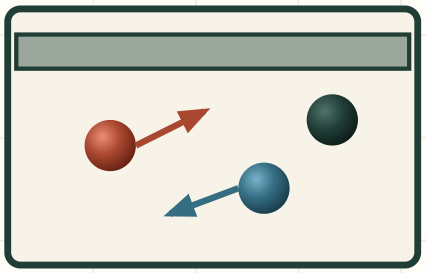


2.1 Element, compound or mixture? ..... Principle 1: Conservation of Count & Charge
2.2 How do charges interact? ..... Principle 2: Electrostatic Balance
2.3 How can atoms share electrons? ..... Principle 2: Electrostatic Balance
2.4 What is a molecule? ..... Principle 1: Conservation of Count & Charge
2.5 A formula without a molecule ..... Principle 2: Electrostatic Balance
2.6 Why composition is not enough ..... Principle 2: Electrostatic Balance
2.7 Unequal sharing and molecular dipoles ..... Principle 2: Electrostatic Balance

CHAPTER 3 · CHAPTER 3 PRELUDE · WHAT ACTUALLY BREAKS WHEN WATER BOILS?

Chapter 3: Why matter has different states

Big Question: When a kettle boils on the stove, bubbles of invisible gas rise through the liquid and escape as steam. A surprisingly common intuition says that boiling splits H_2O into hydrogen gas (H_2) and oxygen gas (O_2)—yet steam never explodes when it touches the stove flame, and it condenses right back into water on a cool mirror! What is the real submicroscopic tug-of-war behind solids, liquids, gases, and solutions?



$$P \cdot V \propto N \cdot k_B\, T$$

$\langle E_k \rangle \propto T$ · Collisions → P

Identity conserved

3.1 One substance, three states ..... Principle 3: Thermal Motion vs. Attraction
3.2 Temperature and particle motion ..... Principle 3: Thermal Motion vs. Attraction
3.3 Changing state is not changing identity ..... Principle 3: Thermal Motion vs. Attraction
3.4 Squeezing a gas ..... Principle 3: Thermal Motion vs. Attraction
3.5 Where did the solute go? ..... Principle 4: Dynamic & Statistical Ratios
3.6 Intermolecular forces vs. thermal motion ..... Principle 3: Thermal Motion vs. Attraction

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FROM PARTICLE THINKER TO QUANTITATIVE ANALYST (CHAPTERS 4–6)

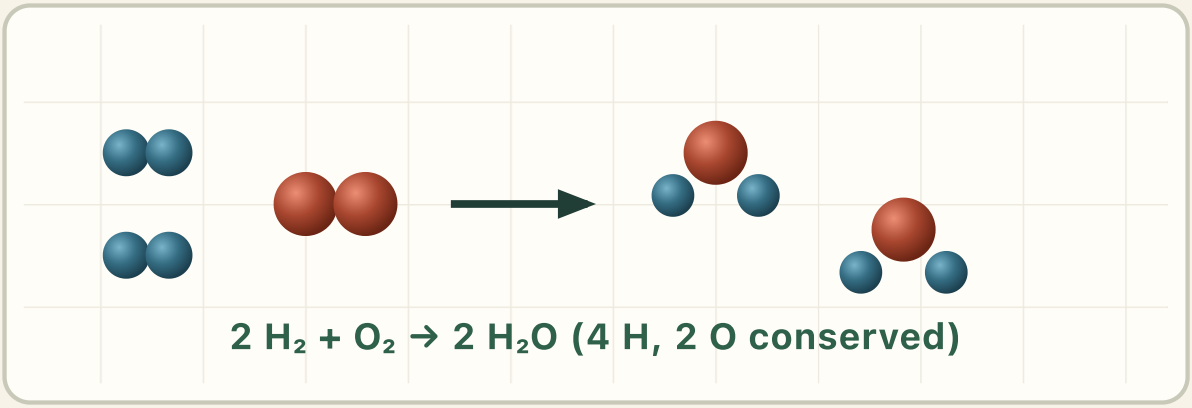
Volume II · Change, Quantities & Driving Forces

Account for every atom, electron, and joule in chemical transformations: balanced equations, bond-breaking versus bond-making energy ledgers, activation barriers, Le Châtelier equilibrium shifts, and the mole.

CHAPTER 4 · CHAPTER 4 PRELUDE · WHERE DOES THE WOOD GO WHEN A CAMPFIRE BURNS?

Chapter 4: What changes in a reaction?

Big Question: When a heavy oak log burns in an open fireplace, only a light pile of grey ash remains—making it look as if matter vanished into thin air. Conversely, when iron rusts in damp air, the flaky red rust weighs MORE than the original iron nail! Once Antoine Lavoisier trapped all the gases in a sealed vessel, he proved that not a single atom is created or destroyed. How do we write and audit the strict atomic ledger of a chemical reaction?

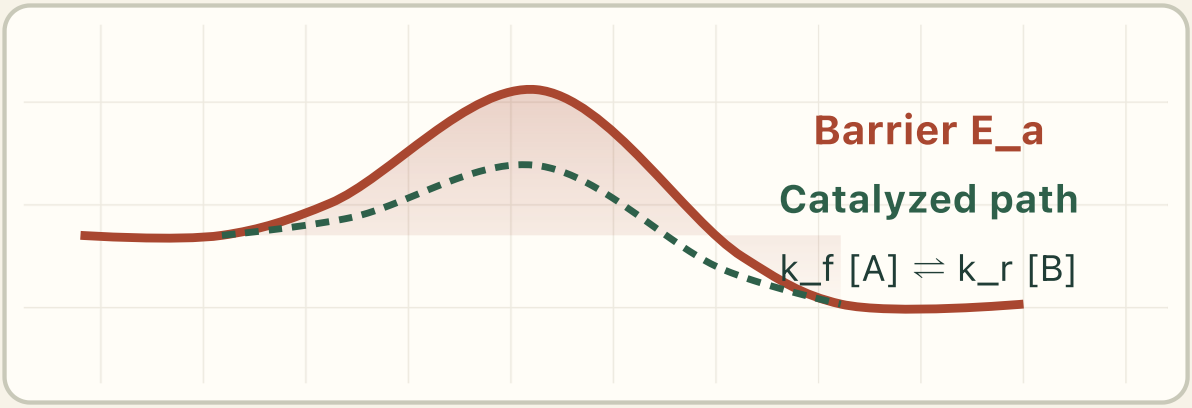


4.1 What counts as a chemical change? — Principle 1: Conservation of Count & Charge
4.2 Reading a chemical formula — Principle 2: Electrostatic Balance
4.3 Where did every atom go? — Principle 1: Conservation of Count & Charge
4.4 What a balanced equation cannot tell us — Principle 1: Conservation of Count & Charge

CHAPTER 5 · CHAPTER 5 PRELUDE · WHY DOESN'T THIS PAGE BURST INTO FLAME RIGHT NOW?

Chapter 5: Energy, rates and equilibrium

Big Question: Paper and oxygen gas react exothermically to form CO_2 and H_2O, releasing a large amount of heat. Yet this book can sit in air for centuries without catching fire! Even more puzzling, some reactions release heat while others freeze water to a beaker, and many reactions refuse to go to 100% completion no matter how long you wait. How do bond energies, activation barriers, catalysts, and dynamic equilibrium govern every chemical transformation?

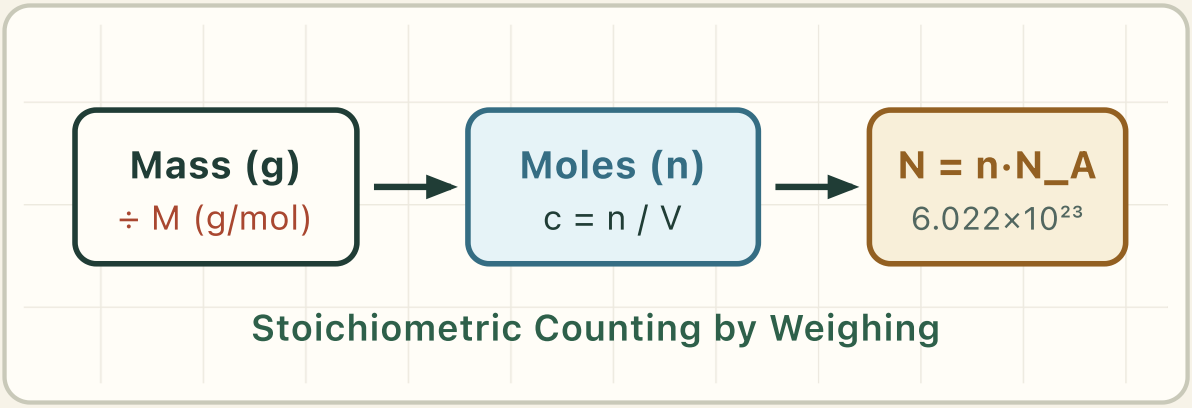


5.1 Where does the energy go? — Principle 3: Thermal Motion vs. Attraction
5.2 Why is there an energy barrier? — Principle 3: Thermal Motion vs. Attraction
5.3 How quickly does change happen? — Principle 4: Dynamic & Statistical Ratios
5.4 A different pathway — Principle 3: Thermal Motion vs. Attraction
5.5 A balance that keeps moving — Principle 4: Dynamic & Statistical Ratios
5.6 Bond-breaking cost vs. bond-making payback — Principle 3: Thermal Motion vs. Attraction
5.7 Perturbing a dynamic equilibrium — Principle 4: Dynamic & Statistical Ratios

CHAPTER 6 · CHAPTER 6 PRELUDE · HOW DO CHEMISTS COUNT 10^{23} REACTING MOLECULES WITH A SCALE AND A FLASK?

Chapter 6: A quantitative language

Big Question: When an airbag inflates in milliseconds or a pharmacist prepares an intravenous saline solution, being off by a factor of two can be catastrophic. We cannot count molecules with tweezers—we weigh powders in grams and measure liquids in litres. How do we translate fluently between grams, moles, limiting reactants, and molar concentration?



6.1 Ratios tell a story ..... Principle 4: Dynamic & Statistical Ratios
6.2 Counting beyond what we can see ..... Principle 4: Dynamic & Statistical Ratios
6.3 From moles to mass ..... Principle 4: Dynamic & Statistical Ratios
6.4 Which reactant runs out first? ..... Principle 1: Conservation of Count & Charge
6.5 How much in each litre? ..... Principle 4: Dynamic & Statistical Ratios
6.6 From laboratory grams to product grams ..... Principle 4: Dynamic & Statistical Ratios

FROM QUANTITATIVE ANALYST TO CHEMISTRY HERO (CHAPTERS 7–11)

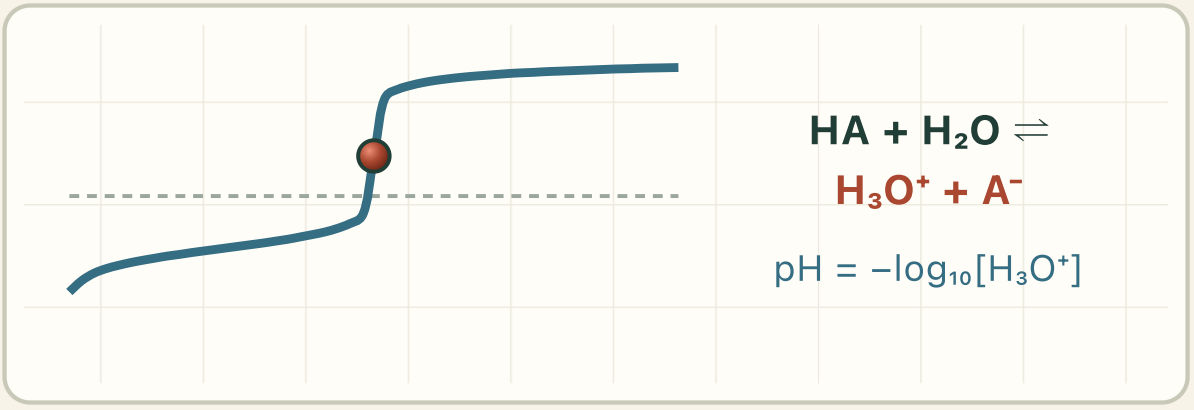
Volume III · Chemical Systems & The Real World

Synthesize proton transfer, titration buffers, electrochemistry, periodic trends, carbon functional groups, and planetary materials into real-world chemical problem solving.

CHAPTER 7 · CHAPTER 7 PRELUDE · WHY DOES LEMON JUICE TASTE SOUR WHILE BLOOD STAYS LOCKED AT PH 7.4?

Chapter 7: Acids, bases and proton transfer

Big Question: Vinegar and battery acid can be prepared at the exact same 0.10 mol/L concentration, yet one goes on your salad and the other corrodes metal. Meanwhile, human blood stays within a razor-thin window of pH 7.35–7.45 even when your muscles produce lactic acid during a sprint. How does proton (H^+) transfer, logarithmic pH, and conjugate acid–base buffering control aqueous chemistry?



7.1 What makes a solution acidic? ..... Principle 2: Electrostatic Balance

7.2 Strong or concentrated? ..... Principle 4: Dynamic & Statistical Ratios

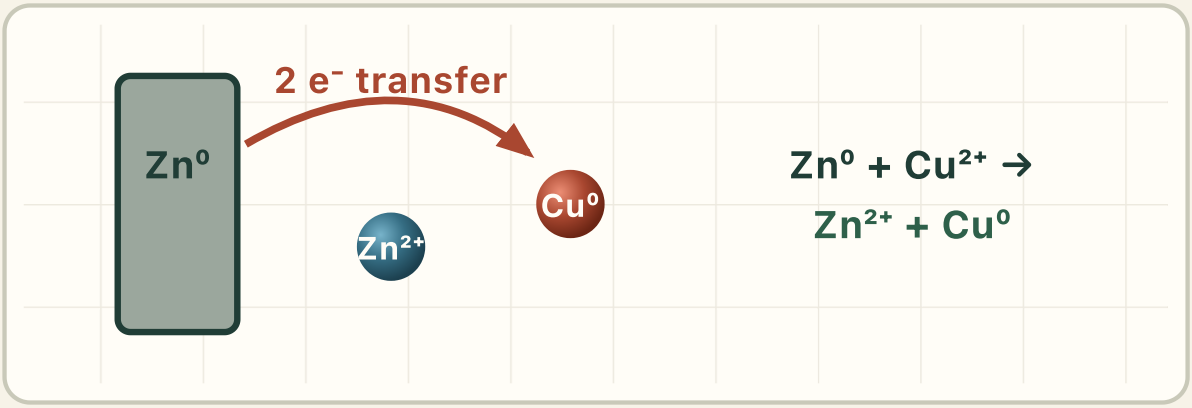
7.3 Every step on the pH scale is tenfold ..... Principle 4: Dynamic & Statistical Ratios

7.4 Titration curves and buffer resistance ..... Principle 4: Dynamic & Statistical Ratios

CHAPTER 8 · CHAPTER 8 PRELUDE · HOW DOES A CHEMICAL REACTION PUSH ELECTRONS THROUGH A WIRE?

Chapter 8: Electron transfer and electrochemistry

Big Question: When you dip a zinc strip into blue copper(II) sulfate, copper metal plates onto the zinc while the solution warms up. If you separate those exact same reactants into two half-cells connected by a wire and a salt bridge, the electron transfer powers a lightbulb or a clock! How do electron conservation, half-reaction stoichiometry, and thermodynamic driving force govern redox chemistry?



8.1 When metals and ions trade electrons ..... Principle 2: Electrostatic Balance

8.2 Balancing electrons as well as atoms ..... Principle 1: Conservation of Count & Charge

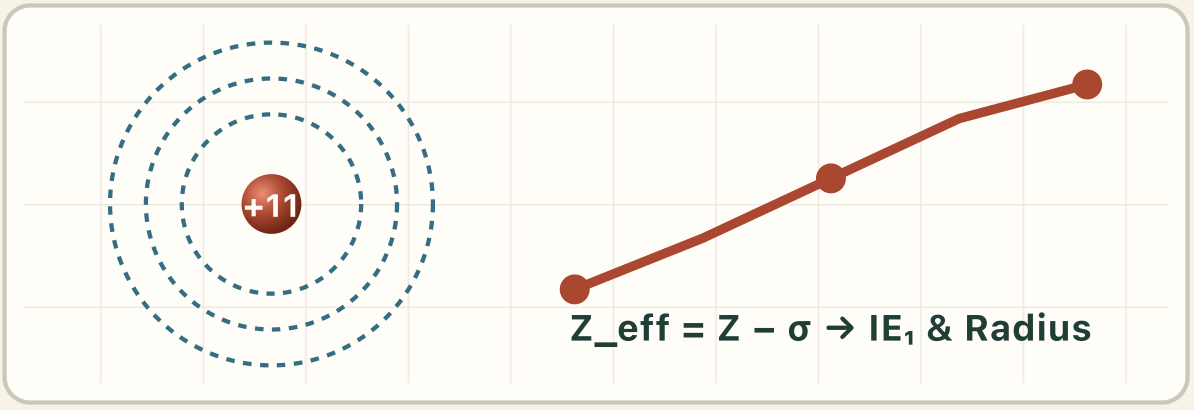
8.3 Why doesn't copper dissolve in zinc sulfate? ..... Principle 2: Electrostatic Balance

8.4 Galvanic cells vs. electrolytic charging ..... Principle 2: Electrostatic Balance

CHAPTER 9 · CHAPTER 9 PRELUDE · WHY DOES THE PERIODIC TABLE HAVE ITS SHAPE?

Chapter 9: Periodic trends and quantum orbitals

Big Question: Dmitri Mendeleev organized the elements by recurring chemical properties decades before anyone knew about protons or electron shells. Today we can explain the entire periodic table from first principles: why do atoms shrink as you move from left to right across a row even though you are adding more protons, neutrons, and electrons? And why does removing an electron ALWAYS cost energy?



9.1 Core shielding and effective nuclear charge

Principle 2: Electrostatic Balance

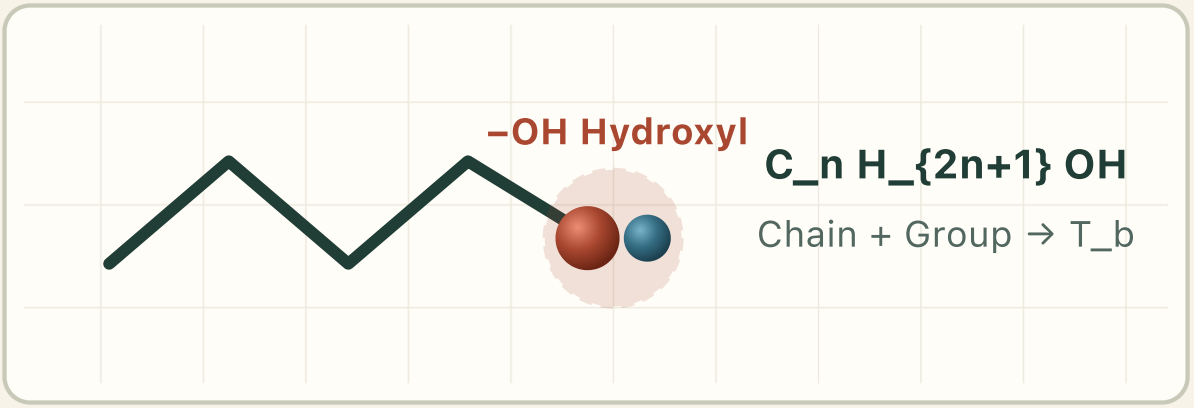
9.2 Why removing an electron costs energy

Principle 3: Thermal Motion vs. Attraction

CHAPTER 10 · CHAPTER 10 PRELUDE · WHY IS LIFE BUILT ON CARBON AND FUNCTIONAL GROUPS?

Chapter 10: Carbon chains and the chemistry of life

Big Question: Over 95% of all known chemical compounds contain carbon! From methane gas and plastic polymers to DNA, glucose, and cell membranes, carbon's four valence electrons snap together into chains, branched skeletons, and rings. Yet a pure carbon–hydrogen chain is greasy and insoluble in water—how does swapping a single hydrogen atom for an –OH or $–NH_2$ group transform a nonpolar gas into a water-soluble molecule of life?



10.1 Carbon chains, rings and 3D networks — Principle 1: Conservation of Count & Charge

10.2 How functional groups transform properties — Principle 2: Electrostatic Balance

10.3 Hydrocarbon chain length and dispersion forces — Principle 3: Thermal Motion vs. Attraction

10.4 Same formula, different molecule: Constitutional isomers — Principle 1: Conservation of Count & Charge

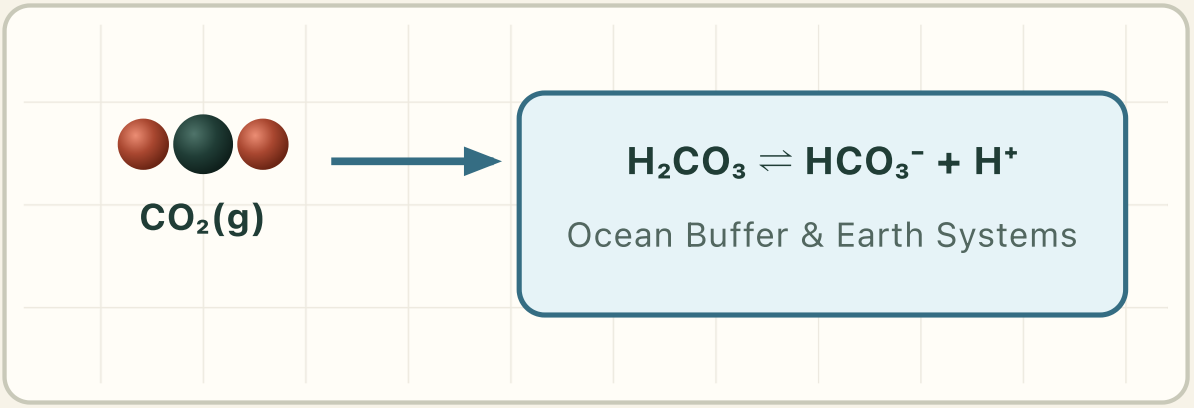
10.5 Double bonds, unsaturation and locked 3D geometry — Principle 2: Electrostatic Balance

10.6 Carboxylic acids, amines and condensation links of life — Principle 2: Electrostatic Balance

CHAPTER 11 · CHAPTER 11 PRELUDE · CAPSTONE: UNITING ALL FOUR FIRST PRINCIPLES ON PLANET EARTH

Chapter 11: Earth, materials and energy systems

Big Question: You have travelled from counting protons in a single atom to analyzing bonds, gases, reaction energy ledgers, dynamic equilibria, moles, buffers, and electrochemical cells. In the real world, these topics never live in separate chapters! When humanity transitions from burning hydrocarbon fuels (releasing CO_2 that perturbs the ocean's carbonate buffer) to storing solar electricity in rechargeable lithium batteries, every single principle you have mastered works together simultaneously.



11.1 Ocean acidification and carbonate buffering ..... Principle 4: Dynamic & Statistical Ratios

11.2 Energy storage in rechargeable cells ..... Principle 2: Electrostatic Balance

ARCHITECTURAL SPINE OF CHEMISTRY

The Four First Principles & Prerequisite Concept Graph

Principle 1: Counting & Conservation (Principle 1: Conservation of Count & Charge)

Protons define the element; nucleons (protons + neutrons) define the mass number; atoms and net electric charge are strictly conserved in chemical reactions.

Cross-chapter spine: Connects Atomic Identity & Notation (Ch. 1), Substance Classification (Ch. 2), Formula Reading & Equation Balancing (Ch. 4), Limiting Reactants (Ch. 6), and Electron-Plus-Atom Accounting in Redox (Ch. 8).

Principle 2: Electrostatic Attraction & Charge Balance (Principle 2: Electrostatic Balance)

Opposite charges attract (+q and –q) and like charges repel; ions, molecules, and extended lattices arrange so net electric charge balances or electrostatic attraction is maximized.

Cross-chapter spine: Connects Ion Charge Arithmetic (Ch. 1), Coulomb Forces, Ionic Lattices & Charge-Balanced Formulas (Ch. 2), Proton (H^+) Transfer in Acids & Bases (Ch. 7), and Electron Transfer in Redox (Ch. 8).

Principle 3: Thermal Motion vs. Interparticle Attraction (Principle 3: Thermal Motion vs. Attraction)

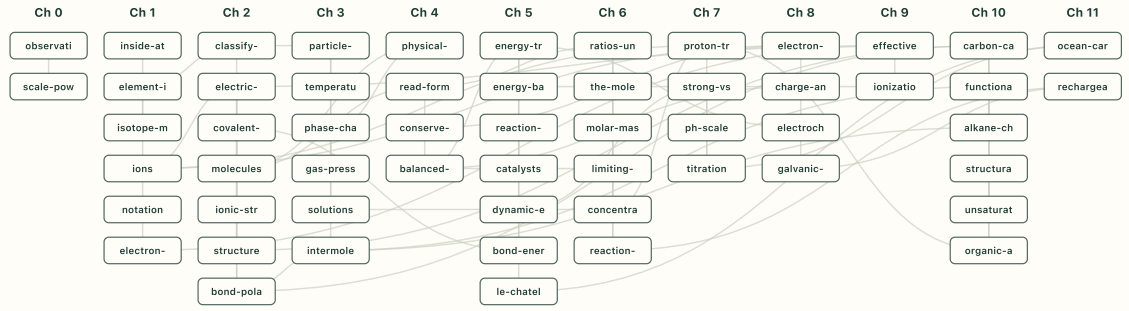
States of matter, phase changes, gas pressure, and reaction barriers are a competition between thermal kinetic energy (temperature T pushing particles apart or over a barrier) and attractions or barriers holding them back.

Cross-chapter spine: Connects Solid/Liquid/Gas States, Phase Transitions & Ideal Gas Pressure (Ch. 3) with Reaction Energy Profiles, Activation Barriers & Catalysis (Ch. 5).

Principle 4: Statistical Ratios & Dynamic Balance (Principle 4: Dynamic & Statistical Ratios)

Isotopic mixtures, reversible reactions, weak-acid ionization, concentration, and the mole bridge single-particle probabilities to macroscopic counts (6.022×10^{23}).

Cross-chapter spine: Connects Isotopic Mixture Averages (Ch. 1), Solution Dilution (Ch. 3), Reversible Rates & Dynamic Equilibrium (Ch. 5), The Mole & Molarity (Ch. 6), and Weak-Acid Equilibrium & Logarithmic pH (Ch. 7).



Curriculum Concept DAG: Directed prerequisite links connecting all 56 investigations from Chapter 0 to Chapter 11.

FROM ZERO TO PARTICLE THINKER (CHAPTERS 0–3)

Volume I · Particles, Identity & Structure

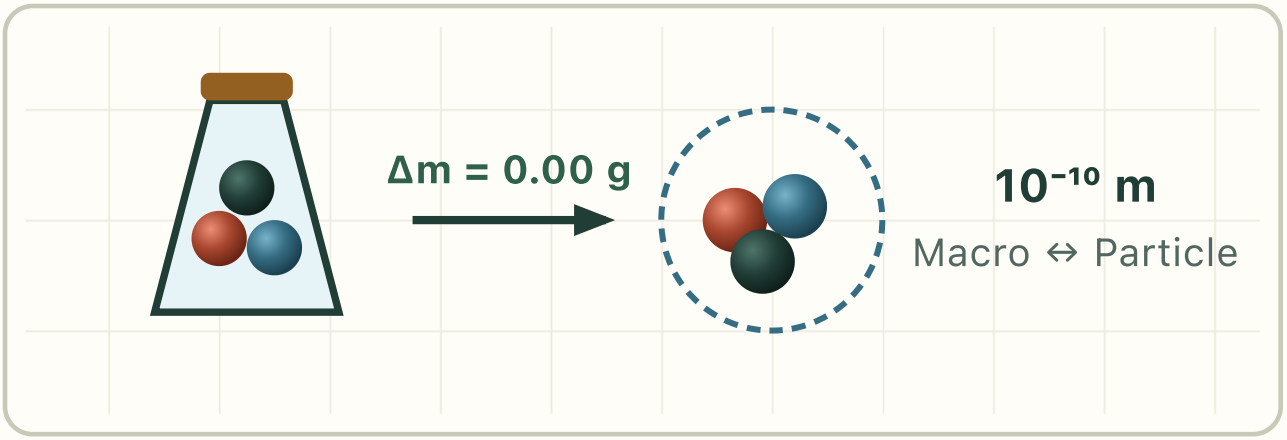
Build the chemist's submicroscopic lens from scratch: observation versus inference, subatomic particle counting, electrostatic bonding, 3D molecular geometry, and intermolecular forces across states of matter.

VOLUME I · PARTICLES, IDENTITY & STRUCTURE · CHAPTER 0

Chapter 0: The chemist's lens: scale, models and conservation

Before counting protons, learn how chemists connect macroscopic measurements in a sealed flask to invisible particles across ten billionfold powers of ten.

Chapter 0 Prelude · How Do We Know Atoms Exist If We Cannot See Them?: A single drop of water weighs about 0.05 grams and looks completely smooth and continuous to your eye. Yet chemists claim that drop holds more than 1.6×10^{21} discrete, jostling H_2O molecules! Before we count protons or balance equations, how do scientists connect what we measure on a laboratory bench to an invisible particle world ten billion times smaller?



Macroscopic Bench: Direct bench measurements: mass on a balance (g), volume in a cylinder (mL), temperature (°C/K), and phase.

Symbolic Notation: Quantitative notation: chemical formulas (H_2O), mole amounts ($n = m / M$), and Avogadro's constant (6.022×10^{23} mol^{-1}).

Submicroscopic Particle Model: Inferred submicroscopic models: discrete atoms, molecules, and ions separated across powers of ten.

INVESTIGATION 0.1 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: COMPOSITION

Observation versus model inference

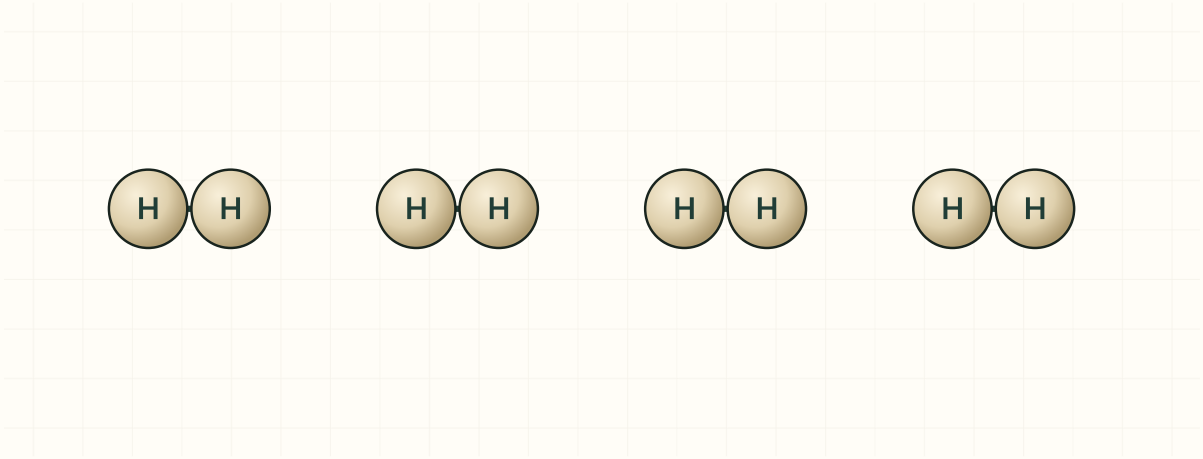
Separating direct laboratory measurements from invisible particle-model inferences

Science begins by separating what we directly observe with our senses or instruments (macroscopic evidence) from the invisible submicroscopic particle stories (inferences) we construct to explain those observations.

Bench Observation: Clear liquid water and an invisible mixture of hydrogen and oxygen gas look completely different in a flask, yet any particle drawing of H_2, O_2, or H_2O is a human-built model rather than a direct photograph.

Investigation Task: Compare the Pure H_2, Pure O_2, Mixture of H_2 and O_2, and Pure H_2O samples. State one macroscopic observation and one particle-level inference.

Observation versus model inference



Grouped atom symbols show molecular composition. Lines mark connectivity, not exact bond order, electron positions or geometry.

Account for the model

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like sorting stock by identity: count element identities, chemical substances and particles in separate columns. Mixing two stocks is not evidence that they reacted.

Current explanatory plate · intro-models-2

Substances represented 1

Model note: H_2 is an element: each molecule contains only hydrogen atoms. A molecule need not be a compound.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.1; 2.4–2.7. Substances, bonding and structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.1–2. Structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1, C-7.1 (pp. 11–12). Compounds, bonding and fixed proportions Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 0.1: Deterministic particle & mathematical plate for *Observation versus model inference*. First Principle: Macroscopic observations record measurable bulk properties (mass, volume, phase, colour), whereas submicroscopic particle models are testable causal inferences connecting those observations to conserved atoms.

Historical & Experimental Evidence: Macroscopic bench measurements versus submicroscopic particle diagrams

When you boil pure water in a glass flask, a thermometer reads 100 °C at 1 atm and colourless steam condenses on a cold watch glass. Those are direct macroscopic observations. Saying that each water droplet contains billions of bent H_2O molecules held by hydrogen bonds is a powerful scientific inference—a model we test against evidence rather than something seen with the naked eye.

Key Terms

Macroscopic observation
A property or change measured directly at the bulk laboratory scale (such as mass, volume, temperature, pressure, phase, or colour).

Submicroscopic inference (model)
A conceptual or mathematical representation of atoms, ions, molecules, and forces constructed to explain and predict macroscopic observations.

Worked Reasoning Trace

Held fixed: Sample mass and temperature in the laboratory.

Changed: Switched from Mixture of H_2 and O_2 (sample 2) to Pure H_2O (sample 3).

Model result: Sample 2 contains separate diatomic H_2 and O_2 molecules (2 substances); Sample 3 contains bonded H_2O molecules (1 compound, 2 elements).

Model boundary: Selecting Sample 3 in the model switches our particle diagram, not a real flask; in the laboratory, turning H_2 + O_2 into H_2O is an exothermic chemical reaction.

Common Misconception Trap · Observation versus model inference

Intuitive Trap: "I think individual h_2o molecules have a bent 104.5° bond angle — because particle models are literal optical photographs taken with an ordinary classroom magnifying glass."

Particle Mechanism Refutation: "Wait — look at the particle ledger: A colourless liquid boils at 100 °C at 1 atm pressure. Observations are directly measured bulk properties (such as boiling point, mass, or colour), whereas particle diagrams and bond angles are theoretical models inferred from experimental evidence."

Why Claim A fails: Atoms (~0.1 nm) are smaller than the wavelength of visible light (400–700 nm) and cannot be seen with an optical magnifying glass. · Why Claim B holds up: Exact mechanism: chemistry bridges direct macroscopic measurements to testable submicroscopic particle models.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Pure H_2 gas (1 element, 1 substance)	Diatomic hydrogen molecules (H_2) with a single proton count (Z = 1).	H_2 is an element: each molecule contains only hydrogen atoms. A molecule need not be a compound.
Mixture of H_2 and O_2 (2 elements, 2 substances)	Unreacted H_2 and O_2 molecules mixed in the same container.	H_2O is a molecular compound: hydrogen and oxygen combined in a fixed atom ratio of 2:1.
Pure H_2O compound (2 elements, 1 substance)	Each oxygen atom is covalently bonded to two hydrogen atoms in a fixed 2:1 ratio.	H_2 and O_2 together form a mixture. Their proportions can vary. Placing them together in a diagram does not show a reaction.

Check Question: Which statement is a direct macroscopic observation rather than a submicroscopic particle inference? · **Transfer Challenge:** Transfer challenge: Look at a glass of clear salt water on a table. State one direct macroscopic observation about the liquid and one submicroscopic particle inference about what is inside.

INVESTIGATION 0.2 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: MOLE

Powers of ten: from flask to atom

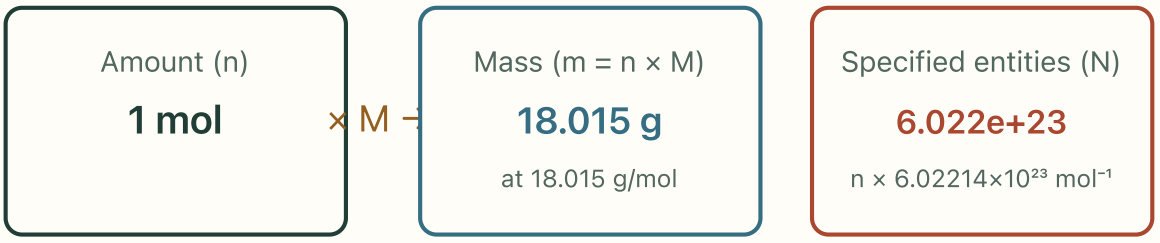
Bridging the 10^{23} scale gap between a single molecule and a laboratory balance

A single water molecule has a diameter near 0.3 nanometres (3×10^{-10} m) and a mass near 3×10^{-23} g—far too small to weigh individually on a laboratory balance. Chemists bridge this ten-billion-trillion-fold scale gap by counting particles in moles.

Bench Observation: An 18.015 g sip of water in a graduated cylinder contains 6.022×10^{23} water molecules.

Investigation Task: Hold molar mass near 18 g/mol and vary the mole amount from 0.25 mol to 2.00 mol. Compare the macroscopic mass in grams with the submicroscopic entity count ($\times 10^{23}$).

Powers of ten: from flask to atom



Unit bridge connecting molar amount (mol), laboratory mass (g) and number of specified entities (atoms, molecules or formula units).

Amount-to-mass account

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like converting a counted order to its mass: n tells how many moles, M gives grams per mole, and n × M gives grams. Counting units does not change the substance.

Current explanatory plate · intro-models-2

Mass / g	18.015 g
Number of specified entities	6.02214076e+23 entities

m = n × M = 1 mol × 18.015 g/mol = 18.015 g

Model note: m = nM; N = n × 6.02214076 × 10²³ mol⁻¹. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.

Scientific inputs · scientific-values-2

Avogadro constant: 6.02214076e+23 mol⁻¹ · defined. Counts specified entities per mole; state whether atoms, molecules or formula units. Exact SI defining constant. Source

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.2; 3.3 Supplement 2–3, 5. Relative masses, moles and reacting quantities Only the named concept is

connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.2.1–2 (HT). Moles, molar masses and reacting amounts Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 0.2: Deterministic particle & mathematical plate for *Powers of ten: from flask to atom*. First Principle: Because single molecules have masses near 10^{-23} g, the mole (6.022×10^{23} entities) scales submicroscopic particle counts up to weighable laboratory gram amounts.

Historical & Experimental Evidence: Perrin's Brownian motion and X-ray crystal counting of Avogadro's constant

How do we know how many atoms sit in a gram of matter if no human can count them one by one? Jean Perrin (tracking microscopic particles jostled by water molecules) and later X-ray crystallography of ultra-pure silicon spheres established that 18.015 g of water contains $6.02214076 \times 10^{23}$ molecules—linking the submicroscopic world (~10^{-10} m) to the macroscopic lab (~10^{-2} m).

Key Terms

Submicroscopic scale (nanometre / picometre)
The length scale of individual atoms and small molecules, roughly 0.1 to 1 nanometre (10^{-10} to 10^{-9} m).

Avogadro's scaling bridge
Using 1 mole = 6.02214076 × 10^{23} entities so that relative atomic masses in u translate directly into laboratory masses in grams.

Worked Reasoning Trace

Held fixed: Molar mass M = 18 g/mol (approximate value for H_2O).

Changed: Increased amount n from 0.50 mol to 1.00 mol.

Model result: Laboratory mass doubles from 9.0 g to 18.0 g while specified entities double from 3.011 × 10^{23} to 6.022 × 10^{23} molecules.

Model boundary: Individual molecules do not grow larger when you increase n from 0.50 mol to 1.00 mol; only the count of molecules increases.

Common Misconception Trap · Powers of ten: from flask to atom

Intuitive Trap: "I think one mole of every substance always has a mass of exactly 1.00 gram — because causes each individual atom to swell until it is visible to the naked eye."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Individual atoms and molecules are far too small to count one by one, so a fixed large counting unit links particle ratios to weighable masses. Multiplies the number of discrete molecules by 6.022 × 10^{23} so that a single molecule's mass in u (10^{-24} g scale) becomes a weighable mass in grams, while each molecule's size stays unchanged."

Why Claim A fails: Individual molecules remain ~10^{-10} m across regardless of sample size. · Why Claim B holds up: Exact mechanism: 1 u × N_A = 1 g/mol, so Avogadro's number bridges atomic mass units to laboratory grams by counting 6.022 × 10^{23} intact particles.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
0.50 mol water (~9 g, 3.01 × 10^{23} molecules)	Half a mole of water is a small 9 mL spoonful containing 3.011 × 10^{23} H_2O molecules.	m = nM; N = n × 6.02214076 × 10^{23} mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.
1.00 mol water (~18 g, 6.02 × 10^{23} molecules)	One mole of water weighs 18 g and contains 6.022 × 10^{23} H_2O molecules.	m = nM; N = n × 6.02214076 × 10^{23} mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.
2.00 mol carbon (~24 g, 1.20 × 10^{24} atoms)	Two moles of carbon-12 weigh 24 g and contain 1.204 × 10^{24} carbon atoms.	m = nM; N = n × 6.02214076 × 10^{23} mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.

Check Question: Why do chemists use the mole (6.022 × 10^{23} entities) when connecting particle models to laboratory measurements? · **Transfer Challenge:** Transfer challenge: If 1.00 mol of water (18 g) contains 6.022 × 10^{23} H_2O molecules, how many total atoms (H + O combined) are in that 18 g sample?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 0 · The Sealed Flask Boil-Off

A student seals 18.0 g of liquid water inside a strong evacuated glass bulb and heats it until every drop turns into invisible vapour. They claim: 'Because the liquid vanished, the water molecules broke apart into hydrogen and oxygen gas, and the bulb now weighs less.'

- Clue 1 (Macroscopic Balance): Before heating, the sealed bulb + liquid weighed 118.0 g; after vaporizing completely, the hot sealed bulb still weighs 118.0 g.
- Clue 2 (Cooling Test): When cooled back to 25 °C, colourless liquid water droplets condense on the glass walls with a normal boiling point of 100 °C.
- Clue 3 (Scale Ledger): 18.0 g of H_2O corresponds to 1.00 mol = 6.022 × 10^{23} molecules.

Resolution: Macroscopic mass conservation (118.0 g) and reversible condensation prove a physical phase change of intact H_2O molecules rather than chemical decomposition.

What our upgraded model now explains: Separates direct macroscopic observations from testable submicroscopic particle inferences and bridges the 10^{23} scale gap using the mole. · **Next boundary:** Treating molecules as whole spheres cannot explain what makes a hydrogen atom

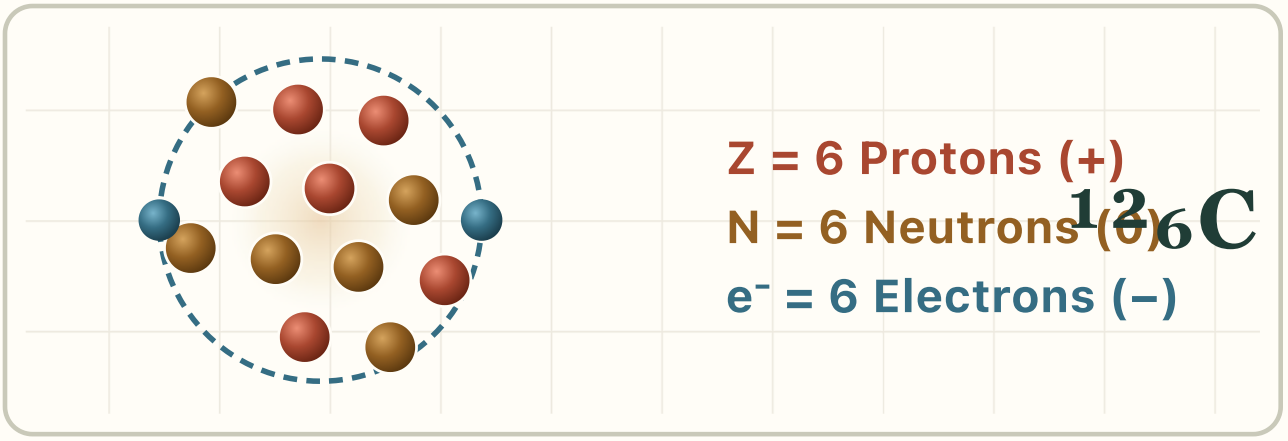
different from an oxygen atom or where electric charge comes from → motivating Chapter 1 (Inside the Atom).

VOLUME I · PARTICLES, IDENTITY & STRUCTURE · CHAPTER 1

Chapter 1: What makes an atom?

Start with evidence, then build a model of an atom. No chemistry vocabulary is assumed.

Chapter 1 Prelude · Why Doesn't Losing an Electron Turn Sodium into Neon?: Gold and mercury sit right next to each other on the periodic table (atomic numbers 79 and 80), yet one is a precious yellow solid and the other is a dense silvery liquid. Meanwhile, table salt contains sodium ions (Na^+) that have the exact same number of electrons (10 e^-) as neon gas—so why doesn't Na^+ turn into neon? To answer, we must open the atom and track protons, neutrons, and electrons separately.



Macroscopic Bench: Rutherford gold-foil scattering flashes, mass-spectrometer isotope peaks, and contrasting element samples.

Symbolic Notation: Nuclide and ion notation $^{A}_{Z}X^{q}$ where Z = protons, A = Z + N, and q = Z − e^-.

Submicroscopic Particle Model: Concentrated positive nucleus (Z protons + N neutrons) surrounded by negative electrons ($-e^-$) in quantized shells.

INVESTIGATION 1.1 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: ATOM

Inside an atom

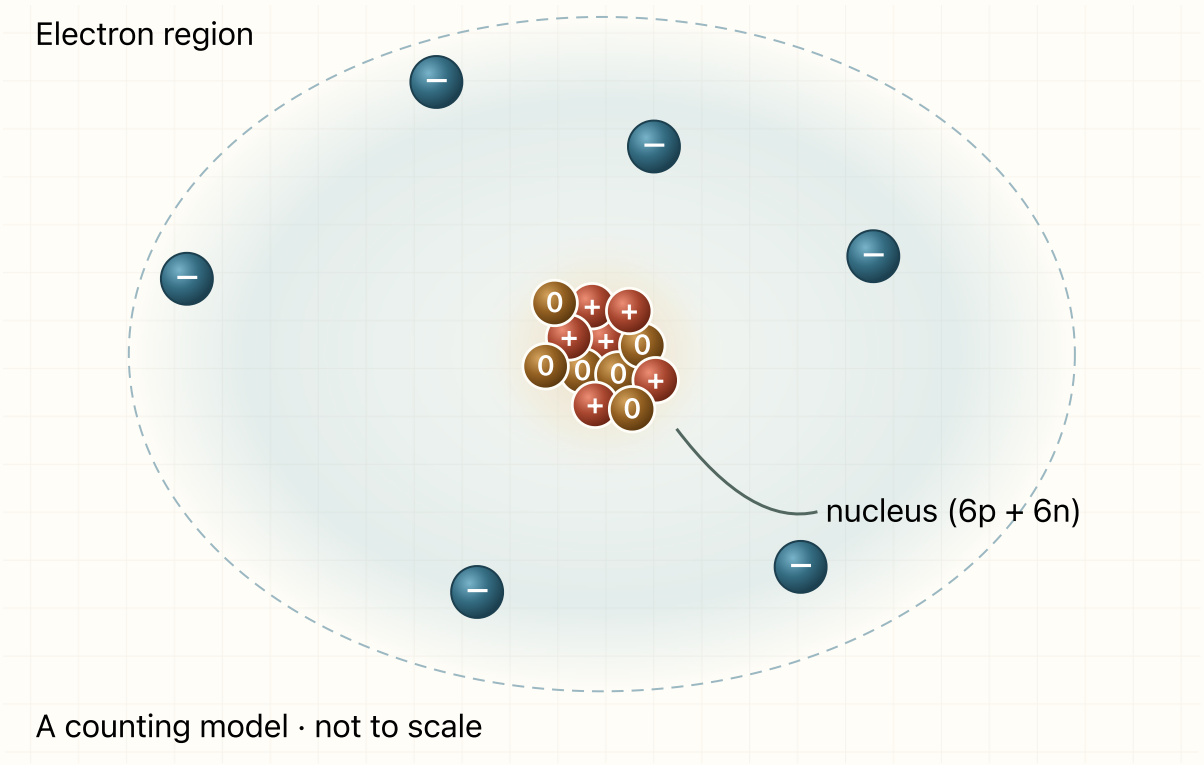
How indirect scattering evidence supports a nuclear model

Chemistry studies substances and how they change. We begin with an atom: a tiny physical entity that we describe using evidence and models. The nucleus contains protons and neutrons; electrons occupy the surrounding region.

Bench Observation: Scattering experiments supported a small concentrated positive nucleus. Our diagram is a representation built from evidence, not a view through a microscope.

Investigation Task: Compare the three particle counts. Draw what the diagram can tell you and name one thing it cannot show.

Inside an atom



Charge Tug-of-War Balance (+q vs. −q) Net charge 0 (Balanced ✓)
6 protons (+6 e)
6 electrons (−6 e)

Charge balanced ✓: +6 nuclear proton charge cancels −6 electron charge (net q = 0).

Tokens count particles. The shading is not a calculated electron cloud, and the tokens are not measured positions.

126C

Particle inventory & signed charge

Element identity follows proton count. Charge is the signed balance of positive protons and negative electrons.

The bookkeeping idea: Like a signed balance sheet: each proton contributes +1 and each electron −1 to q/e; neutrons contribute zero. Keep the nucleon inventory A = p + n in a separate account. Changing an entry edits this model; it does not perform a reaction.

Current explanatory plate · intro-models-2

Protons 6 particles

Neutrons	6 particles
Electrons	6 particles
Mass number A	12 nucleons
Charge q/e	0

A = p + n = 6 + 6 = 12; q/e = p − electrons = 6 − 6 = 0

Model note: Model editing is not a chemical reaction. Counting establishes neither nuclear nor ionic stability.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.2 Core 1–6; 2.3 Core 1–2. Particle counts, electron arrangements and isotope notation Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.1.1.4–4.1.1.7. Particle charge, counts, isotopes and electron arrangements Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Structure of an Atom · C-1.1, C-1.3 (pp. 9–10). Particles, symbols and inventories Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 1.1: Deterministic particle & mathematical plate for *Inside an atom*. First Principle: Atoms are physical entities whose mass is concentrated in a tiny positive nucleus (protons + neutrons) surrounded by negative electrons; our diagram counts particles rather than showing literal scale.

Historical & Experimental Evidence: Alpha-particle scattering and cathode rays

Scientists could not see individual atoms through an optical microscope. In 1897, J. J. Thomson deflected cathode rays with electric and magnetic fields, showing that atoms contain tiny negatively charged electrons. In 1909–1911, Hans Geiger, Ernest Marsden and Ernest Rutherford directed positively charged alpha particles at thin gold foil: almost all passed straight through, while a tiny fraction rebounded sharply—supporting a model where most mass and positive charge sit in a minute central nucleus.

Key Terms

Nucleus
The dense central region of an atom containing positively charged protons and uncharged neutrons.

Electron region
The much larger volume surrounding the nucleus occupied by negatively charged electrons.

Counting model
A simplified diagram designed to track how many protons, neutrons and electrons are present, not their true scale or paths.

Worked Reasoning Trace

Held fixed: 6 protons and 6 neutrons in the nucleus.

Changed: Inspected the diagram versus the particle count table.

Model result: The diagram displays 6 proton tokens (+), 6 neutron tokens (0) and 6 electron tokens (–).

Model boundary: If a real nucleus were the size of a marble, the atom's electron region would span a football stadium—so no readable screen diagram can be drawn to scale.

Common Misconception Trap · Inside an atom

Intuitive Trap: "I think measured electron positions — because a time-lapse microscope image of electrons orbiting on fixed rings."

Particle Mechanism Refutation: "Wait — look at the particle ledger: The numbers of each particle represented. A counting model whose tokens represent integer particle inventories rather than scale distances or classical trajectories."

Why Claim A fails: Electrons do not travel on classical tracks and cannot be photographed in fixed rings. · Why Claim B holds up: Exact mechanism: the diagram is a counting model that encodes integer numbers of protons, neutrons and electrons.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Helium-4 (2p, 2n, 2e)	A small neutral counting example with 2 protons, 2 neutrons and 2 electrons.	Helium-4, atomic number 2, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Helium has introductory shell populations 2; those populations do not describe an arbitrary ion.
Carbon-12 (6p, 6n, 6e)	Six protons and six neutrons in the nucleus, surrounded by six electrons.	Carbon-12, atomic number 6, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Carbon has introductory shell populations 2, 4; those populations do not describe an arbitrary ion.
Neon-20 (10p, 10n, 10e)	Ten of each subatomic particle in a neutral neon-20 model.	Neon-20, atomic number 10, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Neon has introductory shell populations 2, 8; those populations do not describe an arbitrary ion.

Check Question: What can a counting diagram reliably communicate? · **Transfer Challenge:** Transfer challenge: If a classmate claims the blue shading in our diagram is a photograph of an electron cloud, how would you explain the difference between our schematic diagram and experimental scattering evidence?

INVESTIGATION 1.2 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: ATOM

What identifies an element?

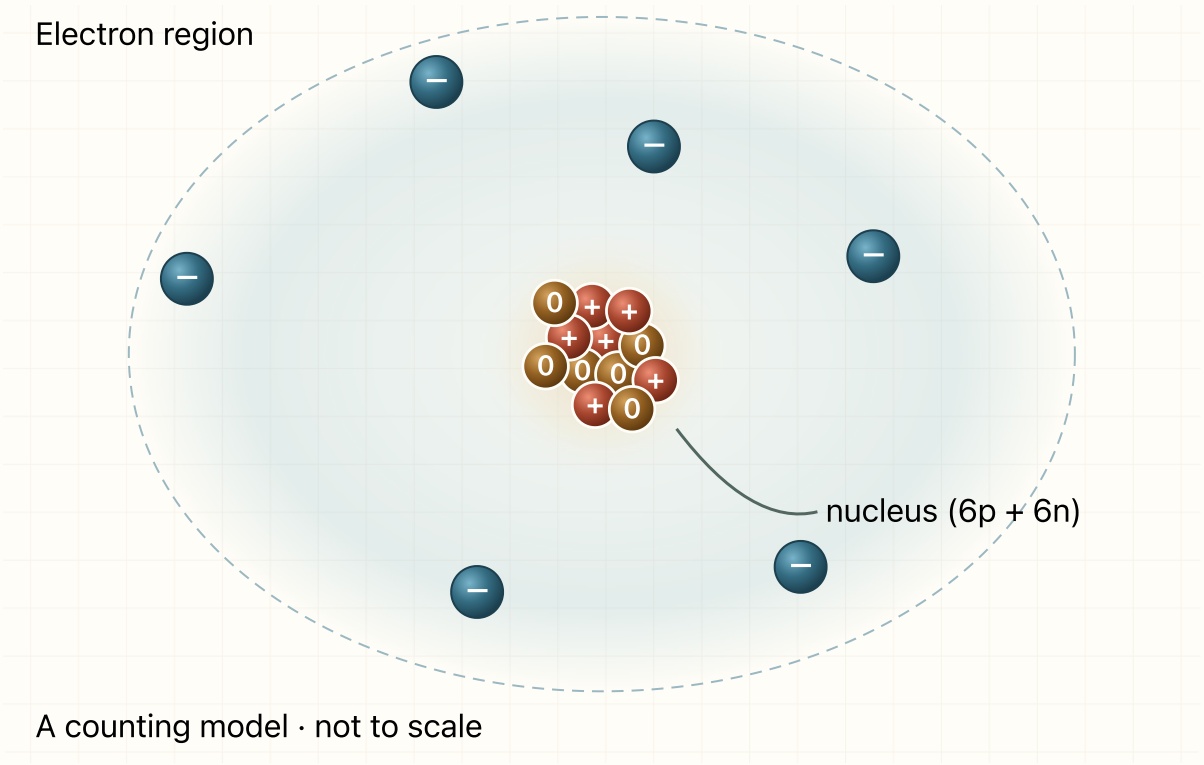
Why proton count—and only proton count—names the element

An element is identified by its proton number. The name stays the same when only the neutron or electron count changes.

Bench Observation: Carbon samples can have different structures and properties while sharing atomic number 6.

Investigation Task: Change electrons first, then neutrons, then protons. Save a pair with the same element and a pair with different elements.

What identifies an element?



Charge Tug-of-War Balance (+q vs. −q) Net charge 0 (Balanced ✓)
6 protons (+6 e)
6 electrons (−6 e)

Charge balanced ✓: +6 nuclear proton charge cancels −6 electron charge (net q = 0).

Tokens count particles. The shading is not a calculated electron cloud, and the tokens are not measured positions.

126C

Particle inventory & signed charge

Element identity follows proton count. Charge is the signed balance of positive protons and negative electrons.

The bookkeeping idea: Like a signed balance sheet: each proton contributes +1 and each electron −1 to q/e; neutrons contribute zero. Keep the nucleon inventory A = p + n in a separate account. Changing an entry edits this model; it does not perform a reaction.

Current explanatory plate · intro-models-2

Protons 6 particles

Neutrons	6 particles
Electrons	6 particles
Mass number A	12 nucleons
Charge q/e	0

A = p + n = 6 + 6 = 12; q/e = p − electrons = 6 − 6 = 0

Model note: Model editing is not a chemical reaction. Counting establishes neither nuclear nor ionic stability.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.2 Core 1–6; 2.3 Core 1–2. Particle counts, electron arrangements and isotope notation Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.1.1.4–4.1.1.7. Particle charge, counts, isotopes and electron arrangements Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Structure of an Atom · C-1.1, C-1.3 (pp. 9–10). Particles, symbols and inventories Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 1.2: Deterministic particle & mathematical plate for *What identifies an element?*. First Principle: Proton count (atomic number Z) alone defines which element an atom is; changing neutrons or electrons never turns one element into another.

Historical & Experimental Evidence: Nuclear charge and X-ray spectra (Moseley)

In 1913, Henry Moseley measured the X-ray frequencies emitted by different elements and showed they scaled directly with an integer nuclear charge Z—the number of protons in the nucleus. Whether carbon is found in soft dark graphite, transparent diamond, or carbon dioxide gas, every carbon nucleus has exactly Z = 6 protons.

Key Terms

Atomic number (Z)
The number of protons in an atom's nucleus; this integer uniquely identifies the chemical element.

Element
A class of atoms that all share the same proton count (atomic number Z).

Worked Reasoning Trace

Held fixed: Protons fixed at 6 (Carbon) and neutrons at 6.

Changed: Reduced electrons from 6 to 5, then changed protons from 6 to 7.

Model result: Changing electrons kept Carbon (Z = 6). Changing protons to 7 immediately changed the element to Nitrogen (Z = 7).

Model boundary: Clicking '+' on protons edits a computer model; in nature, changing proton count is a nuclear transformation, not an ordinary chemical reaction.

Common Misconception Trap · What identifies an element?

Intuitive Trap: "I think neon — because the total number of electrons occupying its outer shell."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Oxygen. The integer number of positively charged protons in its nucleus (atomic number Z)."

Why Claim A fails: Ions gain or lose electrons without changing which element they belong to. · Why Claim B holds up: Exact mechanism: nuclear charge Z (proton count) defines the element across all its ions and isotopes.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Neutral Carbon (6p, 6n, 6e)	Atomic number Z = 6 identifies carbon.	Carbon-12, atomic number 6, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Carbon has introductory shell populations 2, 4; those populations do not describe an arbitrary ion.
Oxide ion model (8p, 8n, 10e)	8 protons means oxygen, even though 10 electrons matches neon's electron count.	Oxygen-16, atomic number 8, charge minus 2. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Oxygen has introductory shell populations 2, 6; those populations do not describe an arbitrary ion.
Neutral Neon (10p, 10n, 10e)	10 protons identifies neon (Z = 10).	Neon-20, atomic number 10, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Neon has introductory shell populations 2, 8; those populations do not describe an arbitrary ion.

Check Question: An inventory has 8 protons and 10 electrons. Which element is it? · **Transfer Challenge:** Transfer challenge: A particle has 12 protons, 12 neutrons and 10 electrons. Which element is it, and which particle count did you ignore when naming the element?

INVESTIGATION 1.3 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: ISOTOPES

Same element, different isotope

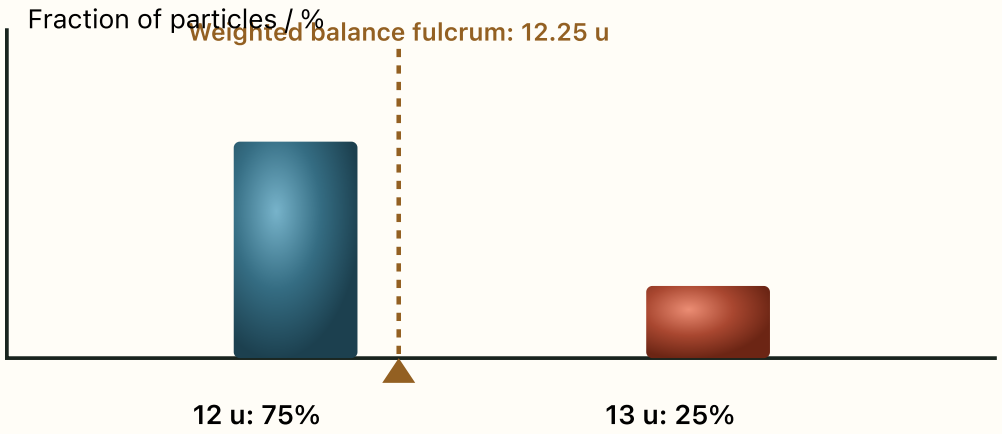
Separating integer mass numbers from weighted mixture averages

Isotopes share a proton count but have different neutron counts. Mass number counts protons plus neutrons. An average for a mixture is a different kind of number.

Bench Observation: Measured isotope masses and abundances can be used to find an average atomic mass. The values here are deliberately simplified synthetic data.

Investigation Task: Predict the average for an even mixture. Move the percentage control and explain why the average stays between the two input masses.

Same element, different isotope



Weighted isotopic distribution for Synthetic 12 u / 13 u pair. No individual atom has the intermediate decimal mass 12.25 u.

Weighted contributions

Each isotope contributes its mass multiplied by its fraction. The average is not the mass of an individual atom.

The bookkeeping idea: Like a weighted stock valuation: each isotope's mass is multiplied by its share, then the contributions are added. The average describes the mixture, not a single atom.

Current explanatory plate · intro-models-2

Illustrative average mass / u 12.25 u

$\bar{m} = (1 - f)m_{light} + f\, m_{heavy} = 0.75 \times 12\ u + 0.25 \times 13\ u = 12.25\ u$

Model note: A synthetic mixture with masses defined as 12 u and 13 u. These simplified values are not measured carbon isotope masses or a measured mass spectrum. No individual particle has the mixture average mass.

Scientific inputs · scientific-values-2

Synthetic 12 u / 13 u pair: light teaching mass: 12 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture. Source

Synthetic 12 u / 13 u pair: heavy teaching mass: 13 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the

explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture. Source

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.3 Core 1; Supplement 4. Isotope identity and weighted relative atomic mass Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.1.1.5–4.1.1.6. Isotopes and abundance-weighted relative mass Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Structure of an Atom · C-1.1 (pp. 9–10). Isotopes; abundance calculations extend the outcome Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 1.3: Deterministic particle & mathematical plate for *Same element, different isotope*. First Principle: Every single atom has an integer nucleon count (A = Z + N), whereas a sample's average atomic mass is a statistical weighted average across billions of atoms.

Historical & Experimental Evidence: Mass spectrometry separates isotopes by mass-to-charge

When a pure element sample is ionized and passed through a magnetic field in a mass spectrometer, atoms of the same element can land at distinct positions corresponding to different masses. Natural chlorine, for example, produces peaks near 35 u (about 75.8%) and 37 u (about 24.2%), giving a weighted average near 35.45 u—even though no single chlorine atom has a mass of 35.45 u.

Key Terms

Isotopes
Atoms of the same element (equal proton count Z) that have different numbers of neutrons, and therefore different mass numbers A.

Mass number (A)
The exact integer total of protons plus neutrons in a single nucleus (A = Z + N).

Weighted average mass
The average mass of a mixture of isotopes weighted by their relative abundance fractions.

Worked Reasoning Trace

Held fixed: Two synthetic species defined at 12 u and 13 u.

Changed: Increased the percentage of 13 u particles from 0% to 25%.

Model result: Average mass shifts from 12.00 u to 0.75 × 12 + 0.25 × 13 = 12.25 u.

Model boundary: No single particle weighs 12.25 u, and real isotope masses (like ^{13}C = 13.003355 u) differ slightly from integer mass numbers due to nuclear binding energy.

Common Misconception Trap · Same element, different isotope

Intuitive Trap: "I think 42 u — because each individual atom containing a fractional number of neutrons."

Particle Mechanism Refutation: "Wait — look at the particle ledger: 21 u. Weighting whole atoms of different integer mass numbers by their abundance fractions in the sample."

Why Claim A fails: Neutrons are discrete particles; a nucleus never contains half a neutron. · Why Claim B holds up: Exact mechanism: each individual nucleus has an integer nucleon count A, while the macroscopic sample is a weighted mixture of isotopes.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Pure 12 u (0% mass-13)	Only mass-12 particles are present; average is 12 u.	A synthetic mixture with masses defined as 12 u and 13 u. These simplified values are not measured carbon isotope masses or a measured mass spectrum. No individual particle has the mixture average mass.
Chlorine ~3:1 signature (25% ^{37}Cl)	75% ^{35}Cl (35 u) and 25% ^{37}Cl (37 u) gives a weighted average of 35.5 u.	Chlorine isotopic model (^{35}Cl at 35 u and ^{37}Cl at 37 u). Natural chlorine is roughly 75% ^{35}Cl and 25% ^{37}Cl (~3:1 ratio), giving a weighted average near 35.5 u even though no single chlorine nucleus has mass 35.5 u.
Bromine ~1:1 signature (50% ^{81}Br)	50% ^{79}Br (79 u) and 50% ^{81}Br (81 u) places the average at 80.0 u with zero Br-80 atoms.	Bromine isotopic model (^{79}Br at 79 u and ^{81}Br at 81 u). Natural bromine is nearly a 50% : 50% (1:1) mixture, placing its weighted average near 80.0 u even though bromine contains zero Br-80 atoms here.

Check Question: A synthetic mixture has equal numbers of particles of mass 20 u and 22 u. What is its average? · **Transfer Challenge:** Transfer challenge: If a synthetic mixture is 75% particles of mass 10 u and 25% particles of mass 14 u, what is the average mass, and why is no single particle 11 u?

INVESTIGATION 1.4 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: ATOM

Where does charge come from?

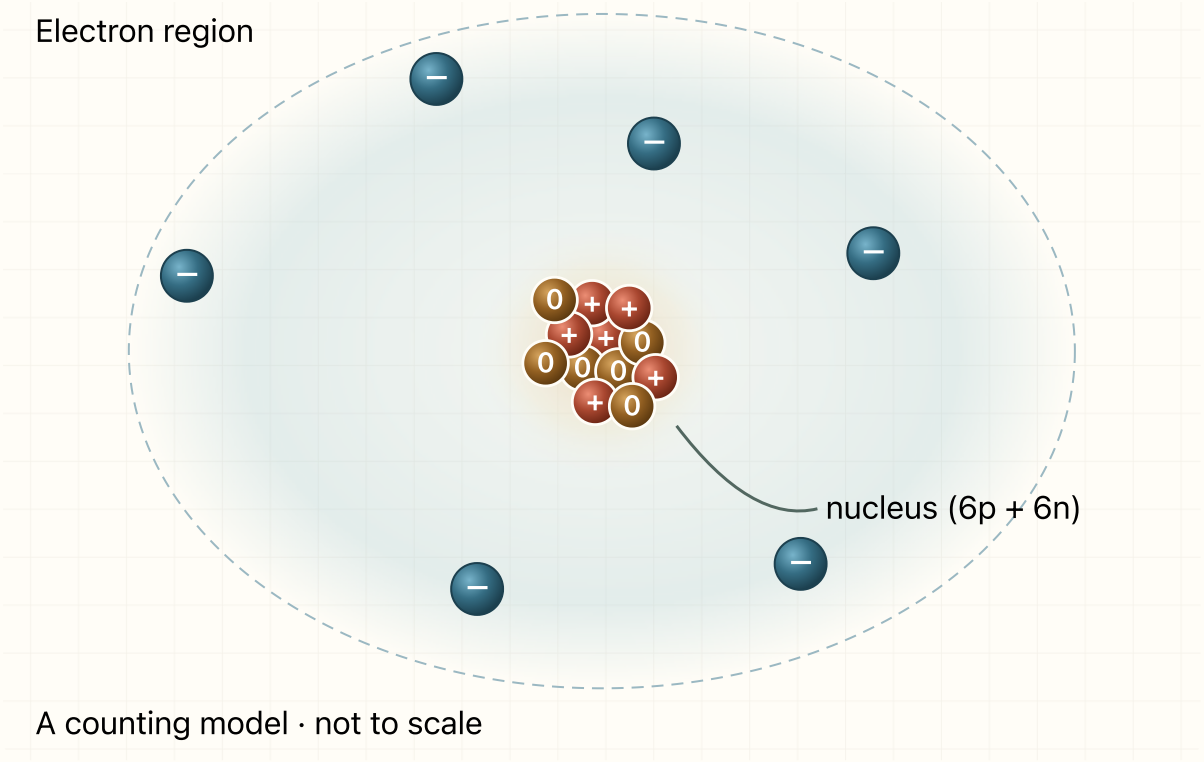
How imbalance between protons and electrons creates net electric charge

A proton contributes one positive elementary charge and an electron one negative charge. A neutron contributes no net electric charge.

Bench Observation: Ions are charged particles. Whether a particular ion persists depends on its environment, not just this subtraction.

Investigation Task: Keep protons and neutrons fixed. Remove one electron, save, then add two electrons and compare.

Where does charge come from?



Charge Tug-of-War Balance (+q vs. −q) Net charge 0 (Balanced ✓)
6 protons (+6 e)
6 electrons (−6 e)

Charge balanced ✓: +6 nuclear proton charge cancels −6 electron charge (net q = 0).

Tokens count particles. The shading is not a calculated electron cloud, and the tokens are not measured positions.

126C

Particle inventory & signed charge

Element identity follows proton count. Charge is the signed balance of positive protons and negative electrons.

The bookkeeping idea: Like a signed balance sheet: each proton contributes +1 and each electron −1 to q/e; neutrons contribute zero. Keep the nucleon inventory A = p + n in a separate account. Changing an entry edits this model; it does not perform a reaction.

Current explanatory plate · intro-models-2

Protons 6 particles

Neutrons	6 particles
Electrons	6 particles
Mass number A	12 nucleons
Charge q/e	0

A = p + n = 6 + 6 = 12; q/e = p − electrons = 6 − 6 = 0

Model note: Model editing is not a chemical reaction. Counting establishes neither nuclear nor ionic stability.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.2 Core 1–6; 2.3 Core 1–2. Particle counts, electron arrangements and isotope notation Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.1.1.4–4.1.1.7. Particle charge, counts, isotopes and electron arrangements Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Structure of an Atom · C-1.1, C-1.3 (pp. 9–10). Particles, symbols and inventories Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 1.4: Deterministic particle & mathematical plate for *Where does charge come from?*. First Principle: Net electric charge comes from the exact tug-of-war between positive protons (+1 each) and negative electrons (–1 each): q = Z – e^-, while the nucleus stays unchanged.

Historical & Experimental Evidence: Electrolytic conductivity of mobile ions

Michael Faraday and Svante Arrhenius studied how dissolved salts and molten compounds conduct electric current. Pure solid sodium chloride does not conduct well, but when melted or dissolved in water, positively charged sodium ions (Na^+) and negatively charged chloride ions (Cl^-) move in opposite directions between electrodes.

Key Terms

Elementary charge (e)
The magnitude of electric charge carried by a single proton (+1 e) or a single electron (–1 e).

Ion
An atom or molecule with a non-zero net electric charge because its proton count and electron count are unequal.

Worked Reasoning Trace

Held fixed: 6 protons (+6) and 6 neutrons (0) in the nucleus.

Changed: Decreased electrons from 6 (–6) to 5 (–5).

Model result: Net charge becomes 6 – 5 = +1 elementary charge, while the element remains Carbon (Z = 6).

Model boundary: We can subtract any two integers in this model, but whether an ion like C^+ exists stably depends on chemical environment (such as a high-vacuum gas beam versus water).

Common Misconception Trap · Where does charge come from?

Intuitive Trap: "I think –1 elementary charge — because subtracting neutron count from proton count in the nucleus."

Particle Mechanism Refutation: "Wait — look at the particle ledger: +1 elementary charge. Adding (+1 e) per proton and (–1 e) per electron, while neutrons carry 0 charge."

Why Claim A fails: Neutrons affect mass number A, not electric charge. · Why Claim B holds up: Exact mechanism: q = protons – electrons because each proton is +1 e, each electron is –1 e, and neutrons are electrically neutral.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Neutral Sodium-23 (11p, 12n, 11e)	11 positive protons balance 11 negative electrons: net charge 0.	Sodium-23, atomic number 11, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Sodium has introductory shell populations 2, 8, 1; those populations do not describe an arbitrary ion.
Sodium ion Na^+ (11p, 12n, 10e)	One fewer electron leaves a net charge of 11 – 10 = +1.	Sodium-23, atomic number 11, charge plus 1. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Sodium has introductory shell populations 2, 8, 1; those populations do not describe an arbitrary ion.
Oxide ion O^{2-} (8p, 8n, 10e)	Two extra electrons give 8 – 10 = –2 elementary charges.	Oxygen-16, atomic number 8, charge minus 2. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Oxygen has introductory shell populations 2, 6; those populations do not describe an arbitrary ion.

Check Question: What is the net charge for 11 protons and 10 electrons? · **Transfer Challenge:** Transfer challenge: A sulfide ion has 16 protons, 16 neutrons and 18 electrons. What is its net charge, and why do the 16 neutrons contribute zero to that charge?

INVESTIGATION 1.5 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: ATOM

Reading an atom’s identity

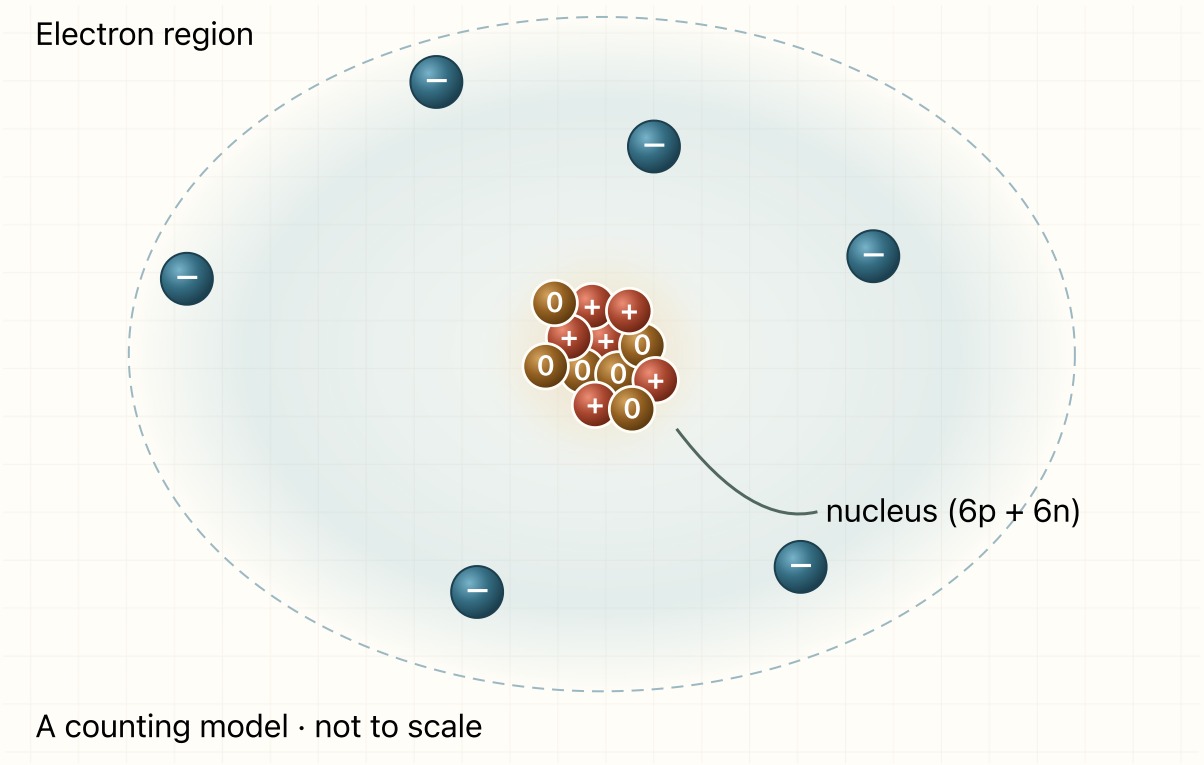
Reading and writing compact nuclear and ionic symbols

Atomic number Z is the proton count; mass number A is protons plus neutrons. Ionic charge is written separately from both.

Bench Observation: Symbols compactly communicate an inventory, but do not establish its stability.

Investigation Task: Make an inventory with 8 protons, 9 neutrons and 10 electrons. Explain each label before changing a count.

Reading an atom's identity



Charge Tug-of-War Balance (+q vs. −q) Net charge 0 (Balanced ✓)
6 protons (+6 e)
6 electrons (−6 e)

Charge balanced ✓: +6 nuclear proton charge cancels −6 electron charge (net q = 0).

Tokens count particles. The shading is not a calculated electron cloud, and the tokens are not measured positions.

126C

Particle inventory & signed charge

Element identity follows proton count. Charge is the signed balance of positive protons and negative electrons.

The bookkeeping idea: Like a signed balance sheet: each proton contributes +1 and each electron −1 to q/e; neutrons contribute zero. Keep the nucleon inventory A = p + n in a separate account. Changing an entry edits this model; it does not perform a reaction.

Current explanatory plate · intro-models-2

Protons 6 particles

Neutrons	6 particles
Electrons	6 particles
Mass number A	12 nucleons
Charge q/e	0

A = p + n = 6 + 6 = 12; q/e = p − electrons = 6 − 6 = 0

Model note: Model editing is not a chemical reaction. Counting establishes neither nuclear nor ionic stability.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.2 Core 1–6; 2.3 Core 1–2. Particle counts, electron arrangements and isotope notation Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.1.1.4–4.1.1.7. Particle charge, counts, isotopes and electron arrangements Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Structure of an Atom · C-1.1, C-1.3 (pp. 9–10). Particles, symbols and inventories Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 1.5: Deterministic particle & mathematical plate for *Reading an atom's identity*. First Principle: Nuclear notation (ᴬᶻX^q) maps each corner directly to a particle count: upper-left A = protons + neutrons, lower-left Z = protons, and upper-right q = protons − electrons.

Historical & Experimental Evidence: International IUPAC nuclear notation convention

To avoid confusion across languages and laboratories, chemists and physicists standardized nuclear symbol positions around the element symbol X: mass number A at the upper left (^{A}X), atomic number Z at the lower left ($_{Z}X$), and ionic charge q at the upper right (X^{q±}).

Key Terms

Nuclear notation ($^{A}_{Z}X^{q}$)
Standard symbol placing mass number A top-left, atomic number Z bottom-left, and net charge top-right.

Neutron count (N = A − Z)
Because mass number A counts protons + neutrons and Z counts protons, subtracting A − Z gives the neutron count.

Worked Reasoning Trace

Held fixed: 8 protons (Z = 8, Oxygen) and 9 neutrons.

Changed: Set electrons to 10.

Model result: Mass number A = 8 + 9 = 17 (upper left); atomic number Z = 8 (lower left); charge q = 8 − 10 = 2− (upper right): $^{17}_{8}O^{2-}$.

Model boundary: The integer A = 17 in nuclear notation is a nucleon count, whereas the decimal 15.999 on a periodic table is a reference average atomic mass.

Common Misconception Trap · Reading an atom's identity

Intuitive Trap: "I think 15 — because the number of neutrons alone, ignoring protons."

Particle Mechanism Refutation: "Wait — look at the particle ledger: 8. The sum of protons (Z) plus neutrons (N) in that single nucleus, so N = A − Z."

Why Claim A fails: The upper-left superscript is the total nucleon count A, not N alone. · Why Claim B holds up: Exact mechanism: A counts total nucleons (Z + N), so subtracting the lower-left proton count Z leaves the neutron count N.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Carbon-14 neutral (6p, 8n, 6e)	A = 14 upper-left, Z = 6 lower-left, no charge superscript.	Carbon-14, atomic number 6, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Carbon has introductory shell populations 2, 4; those populations do not describe an arbitrary ion.
Oxygen-17 2− ion (8p, 9n, 10e)	A = 17, Z = 8, charge = 2−.	Oxygen-17, atomic number 8, charge minus 2. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Oxygen has introductory shell populations 2, 6; those populations do not describe an arbitrary ion.
Magnesium-24 2+ ion (12p, 12n, 10e)	A = 24, Z = 12, charge = 2+.	Magnesium-24, atomic number 12, charge plus 2. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Magnesium has introductory shell populations 2, 8, 2; those populations do not describe an arbitrary ion.

Check Question: An atom has Z = 7 and A = 15. How many neutrons does it have? · **Transfer Challenge:** Transfer challenge: Write down or describe the full nuclear notation for an inventory with 13 protons, 14 neutrons and 10 electrons. Which number goes in each corner?

INVESTIGATION 1.6 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: ATOM

What might explain patterns?

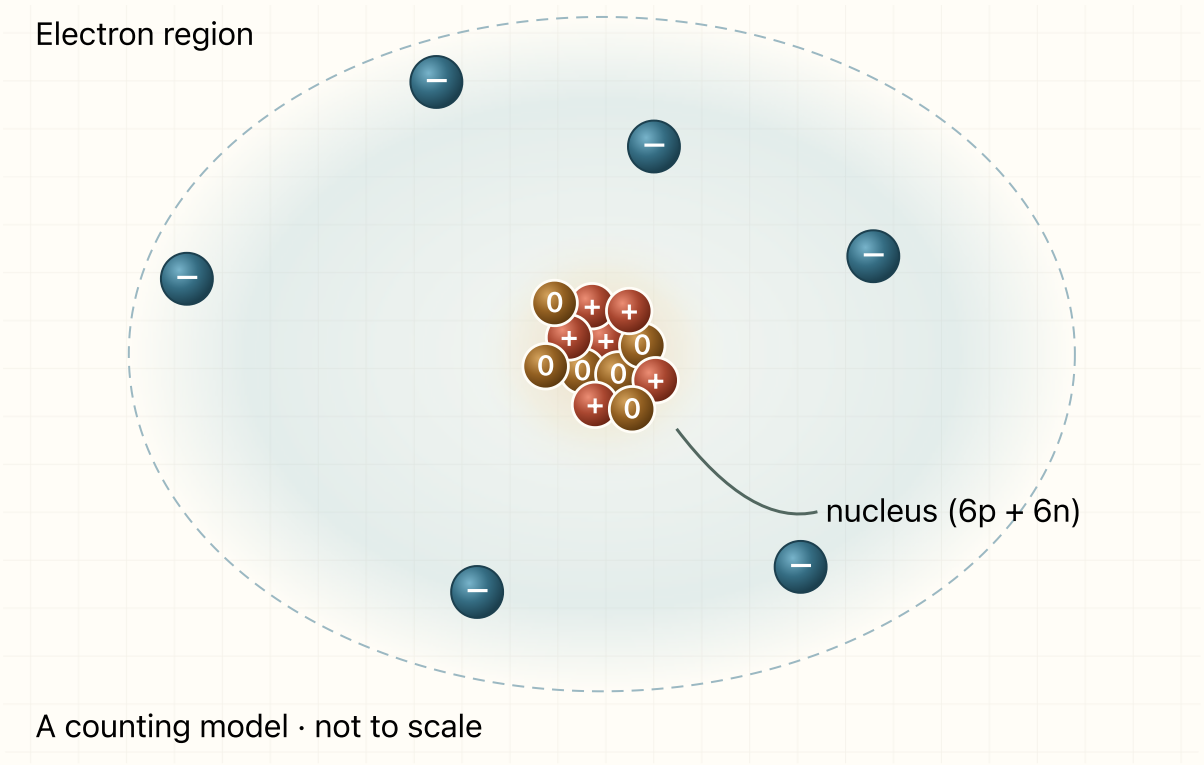
Recurring shell populations across neutral elements 1–18

Electron arrangements help us explain recurring chemical patterns. The simple shell-population lists here apply to neutral ground-state elements 1–18.

Bench Observation: Similar chemical behaviour motivates models of electron structure; a completed shell is not a promise of stability in every situation.

Investigation Task: Compare neutral lithium and sodium. Identify a shared pattern, then say what would need evidence beyond a counting model.

What might explain patterns?



Charge Tug-of-War Balance (+q vs. −q) Net charge 0 (Balanced ✓)
6 protons (+6 e)
6 electrons (−6 e)

Charge balanced ✓: +6 nuclear proton charge cancels −6 electron charge (net q = 0).

Tokens count particles. The shading is not a calculated electron cloud, and the tokens are not measured positions.

126C

Particle inventory & signed charge

Element identity follows proton count. Charge is the signed balance of positive protons and negative electrons.

The bookkeeping idea: Like a signed balance sheet: each proton contributes +1 and each electron −1 to q/e; neutrons contribute zero. Keep the nucleon inventory A = p + n in a separate account. Changing an entry edits this model; it does not perform a reaction.

Current explanatory plate · intro-models-2

Protons 6 particles

Neutrons	6 particles
Electrons	6 particles
Mass number A	12 nucleons
Charge q/e	0

A = p + n = 6 + 6 = 12; q/e = p − electrons = 6 − 6 = 0

Model note: Model editing is not a chemical reaction. Counting establishes neither nuclear nor ionic stability.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.2 Core 5–6. Electron arrangements Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.1.1.7. Electron arrangements Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Structure of an Atom · C-1.1, C-1.3 (pp. 9–10). Particles, symbols and inventories Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 1.6: Deterministic particle & mathematical plate for *What might explain patterns?*. First Principle: Electrons occupy grouped energy shells due to electrostatic attraction to the nucleus and repulsion from other electrons—not because atoms 'want' full shells.

Historical & Experimental Evidence: Successive ionization energies and periodic families

When chemists measured the energy required to remove one electron from neutral gaseous atoms across the first 18 elements, they saw sharp peaks at Helium (Z = 2), Neon (Z = 10), and Argon (Z = 18), followed by sharp drops at Lithium (Z = 3) and Sodium (Z = 11). These energy jumps support a model where electrons populate grouped energy shells (2, then up to 8, then up to 8 for Z ≤ 18).

Key Terms

Ground-state shell population
The count of electrons in each principal energy shell of a neutral atom in its lowest-energy state (e.g., 2 · 8 · 1 for neutral sodium).

Valence (outer-shell) electrons
Electrons in the outermost occupied shell, which strongly influence how an atom interacts with other atoms.

Worked Reasoning Trace

Held fixed: Neutral ground-state condition (electrons = protons).

Changed: Compared Lithium (Z = 3) with Sodium (Z = 11).

Model result: Lithium has shell population 2 · 1; Sodium has 2 · 8 · 1. Both have 1 electron in their outermost shell.

Model boundary: Shell populations are counts of electrons in energy levels, not circular planetary tracks, and a 'full shell' is not a magical wish or universal guarantee of stability.

Common Misconception Trap · What might explain patterns?

Intuitive Trap: "I think literal circular paths — because how many electrons the atom 'wants' to borrow from a friend to feel happy."

Particle Mechanism Refutation: "Wait — look at the particle ledger: A simplified neutral ground-state arrangement. How electrons are distributed across quantized energy levels in a neutral ground-state model, without implying atoms have personal desires."

Why Claim A fails: Atoms do not have desires; chemical changes are governed by electrostatic forces and energy states. · Why Claim B holds up: Exact mechanism: shell populations summarize quantized energy-level occupancy supported by ionization energy evidence.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Lithium (Z = 3, shells 2 · 1)	Neutral lithium: 2 inner-shell electrons and 1 outer-shell electron.	Lithium-7, atomic number 3, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Lithium has introductory shell populations 2, 1; those populations do not describe an arbitrary ion.
Sodium (Z = 11, shells 2 · 8 · 1)	Neutral sodium: shares the single outer-shell electron pattern with lithium.	Sodium-23, atomic number 11, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Sodium has introductory shell populations 2, 8, 1; those populations do not describe an arbitrary ion.
Neon (Z = 10, shells 2 · 8)	Neutral neon: closed second shell with 8 electrons.	Neon-20, atomic number 10, neutral. Editing counts does not demonstrate a chemical reaction or stability. Neutral ground-state Neon has introductory shell populations 2, 8; those populations do not describe an arbitrary ion.

Check Question: What does a shell-population list represent here? · **Transfer Challenge:** Transfer challenge: Compare neutral Fluorine (Z = 9) and neutral Chlorine (Z = 17). What shell-population pattern do they share, and why must we still avoid saying fluorine 'wants' an electron?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 1 · The Three Mystery Beams

An ion-beam analyzer characterizes three subatomic species X, Y, and Z: Species X has 8 protons, 8 neutrons, and 10 electrons; Species Y has 8 protons, 10 neutrons, and 8 electrons; Species Z has 10 protons, 10 neutrons, and 10 electrons.

- Clue 1 (Proton Identity): Species X and Y both have Z = 8, whereas Species Z has Z = 10.
- Clue 2 (Nucleon Count): Species X has A = 16, whereas Species Y and Z both have A = 20.
- Clue 3 (Charge Balance): Species X has 8p and 10e^-; Species Y and Z have equal protons and electrons.

Resolution: Proton count Z alone determines the element: Z = 8 is oxygen (X is $^{16}_{8}O^{2-}$, Y is $^{18}_{8}O$), while Z = 10 is neon ($^{20}_{10}Ne$), regardless of matching electron count (10 e^-) or mass number (A = 20).

What our upgraded model now explains: Explains elemental identity (Z), isotopes and weighted atomic mass (A vs. mixture average), ion charge (q = Z – e^-), and neutral ground-state shell populations. · **Next boundary:** Counting particles inside one isolated atom cannot explain how

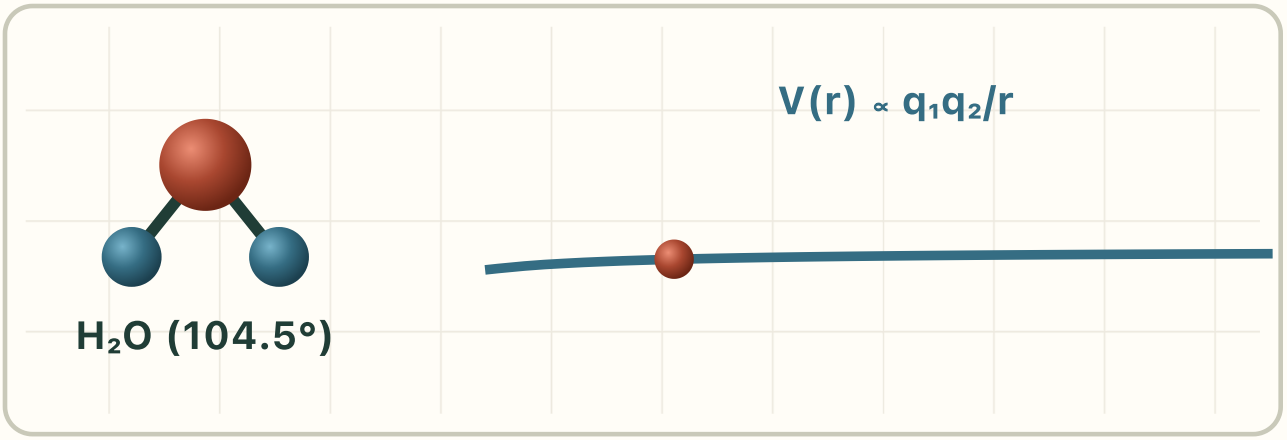
atoms stick together into H_2O molecules, NaCl crystals, or diamond networks → motivating Chapter 2 (How Atoms Form Substances).

VOLUME I · PARTICLES, IDENTITY & STRUCTURE · CHAPTER 2

Chapter 2: How atoms form substances

Distinguish what a substance contains from how its particles are connected.

Chapter 2 Prelude · How Do Toxic Sodium and Chlorine Gas Become Edible Salt?: Pure carbon can be soft grey pencil lead (graphite) or the hardest gemstone on Earth (diamond). Meanwhile, water (H_2O) is a liquid made of discrete 3-atom molecules, whereas table salt (NaCl) forms hard cubic crystals with no individual 'NaCl molecules' at all. How do electrostatic forces and 3D geometry turn the same set of elements into radically different substances?



Macroscopic Bench: Hardness of diamond vs. graphite, conductivity of molten/dissolved salts, and deflection of polar water streams.

Symbolic Notation: Molecular formulas (H_2O, CO_2), empirical ionic formula units (NaCl, $MgCl_2$), and dipole vectors ($\delta^+ \rightarrow \delta^-$).

Submicroscopic Particle Model: Discrete covalent molecules (polar vs. nonpolar geometries), extended 3D ionic lattices, and covalent networks.

INVESTIGATION 2.1 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: COMPOSITION

Element, compound or mixture?

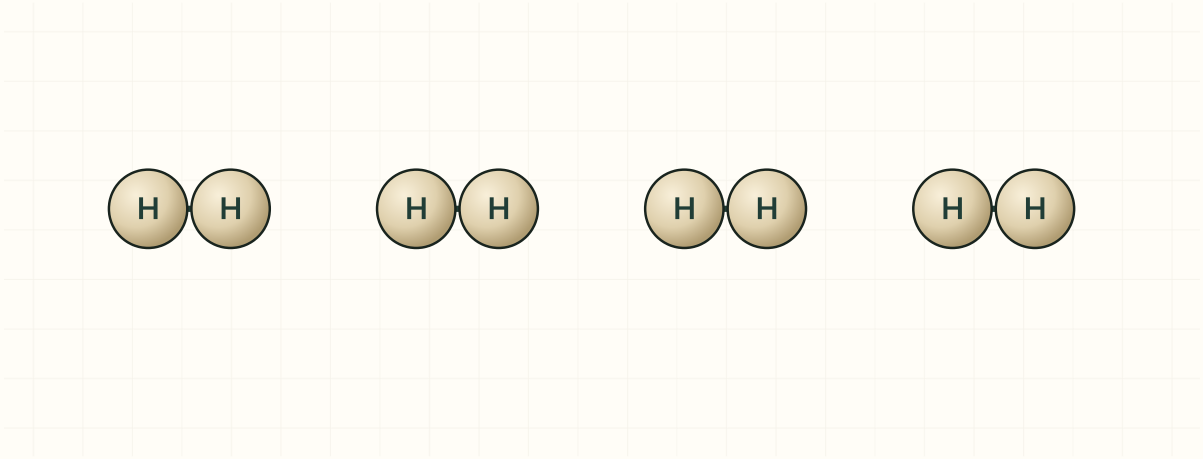
Separating elements, compounds and variable-composition mixtures

A substance has a characteristic composition. A compound contains more than one element chemically combined; a mixture contains more than one substance.

Bench Observation: Pure water has a fixed H:O atom ratio. The amount of salt mixed with water can vary.

Investigation Task: Compare all six samples. Explain why oxygen molecules are still an element and why a mixture is not a new formula.

Element, compound or mixture?



Grouped atom symbols show molecular composition. Lines mark connectivity, not exact bond order, electron positions or geometry.

Account for the model

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like sorting stock by identity: count element identities, chemical substances and particles in separate columns. Mixing two stocks is not evidence that they reacted.

Current explanatory plate · intro-models-2

Substances represented 1

Model note: H_2 is an element: each molecule contains only hydrogen atoms. A molecule need not be a compound.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.1; 2.4–2.7. Substances, bonding and structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.1–2. Structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1, C-7.1 (pp. 11–12). Compounds, bonding and fixed proportions Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 2.1: Deterministic particle & mathematical plate for *Element, compound or mixture?*. First Principle: A pure compound has a fixed atom ratio locked by chemical bonds, whereas a mixture contains separate substances whose proportions can vary freely.

Historical & Experimental Evidence: Law of definite proportions (Proust) vs. physical mixtures

Joseph Proust demonstrated that pure chemical compounds always contain the same elements in the exact same mass and atom ratio—pure water from any source has 2 hydrogen atoms per 1 oxygen atom. By contrast, a mixture of hydrogen gas and oxygen gas can be blended in any ratio (1:1, 5:1, 10:1) and separated physically without breaking chemical bonds.

Key Terms

Pure substance
Matter with a uniform, definite chemical composition—either a single element (like H_2 or diamond C) or a single compound (like H_2O or NaCl).

Compound
A substance composed of two or more different elements chemically bonded in a fixed atomic ratio.

Mixture
Two or more distinct substances physically intermingled in proportions that can vary.

Worked Reasoning Trace

Held fixed: Hydrogen and oxygen atoms present in the sample.

Changed: Switched from 'Water H_2O' (sample 2) to 'Hydrogen and oxygen mixture' (sample 3).

Model result: Water is 1 pure compound with bonded H_2O molecules; the H_2 + O_2 mixture contains 2 separate substances whose ratio can vary.

Model boundary: Drawing H_2 and O_2 molecules in the same box represents an unreacted mixture; it does not show a chemical reaction occurring.

Common Misconception Trap · Element, compound or mixture?

Intuitive Trap: "I think it always has a single fixed formula — because every atom in a mixture is bonded into a single giant molecule."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Their proportions can vary. A mixture contains two or more distinct chemical species that are not chemically bonded to each other in a fixed ratio."

Why Claim A fails: Fixed chemical bonding between different elements defines a compound, not a mixture. · Why Claim B holds up: Exact mechanism: independent molecules or substances in a mixture retain their separate identities and can be present in varying proportions.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Oxygen O_2 (Molecular element)	One substance, one element: each molecule has two bonded oxygen atoms.	O_2 is an element: two oxygen atoms per molecule, one elemental identity.
Water H_2O (Molecular compound)	One substance, two elements bonded in a fixed 2:1 ratio.	H_2O is a molecular compound: hydrogen and oxygen combined in a fixed atom ratio of 2:1.
H_2 and O_2 mixture (Two substances)	Two separate elemental substances mixed together without reacting.	H_2 and O_2 together form a mixture. Their proportions can vary. Placing them together in a diagram does not show a reaction.

Check Question: Which description fits a mixture of two substances? · **Transfer Challenge:** Transfer challenge: Explain why oxygen gas (O_2) is classified as an element even though it consists of molecules, whereas carbon dioxide (CO_2) is both molecular and a compound.

INVESTIGATION 2.2 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: FORCES

How do charges interact?

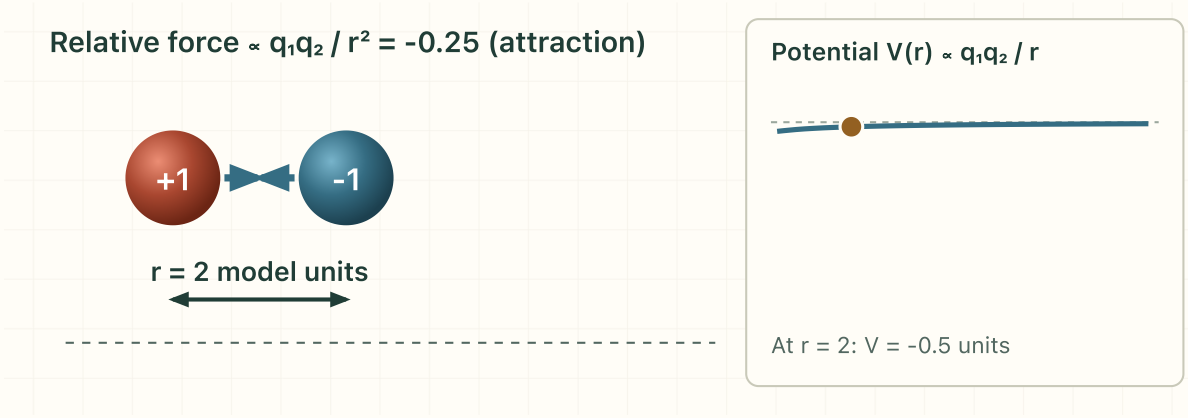
How electrostatic attraction and repulsion scale with charge and distance

Charges interact: opposite signs attract and like signs repel. Chemical bonding needs a richer model than isolated point charges.

Bench Observation: Electrostatic effects can be observed with charged objects. The numerical force here is relative and uses an idealized rule.

Investigation Task: Change one sign while holding separation fixed. Then double separation and compare the magnitude.

How do charges interact?



Idealized point-charge schematic with live Coulomb potential curve V(r). Vector arrow lengths scale with $|q_1q_2|/r^2$.

Signed interaction account

The charge product determines the sign of this ideal electrostatic contribution; separation controls its magnitude.

The bookkeeping idea: Use a signed calculation with a declared reference: charge signs set attraction or repulsion, and separation changes the magnitude. Relative model units are not newtons, and the result does not establish a stable bond.

Current explanatory plate · intro-models-2

Signed relative electrostatic force -0.25 relative model units

$F_{rel} = q_1q_2/r^2 = (1 \times -1) / 2^2 = -0.25$ model units (not newtons)

Model note: Opposite charges attract. Point-charge model proportional to q_1q_2/r^2; negative denotes attraction. It omits quantum structure, screening and other interactions, so cannot alone predict a chemical bond.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Enrichment: No direct topic objective assigned. Model exploration beyond the mapped chemistry introduction Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Enrichment: No direct topic objective assigned. Model exploration beyond the mapped chemistry introduction Only the named

concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 2.2: Deterministic particle & mathematical plate for *How do charges interact?*. First Principle: Opposite charges attract ($q_1q_2 < 0$) and like charges repel ($q_1q_2 > 0$), with force scaling as $1/r^2$—the fundamental physical driver behind ionic and covalent interactions.

Historical & Experimental Evidence: Torsion-balance force measurements (Coulomb)

In 1785, Charles-Augustin de Coulomb used a sensitive torsion balance to measure the force between charged spheres. He showed that the electrostatic force is proportional to the product of the two charges (q_1q_2) and inversely proportional to the square of the distance between their centres ($1/r^2$).

Key Terms

Inverse-square relationship (1/r²)
When separation r is multiplied by 2, the force magnitude divides by $2^2 = 4$; when r is multiplied by 3, force divides by $3^2 = 9$.

Signed electrostatic force
In this model, a negative value ($q_1q_2 < 0$) indicates attraction between opposite charges, and a positive value ($q_1q_2 > 0$) indicates repulsion between like charges.

Worked Reasoning Trace

Held fixed: First charge $q_1 = +1$ e and second charge $q_2 = -1$ e.

Changed: Doubled separation r from 2 model units to 4 model units.

Model result: Relative force changed from $-1/4 = -0.25$ to $-1/16 = -0.0625$ (one quarter the magnitude, still attractive).

Model boundary: Real atoms and ions are not mathematical points: at very short distances, electron clouds overlap and quantum repulsion balances attraction.

Common Misconception Trap · How do charges interact?

Intuitive Trap: "I think twice as much — because electrostatic force decreases linearly with distance (1 / r)."

Particle Mechanism Refutation: "Wait — look at the particle ledger: One quarter. Electrostatic force scales with q_1q_2 / r^2, so multiplying distance r by 2 divides force magnitude by $2^2 = 4$."

Why Claim A fails: A 1/r relationship is electric potential energy, whereas force scales as $1/r^2$. · Why Claim B holds up: Exact mechanism: the $1/r^2$ dependence causes doubling the separation to reduce the force magnitude to one quarter.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
+1 and –1 at distance 2	Opposite charges attract: relative force = –0.25.	Opposite charges attract. Point-charge model proportional to q_1q_2/r^2; negative denotes attraction. It omits quantum structure, screening and other interactions, so cannot alone predict a chemical bond.
+1 and –1 at distance 4 (doubled r)	Doubling distance reduces force magnitude to one quarter (–0.0625).	Opposite charges attract. Point-charge model proportional to q_1q_2/r^2; negative denotes attraction. It omits quantum structure, screening and other interactions, so cannot alone predict a chemical bond.
+2 and +2 at distance 2 (like charges)	Like charges repel (+1.00).	Like charges repel. Point-charge model proportional to q_1q_2/r^2; negative denotes attraction. It omits quantum structure, screening and other interactions, so cannot alone predict a chemical bond.

Check Question: In this point-charge model, doubling separation changes force magnitude to… · **Transfer Challenge:** Transfer challenge: Hold separation r fixed at 2 units. What happens to the sign and magnitude of the force if you change charges from (+1, –1) to (+2, –2)?

INVESTIGATION 2.3 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: COMPOSITION

How can atoms share electrons?

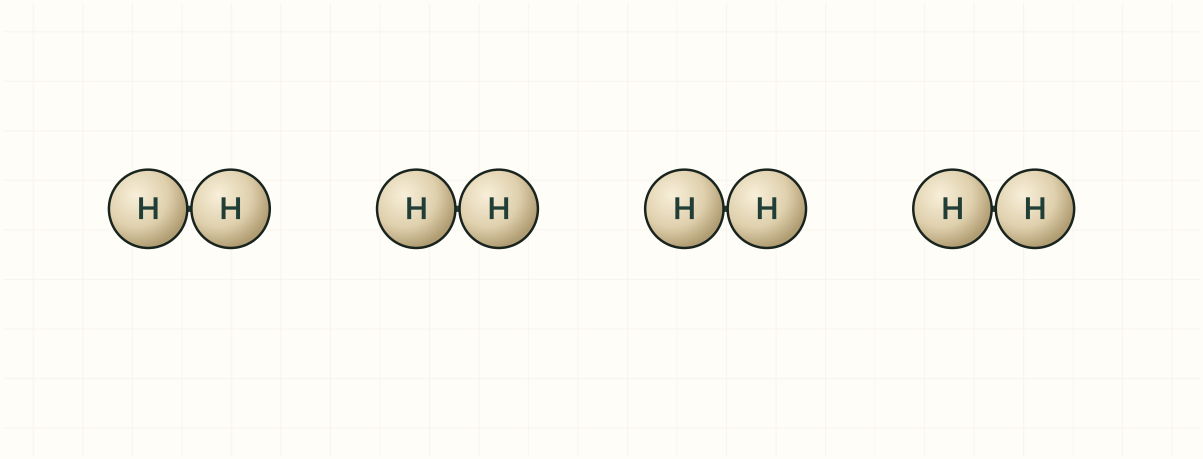
Shared electron density and electrostatic balance without teleology

In a covalent bond, electron density is shared between atoms. Attraction between nuclei and electrons contributes to bonding, while other interactions also matter. Atoms do not decide or want to fill shells.

Bench Observation: Hydrogen commonly occurs as H_2 molecules. Evidence about molecules supports bonding models; selecting H_2 in this app does not create a bond experimentally.

Investigation Task: Compare H_2, O_2 and H_2O. Sketch which atoms belong to one molecule, then explain why a formula alone does not specify the electron distribution or molecular shape.

How can atoms share electrons?



Grouped atom symbols show molecular composition. Lines mark connectivity, not exact bond order, electron positions or geometry.

Account for the model

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like sorting stock by identity: count element identities, chemical substances and particles in separate columns. Mixing two stocks is not evidence that they reacted.

Current explanatory plate · intro-models-2

Substances represented 1

Model note: H_2 is an element: each molecule contains only hydrogen atoms. A molecule need not be a compound.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.5 Core 1–3; Supplement 4. Covalent structures Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.1.4. Covalent bonding Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1, C-7.1 (pp. 11–12). Compounds, bonding and fixed proportions Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 2.3: Deterministic particle & mathematical plate for *How can atoms share electrons?*. First Principle: Covalent bonding arises when shared electron density simultaneously attracts both positively charged nuclei, lowering potential energy without any atom having intentions.

Historical & Experimental Evidence: Bond lengths, dissociation energies and electron density maps

X-ray diffraction and spectroscopy show that in a hydrogen molecule (H_2), the two nuclei sit at an equilibrium internuclear distance of 74 pm with shared electron density concentrated between them. Breaking one mole of H–H bonds requires 436 kJ of energy —empirical proof of a stabilizing electrostatic and quantum interaction, not a mystical desire for 'full shells'.

Key Terms

Covalent bond
A chemical bond formed when electron density is shared between two nuclei, balancing nuclear-electron attractions against electron-electron and nuclear-nuclear repulsions.

Teleological fallacy
Speaking as if atoms have desires or intentions (e.g., 'atoms want 8 electrons') instead of describing physical forces and energy.

Worked Reasoning Trace

Held fixed: Molecular representations in the composition model.

Changed: Compared H_2 (element) with H_2O (compound).

Model result: In both H_2 and H_2O, lines connect atoms within discrete molecules where electron density is shared.

Model boundary: A flat structural line (H–O–H) shows which atoms are bonded, not the continuous 3D electron density distribution.

Common Misconception Trap · How can atoms share electrons?

Intuitive Trap: "I think atoms choose to complete their shells — because each atom consciously decides to share electrons so both can obey a rule."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Shared electron density and interactions involving nuclei and electrons. Shared electron density between the two positively charged nuclei lowers the system's potential energy through electrostatic and quantum interactions."

Why Claim A fails: Teleological language ('decides', 'wants') confuses a human metaphor with physical forces. · Why Claim B holds up: Exact mechanism: attraction of both nuclei to the shared electron density between them stabilizes the bonded state relative to separated atoms.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Hydrogen H_2 molecules	Pairs of hydrogen atoms sharing electron density in discrete H_2 molecules.	H_2 is an element: each molecule contains only hydrogen atoms. A molecule need not be a compound.
Oxygen O_2 molecules	Pairs of oxygen atoms covalently bonded in O_2 molecules.	O_2 is an element: two oxygen atoms per molecule, one elemental identity.
Water H_2O molecules	Each oxygen atom shares electron density with two hydrogen atoms.	H_2O is a molecular compound: hydrogen and oxygen combined in a fixed atom ratio of 2:1.

Check Question: Which is a useful starting description of covalent bonding? · **Transfer Challenge:** Transfer challenge: Rewrite the sentence 'Hydrogen atoms bond because each atom wants two electrons' into a scientifically accurate explanation using nuclei, electrons and attraction.

INVESTIGATION 2.4 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: FORMULA

What is a molecule?

Counting atoms inside discrete molecular groups

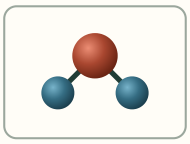
A molecule is a discrete group of bonded atoms. A formula tells us the kinds and numbers of atoms, but not every detail of their arrangement.

Bench Observation: Water and carbon dioxide contain different atom ratios and have different molecular shapes.

Investigation Task: Count atoms in three water molecules and then three carbon dioxide molecules. Draw groups rather than an ungrouped total.

What is a molecule?

Symbolic multiplier: 1 × H_2O



Solid boxes group the 1 discrete H_2O molecules; subscripts stay fixed inside each molecule.

Inventory per molecule or formula unit

Subscripts specify composition; a coefficient multiplies the whole inventory.

The bookkeeping idea: Like an itemized packing list: subscripts count atoms inside one molecule or formula unit; a coefficient multiplies the whole list. For an ionic formula, the signed charge entries must also reconcile.

Current explanatory plate · intro-models-2

H atoms 2

O atoms 1

Model note: 1 × H_2O. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.1 Core 1–4. Reading formulae and counting atoms Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.1.1–2. Atom inventories and relative formula mass Only the named concept is connected. Practical work and the

complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1 (pp. 11–12). Formulae and formula-unit masses Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 2.4: Deterministic particle & mathematical plate for *What is a molecule?*. First Principle: A molecular subscript counts atoms bonded inside one discrete molecule, whereas a coefficient counts how many separate molecules are present.

Historical & Experimental Evidence: Gas-volume ratios (Gay-Lussac & Avogadro) and molecular formulas

Amedeo Avogadro proposed in 1811 that equal volumes of gases at the same temperature and pressure contain equal numbers of discrete particles—which he called molecules. That insight explained why 2 volumes of hydrogen gas react with 1 volume of oxygen gas to yield 2 volumes of water vapour: oxygen and hydrogen gases exist as diatomic molecules (H_2 and O_2), and each water molecule is H_2O.

Key Terms

Molecule
A discrete, electrically neutral group of two or more atoms held together by covalent bonds.

Subscript
The small lowered number in a formula (such as the 2 in CO_2) that counts how many atoms of that element belong to one molecule.

Worked Reasoning Trace

Held fixed: Formula selected as CO_2 (1 carbon atom and 2 oxygen atoms per molecule).

Changed: Increased the number of molecules from 1 to 3.

Model result: $3 \times CO_2$ gives 3 carbon atoms and $3 \times 2 = 6$ oxygen atoms arranged in 3 separate molecular groups.

Model boundary: Writing '3 C and 6 O' as an ungrouped total loses the crucial structural fact that the atoms are packaged into three discrete CO_2 molecules.

Common Misconception Trap · What is a molecule?

Intuitive Trap: "I think 4 — because 1 giant molecule containing 4 carbon atoms and 2 oxygen atoms."

Particle Mechanism Refutation: "Wait — look at the particle ledger: 8. 4 discrete CO_2 molecules, each containing 1 carbon atom covalently bonded to 2 oxygen atoms ($4 \times 2 = 8$ O atoms)."

Why Claim A fails: A coefficient in front of a formula scales the number of separate molecules rather than merging them into one C_4O_2 molecule. · Why Claim B holds up: Exact mechanism: the coefficient 4 counts complete CO_2 molecules, so every subscript inside CO_2 is multiplied by 4.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
3 × H_2O (Water molecules)	3 discrete water molecules: 6 H atoms and 3 O atoms in total.	3 × H_2O. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.
3 × CO_2 (Carbon dioxide molecules)	3 discrete carbon dioxide molecules: 3 C atoms and 6 O atoms.	3 × CO_2. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.
4 × CO_2 (Check case)	4 molecules × 2 oxygen atoms per molecule = 8 oxygen atoms.	4 × CO_2. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.

Check Question: How many oxygen atoms are in four CO_2 molecules? · **Transfer Challenge:** Transfer challenge: Compare 3 molecules of H_2O with 3 molecules of CO_2. Which element count is equal in both samples, and why are the molecules still completely different substances?

INVESTIGATION 2.5 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: COMPOSITION

A formula without a molecule

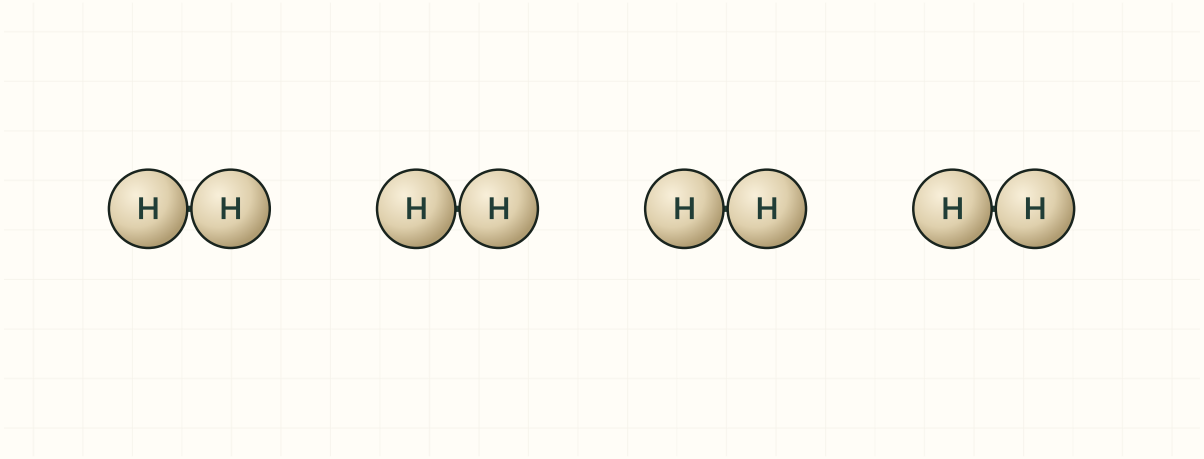
Why NaCl represents an extended crystal lattice ratio, not a discrete molecule

Ionic solids contain extended arrangements of positive and negative ions. A formula unit describes their ratio.

Bench Observation: Solid sodium chloride is not made of separate little NaCl molecules. Its structure extends through a crystal.

Investigation Task: Compare sodium chloride with water. Sketch a repeated ionic arrangement and a separate molecule; label the distinction.

A formula without a molecule



Grouped atom symbols show molecular composition. Lines mark connectivity, not exact bond order, electron positions or geometry.

Account for the model

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like sorting stock by identity: count element identities, chemical substances and particles in separate columns. Mixing two stocks is not evidence that they reacted.

Current explanatory plate · intro-models-2

Substances represented 1

Model note: H_2 is an element: each molecule contains only hydrogen atoms. A molecule need not be a compound.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.4 Core 1–4; Supplement 5. Ionic structures Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.1.2–3. Ionic structures Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1, C-7.1 (pp. 11–12). Compounds, bonding and fixed proportions Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 2.5: Deterministic particle & mathematical plate for *A formula without a molecule*. First Principle: Because electrostatic attraction acts in all directions, positive and negative ions pack into continuous 3D alternating lattices where the formula (like NaCl) states the neutral charge ratio.

Historical & Experimental Evidence: X-ray crystallography of rock salt (W. H. & W. L. Bragg, 1913)

When William Henry Bragg and William Lawrence Bragg directed X-rays at a crystal of sodium chloride (NaCl), the diffraction pattern revealed a continuous three-dimensional alternating grid where every Na^+ ion is surrounded by six equidistant Cl^- neighbours, and every Cl^- ion is surrounded by six Na^+ neighbours. No Na^+ ion is paired exclusively with a single Cl^- partner.

Key Terms

Ionic lattice
A repeating three-dimensional arrangement of positively charged cations and negatively charged anions held together by electrostatic attractions.

Formula unit
The simplest whole-number ratio of ions in an extended ionic compound (e.g., 1 Na^+ : 1 Cl^- in NaCl).

Worked Reasoning Trace

Held fixed: Pure compound classification.

Changed: Compared Water (H_2O, sample 2) with Solid sodium chloride (NaCl, sample 4).

Model result: Water displays separate 3-atom molecules; NaCl displays an alternating grid of Na^+ and Cl^- ions in a 1:1 ratio.

Model boundary: Our 2D screen drawing shows a 12-ion slice; a real grain of table salt extends in three dimensions for trillions of ions.

Common Misconception Trap · A formula without a molecule

Intuitive Trap: "I think one isolated nacl molecule per crystal — because isolated diatomic Na–Cl gas molecules floating independently."

Particle Mechanism Refutation: "Wait — look at the particle ledger: A 1:1 ion ratio. A continuous 3D alternating crystal lattice where each Na^+ attracts six surrounding Cl^- neighbours in a 1:1 empirical ratio."

Why Claim A fails: Solid table salt is an extended ionic lattice, not a collection of discrete molecules. · Why Claim B holds up: Exact mechanism: X-ray crystallography shows an extended ionic lattice with no exclusive one-to-one molecular pairs; NaCl is the simplest 1:1 ion ratio.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Water H_2O (Discrete molecules)	Separate H_2O molecules with strong covalent bonds inside each molecule.	H_2O is a molecular compound: hydrogen and oxygen combined in a fixed atom ratio of 2:1.
Sodium chloride solid (Extended ionic lattice)	Alternating Na^+ and Cl^- ions in a continuous 1:1 ratio.	Solid NaCl is an ionic compound with an extended arrangement of Na^+ and Cl^- ions. NaCl gives a 1:1 ratio, not a discrete molecule in the solid.

Check Question: What does NaCl describe for solid sodium chloride? · **Transfer Challenge:** Transfer challenge: Magnesium chloride is an ionic solid with formula $MgCl_2$. Based on what you learned about NaCl, what does the '2' in $MgCl_2$ mean in terms of ions in the crystal?

INVESTIGATION 2.6 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: COMPOSITION

Why composition is not enough

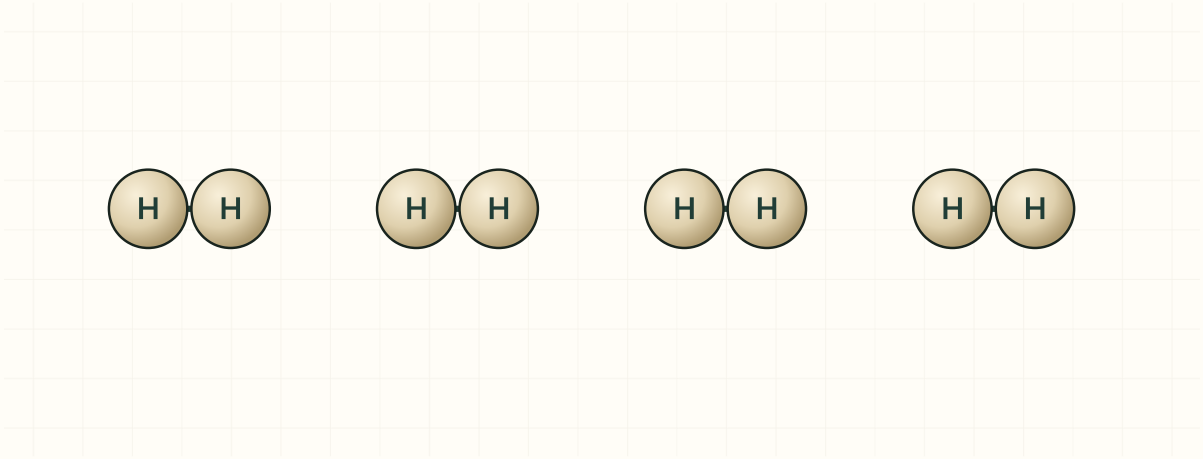
How extended 3D bonding architecture governs material behaviour

Materials with the same element can have different structures. A property explanation must connect bonding and arrangement with observations.

Bench Observation: Graphite marks paper easily; diamond is much harder. Both are carbon, but their structures differ.

Investigation Task: Select diamond. Draw a question about structure that the formula C cannot answer. Compare with the opening graphite observation.

Why composition is not enough



Grouped atom symbols show molecular composition. Lines mark connectivity, not exact bond order, electron positions or geometry.

Account for the model

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like sorting stock by identity: count element identities, chemical substances and particles in separate columns. Mixing two stocks is not evidence that they reacted.

Current explanatory plate · intro-models-2

Substances represented 1

Model note: H_2 is an element: each molecule contains only hydrogen atoms. A molecule need not be a compound.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.1; 2.4–2.7. Substances, bonding and structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.1–2. Structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1, C-7.1 (pp. 11–12). Compounds, bonding and fixed proportions Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 2.6: Deterministic particle & mathematical plate for *Why composition is not enough*. First Principle: Macroscopic material properties (like soft graphite vs. hard diamond) depend on how atoms are connected in 3D space, not merely which element is present.

Historical & Experimental Evidence: Layered sheets in graphite vs. tetrahedral network in diamond

Combustion experiments proved as early as 1772–1797 (Lavoisier and Tennant) that burning equal masses of pure diamond and pure graphite in oxygen produces the exact same mass of carbon dioxide—proving both are pure carbon (Z = 6). X-ray diffraction later explained their drastically different hardness: diamond links every carbon atom into a rigid 3D tetrahedral network, while graphite forms flat hexagonal layers that slide past one another easily.

Key Terms

Allotropes
Different structural forms of the same element in the same physical state (such as graphite and diamond, or O_2 and O_3).

Extended covalent network
A giant structure in which covalent bonds link atoms continuously throughout the crystal rather than forming small discrete molecules.

Worked Reasoning Trace

Held fixed: Elemental composition: 100% Carbon (formula C).

Changed: Selected Diamond carbon (sample 5) and compared with the opening Graphite observation.

Model result: Both have elemental formula C, so composition alone cannot distinguish them; bonding connectivity explains why diamond is hard and graphite marks paper.

Model boundary: Our 2D connectivity sketch shows continuous bonding links, but cannot show the 109.5° three-dimensional tetrahedral angles of a real diamond crystal.

Common Misconception Trap · Why composition is not enough

Intuitive Trap: "I think yes; every carbon sample is equally hard — because diamond atoms have more protons in their nuclei than graphite atoms."

Particle Mechanism Refutation: "Wait — look at the particle ledger: No; structure information is also needed. Diamond links carbon atoms in a rigid 3D tetrahedral covalent network, whereas graphite forms flat sheets with weak attractions between layers."

Why Claim A fails: Every carbon atom in both diamond and graphite has exactly 6 protons. · Why Claim B holds up: Exact mechanism: both are pure carbon (Z = 6), so their contrasting macroscopic properties come from their 3D bonding architecture.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Diamond carbon (Extended network)	Continuous covalent network of carbon atoms.	Diamond is an element with an extended covalent structure of carbon. The formula C alone does not specify structure or hardness.
Oxygen O_2 (Discrete diatomic molecules)	Contrast: an element made of small two-atom molecules rather than an extended solid network.	O_2 is an element: two oxygen atoms per molecule, one elemental identity.

Check Question: Can the formula C alone tell you whether a sample is graphite or diamond? · **Transfer Challenge:** Transfer challenge: Oxygen exists as dioxygen (O_2, the gas we breathe) and ozone (O_3, an ultraviolet-absorbing gas with a sharp odour). Why isn't knowing 'this sample contains only the element oxygen' enough to predict its properties?

INVESTIGATION 2.7 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: POLARITY

Unequal sharing and molecular dipoles

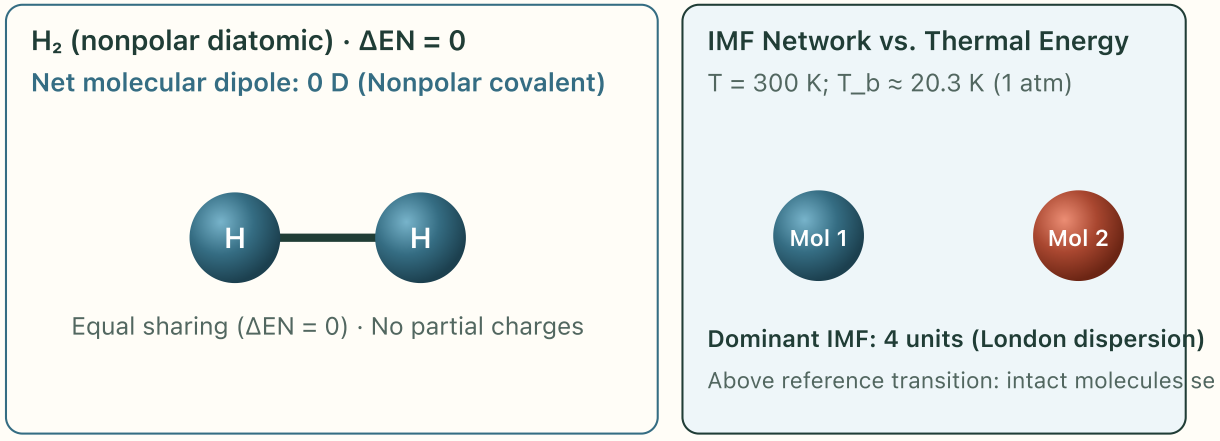
Why unequal electron sharing (ΔEN) and 3D molecular shape determine net polarity

When two bonded atoms have different electronegativities (ΔEN > 0), shared electron density shifts toward the more electronegative atom, creating partial charges (δ^+ and δ^-). Whether the whole molecule has a net dipole moment depends on both bond polarity and 3D molecular geometry.

Bench Observation: A thin stream of liquid water (H_2O) deflects strongly toward a charged rod, whereas liquid carbon tetrachloride or carbon dioxide (CO_2) has zero net dipole despite having polar covalent bonds.

Investigation Task: Compare H_2O (bent, 104.5°) with CO_2 (linear, 180°) and CH_4 (tetrahedral, 109.5°). Explain why CO_2 has polar C=O bonds (ΔEN = 0.89) yet zero net molecular dipole.

Unequal sharing and molecular dipoles



Left: 3D bond dipole addition (δ⁺ → δ⁻). Right: Intermolecular attraction dashed bridge vs. thermal kinetic motion at 300 K.

Vector contributions

Bond dipoles add as vectors in space. Equal scalar magnitudes can cancel when their directions oppose.

The bookkeeping idea: This account needs arrows, not just totals: dipole contributions add with direction. Opposing vectors can cancel even though every bond contribution is nonzero.

Current explanatory plate · intro-models-2	
Bond electronegativity difference ΔEN	0 dimensionless
Net molecular dipole moment / D	0 D
Relative intermolecular attraction / model units	4 relative model units
Reference 1 atm boiling/sublimation point / K	20.3 K

Model note: Hydrogen (H_2) at 300 K (Above reference transition at 1 atm via weak London dispersion only): Two identical H nuclei share electrons equally (ΔEN = 0, dipole = 0 D). Weak dispersion attractions are easily overcome by thermal motion above 20.3 K.

Scientific inputs · scientific-values-2

Hydrogen (H_2): deltaEN: 0 Pauling scale · reference. Difference of rounded Pauling electronegativities; not a measured bond charge. Introductory approximation. Checked 2026-10-28. Source

Hydrogen (H_2): dipole: 0 D · reference. Rounded gas-phase dipole; zero values reflect the geometry shown. Introductory approximation. Checked 2026-10-28. Source

Hydrogen (H_2): boiling: 20.3 K · reference. Approximate pure-substance normal boiling point at 1 atm. Introductory approximation. Checked 2026-10-28. Source

Hydrogen (H_2): attraction: 4 model units · illustrative. Author-selected comparison score, not an energy. Introductory approximation. Checked 2026-10-28.

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.5 Supplement 5. Qualitative intermolecular attractions in small molecules Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.2.4. Molecular attractions; dipoles are enrichment Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 2.7: Deterministic particle & mathematical plate for *Unequal sharing and molecular dipoles*. First Principle: Electronegativity differences (ΔEN) create bond dipoles between δ^+ and δ^- atoms, and 3D molecular geometry governs whether those vectors cancel (CO_2, CH_4) or produce a net molecular dipole (H_2O, NH_3).

Historical & Experimental Evidence: Electric-field deflection of polar liquid streams and dielectric measurements

When a thin stream of water (H_2O) falls past an electrically charged rod, the stream bends markedly toward the rod—regardless of whether the rod is positively or negatively charged! In contrast, a stream of nonpolar hexane or liquid carbon tetrachloride falls straight down. Gas-phase microwave spectroscopy confirms that bent H_2O has a permanent dipole moment of 1.85 D, whereas linear CO_2 has 0.00 D because its two polar C=O bonds pull in opposite directions.

Key Terms

Electronegativity difference (ΔEN)
A measure of how unequally two bonded atoms attract shared electron density; larger ΔEN produces larger partial charges δ^+ and δ^-.

Molecular dipole moment
The 3D vector sum of all individual bond dipoles and lone-pair asymmetries in a molecule, measured in debyes (D).

Worked Reasoning Trace

Held fixed: Thermal energy T = 300 K.

Changed: Switched from System 2 (H_2O bent, 104.5°) to System 3 (CO_2 linear, 180°).

Model result: Both molecules contain polar covalent bonds (ΔEN = 1.24 for O–H, 0.89 for C=O), but bent H_2O has a net dipole of 1.85 D whereas linear CO_2 cancels to 0.00 D.

Model boundary: Partial charges δ^+ and δ^- are fractions of an elementary charge arising from unequal electron sharing, not full +1 or –1 ionic charges.

Common Misconception Trap · Unequal sharing and molecular dipoles

Intuitive Trap: "I think carbon and oxygen have identical electronegativities so δen is zero — because the total number of protons and neutrons in the molecule's nuclei."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Its two equal C=O bond dipoles point in exactly opposite directions (180° linear geometry) and cancel vectorially. The 3D vector sum of its individual bond dipoles (from ΔEN) and lone-pair asymmetries—which cancel to zero in symmetric linear CO_2 or tetrahedral CH_4 but add up in bent H_2O."

Why Claim A fails: Nuclear mass numbers do not create electric dipole moments. · Why Claim B holds up: Exact mechanism: dipole moments are vectors with both magnitude (from ΔEN) and 3D direction (from molecular geometry).

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Nonpolar diatomic: H_2 (ΔEN = 0, 0.00 D)	Two identical hydrogen nuclei share electrons equally.	Hydrogen (H_2) at 300 K (Above reference transition at 1 atm via weak London dispersion only): Two identical H nuclei share electrons equally (ΔEN = 0, dipole = 0 D). Weak dispersion attractions are easily overcome by thermal motion above 20.3 K.
Polar bent molecule: H_2O (ΔEN = 1.24, 1.85 D)	Two polar O–H bond dipoles add constructively in a 104.5° bent geometry.	Water (H_2O) at 300 K (Below reference transition; solid/liquid not distinguished via strong hydrogen bonding + dipole + dispersion): Bent 104.5° geometry prevents the two polar O–H bond dipoles from cancelling (net dipole = 1.85 D), enabling directional O–H···O hydrogen bonding and an unusually high boiling point (373.15 K).
Polar bonds that cancel: CO_2 (ΔEN = 0.89, 0.00 D)	Equal C=O dipoles point 180° apart and cancel to zero net dipole.	Carbon dioxide (CO_2) at 300 K (Above reference transition at 1 atm via London dispersion (polar bonds cancel by 180° symmetry)): Each C=O bond is polar (ΔEN = 0.89), but linear 180° geometry points the two bond dipoles in exact opposite directions so they cancel to net dipole = 0 D.

Check Question: Why does a carbon dioxide (CO_2) molecule have zero net dipole moment even though each C=O bond is polar? · **Transfer Challenge:** Transfer challenge: Ammonia (NH_3, trigonal pyramidal) and boron trifluoride (BF_3, flat trigonal planar) both have three polar bonds around a central atom. Why does NH_3 have a net dipole moment while BF_3 has zero net dipole?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 2 · The Two Clear Liquids & The White Crystal

You are given three unlabelled vials: Vial 1 contains a clear liquid whose thin stream bends strongly toward a charged rod; Vial 2 contains a volatile liquid with polar bonds whose stream does NOT bend toward a charged rod; Vial 3 is a brittle white solid that melts only above 800 °C.

- Clue 1 (Dipole Test): Vial 1 has a permanent molecular dipole (1.85 D), whereas Vial 2 has polar covalent bonds that cancel to 0.00 D.
- Clue 2 (Lattice Test): Vial 3 conducts electricity when dissolved in water and has a 1:1 cation-to-anion ratio.
- Clue 3 (Candidate List): The three samples are H_2O (bent), CO_2/CCl_4 (symmetric), and NaCl (ionic lattice).

Resolution: Bent H_2O has a net 1.85 D dipole (Vial 1); symmetric linear/tetrahedral molecules cancel their bond dipoles to 0 D (Vial 2); NaCl is an extended 3D ionic lattice (Vial 3).

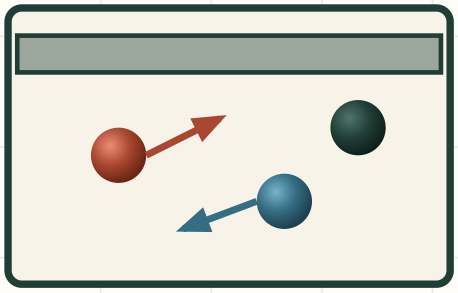
What our upgraded model now explains: Explains elements vs. compounds vs. mixtures, Coulomb attraction/repulsion ($F \propto q_1q_2/r^2$), covalent electron sharing, ionic lattices, allotropes, and 3D molecular dipoles. · **Next boundary:** Knowing the bonding inside a substance does not yet explain why heating ice turns it into liquid water and steam without breaking a single O–H bond → motivating Chapter 3 (States of Matter).

VOLUME I · PARTICLES, IDENTITY & STRUCTURE · CHAPTER 3

Chapter 3: Why matter has different states

Follow particles, motion and spacing while keeping track of chemical identity.

Chapter 3 Prelude · What Actually Breaks When Water Boils?: When a kettle boils on the stove, bubbles of invisible gas rise through the liquid and escape as steam. A surprisingly common intuition says that boiling splits H_2O into hydrogen gas (H_2) and oxygen gas (O_2)—yet steam never explodes when it touches the stove flame, and it condenses right back into water on a cool mirror! What is the real submicroscopic tug-of-war behind solids, liquids, gases, and solutions?



$$P \cdot V \propto N \cdot k_B\, T$$

$\langle E_k \rangle \propto T$ · Collisions → P

Identity conserved

Macroscopic Bench: Phase transitions (melting/boiling points), gas manometer pressure P(T, V), and solution dilution.

Submicroscopic Particle Model: Competition between thermal kinetic motion (k_B T) and intermolecular forces (dispersion, dipole–dipole, H-bonding); wall collisions.

Symbolic Notation: State symbols (s), (l), (g), (aq); ideal gas law PV = nRT; molar concentration c = n / V.

INVESTIGATION 3.1 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: STATE

One substance, three states

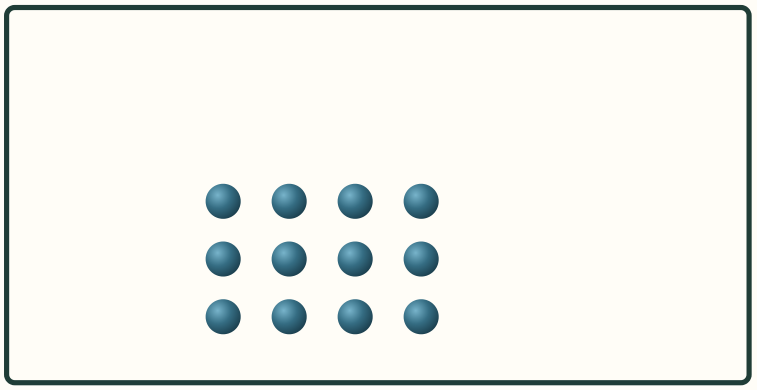
Connecting macroscopic shape and volume to particle spacing and motion

Matter can be solid, liquid or gas. Particle models connect what we observe to motion, spacing and interactions.

Bench Observation: A liquid takes the shape of its container while retaining a nearly fixed volume. A gas expands to fill available space.

Investigation Task: Change state and compare particle spacing. Explain why the particles themselves have not grown or changed element.

One substance, three states



Chamber Gauges

Motion ∝ √T

Phase: Solid

Attraction = undefined units

Red vectors show relative th

Schematic piston chamber with relative velocity vectors. Toggle 3D Spatial View below to rotate the 3D particle chamber.

Same particles, different arrangement

A physical state change preserves chemical identity. Spacing and arrangement change; the drawn symbols are a representative sample.

The bookkeeping idea: Like rearranging the same inventory: changing a physical state changes arrangement and motion, not particle identity. The 12 symbols are a drawn sample, not the amount of a real beaker.

Current explanatory plate · intro-models-2

Particles represented in every panel 12 representative particles

Model note: Solid: particles remain near neighbours and vibrate about positions. This simplified ordered layout does not represent every solid. Positions are schematic, not measured. Selecting a state does not calculate a phase boundary; temperature and pressure must be considered for a real substance.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 1.1 Core 1–4. States, particle arrangements and changes of state Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.2.1–2. Particle descriptions and state changes Only the named concept is connected. Practical work and the

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Figure 3.1: Deterministic particle & mathematical plate for *One substance, three states*. First Principle: Every state of matter is a competition between two things: thermal motion (temperature pushing particles apart) and interparticle attraction (pulling them together).

Historical & Experimental Evidence: Compressibility and diffusion across solid, liquid and gas

Macroscopically, a solid keeps its shape and volume, a liquid flows to take its container's shape while resisting compression, and a gas expands to fill any container and is easily compressed by a piston. The kinetic-molecular particle model explains all three observations without changing the particles themselves: only their spacing, arrangement and freedom of motion change.

Key Terms

Solid state
Particles are closely packed and vibrate continuously about fixed average positions.

Liquid state
Particles remain close together (so volume stays nearly constant) but have enough kinetic energy to slip past neighbours.

Gas state
Particles are far apart compared with their own size and move rapidly in straight lines between collisions.

Worked Reasoning Trace

Held fixed: 12 particles of the same substance in the model panel.

Changed: Switched state from Solid (0) to Liquid (1) to Gas (2).

Model result: All 12 particles remain identical in size and elemental identity; only their spacing and arrangement change.

Model boundary: A common beginner misconception is that particles themselves expand or melt when a solid melts—in reality, individual atoms/molecules do not swell.

Common Misconception Trap · One substance, three states

Intuitive Trap: "I think its elemental identities — because causes each individual atom or molecule to swell to twice its diameter."

Particle Mechanism Refutation: "Wait — look at the particle ledger: The arrangement and motion of its particles. Increases particle freedom to slip past neighbours while keeping each particle's size and chemical formula unchanged."

Why Claim A fails: Individual atoms and molecules retain their dimensions during melting; only their packing arrangement changes. · Why Claim B holds up: Exact mechanism: phase changes alter interpersonal spacing and motion between particles without altering the particles themselves.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Solid state	Closely packed arrangement; particles vibrate in place.	Solid: particles remain near neighbours and vibrate about positions. This simplified ordered layout does not represent every solid. Positions are schematic, not measured. Selecting a state does not calculate a phase boundary; temperature and pressure must be considered for a real substance.
Liquid state	Close spacing preserved, but particles rearrange freely.	Liquid: particles remain close and rearrange. They are not fixed in a lattice. Positions are schematic, not measured. Selecting a state does not calculate a phase boundary; temperature and pressure must be considered for a real substance.
Gas state	Wide spacing between particles filling the container.	Gas: particles are widely separated relative to their size and move between collisions. Particle identity has not changed. Positions are schematic, not measured. Selecting a state does not calculate a phase boundary; temperature and pressure must be considered for a real substance.

Check Question: Which changes when a substance melts without reacting? · **Transfer Challenge:** Transfer challenge: Why is a gas easy to compress into a smaller volume with a bicycle pump, whereas liquid water is nearly incompressible, even though both are made of H_2O or air molecules?

INVESTIGATION 3.2 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: GAS

Temperature and particle motion

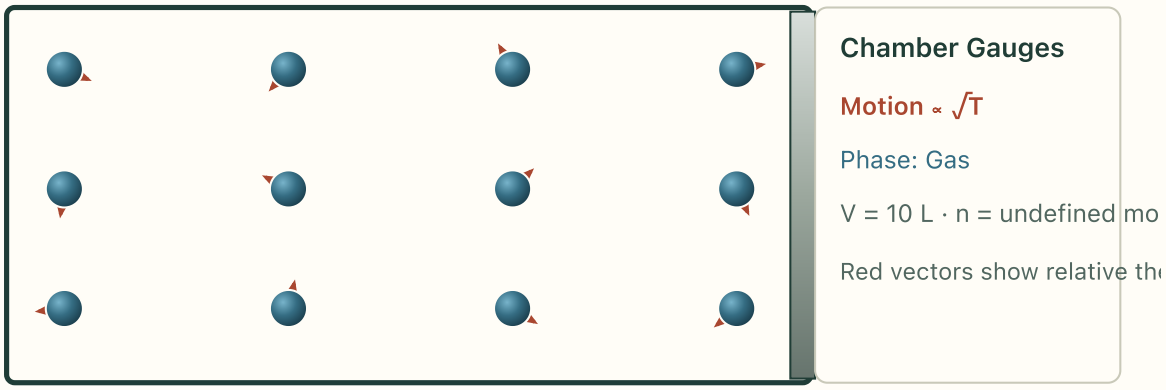
How kelvin temperature drives average kinetic energy and wall collisions

Temperature in kelvin connects to average translational kinetic energy in an ideal gas. Individual particles do not all have the same speed.

Bench Observation: A confined gas commonly exerts greater pressure when warmed while its volume and amount stay fixed.

Investigation Task: Hold volume at 10 L. Compare 200 K and 400 K. Explain the pressure change without saying the particles got larger.

Temperature and particle motion



Schematic piston chamber with relative velocity vectors. Toggle 3D Spatial View below to rotate the 3D particle chamber.

Pressure account · fixed 1 mol

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: The amount account stays at 1 mol. Pressure is a calculated response to temperature and volume, not another substance to add to the inventory. Use kelvin and the units attached to R.

Current explanatory plate · intro-models-2

Calculated pressure / kPa 249.43387854000002 kPa

P = nRT/V = 1 mol × 8.314462618 kPa·L/(mol·K) × 300 K ÷ 10 L = 249.434 kPa

Model note: P = nRT/V, with n = 1 mol and R = 8.314462618 kPa L mol^{-1} K^{-1}. Ideal gas model: ignores particle size and attractions. It does not predict condensation.

Scientific inputs · scientific-values-2

Molar gas constant R, rounded for the local model: 8.314462618 kPa·L·mol^{-1}·K^{-1} · reference. Absolute temperature in kelvin. The ideal gas model ignores particle size and attractions. Rounded from the exact SI-derived gas constant. Source

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 1.1 Core 4; Supplement 6. Gas temperature, pressure and volume relationships Only the named concept is

connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.5 (chemistry, HT). Gas quantities; the ideal-gas equation is enrichment Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

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Figure 3.2: Deterministic particle & mathematical plate for *Temperature and particle motion*. First Principle: Absolute temperature (T in kelvin) measures average translational kinetic energy—warming a confined gas speeds up particles so they strike walls harder and more often.

Historical & Experimental Evidence: Constant-volume gas thermometry (Amontons & Gay-Lussac)

When a fixed amount of gas is sealed in a rigid bulb connected to a pressure gauge (manometer) and warmed, its pressure rises linearly with temperature. Extrapolating this pressure-temperature line backward for different low-density gases points to a single absolute zero near −273.15 °C (0 K), the foundation of the Kelvin temperature scale.

Key Terms

Kelvin scale (K)
The absolute thermodynamic temperature scale starting at 0 K (absolute zero), where T(K) = T(°C) + 273.15.

Average translational kinetic energy
The average energy of straight-line motion of gas particles, which is directly proportional to temperature in kelvin.

Worked Reasoning Trace

Held fixed: Amount n = 1 mol and Volume V = 10 L.

Changed: Doubled kelvin temperature T from 200 K to 400 K.

Model result: Calculated ideal gas pressure doubles from 166.29 kPa to 332.58 kPa (P = nRT/V).

Model boundary: At 400 K, not every particle moves at the exact same speed; particles have a broad Maxwell–Boltzmann distribution of speeds whose average kinetic energy doubled.

Common Misconception Trap · Temperature and particle motion

Intuitive Trap: "I think halves pressure — because doubles the number of gas molecules inside the sealed container."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Doubles pressure. Doubles the average translational kinetic energy of the gas particles, so they strike the container walls more frequently and with greater momentum."

Why Claim A fails: In a sealed rigid container with no reaction, the number of gas particles n stays constant. · Why Claim B holds up: Exact mechanism: kelvin temperature is proportional to average translational kinetic energy, driving wall-collision rate and impulse.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Cooler gas: 200 K in 10 L	P = 166.3 kPa at 200 K and 10 L.	P = nRT/V, with n = 1 mol and R = 8.314462618 kPa L mol^{-1} K^{-1}. Ideal gas model: ignores particle size and attractions. It does not predict condensation.
Baseline: 300 K in 10 L	P = 249.4 kPa at 300 K and 10 L.	P = nRT/V, with n = 1 mol and R = 8.314462618 kPa L mol^{-1} K^{-1}. Ideal gas model: ignores particle size and attractions. It does not predict condensation.
Doubled T: 400 K in 10 L	P = 332.6 kPa—double the 200 K pressure at fixed 10 L volume.	P = nRT/V, with n = 1 mol and R = 8.314462618 kPa L mol^{-1} K^{-1}. Ideal gas model: ignores particle size and attractions. It does not predict condensation.

Check Question: For a fixed amount of ideal gas at fixed volume, doubling kelvin temperature does what? · **Transfer Challenge:** Transfer challenge: Why must we measure temperature in kelvin (K) rather than degrees Celsius (°C) when using the proportional rule 'doubling temperature doubles ideal gas pressure'?

INVESTIGATION 3.3 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: STATE

Changing state is not changing identity

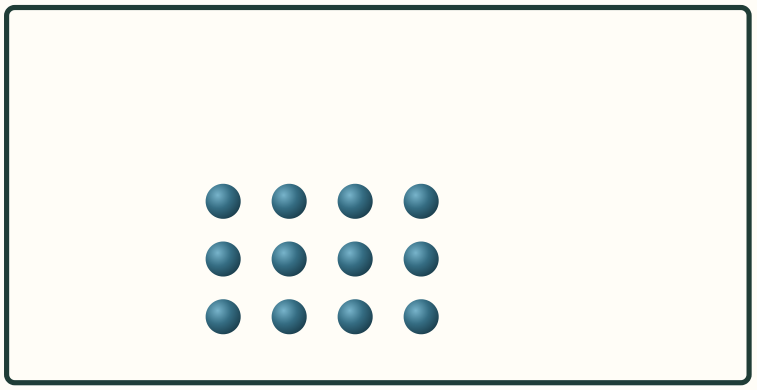
Separating intermolecular attractions from covalent bonds inside molecules

Melting, freezing, vaporization and condensation change physical state. The temperature at which a phase change occurs depends on the substance and pressure.

Bench Observation: Liquid water and water vapour both contain H_2O. Boiling is not the splitting of water into hydrogen and oxygen.

Investigation Task: Sketch before and after a phase change. Say what this state selector cannot predict about a real sample.

Changing state is not changing identity



Chamber Gauges

Motion ∝ √T

Phase: Solid

Attraction = undefined units

Red vectors show relative th

Schematic piston chamber with relative velocity vectors. Toggle 3D Spatial View below to rotate the 3D particle chamber.

Same particles, different arrangement

A physical state change preserves chemical identity. Spacing and arrangement change; the drawn symbols are a representative sample.

The bookkeeping idea: Like rearranging the same inventory: changing a physical state changes arrangement and motion, not particle identity. The 12 symbols are a drawn sample, not the amount of a real beaker.

Current explanatory plate · intro-models-2

Particles represented in every panel 12 representative particles

Model note: Solid: particles remain near neighbours and vibrate about positions. This simplified ordered layout does not represent every solid. Positions are schematic, not measured. Selecting a state does not calculate a phase boundary; temperature and pressure must be considered for a real substance.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

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AQA GCSE Chemistry 8462 · Partial connection: 4.2.2.1–2. Particle descriptions and state changes Only the named concept is connected. Practical work and the

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Figure 3.3: Deterministic particle & mathematical plate for *Changing state is not changing identity*. First Principle: Boiling or melting overcomes attractions between molecules while leaving covalent bonds inside each molecule completely intact.

Historical & Experimental Evidence: Heating curves and vapour analysis of boiling water

When liquid water boils at 100 °C (at 1 atm), the steam rising from the kettle condenses back into pure liquid water on a cool surface—unlike electrolysis, which uses electric current to split water into flammable hydrogen gas (H_2) and oxygen gas (O_2). Boiling overcomes attractions between H_2O molecules while leaving the O–H covalent bonds inside each molecule intact.

Key Terms

Phase transition
A physical change from one state of matter to another (melting, freezing, vaporization, condensation) without changing chemical identity.

Intermolecular attractions vs. Intramolecular bonds
Attractions between separate H_2O molecules are much weaker than the covalent O–H bonds holding each H_2O molecule together.

Worked Reasoning Trace

Held fixed: Substance identity (pure H_2O).

Changed: Switched state selector from Liquid (1) to Gas (2).

Model result: The 12 particles move much farther apart, representing water vapour, while every particle remains an intact H_2O unit.

Model boundary: Our simple state selector does not compute the boiling point (e.g., 100 °C at 101.3 kPa for water vs. –196 °C for nitrogen); phase boundaries require substance-specific thermodynamic data.

Common Misconception Trap · Changing state is not changing identity

Intuitive Trap: "I think each becomes h_2 and o_2 — because cleaves the covalent O–H bonds inside each H_2O molecule to produce H_2 and O_2 gases."

Particle Mechanism Refutation: "Wait — look at the particle ledger: They remain H_2O while changing arrangement and separation. Overcomes intermolecular attractions between intact H_2O molecules so they separate widely into the gas phase."

Why Claim A fails: Splitting O–H bonds into H_2 and O_2 requires chemical decomposition (such as electrolysis), not boiling. · Why Claim B holds up: Exact mechanism: vaporization separates intact H_2O molecules against intermolecular attractions while leaving intramolecular O–H bonds unbroken.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Liquid (Close, rearranging)	Molecules are close enough to attract neighbours strongly while sliding past each other.	Liquid: particles remain close and rearrange. They are not fixed in a lattice. Positions are schematic, not measured. Selecting a state does not calculate a phase boundary; temperature and pressure must be considered for a real substance.
Gas / Vapour (Widely separated)	Molecules have overcome neighbour attractions and separated widely, still intact.	Gas: particles are widely separated relative to their size and move between collisions. Particle identity has not changed. Positions are schematic, not measured. Selecting a state does not calculate a phase boundary; temperature and pressure must be considered for a real substance.

Check Question: When pure water boils without reacting, what happens to its molecules? · **Transfer Challenge:** Transfer challenge: What observation in a kitchen proves that steam condensing on a cold pot lid is still water (H_2O) rather than hydrogen and oxygen gas?

INVESTIGATION 3.4 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: GAS

Squeezing a gas

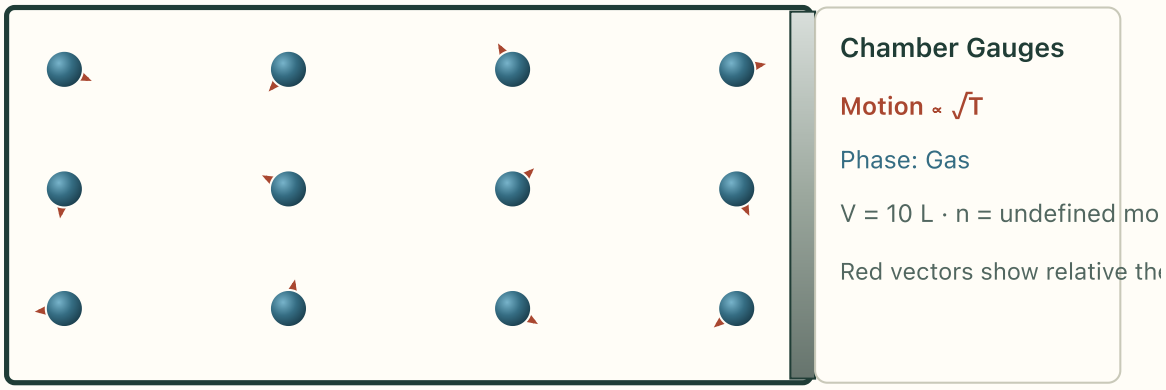
Inverse proportionality between volume and pressure at constant temperature

Gas pressure comes from interactions with the container walls. The ideal gas law links pressure, amount, volume and temperature.

Bench Observation: Reducing a gas volume at fixed temperature and amount generally raises its pressure in the ideal-gas regime.

Investigation Task: Hold temperature at 300 K. Compare 5 L, 10 L and 20 L and save two results.

Squeezing a gas



Schematic piston chamber with relative velocity vectors. Toggle 3D Spatial View below to rotate the 3D particle chamber.

Pressure account · fixed 1 mol

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: The amount account stays at 1 mol. Pressure is a calculated response to temperature and volume, not another substance to add to the inventory. Use kelvin and the units attached to R.

Current explanatory plate · intro-models-2

Calculated pressure / kPa 249.43387854000002 kPa

P = nRT/V = 1 mol × 8.314462618 kPa·L/(mol·K) × 300 K ÷ 10 L = 249.434 kPa

Model note: P = nRT/V, with n = 1 mol and R = 8.314462618 kPa L mol^{-1} K^{-1}. Ideal gas model: ignores particle size and attractions. It does not predict condensation.

Scientific inputs · scientific-values-2

Molar gas constant R, rounded for the local model: 8.314462618 kPa·L·mol^{-1}·K^{-1} · reference. Absolute temperature in kelvin. The ideal gas model ignores particle size and attractions. Rounded from the exact SI-derived gas constant. Source

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Figure 3.4: Deterministic particle & mathematical plate for *Squeezing a gas*. First Principle: Halving the volume of a fixed gas sample crowds the same moving particles into half the space, doubling wall collisions per second and doubling pressure ($P \propto 1/V$).

Historical & Experimental Evidence: Boyle's J-tube mercury compression experiment (1662)

Robert Boyle trapped a fixed pocket of air in the closed short end of a J-shaped glass tube and poured mercury into the long open end. When he doubled the total pressure on the trapped air at constant room temperature, the air volume compressed to almost exactly half—establishing $P \propto 1/V$ (PV = constant at fixed n and T).

Key Terms

Ideal gas law (P = nRT/V)
A model equation relating pressure P, molar amount n, gas constant R, kelvin temperature T, and container volume V.

Inverse proportion (P ∝ 1/V)
At fixed amount and temperature, multiplying volume V by 1/2 multiplies pressure P by 2 because the same particles collide with a smaller wall area more frequently.

Worked Reasoning Trace

Held fixed: Amount n = 1 mol and Temperature T = 300 K.

Changed: Halved Volume V from 10 L to 5 L, then doubled V from 10 L to 20 L.

Model result: At 10 L, P = 249.4 kPa; at 5 L, P doubles to 498.9 kPa; at 20 L, P halves to 124.7 kPa.

Model boundary: At very high pressures or low temperatures, real gases deviate from P = nRT/V because real molecules have finite volume and attract one another (and eventually condense).

Common Misconception Trap · Squeezing a gas

Intuitive Trap: "I think half the pressure — because makes each particle move twice as fast by raising its temperature."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Twice the pressure. Packs the same number of particles into half the space at the same average speed, doubling the frequency of wall collisions per unit area."

Why Claim A fails: Constant temperature means average particle speed and kinetic energy remain constant. · Why Claim B holds up: Exact mechanism: with T fixed, average kinetic energy is unchanged, but halving V doubles particle number density (N/V) and wall collision frequency.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Compressed: 300 K in 5 L	Halving volume from 10 L to 5 L doubles pressure to 498.9 kPa.	P = nRT/V, with n = 1 mol and R = 8.314462618 kPa L mol^{-1} K^{-1}. Ideal gas model: ignores particle size and attractions. It does not predict condensation.
Baseline: 300 K in 10 L	Reference state: 249.4 kPa.	P = nRT/V, with n = 1 mol and R = 8.314462618 kPa L mol^{-1} K^{-1}. Ideal gas model: ignores particle size and attractions. It does not predict condensation.
Expanded: 300 K in 20 L	Doubling volume to 20 L halves pressure to 124.7 kPa.	P = nRT/V, with n = 1 mol and R = 8.314462618 kPa L mol^{-1} K^{-1}. Ideal gas model: ignores particle size and attractions. It does not predict condensation.

Check Question: At fixed temperature and amount, halving ideal gas volume gives… · **Transfer Challenge:** Transfer challenge: If 1 mol of ideal gas at 300 K is compressed from 12 L to 4 L (one third of its initial volume), by what factor does its pressure change, and what assumption of the ideal gas model might break down if you keep squeezing to 0.05 L?

INVESTIGATION 3.5 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: SOLUTION

Where did the solute go?

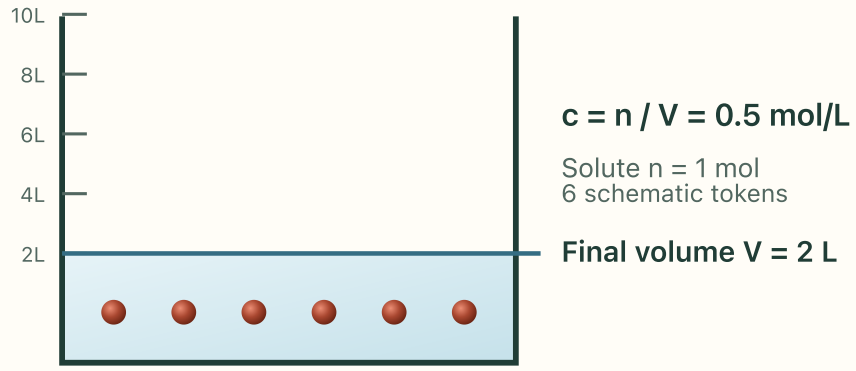
Conservation of dissolved solute during dilution and evaporation

A dissolved solute remains present even if you cannot see separate pieces. Concentration describes how much solute is present per volume of solution.

Bench Observation: Dissolved salt can be recovered by removing water. Ordinary filtration does not remove dissolved ions from water.

Investigation Task: Keep the dissolved amount fixed and increase final solution volume. Explain what changes and what remains.

Where did the solute go?



Schematic beaker showing final solution volume V and dispersed solute tokens (6 tokens per mole).

Same stock, more space

Adding solvent increases final solution volume. In this ideal dilution model, solute amount remains in the account.

The bookkeeping idea: Like spreading the same stock across more shelves: dilution changes the space available, not the amount owned. Count solute in mol, volume in L, and stock per litre in mol/L. Never add these unlike units together.

Current explanatory plate · intro-models-2

Solute amount	1 mol
Solution volume	2 L
Concentration	0.5 mol/L

c = n / V = 1 mol ÷ 2 L = 0.5 mol/L

Model note: Final solution volume is used, not solvent volume. No reaction, solute loss or nonideal volume mixing is simulated.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.3 Core 1; Supplement 5. Concentration and solution quantities Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.2.5; 4.3.4 (chemistry, HT). Mass and molar concentrations Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Mixtures and their Separation · C-1.2 (pp. 7–8). Concentration; molarity is enrichment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 3.5: Deterministic particle & mathematical plate for *Where did the solute go?*. First Principle: Dissolving disperses conserved solute particles uniformly among solvent molecules; adding solvent increases volume V and lowers concentration c = n/V while solute count n stays constant.

Historical & Experimental Evidence: Gravimetric recovery of dissolved salt by evaporation

If you dissolve 5.00 g of sodium chloride in water, the clear liquid passes completely through ordinary filter paper—showing the dispersed Na^+ and Cl^- ions are far smaller than suspended sand grains. Yet if you gently evaporate all the water, you recover the original 5.00 g of solid salt, confirming the solute remained conserved throughout dilution.

Key Terms

Solute and Solvent
The solute is the dissolved substance; the solvent (such as water) is the medium in which the solute is dispersed.

Concentration (c = n / V)
The amount of dissolved solute (in moles) divided by the total final volume of the solution (in litres).

Worked Reasoning Trace

Held fixed: Dissolved solute amount n = 1.0 mol.

Changed: Increased final solution volume V from 2.0 L to 4.0 L (dilution).

Model result: Concentration c = n/V drops from 0.50 mol/L to 0.25 mol/L while total solute stays 1.0 mol.

Model boundary: In c = n/V, V is the final solution volume in a calibrated flask, not simply the volume of pure water added (since volumes do not always add strictly linearly).

Common Misconception Trap · Where did the solute go?

Intuitive Trap: "I think amount of solute necessarily increases — because destroys half of the dissolved solute ions so the colour fades."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Concentration decreases. Increases the total solution volume V while the number of dissolved solute particles n stays conserved, lowering the ratio c = n / V."

Why Claim A fails: Solute particles are conserved during dilution and can be recovered by evaporating the water. · Why Claim B holds up: Exact mechanism: dilution spreads the same conserved number of solute particles across a larger volume of solvent.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
1.0 mol in 2.0 L (0.50 mol/L)	Baseline concentration of 0.50 mol/L.	c = n/V. V is the final solution volume, not the volume of water poured in. Assumes all of the specified solute is dissolved and no reaction occurs.
Diluted: 1.0 mol in 4.0 L (0.25 mol/L)	Same 1.0 mol of solute spread across twice the final volume halves concentration.	c = n/V. V is the final solution volume, not the volume of water poured in. Assumes all of the specified solute is dissolved and no reaction occurs.
Concentrated: 2.0 mol in 1.0 L (2.0 mol/L)	More solute in a smaller final volume increases concentration.	c = n/V. V is the final solution volume, not the volume of water poured in. Assumes all of the specified solute is dissolved and no reaction occurs.

Check Question: Adding water without losing solute changes which quantity? · **Transfer Challenge:** Transfer challenge: Explain why pouring salt water through a coffee filter does not separate the salt from the water, and state what physical process would recover the solid salt.

INVESTIGATION 3.6 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: POLARITY

Intermolecular forces vs. thermal motion

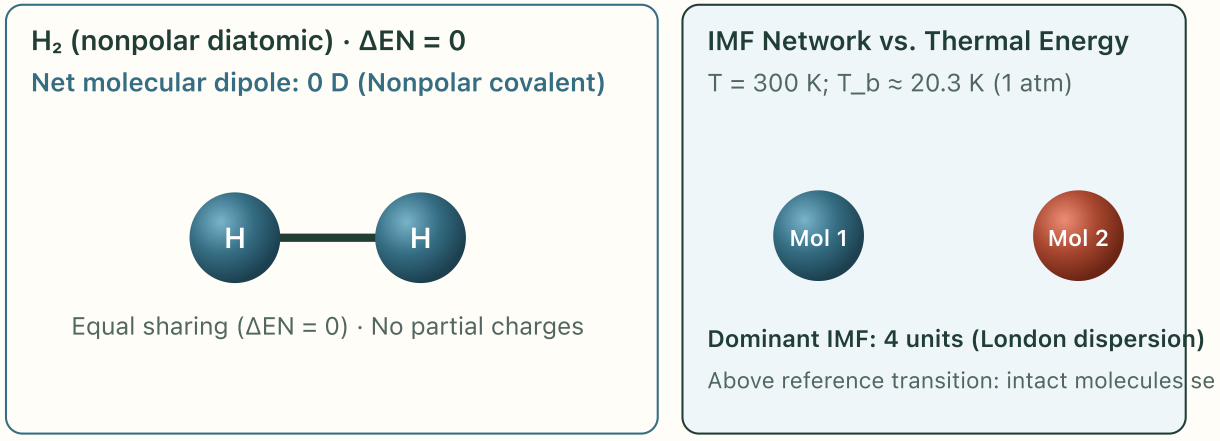
How dispersion forces, dipole interactions, and hydrogen bonding compete with thermal motion

Whether a molecular substance is a solid, liquid, or gas at a given temperature depends on the competition between intermolecular attractions (dispersion forces, dipole–dipole interactions, and hydrogen bonding) and average thermal kinetic energy (k_B T).

Bench Observation: At 1 atm and 298 K, nonpolar H_2 (boiling point 20 K) and CH_4 (112 K) are gases, whereas hydrogen-bonded H_2O (373 K) is a liquid—even though H_2O has a low molar mass of 18 g/mol.

Investigation Task: Select H_2O (bent) and raise thermal energy from 250 K to 420 K, then compare with CH_4 and CO_2 at 300 K. Distinguish breaking intermolecular hydrogen bonds from breaking covalent O–H bonds.

Intermolecular forces vs. thermal motion



Left: 3D bond dipole addition (δ⁺ → δ⁻). Right: Intermolecular attraction dashed bridge vs. thermal kinetic motion at 300 K.

Vector contributions

Bond dipoles add as vectors in space. Equal scalar magnitudes can cancel when their directions oppose.

The bookkeeping idea: This account needs arrows, not just totals: dipole contributions add with direction. Opposing vectors can cancel even though every bond contribution is nonzero.

Current explanatory plate · intro-models-2

Bond electronegativity difference ΔEN	0 dimensionless
Net molecular dipole moment / D	0 D
Relative intermolecular attraction / model units	4 relative model units
Reference 1 atm boiling/sublimation point / K	20.3 K

Model note: Hydrogen (H_2) at 300 K (Above reference transition at 1 atm via weak London dispersion only): Two identical H nuclei share electrons equally (ΔEN = 0, dipole = 0 D). Weak dispersion attractions are easily overcome by thermal motion above 20.3 K.

Scientific inputs · scientific-values-2

Hydrogen (H_2): deltaEN: 0 Pauling scale · reference. Difference of rounded Pauling electronegativities; not a measured bond charge. Introductory approximation. Checked 2026-10-28. Source

Hydrogen (H_2): dipole: 0 D · reference. Rounded gas-phase dipole; zero values reflect the geometry shown. Introductory approximation. Checked 2026-10-28. Source

Hydrogen (H_2): boiling: 20.3 K · reference. Approximate pure-substance normal boiling point at 1 atm. Introductory approximation. Checked 2026-10-28. Source

Hydrogen (H_2): attraction: 4 model units · illustrative. Author-selected comparison score, not an energy. Introductory approximation. Checked 2026-10-28.

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.5 Supplement 5. Qualitative intermolecular attractions in small molecules Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.2.4. Molecular attractions; dipoles are enrichment Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 3.6: Deterministic particle & mathematical plate for *Intermolecular forces vs. thermal motion*. First Principle: Boiling point reflects the temperature where average thermal kinetic energy (k_B T) overcomes intermolecular attractions (dispersion, dipole–dipole, or hydrogen bonding) without breaking covalent bonds.

Historical & Experimental Evidence: Boiling point anomalies of hydrides (CH_4 vs. NH_3 vs. H_2O)

Methane (CH_4, 16 g/mol), ammonia (NH_3, 17 g/mol), and water (H_2O, 18 g/mol) have nearly identical molar masses and 10 electrons each. Yet at 1 atm, CH_4 boils at a frigid 112 K (−161 °C), NH_3 boils at 240 K (−33 °C), and H_2O boils at 373 K (+100 °C)! Because polar O–H and N–H groups form directional intermolecular hydrogen bonds between δ^+ H and δ^- lone pairs on neighbouring molecules, much greater thermal kinetic energy (k_B T) is needed to separate molecules into the gas phase.

Key Terms

Intermolecular forces (IMFs)
Electrostatic attractions between separate molecules (dispersion forces, dipole–dipole attractions, and hydrogen bonding)—always weaker than the covalent bonds inside a molecule.

Hydrogen bond
A strong directional dipole–dipole attraction between a δ^+ hydrogen atom (covalently bonded to N, O, or F) and a lone pair on a neighbouring electronegative N, O, or F atom.

Worked Reasoning Trace

Held fixed: Thermal energy T = 300 K (room temperature).

Changed: Switched from System 4 (CH_4 tetrahedral, 0 D) to System 2 (H_2O bent, 1.85 D).

Model result: IMF strength jumps from 4.5 units (dispersion only, T_b = 112 K, gas at 300 K) to 22.0 units (hydrogen bonding, T_b = 373 K, liquid at 300 K).

Model boundary: Raising T above 373 K separates intact H_2O molecules against intermolecular hydrogen bonds; the intramolecular O–H covalent bonds (~463 kJ/mol) do not break during boiling.

Common Misconception Trap · Intermolecular forces vs. thermal motion

Intuitive Trap: "I think boiling water breaks the covalent o–h bonds inside each h_2o molecule into h_2 and o_2 — because breaks the covalent bonds inside each molecule to turn H_2O into H_2 and O_2 gases."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Strong intermolecular hydrogen-bonding attractions between polar H_2O molecules require much more thermal kinetic energy to overcome than weak dispersion forces between nonpolar CH_4 molecules. Supplies enough average thermal kinetic energy (k_B T) for intact molecules to overcome intermolecular attractions (such as hydrogen bonds or dispersion forces) and separate into the gas phase."

Why Claim A fails: Intramolecular covalent O–H bonds (~463 kJ/mol) are roughly 20 times stronger than intermolecular hydrogen bonds (~20 kJ/mol) and remain intact in steam. · Why Claim B holds up: Exact mechanism: boiling is a contest between intermolecular attractions holding intact molecules together and thermal kinetic energy pulling them apart.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Methane CH_4 at 300 K (Dispersion only, gas)	T = 300 K is far above T_b = 112 K, so thermal motion easily overcomes weak dispersion forces.	Methane (CH_4) at 300 K (Above reference transition at 1 atm via weak London dispersion (symmetric tetrahedron)): Four weakly polar C–H bonds arranged symmetrically in a 109.5° tetrahedron cancel to net dipole = 0 D, leaving only weak dispersion attractions (boiling point 111.7 K).
Water H_2O at 300 K (H-bonding, liquid)	Strong intermolecular hydrogen bonding (T_b = 373 K) holds H_2O in the liquid state at 300 K.	Water (H_2O) at 300 K (Below reference transition; solid/liquid not distinguished via strong hydrogen bonding + dipole + dispersion): Bent 104.5° geometry prevents the two polar O–H bond dipoles from cancelling (net dipole = 1.85 D), enabling directional O–H···O hydrogen bonding and an unusually high boiling point (373.15 K).
Water H_2O at 425 K (Steam, intact H_2O gas)	Above 373 K, thermal energy overcomes intermolecular hydrogen bonds while O–H covalent bonds stay intact.	Water (H_2O) at 425 K (Above reference transition at 1 atm via strong hydrogen bonding + dipole + dispersion): Bent 104.5° geometry prevents the two polar O–H bond dipoles from cancelling (net dipole = 1.85 D), enabling directional O–H···O hydrogen bonding and an unusually high boiling point (373.15 K).

Check Question: Why does water (H_2O, 18 g/mol) have a much higher normal boiling point (373 K) than methane (CH_4, 16 g/mol, 112 K)? · **Transfer Challenge:** Transfer challenge: At 300 K, CO_2 (44 g/mol, 0 D) has stronger dispersion forces than H_2 (2 g/mol, 0 D), yet both are gases at room temperature. Why does CO_2 have a higher boiling/sublimation temperature (195 K) than H_2 (20 K)?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 3 · The Rigid Cylinder Pressure Anomaly

An engineer seals 1.0 mol of methane gas (CH_4, T_b = 112 K) in a rigid 10 L steel cylinder at 200 K and measures pressure P_1. They then warm the cylinder to 400 K and transfer the gas into a 5 L cylinder at 400 K.

- Clue 1 (Temperature Step): Warming from 200 K to 400 K at fixed 10 L doubles absolute kelvin temperature.
- Clue 2 (Volume Step): Compressing from 10 L to 5 L at fixed 400 K halves the container volume.
- Clue 3 (Identity Check): At both 200 K and 400 K (well above T_b = 112 K), CH_4 molecules remain intact gas particles.

Resolution: By P = nRT/V, doubling T (200 K → 400 K) multiplies pressure by 2, and halving V (10 L → 5 L) multiplies pressure by another factor of 2, giving 4 × P_1.

What our upgraded model now explains: Explains solid/liquid/gas spacing, why boiling preserves molecular identity, why H_2O boils higher than CH_4, gas pressure $P \propto T/V$, and solute conservation during dilution. · **Next boundary:** Physical state changes keep every molecule intact; to explain how new substances form when bonds actually break and rearrange, we need Chapter 4 (Chemical Reactions).

FROM PARTICLE THINKER TO QUANTITATIVE ANALYST (CHAPTERS 4–6)

Volume II · Change, Quantities & Driving Forces

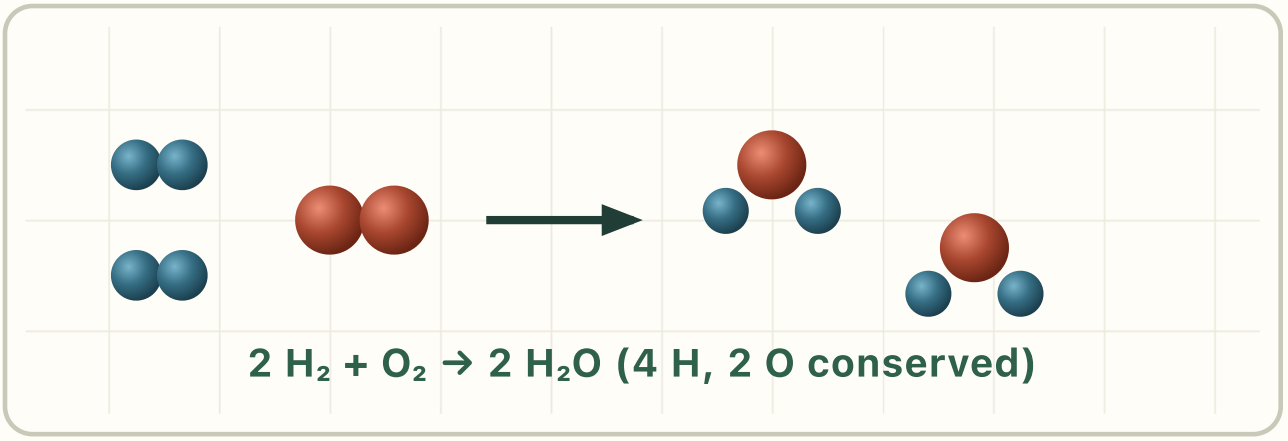
Account for every atom, electron, and joule in chemical transformations: balanced equations, bond-breaking versus bond-making energy ledgers, activation barriers, Le Châtelier equilibrium shifts, and the mole.

VOLUME II · CHANGE, QUANTITIES & DRIVING FORCES · CHAPTER 4

Chapter 4: What changes in a reaction?

Read formulae and account for every atom. A drawing or equation alone cannot prove a reaction occurs.

Chapter 4 Prelude · Where Does the Wood Go When a Campfire Burns?: When a heavy oak log burns in an open fireplace, only a light pile of grey ash remains—making it look as if matter vanished into thin air. Conversely, when iron rusts in damp air, the flaky red rust weighs MORE than the original iron nail! Once Antoine Lavoisier trapped all the gases in a sealed vessel, he proved that not a single atom is created or destroyed. How do we write and audit the strict atomic ledger of a chemical reaction?



Macroscopic Bench: New substance formation (precipitates, gas evolution, mass conservation in sealed vessels).

Submicroscopic Particle Model: Rearrangement of atomic bonding connectivity while conserving the exact count of each element's nuclei.

Symbolic Notation: Balanced chemical equations with stoichiometric coefficients multiplying immutable formula subscripts.

INVESTIGATION 4.1 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: COMPOSITION

What counts as a chemical change?

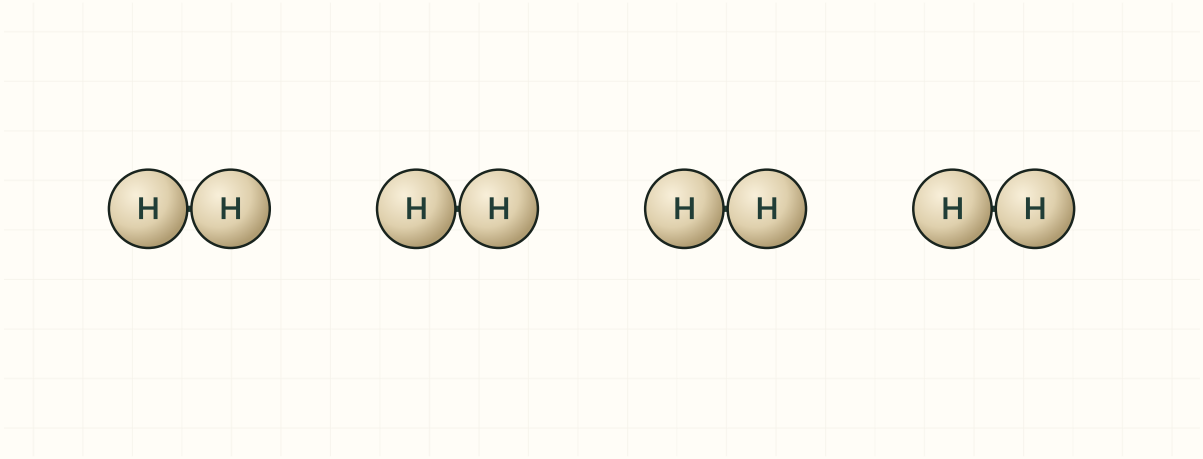
Distinguishing a rearrangement of existing molecules from the formation of new substances

A chemical change produces different substances. A physical change can alter state, shape or arrangement without that change of identity.

Bench Observation: Ice melting and iron rusting differ: one changes state, the other produces new substances.

Investigation Task: Compare a mixture of H_2 and O_2 with H_2O. Explain why selecting another diagram is not itself experimental evidence of a reaction.

What counts as a chemical change?



Grouped atom symbols show molecular composition. Lines mark connectivity, not exact bond order, electron positions or geometry.

Account for the model

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like sorting stock by identity: count element identities, chemical substances and particles in separate columns. Mixing two stocks is not evidence that they reacted.

Current explanatory plate · intro-models-2

Substances represented 1

Model note: H_2 is an element: each molecule contains only hydrogen atoms. A molecule need not be a compound.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.1; 2.4–2.7. Substances, bonding and structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.1–2. Structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1, C-7.1 (pp. 11–12). Compounds, bonding and fixed proportions Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 4.1: Deterministic particle & mathematical plate for *What counts as a chemical change?*. First Principle: Physical changes rearrange spacing between intact molecules, whereas chemical changes break and form bonds to create new substances while conserving every element's atoms.

Historical & Experimental Evidence: Testing substance properties before and after a change

When solid ice warms above 0 °C, it turns into liquid water, which still freezes at 0 °C, boils at 100 °C, has density 1.00 g/cm³ and formula H_2O. When iron rusts in moist air, the flaky reddish-brown solid (hydrated iron(III) oxide) is non-magnetic, brittle, heavier than the starting iron alone (because oxygen atoms combined with it), and has completely different chemical properties.

Key Terms

Physical change
A change in physical state, shape or mixing in which no new chemical substances are formed.

Chemical change (chemical reaction)
A transformation in which atoms rearrange their bonding to form one or more new substances with distinct compositions and properties.

Worked Reasoning Trace

Held fixed: Total hydrogen and oxygen atoms represented.

Changed: Compared 'H_2 and O_2 mixture' (sample 3) with 'Water H_2O' (sample 2).

Model result: In the H_2 + O_2 mixture, H–H and O–O bonds are intact; in H_2O, atoms have rearranged into a new compound with O–H bonds.

Model boundary: Switching a dropdown menu in our app only switches which diagram is displayed—real chemical claims require experimental evidence that a new substance formed.

Common Misconception Trap · What counts as a chemical change?

Intuitive Trap: "I think the picture on the screen changed — because changing the spacing between intact molecules without changing their bonding."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Evidence that a new substance formed. Breaking and forming chemical bonds so atoms rearrange into new substances with different particle structures and properties."

Why Claim A fails: Changing spacing between intact molecules is a physical phase change. · Why Claim B holds up: Exact mechanism: chemical change rearranges atomic connectivity to produce new chemical species, verified experimentally by new substance properties.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Unreacted H_2 and O_2 mixture	Two separate elemental gases mixed physically.	H_2 and O_2 together form a mixture. Their proportions can vary. Placing them together in a diagram does not show a reaction.
Water H_2O compound	A single compound with new O–H bonding and distinct properties.	H_2O is a molecular compound: hydrogen and oxygen combined in a fixed atom ratio of 2:1.

Check Question: Which observation would best support a chemical-change claim? · **Transfer Challenge:** Transfer challenge: When a fizzy tablet dissolves in water, bubbles of gas appear and the water cools slightly. What measurement or test would you need to verify whether a new gas substance formed versus trapped air escaping?

INVESTIGATION 4.2 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: FORMULA

Reading a chemical formula

Why changing a subscript changes the substance, while a coefficient scales the amount

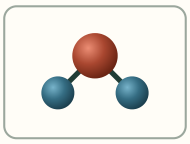
Subscripts count atoms in a molecule or specify a ratio in a formula unit. A coefficient multiplies the whole formula.

Bench Observation: H_2O and H_2O_2 describe different compositions; changing a subscript is not a way to balance an equation.

Investigation Task: Change the number of water molecules. Explain how multiplying a formula differs from changing a subscript.

Reading a chemical formula

Symbolic multiplier: 1 × H_2O



Solid boxes group the 1 discrete H_2O molecules; subscripts stay fixed inside each molecule.

Inventory per molecule or formula unit

Subscripts specify composition; a coefficient multiplies the whole inventory.

The bookkeeping idea: Like an itemized packing list: subscripts count atoms inside one molecule or formula unit; a coefficient multiplies the whole list. For an ionic formula, the signed charge entries must also reconcile.

Current explanatory plate · intro-models-2

H atoms 2

O atoms 1

Model note: 1 × H_2O. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.1 Core 1–4. Reading formulae and counting atoms Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.1.1–2. Atom inventories and relative formula mass Only the named concept is connected. Practical work and the

complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1 (pp. 11–12). Formulae and formula-unit masses Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 4.2: Deterministic particle & mathematical plate for *Reading a chemical formula*. First Principle: Bulk matter cannot accumulate unbalanced electric charge. Therefore, positive and negative ions must combine in the smallest ratio where total positive charge + total negative charge = 0.

Historical & Experimental Evidence: Water (H_2O) vs. Hydrogen peroxide (H_2O_2)

Water (H_2O) is essential for life and extinguishes fires; pure hydrogen peroxide (H_2O_2) is a reactive oxidizer that decomposes into water and oxygen gas. That single subscript change—from 1 oxygen atom per molecule to 2 oxygen atoms per molecule—represents a completely different molecular structure and chemical substance.

Key Terms

Chemical coefficient
The multiplier written in front of a formula (like the 3 in 3 H_2O) that counts how many whole molecules or formula units are present.

Chemical subscript
The fixed atom count inside a formula (like the 2 in H_2O) that defines the substance's identity.

Worked Reasoning Trace

Held fixed: Formula H_2O (2 H atoms and 1 O atom per molecule).

Changed: Increased the multiplier (coefficient) from 1 to 3.

Model result: 3 × H_2O contains 3 × 2 = 6 H atoms and 3 × 1 = 3 O atoms across 3 water molecules.

Model boundary: You can never balance an equation by changing H_2O into H_2O_2, because H_2O_2 is hydrogen peroxide, not water!

Common Misconception Trap · Reading a chemical formula

Intuitive Trap: "I think 3 — because changes the formula of water into H_6O_3 molecules."

Particle Mechanism Refutation: "Wait — look at the particle ledger: 6. Multiplies the number of whole H_2O molecules by 3 (giving 3 × 2 = 6 H atoms and 3 × 1 = 3 O atoms) without altering the H_2O molecule itself."

Why Claim A fails: 3 H_2O means three separate H_2O molecules, not a single bonded H_6O_3 molecule. · Why Claim B holds up: Exact mechanism: a stoichiometric coefficient counts complete formula units or molecules, multiplying every subscript inside the formula.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
1 × H_2O (2 H, 1 O)	A single water molecule: 2 hydrogen atoms and 1 oxygen atom.	1 × H_2O. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.
3 × H_2O (6 H, 3 O)	Three water molecules: coefficient 3 multiplies every subscript in H_2O.	3 × H_2O. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.
1 × $MgCl_2$ via Mg^{2+} + Cl^- combiner	Combining Mg^{2+} (+2) with Cl^- (–1) requires 1 Mg^{2+} : 2 Cl^- ($MgCl_2$) for zero net charge.	1 × $MgCl_2$: 1 Mg^{2+} (+2 e) and 2 Cl^- (–2 e) balance to net charge 0. $MgCl_2$ is the electrically neutral formula-unit ratio in an extended ionic solid, not a discrete molecule.
1 × Al_2O_3 via Al^{3+} + O^{2-} combiner	Combining Al^{3+} (+3) with O^{2-} (–2) requires 2 Al^{3+} : 3 O^{2-} (Al_2O_3) so +6 and –6 balance.	1 × Al_2O_3: 2 Al^{3+} (+6 e) and 3 O^{2-} (–6 e) balance to net charge 0. Al_2O_3 is the electrically neutral formula-unit ratio in an extended ionic solid, not a discrete molecule.

Check Question: Three H_2O molecules contain how many H atoms? · **Transfer Challenge:** Transfer challenge: Explain to a younger student why '2 H_2O' and 'H_4O_2' have the exact same total count of H and O atoms, yet mean different things about how the atoms are grouped.

INVESTIGATION 4.3 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: BALANCE

Where did every atom go?

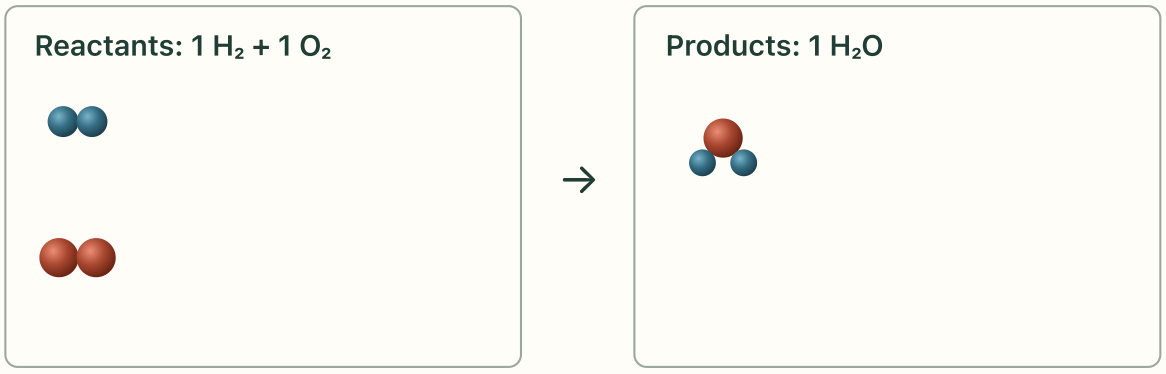
Accounting for every atom of every element across a reaction arrow

Ordinary chemical reactions rearrange atoms. Atom counts for each element must agree on the two sides of a balanced equation.

Bench Observation: A closed-system chemical reaction conserves mass to the accuracy normally used in introductory chemistry.

Investigation Task: Adjust coefficients to match both hydrogen and oxygen counts. Save an unbalanced and a balanced example.

Where did every atom go?



H atoms before: 2 vs. 2 · O atoms before: 2 vs. 1 ·

Atom accounting diagram across reaction systems. Matching counts conserve each element, which does not by itself prove reaction rate or feasibility.

Reconcile every element

Ordinary chemical reactions rearrange atoms. Each element has its own inventory.

The bookkeeping idea: Like reconciling a warehouse inventory: give each element its own row. Atoms before must equal atoms after. A coefficient orders whole bundles; changing a subscript changes what is inside the bundle.

Current explanatory plate
· intro-models-2

H atoms before 2 atoms
H atoms after 2 atoms
O atoms before 2 atoms
O atoms after 1 atoms

H: 2 before ↔ 2 after; difference 0 · O: 2 before ↔ 1 after; difference -1

Model note: Matching atom counts is necessary for a balanced equation. It does not establish that the reaction occurs.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.1 Supplement 5–8. Conserving atoms in symbol equations Only the named concept is connected.

Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.1.1. Conservation of mass and balanced equations Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Chemical Reactions and Equations (p. 4). Balanced equations Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 4.3: Deterministic particle & mathematical plate for *Where did every atom go?*. First Principle: Because chemical reactions only reconnect existing nuclei and electrons, every element must have the exact same atom count before and after reaction.

Historical & Experimental Evidence: Lavoisier's sealed-flask calcination experiments (1774–1789)

Antoine Lavoisier heated tin and mercury in sealed glass retorts and weighed the entire closed apparatus before and after reaction. Although a solid oxide crust formed and air inside was consumed, the total mass of the sealed vessel remained unchanged—demonstrating that chemical reactions neither create nor destroy atoms, but rearrange them.

Key Terms

Conservation of atoms
In an ordinary chemical reaction, the number of atoms of each element in the reactants equals the number of atoms of that same element in the products.

Balanced chemical equation
A symbolic equation whose stoichiometric coefficients give equal atom counts for every element on both sides of the reaction arrow.

Worked Reasoning Trace

Held fixed: Reaction species H_2, O_2 and H_2O.

Changed: Adjusted coefficients from (1, 1, 1) to (2, 1, 2).

Model result: At (1, 1, 1), O atoms are 2 before vs. 1 after (unbalanced). At (2, 1, 2), H atoms are 4 = 4 and O atoms are 2 = 2 (balanced).

Model boundary: Matching total atoms regardless of element (e.g., 6 atoms before = 6 atoms after) is invalid; each element (H and O) must balance individually.

Common Misconception Trap · Where did every atom go?

Intuitive Trap: "I think subscripts inside the given formulae — because changing subscripts (like turning H_2O into H_2O_2) until the numbers match."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Coefficients in front of formulae. Adjusting coefficients so every element has the exact same number of atoms before and after the reaction, without altering any substance's formula subscripts."

Why Claim A fails: Changing subscripts changes the chemical substance itself (H_2O_2 is hydrogen peroxide, not water). · Why Claim B holds up: Exact mechanism: atoms of each element are conserved in chemical reactions, and substance identities are fixed by their subscripts.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Unbalanced: 1 H_2 + 1 O_2 → 1 H_2O	H matches (2 = 2), but O is unbalanced (2 before vs. 1 after).	1 H_2 + 1 O_2 → 1 H_2O is not balanced. This is atom accounting, not evidence that a reaction occurs.
Balanced water: 2 H_2 + 1 O_2 → 2 H_2O	Lowest whole-number balance: 4 H and 2 O on both sides.	2 H_2 + 1 O_2 → 2 H_2O is balanced. This is atom accounting, not evidence that a reaction occurs.
Methane combustion: 1 CH_4 + 2 O_2 → 1 CO_2 + 2 H_2O	1 C, 4 H, and 4 O conserved across both products.	1 CH_4 + 2 O_2 → 1 CO_2 + 2 H_2O is balanced (C atoms: 1 before = 1 after). This is atom accounting, not evidence that a reaction occurs.
Ammonia synthesis: 1 N_2 + 3 H_2 → 2 NH_3	2 N and 6 H atoms conserved on both sides.	1 N_2 + 3 H_2 → 2 NH_3 is balanced. This is atom accounting, not evidence that a reaction occurs.

Check Question: Which may you change when balancing a given reaction? · **Transfer Challenge:** Transfer challenge: Nitrogen (N_2) and hydrogen (H_2) react to form ammonia (NH_3). Why is '1 N_2 + 1 H_2 → 1 NH_3' unbalanced, and what coefficients balance both N and H?

INVESTIGATION 4.4 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: BALANCE

What a balanced equation cannot tell us

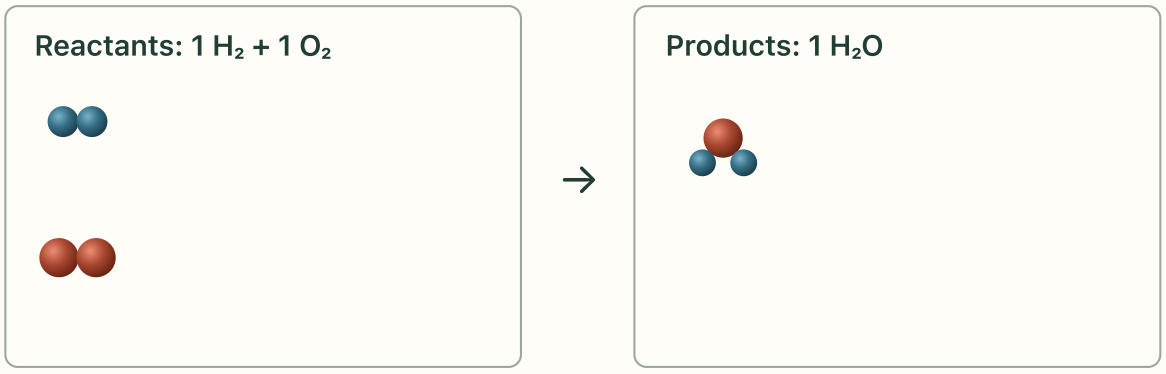
Separating stoichiometric accounting from reaction feasibility and speed

Balancing expresses atom accounting. It does not establish speed, conditions, pathway or whether the proposed reaction occurs.

Bench Observation: Real reaction claims require observations and knowledge of the substances and conditions.

Investigation Task: Find two balanced coefficient sets. Explain the smallest whole-number ratio and one question the equation cannot answer.

What a balanced equation cannot tell us



H atoms before: 2 vs. 2 · O atoms before: 2 vs. 1 ·

Atom accounting diagram across reaction systems. Matching counts conserve each element, which does not by itself prove reaction rate or feasibility.

Reconcile every element

Ordinary chemical reactions rearrange atoms. Each element has its own inventory.

The bookkeeping idea: Like reconciling a warehouse inventory: give each element its own row. Atoms before must equal atoms after. A coefficient orders whole bundles; changing a subscript changes what is inside the bundle.

Current explanatory plate
· intro-models-2

H atoms before 2 atoms
H atoms after 2 atoms
O atoms before 2 atoms
O atoms after 1 atoms

H: 2 before ↔ 2 after; difference 0 · O: 2 before ↔ 1 after; difference -1

Model note: Matching atom counts is necessary for a balanced equation. It does not establish that the reaction occurs.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.1 Supplement 5–8. Conserving atoms in symbol equations Only the named concept is connected.

Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.1.1. Conservation of mass and balanced equations Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Chemical Reactions and Equations (p. 4). Balanced equations Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 4.4: Deterministic particle & mathematical plate for *What a balanced equation cannot tell us*. First Principle: A balanced equation verifies that atom conservation is satisfied on paper, but experimental evidence and energetics are required to know whether the reaction actually proceeds.

Historical & Experimental Evidence: Room-temperature stability of H_2 + O_2 mixtures vs. sparked ignition

You can write the perfectly balanced equation $2\ H_2 + O_2 \rightarrow 2\ H_2O$ on paper in seconds. Yet if you mix pure hydrogen and oxygen gases in a dark glass bulb at room temperature, they can sit for years with no detectable water formation until a spark or platinum catalyst provides an accessible pathway. Furthermore, you can write balanced equations for reactions that are thermodynamically uphill and never proceed spontaneously.

Key Terms

Stoichiometric ratio
The simplest whole-number ratio of reactant and product molecules or moles in a balanced equation (such as 2 : 1 : 2 for H_2 : O_2 : H_2O).

Reaction feasibility & kinetics
Whether a reaction actually occurs under given conditions (thermodynamics) and how fast it proceeds (kinetics)—neither of which is shown by atom counting alone.

Worked Reasoning Trace

Held fixed: Balanced ratio 2 : 1 : 2.

Changed: Compared coefficients (2, 1, 2) with (4, 2, 4) and (6, 3, 6).

Model result: All three conserve H and O atoms in the exact 2 : 1 : 2 ratio.

Model boundary: The balanced equation says 'if H_2 and O_2 react completely to form only H_2O, they do so in a 2:1:2 molecule ratio'—it does not prove the reaction is fast or happens without a spark.

Common Misconception Trap · What a balanced equation cannot tell us

Intuitive Trap: "I think a fast reaction at room temperature — because guarantees that collisions between reactants have zero activation barrier."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Consistent atom accounting. Verifies only that atom counts of each element are conserved between reactants and products, saying nothing about activation barriers or thermodynamic feasibility."

Why Claim A fails: Even balanced, thermodynamically favourable reactions (like 2 H_2 + O_2 → 2 H_2O) can have high activation barriers at room temperature. · Why Claim B holds up: Exact mechanism: stoichiometric balance is a conservation accounting constraint, not a measure of reaction rate or free-energy driving force.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Simplest ratio: 2 H_2 + 1 O_2 → 2 H_2O	4 H and 2 O on each side.	2 H_2 + 1 O_2 → 2 H_2O is balanced. This is atom accounting, not evidence that a reaction occurs.
Doubled: 4 H_2 + 2 O_2 → 4 H_2O	8 H and 4 O on each side—the same 2:1:2 stoichiometric proportion.	4 H_2 + 2 O_2 → 4 H_2O is balanced. This is atom accounting, not evidence that a reaction occurs.
Tripled: 6 H_2 + 3 O_2 → 6 H_2O	12 H and 6 O on each side.	6 H_2 + 3 O_2 → 6 H_2O is balanced. This is atom accounting, not evidence that a reaction occurs.

Check Question: A balanced equation is evidence of… · **Transfer Challenge:** Transfer challenge: Someone writes '2 CO_2 → 2 CO + O_2' and claims 'Because the equation is balanced, carbon dioxide in our classroom air must be decomposing into toxic carbon monoxide right now.' How do you refute this claim?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 4 · The Forbidden Subscript Fix

To balance the combustion of methane (CH_4 + O_2 → CO_2 + H_2O), a student writes: 'CH_4 + O_3 → CO_2 + H_4O' so both sides have 1 C, 4 H, and 3 O!

- Clue 1 (Substance Identity): Water is H_2O (2 H bonded to 1 O) and dioxygen is O_2; O_3 is ozone and H_4O is not water.
- Clue 2 (Coefficient Rule): Balancing may only adjust whole-molecule multipliers (coefficients) in front of true chemical formulas.
- Clue 3 (Atom Audit): With 1 CH_4 on the left (1 C, 4 H), forming 1 CO_2 and 2 H_2O on the right requires 1 C, 4 H, and 4 O atoms.

Resolution: Subscripts define the chemical identity of each molecule (O_2, H_2O, CO_2); balancing adjusts only the stoichiometric coefficients (1 CH_4 + 2 O_2 → 1 CO_2 + 2 H_2O).

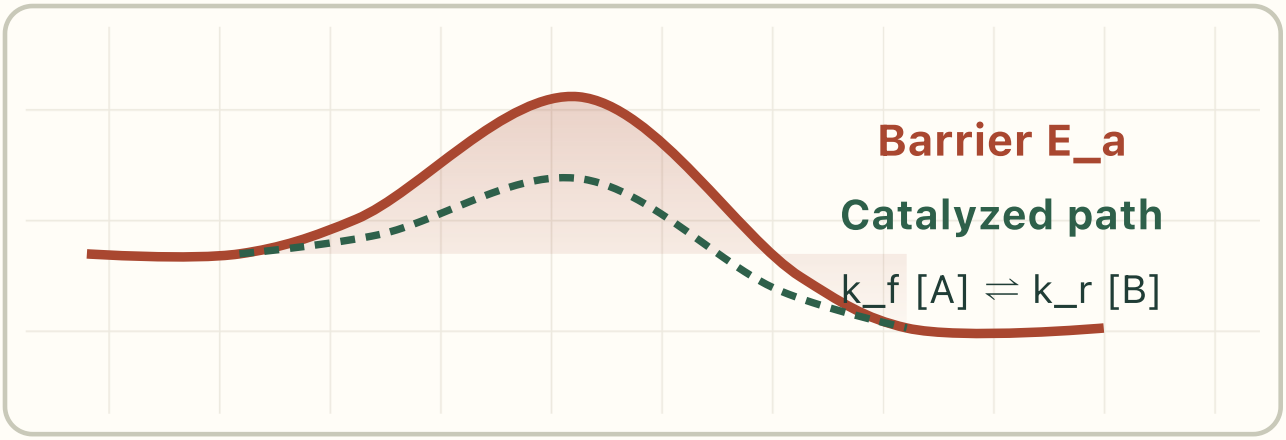
What our upgraded model now explains: Distinguishes chemical bond rearrangement from physical change, separates coefficients from subscripts, and audits atom conservation across balanced equations. · **Next boundary:** A balanced equation on paper cannot tell us whether a reaction releases heat, how high its activation barrier is, or whether it stops at a reversible equilibrium → motivating Chapter 5 (Transformations).

VOLUME II · CHANGE, QUANTITIES & DRIVING FORCES · CHAPTER 5

Chapter 5: Energy, rates and equilibrium

Separate how much energy changes from how quickly a change happens.

Chapter 5 Prelude · Why Doesn't This Page Burst into Flame Right Now?: Paper and oxygen gas react exothermically to form CO_2 and H_2O, releasing a large amount of heat. Yet this book can sit in air for centuries without catching fire! Even more puzzling, some reactions release heat while others freeze water to a beaker, and many reactions refuse to go to 100% completion no matter how long you wait. How do bond energies, activation barriers, catalysts, and dynamic equilibrium govern every chemical transformation?



Macroscopic Bench: Calorimeter temperature changes (ΔT), reaction rates over time, catalyst acceleration, and steady equilibrium mixtures.

Submicroscopic Particle Model: Bond-breaking cost (+E_break) vs. bond-making payback (–E_make), collision activation barrier (E_a), and equal opposing fluxes (k_f[A] = k_r[B]).

Symbolic Notation: Enthalpy ΔH = E_break – E_make, activation energy E_a, rate laws, and equilibrium constant K = k_f / k_r.

INVESTIGATION 5.1 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: ENERGY

Where does the energy go?

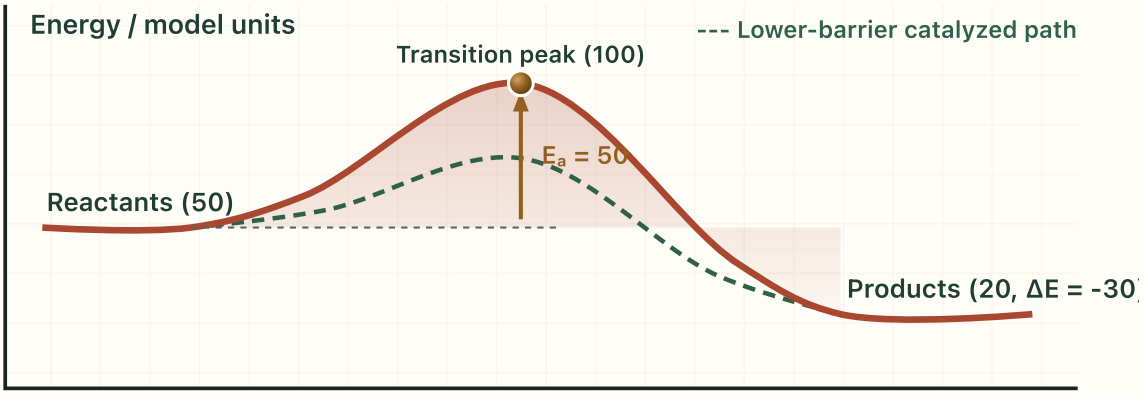
Tracking energy conservation between the reacting chemical system and its surroundings

Choose a system and its surroundings. Energy leaving the system can warm the surroundings; energy entering it can come from the surroundings.

Bench Observation: Some reactions warm their surroundings while others absorb energy. The system boundary matters to the sign of the change.

Investigation Task: Compare products below and above reactants. Describe the direction of energy transfer in each schematic case.

Where does the energy go?



Smooth reaction-coordinate energy profile showing activation barrier E_a, net energy change ΔE, and a lower-barrier catalyzed pathway.

Energy levels & barrier account

Compare levels using one reference. An activation barrier is not the net energy change.

The bookkeeping idea: Like comparing opening and closing balances in the same currency: net change = final level – initial level. The temporary amount needed to cross the barrier is a separate requirement, not the net change.

Current explanatory plate · intro-models-2	
Product minus reactant energy	-30 model energy units
Forward barrier	50 model energy units
Reverse barrier	80 model energy units

ΔE = products – reactants = 20 – 50 = -30 model units; E_a = peak – reactants = 100 – 50 = 50 model units

Model note: Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 5.1 Core 1–3; Supplement 4–6. Energy changes, profiles and activation barriers Only the named concept is

connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.5.1.1–2. Energy changes and reaction profiles Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Chemical Reactions and Equations (p. 4). Exothermic/endothermic changes; barriers are enrichment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 5.1: Deterministic particle & mathematical plate for *Where does the energy go?*. First Principle: Energy is strictly conserved between the chemical system and its surroundings—if products sit at lower potential energy than reactants, the difference warms the surroundings.

Historical & Experimental Evidence: Calorimetry measures temperature changes in the surroundings

In a calorimeter, a chemical reaction takes place inside a chamber surrounded by a known mass of water and a thermometer. Because we cannot put a thermometer inside a chemical bond, we measure the temperature change of the surrounding water: if the water warms up, potential energy left the reacting chemical system and transferred into the thermal energy of the surroundings.

Key Terms

Chemical system vs. Surroundings
The system is the specific set of reactant and product particles being studied; the surroundings are everything else (the solvent water, beaker, thermometer and air) that exchanges energy with it.

Exothermic vs. Endothermic
An energy-releasing (exothermic) process lowers the system's energy and transfers energy to the surroundings; an energy-absorbing (endothermic) process takes energy in from the surroundings.

Worked Reasoning Trace

Held fixed: Reactant energy at 50 model units and peak at 100 model units.

Changed: Compared Product energy at 20 units (below reactants) vs. 70 units (above reactants).

Model result: When products are at 20 units, $\Delta E = 20 - 50 = -30$ units (energy released to surroundings). When products are at 70 units, $\Delta E = 70 - 50 = +20$ units (energy absorbed from surroundings).

Model boundary: Total energy (system + surroundings) is conserved; an exothermic reaction does not 'create' energy out of nothing.

Common Misconception Trap · Where does the energy go?

Intuitive Trap: "I think lose all energy because it is destroyed — because energy disappears from the universe so the products can rest."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Gain energy from the reacting system. Total energy is conserved, so the decrease in chemical potential energy transfers into the kinetic/thermal energy of the surroundings."

Why Claim A fails: Energy is conserved, never destroyed. · Why Claim B holds up: Exact mechanism: by the first law of thermodynamics, potential energy lost by the chemical system is gained by the surroundings (warming the calorimeter).

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Energy-releasing (Reactants 50 → Products 20)	Products sit 30 units lower than reactants; net energy transfers to the surroundings.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.
Energy-absorbing (Reactants 30 → Products 70)	Products sit 40 units higher than reactants; net energy is absorbed from the surroundings.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.
Equal levels (Reactants 50 → Products 50)	Zero net difference between reactants and products, though a 50-unit barrier remains.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.

Check Question: In an energy-releasing reaction, the surroundings can… · **Transfer Challenge:** Transfer challenge: When solid ammonium nitrate dissolves in a cold pack, the water in the pack becomes cold to the touch. Which has higher energy in a schematic profile—the starting state or the dissolved state—and which way did energy flow?

INVESTIGATION 5.2 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: ENERGY

Why is there an energy barrier?

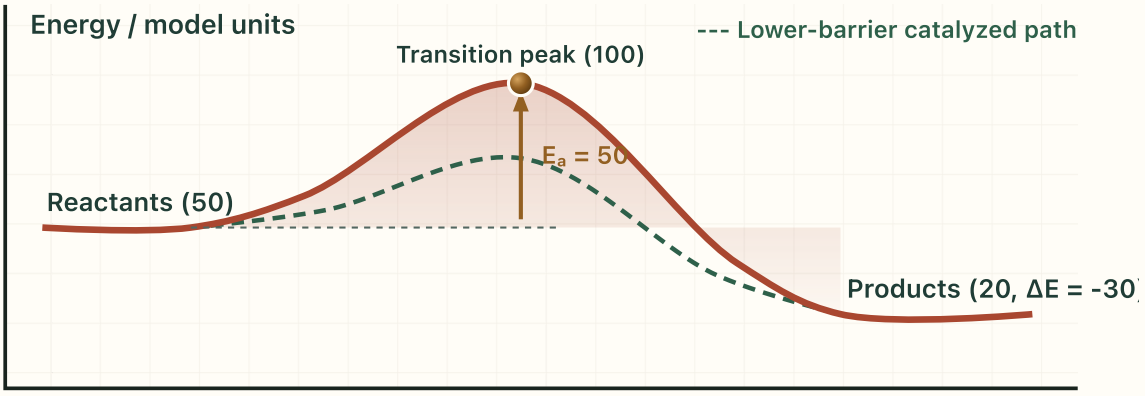
Why an energy-releasing reaction still requires crossing an activation barrier

A pathway can pass through a higher-energy arrangement even when products are lower in energy than reactants. Overall energy change and barrier height answer different questions.

Bench Observation: A reaction that releases energy need not proceed rapidly under every condition.

Investigation Task: Hold reactants and products fixed while changing the peak. Compare the forward and reverse barriers.

Why is there an energy barrier?



Smooth reaction-coordinate energy profile showing activation barrier E_a, net energy change ΔE, and a lower-barrier catalyzed pathway.

Energy levels & barrier account

Compare levels using one reference. An activation barrier is not the net energy change.

The bookkeeping idea: Like comparing opening and closing balances in the same currency: net change = final level – initial level. The temporary amount needed to cross the barrier is a separate requirement, not the net change.

Current explanatory plate · intro-models-2	
Product minus reactant energy	-30 model energy units
Forward barrier	50 model energy units
Reverse barrier	80 model energy units

ΔE = products – reactants = 20 – 50 = -30 model units; E_a = peak – reactants = 100 – 50 = 50 model units

Model note: Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 5.1 Core 1–3; Supplement 4–6. Energy changes, profiles and activation barriers Only the named concept is

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AQA GCSE Chemistry 8462 · Partial connection: 4.5.1.1–2. Energy changes and reaction profiles Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Chemical Reactions and Equations (p. 4). Exothermic/endothermic changes; barriers are enrichment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

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Figure 5.2: Deterministic particle & mathematical plate for *Why is there an energy barrier?*. First Principle: Colliding particles need enough thermal kinetic energy to cross the activation energy barrier (peak minus start), which is independent of the net energy difference between reactants and products.

Historical & Experimental Evidence: Collision energy thresholds and temperature sensitivity (Arrhenius)

Paper and wood in your room are surrounded by oxygen gas, and burning them to CO_2 and H_2O releases a large amount of energy. Why doesn't this textbook burst into flames at room temperature? Svante Arrhenius explained in 1889 that reactant molecules must collide with enough kinetic energy to stretch and distort existing bonds through a high-energy transition state—the activation energy barrier.

Key Terms

Forward activation barrier
The energy difference from the reactant level up to the peak of the pathway (E_peak – E_reactants).

Reverse activation barrier
The energy difference from the product level up to the peak of the pathway (E_peak – E_products).

Worked Reasoning Trace

Held fixed: Reactant energy = 50 units and Product energy = 20 units (overall ΔE = –30 units).

Changed: Raised Peak energy from 100 units to 140 units.

Model result: Forward barrier increases from 50 to 90 units and reverse barrier increases from 80 to 120 units, while overall change stays fixed at –30 units.

Model boundary: The horizontal axis of an energy profile is 'reaction progress' (bond stretching/forming geometry), not clock time in seconds.

Common Misconception Trap · Why is there an energy barrier?

Intuitive Trap: "I think the identity of every product — because changes the net energy released between reactants and products."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Barrier heights but not the overall energy difference. Increases the minimum collision energy required to cross the barrier in both directions without changing the net energy difference ΔE between endpoints."

Why Claim A fails: Endpoint energies were held constant, so ΔE stays constant. · Why Claim B holds up: Exact mechanism: ΔE = E_products – E_reactants depends only on the initial and final states, whereas the activation barrier depends on the peak height.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Low barrier (Peak 90, R 50, P 20)	Forward barrier = 40 units; overall difference = −30 units.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.
Medium barrier (Peak 110, R 50, P 20)	Forward barrier = 60 units; overall difference still −30 units.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.
High barrier (Peak 150, R 50, P 20)	Forward barrier = 100 units; overall difference still −30 units.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.

Check Question: Lowering only the peak changes… · **Transfer Challenge:** Transfer challenge: In a reaction where reactants are at 40 units, products at 10 units and the peak at 100 units, calculate both the forward barrier and the reverse barrier. Why is the reverse barrier larger than the forward barrier?

INVESTIGATION 5.3 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: EQUILIBRIUM

How quickly does change happen?

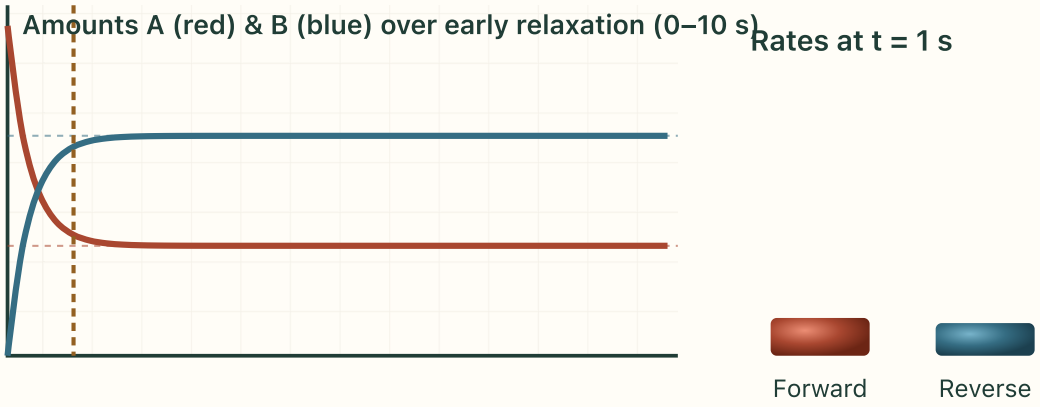
Separating how fast concentrations change from where the system settles

Rate describes change per time. It is different from the eventual amount present. This synthetic first-order system lets us compare amounts with forward and reverse rates.

Bench Observation: In real systems, rates can depend on temperature, concentration, surface area and pathway. This model only changes its stated rate constants.

Investigation Task: Compare t = 0 and later times. Then multiply both rate constants by two and compare the approach to the same ratio.

How quickly does change happen?



Synthetic closed A ⇌ B relaxation. At dynamic equilibrium, forward and reverse rate bars are equal even when amounts A and B differ.

Stock versus flow rate

At equilibrium the opposing rates match. The inventories need not be equal. Added or removed material crosses the system boundary.

The bookkeeping idea: Like a stockroom with deliveries and dispatches: stock is how much is present; flow rate is how much changes per second. Equal opposing rates hold stocks steady without making the two stocks equal.

Current explanatory plate · intro-models-2

Amount A / model units	3.665247122452426 model units
Amount B / model units	6.334752877547574 model units
Forward rate / model units per s	7.330494244904852 model units/s
Reverse rate / model units per s	6.334752877547574 model units/s

Stock A + stock B = 3.66525 + 6.33475 = 10 model units. Forward rate 7.33049; reverse rate 6.33475 model units/s.

Model note: Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 6.2 Core 1–3. Rate factors and changes over time A two-state relaxation analogy cannot establish a real reaction rate law. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.6.1.1–3. Rates and collisions A two-state relaxation analogy cannot establish a real reaction rate law. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 5.3: Deterministic particle & mathematical plate for *How quickly does change happen?*. First Principle: Reaction rate measures how many particles transform per second (collision frequency × barrier-crossing probability), which is distinct from the final equilibrium ratio.

Historical & Experimental Evidence: Spectrophotometric and pressure tracking of concentration over time

Chemists measure reaction rates by recording a physical property—such as colour absorbance in a spectrophotometer or gas pressure—at regular time intervals (e.g., every second). Near t = 0, when reactant A is plentiful and product B is absent, the forward rate is steepest; as A is consumed and B accumulates, the net rate of change slows down.

Key Terms

Reaction rate
The speed of a chemical change, measured as the change in amount or concentration per unit time (e.g., model units per second).

Rate constant (k)
A proportionality factor linking reaction rate to the current amount of reactant at fixed conditions.

Worked Reasoning Trace

Held fixed: Rate constants $k_f = 2\ s^{-1}$ and $k_r = 1\ s^{-1}$.

Changed: Moved Time t from 0 s to 1 s to 20 s.

Model result: At t = 0 s, Amount A = 10, Amount B = 0, forward rate = 20 s^{-1}, reverse rate = 0 s^{-1}. By t = 20 s, A = 3.33, B = 6.67, and both rates equal 6.67 s^{-1}.

Model boundary: This model is a closed synthetic first-order reversible system (A ⇌ B); real reactions can follow more complex rate laws and mechanisms.

Common Misconception Trap · How quickly does change happen?

Intuitive Trap: "I think only the final product amount — because the final equilibrium ratio of products to reactants at infinite time."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Change per unit time. How many reactant-to-product transformations occur per unit time (change in amount divided by change in time)."

Why Claim A fails: Equilibrium extent is governed by the ratio of forward and reverse rate constants, not speed alone. · Why Claim B holds up: Exact mechanism: rate is a derivative/ratio with respect to time (units per second), distinct from equilibrium extent.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Initial start (t = 0 s, k_f=2, k_r=1)	All 10 units are A; forward rate is 20 units/s and reverse rate is 0.	Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.
Early change (t = 1 s, k_f=2, k_r=1)	Amounts are still shifting; forward rate (7.33/s) exceeds reverse rate (6.33/s).	Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.
Faster constants (t = 1 s, k_f=4, k_r=2)	Doubling both rate constants reaches the same 1:2 equilibrium balance much sooner.	Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.

Check Question: Reaction rate describes… · **Transfer Challenge:** Transfer challenge: Compare (k_f = 2, k_r = 1) with (k_f = 4, k_r = 2) at t = 100 s. Why did doubling both rate constants leave the final amounts of A and B unchanged?

INVESTIGATION 5.4 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: ENERGY

A different pathway

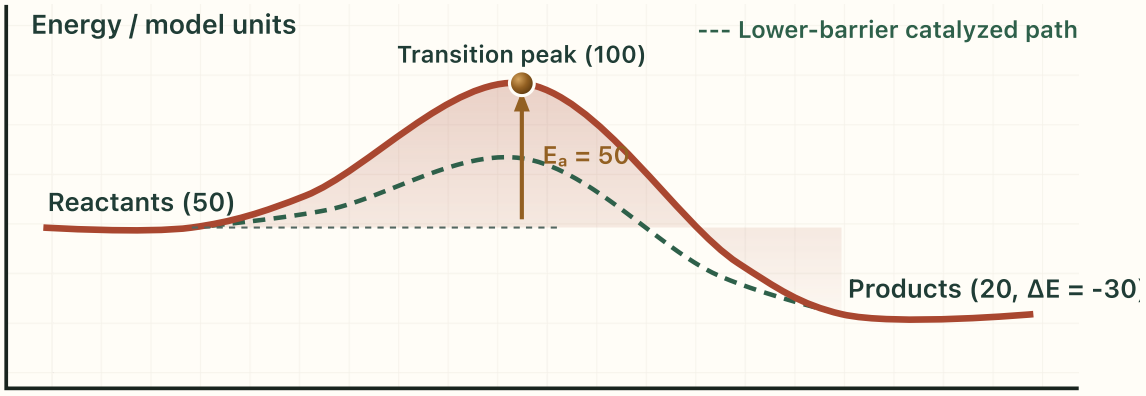
How an alternative lower-barrier pathway speeds both forward and reverse rates

A catalyst offers a different reaction pathway and is regenerated in the overall process. It can increase rates without changing the initial and final energy levels.

Bench Observation: At fixed conditions, a catalyst does not change the equilibrium constant or the equilibrium composition.

Investigation Task: Lower the schematic peak without moving either endpoint. Explain both directions, then compare with the reversible-rate model.

A different pathway



Smooth reaction-coordinate energy profile showing activation barrier E_a, net energy change ΔE, and a lower-barrier catalyzed pathway.

Energy levels & barrier account

Compare levels using one reference. An activation barrier is not the net energy change.

The bookkeeping idea: Like comparing opening and closing balances in the same currency: net change = final level – initial level. The temporary amount needed to cross the barrier is a separate requirement, not the net change.

Current explanatory plate · intro-models-2	
Product minus reactant energy	-30 model energy units
Forward barrier	50 model energy units
Reverse barrier	80 model energy units

ΔE = products – reactants = 20 – 50 = -30 model units; E_a = peak – reactants = 100 – 50 = 50 model units

Model note: Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 6.2 Core 2; Supplement 7. Catalysts and activation barriers Only the named concept is connected. Practical

work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.6.1.4. Catalysts Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Chemical Reactions and Equations (p. 4). Exothermic/endothermic changes; barriers are enrichment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 5.4: Deterministic particle & mathematical plate for *A different pathway*. First Principle: A catalyst provides an alternative pathway with a lower activation peak, speeding up both forward and reverse directions equally without shifting the endpoint energy difference.

Historical & Experimental Evidence: Enzymes and heterogeneous catalysts accelerate attainment of equilibrium

In living cells, the enzyme carbonic anhydrase speeds up the reversible hydration of CO_2 by more than a millionfold without altering the equilibrium ratio of dissolved CO_2 to bicarbonate. Because a catalyst provides a lower-energy transition pathway between the same reactant and product states, it lowers both the forward barrier and the reverse barrier by the exact same amount.

Key Terms

Catalyst
A substance that increases the rate of a chemical reaction by providing an alternative pathway with a lower activation barrier, without being consumed in the overall reaction.

Worked Reasoning Trace

Held fixed: Reactant energy = 50 units and Product energy = 20 units.

Changed: Lowered Peak energy from 130 units (uncatalyzed) to 90 units (catalyzed).

Model result: Forward barrier drops by 40 units (from 80 to 40) AND reverse barrier drops by the same 40 units (from 110 to 70); ΔE stays –30 units.

Model boundary: Because both barriers drop by the same amount, a catalyst cannot make an unfavourable equilibrium produce a higher equilibrium fraction of product.

Common Misconception Trap · A different pathway

Intuitive Trap: "I think the equilibrium constant — because lowering the energy of the products while leaving reactants unchanged."

Particle Mechanism Refutation: "Wait — look at the particle ledger: How quickly equilibrium is approached. Providing an alternative reaction pathway with a lower transition-state peak, which reduces both forward and reverse barriers by the exact same amount."

Why Claim A fails: A catalyst cannot change the energies of the uncatalyzed reactants or products. · Why Claim B holds up: Exact mechanism: lowering the pathway peak equally from both sides speeds up forward and reverse rates by the same factor, reaching the same equilibrium faster.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Uncatalyzed pathway (Peak 130)	Forward barrier = 80 units; reverse barrier = 110 units.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.
Catalyzed pathway (Peak 90)	Both forward and reverse barriers drop by 40 units; ΔE stays −30 units.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.

Check Question: At fixed conditions a catalyst changes… · **Transfer Challenge:** Transfer challenge: Why is it impossible for a catalyst at fixed temperature to lower the forward activation barrier by 30 units without also lowering the reverse activation barrier by 30 units?

INVESTIGATION 5.5 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: EQUILIBRIUM

A balance that keeps moving

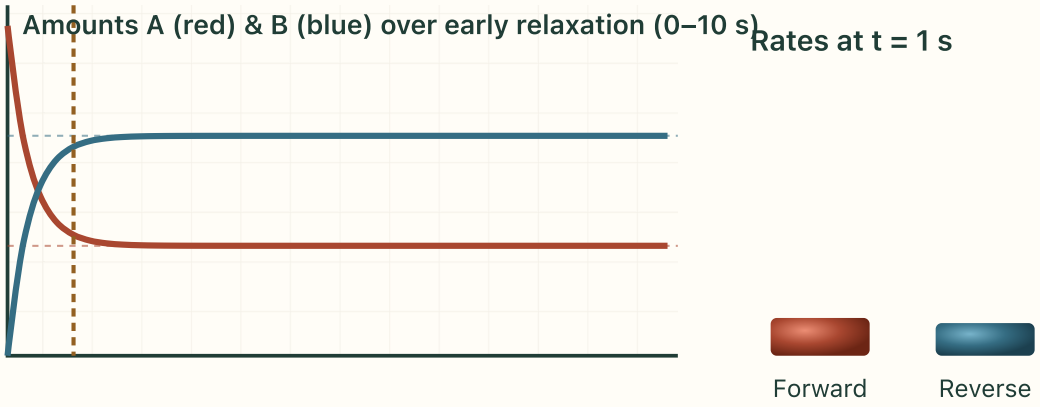
Equal opposing reaction rates coexist with unequal constant amounts

At dynamic equilibrium, forward and reverse reactions continue at equal rates. The amounts can stay constant without being equal.

Bench Observation: A closed reversible system can settle at a stable composition while molecular changes continue.

Investigation Task: Use forward constant 2 and reverse constant 1. Compare amounts and rates after a long time; explain why unequal amounts are consistent with equilibrium.

A balance that keeps moving



Synthetic closed A ⇌ B relaxation. At dynamic equilibrium, forward and reverse rate bars are equal even when amounts A and B differ.

Stock versus flow rate

At equilibrium the opposing rates match. The inventories need not be equal. Added or removed material crosses the system boundary.

The bookkeeping idea: Like a stockroom with deliveries and dispatches: stock is how much is present; flow rate is how much changes per second. Equal opposing rates hold stocks steady without making the two stocks equal.

Current explanatory plate · intro-models-2

Amount A / model units	3.665247122452426 model units
Amount B / model units	6.334752877547574 model units
Forward rate / model units per s	7.330494244904852 model units/s
Reverse rate / model units per s	6.334752877547574 model units/s

Stock A + stock B = 3.66525 + 6.33475 = 10 model units. Forward rate 7.33049; reverse rate 6.33475 model units/s.

Model note: Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 6.3 Supplement 3–4. Equal rates and responses to changed conditions Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.6.2.3–7 (conditions: HT). Equilibrium and changing conditions Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 5.5: Deterministic particle & mathematical plate for *A balance that keeps moving*. First Principle: At dynamic equilibrium, opposing forward and reverse reactions continue ceaselessly at equal rates (k_f · A = k_r · B), locking macroscopic amounts into a steady statistical ratio.

Historical & Experimental Evidence: Isotopic tracer exchange at chemical equilibrium

How do we know a system at chemical equilibrium hasn't simply stopped reacting? When chemists take a mixture already at equilibrium—such as $N_2 + 3\ H_2 \rightleftharpoons 2\ NH_3$—and inject a tiny trace of deuterium-labelled hydrogen (D_2), the macroscopic concentrations of hydrogen, nitrogen and ammonia stay completely constant, yet deuterium atoms rapidly appear inside NH_2D and NHD_2 molecules! Forward and reverse reactions continue ceaselessly at equal rates.

Key Terms

Dynamic equilibrium
The state of a closed reversible system where forward and reverse processes occur at equal non-zero rates, keeping macroscopic amounts constant.

Worked Reasoning Trace

Held fixed: Forward rate constant $k_f = 2\ s^{-1}$ and reverse rate constant $k_r = 1\ s^{-1}$.

Changed: Inspected the system at long time t = 100 s.

Model result: Amount A settles at 3.33 units and Amount B settles at 6.67 units (unequal!), while Forward rate = 2 × 3.33 = 6.67 units/s equals Reverse rate = 1 × 6.67 = 6.67 units/s.

Model boundary: Equal rates at equilibrium almost never mean equal amounts of A and B—amounts are only equal in the special case where $k_f = k_r$.

Common Misconception Trap · A balance that keeps moving

Intuitive Trap: "I think reactant and product amounts in every reaction — because all molecular collisions and chemical reactions come to a complete halt."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Forward and reverse rates. Reactant and product particles continue transforming in both directions at equal non-zero rates, so macroscopic amounts stay constant (and usually unequal)."

Why Claim A fails: Isotopic tracer experiments prove forward and reverse reactions continue ceaselessly at equilibrium. · Why Claim B holds up: Exact mechanism: $k_f \times A = k_r \times B$ means opposing fluxes balance continuously, even when $A \neq B$ (whenever $k_f \neq k_r$).

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
k_f = 2, k_r = 1 at t = 100 s (B favoured)	Amounts are unequal (A = 3.33, B = 6.67), but rates are equal (6.67/s = 6.67/s).	Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.
k_f = 1, k_r = 4 at t = 100 s (A favoured)	Amounts are A = 8.0, B = 2.0; both forward and reverse rates equal 8.0/s.	Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.
Before equilibrium: k_f = 2, k_r = 1 at t = 0 s	Not at equilibrium yet: forward rate (20/s) is greater than reverse rate (0/s).	Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.

Check Question: At dynamic equilibrium, which quantities are equal? · **Transfer Challenge:** Transfer challenge: Set $k_f = 1\ s^{-1}$ and $k_r = 3\ s^{-1}$ at t = 100 s. Explain why Amount A is three times larger than Amount B even though the forward and reverse rates are exactly equal.

INVESTIGATION 5.6 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: BONDENERGY

Bond-breaking cost vs. bond-making payback

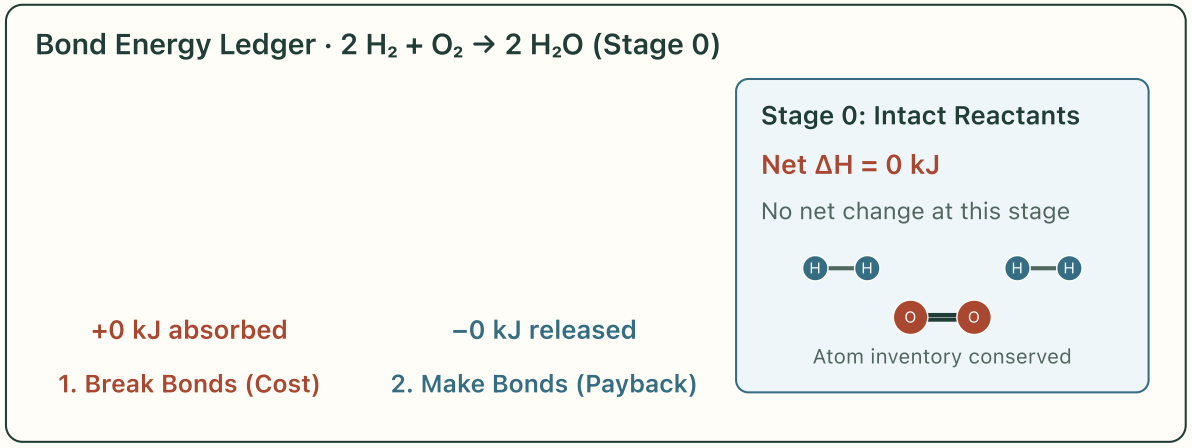
Curing the #1 chemistry misconception: breaking bonds always costs energy; making bonds releases energy

Breaking existing chemical bonds always requires an energy input (+E_break > 0, endothermic) to overcome electrostatic attraction. Forming new chemical bonds always releases energy (–E_make < 0, exothermic). Net reaction enthalpy ΔH is the ledger balance: ΔH = E_break – E_make.

Bench Observation: Combustion of hydrogen ($2\ H_2 + O_2 \rightarrow 2\ H_2O$) releases 484 kJ only because forming four strong O–H bonds releases more energy (1852 kJ) than breaking two H–H and one O=O bond costs (1368 kJ).

Investigation Task: Step through Stage 0 (Reactants), Stage 1 (Separated atoms—bonds broken), and Stage 2 (Products—bonds formed) for $2\ H_2 + O_2 \rightarrow 2\ H_2O$ and $N_2 + O_2 \rightarrow 2\ NO$.

Bond-breaking cost vs. bond-making payback



Breaking bonds always requires energy input (+E_break); forming bonds always releases energy (–E_make). Net ΔH = +E_break – E_make.

Energy input, release & net change

Breaking bonds requires energy; forming bonds releases energy. Use the reaction as written and the stated bond-energy approximation.

The bookkeeping idea: Like recording payments and receipts: breaking bonds requires an energy input; forming bonds releases energy. Net input = energy required – energy released, for the reaction amount stated. The surroundings receive what the reacting system releases.

Current explanatory plate · intro-models-2

Energy absorbed breaking bonds (+E_break) / kJ	0 kJ
Energy released forming bonds (E_make) / kJ	0 kJ
Cumulative enthalpy change ΔH / kJ	0 kJ

Current energy account = input – release = 0 – 0 = 0 kJ, for the reaction amount as written. Bond energies are approximate; this is not an activation barrier.

Model note: 2 H_2 + O_2 → 2 H_2O (Stage 0): Breaking 2 H–H (2×436) + 1 O=O (498) absorbs +1,370 kJ; Forming 4 O–H bonds (4×463.5) releases –1,854 kJ. Breaking chemical bonds ALWAYS requires energy input (+1370 kJ); a reaction is exothermic (ΔH = -484 kJ) only when forming stronger product bonds releases more energy than breaking reactant bonds cost.

Scientific inputs · scientific-values-2

$2\ H_2 + O_2 \rightarrow 2\ H_2O$: bond-break teaching estimate: 1370 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input. Checked 2026-10-28. Source Working: Breaking 2 H–H (2×436) + 1 O=O (498) absorbs +1,370 kJ

$2\ H_2 + O_2 \rightarrow 2\ H_2O$: bond-make teaching estimate: 1854 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input. Checked 2026-10-28. Source Working: Forming 4 O–H bonds (4×463.5) releases –1,854 kJ

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 5.1 Supplement 7–8. Bond breaking, bond making and energy estimates Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.5.1.3 (HT). Reaction estimates from bond energies Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 5.6: Deterministic particle & mathematical plate for *Bond-breaking cost vs. bond-making payback*. First Principle: Breaking chemical bonds always absorbs energy (+E_break > 0) and forming bonds always releases energy (–E_make < 0); net reaction enthalpy is ΔH = E_break – E_make.

Historical & Experimental Evidence: Spectroscopic bond dissociation enthalpies and reaction calorimetry

A widespread everyday myth claims that 'energy is stored inside chemical bonds and released when a bond breaks.' Physics and spectroscopy prove the exact opposite: because a chemical bond is an electrostatic attraction holding nuclei and electrons in a potential energy well, pulling bonded atoms apart ALWAYS requires an energy input (+E_break > 0). Energy is released (–E_make < 0) only when separated atoms snap into stronger, lower-energy product bonds!

Key Terms

Bond dissociation enthalpy (+E_break)
The energy required (always positive / endothermic) to break 1 mole of a given covalent bond in the gas phase into separated atoms.

Bond-energy enthalpy ledger (ΔH)
Net reaction enthalpy estimated as ΔH ≈ ΣE_break(reactant bonds broken) – ΣE_make(product bonds formed).

Worked Reasoning Trace

Held fixed: Reaction system 0: 2 H_2 + O_2 → 2 H_2O.

Changed: Stepped Reaction stage from Stage 1 (Separated atoms, +1368 kJ input) to Stage 2 (Formed 2 H_2O, –1852 kJ payback).

Model result: Breaking two H–H bonds (2 × 436) and one O=O bond (496) absorbs +1368 kJ; forming four O–H bonds (4 × 463) releases –1852 kJ, giving net ΔH = –484 kJ (exothermic).

Model boundary: Average bond enthalpies apply to gas-phase molecules and are approximations when comparing different molecular environments.

Common Misconception Trap · Bond-breaking cost vs. bond-making payback

Intuitive Trap: "I think breaking chemical bonds directly releases stored energy like snapping a glowstick — because reactant bonds break and release the heat that was trapped inside those bonds."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Breaking bonds always absorbs energy (+E_break), and forming bonds always releases energy (–E_make); an exothermic reaction releases more energy forming new bonds than it spent breaking old ones. The energy released when separated atoms form stronger, lower-potential-energy product bonds (–E_make) exceeds the energy absorbed to break the original reactant bonds (+E_break)."

Why Claim A fails: Breaking a bond requires pulling attracted nuclei and electrons apart against electrostatic force, which always absorbs energy. · Why Claim B holds up: Exact mechanism: breaking bonds always costs energy (+E_break > 0) and making bonds always releases energy (–E_make < 0); net ΔH = E_break – E_make < 0 when E_make > E_break.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
$2\ H_2 + O_2 \rightarrow 2\ H_2O$: Stage 1 (Break bonds: +1368 kJ)	Pulling 2 H–H and 1 O=O apart into 4 H and 2 O atoms requires +1368 kJ of energy input.	$2\ H_2 + O_2 \rightarrow 2\ H_2O$ (Stage 1): Breaking 2 H–H (2×436) + 1 O=O (498) absorbs +1,370 kJ; Forming 4 O–H bonds (4×463.5) releases −1,854 kJ. Breaking chemical bonds ALWAYS requires energy input (+1370 kJ); a reaction is exothermic (ΔH = -484 kJ) only when forming stronger product bonds releases more energy than breaking reactant bonds cost.
$2\ H_2 + O_2 \rightarrow 2\ H_2O$: Stage 2 (Make 4 O–H: −1852 kJ, ΔH = −484 kJ)	Forming 4 strong O–H bonds releases 1852 kJ, yielding a net exothermic release of −484 kJ.	$2\ H_2 + O_2 \rightarrow 2\ H_2O$ (Stage 2): Breaking 2 H–H (2×436) + 1 O=O (498) absorbs +1,370 kJ; Forming 4 O–H bonds (4×463.5) releases −1,854 kJ. Breaking chemical bonds ALWAYS requires energy input (+1370 kJ); a reaction is exothermic (ΔH = -484 kJ) only when forming stronger product bonds releases more energy than breaking reactant bonds cost.
$N_2 + O_2 \rightarrow 2\ NO$: Endothermic (+1444 break vs. −1260 make, ΔH = +184 kJ)	Breaking the ultra-strong N≡N triple bond (945 kJ) and O=O (499 kJ) costs more than forming 2 N=O bonds releases!	$N_2 + O_2 \rightarrow 2\ NO$ (Stage 2): Breaking 1 N≡N (945) + 1 O=O (498) absorbs +1,443 kJ; Forming 2 N=O bonds (2×631.5) releases −1,263 kJ. Breaking chemical bonds ALWAYS requires energy input (+1443 kJ); a reaction is exothermic (ΔH = 180 kJ) only when forming stronger product bonds releases more energy than breaking reactant bonds cost.

Check Question: Which statement accurately describes the energetics of breaking and forming chemical bonds? · **Transfer Challenge:** Transfer challenge: Compare methane combustion ($CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$, ΔH = −802 kJ) with $N_2 + O_2 \rightarrow 2\ NO$ (ΔH = +184 kJ). In which reaction are the product bonds collectively stronger (lower in potential energy) than the reactant bonds?

INVESTIGATION 5.7 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: EQUILIBRIUM

Perturbing a dynamic equilibrium

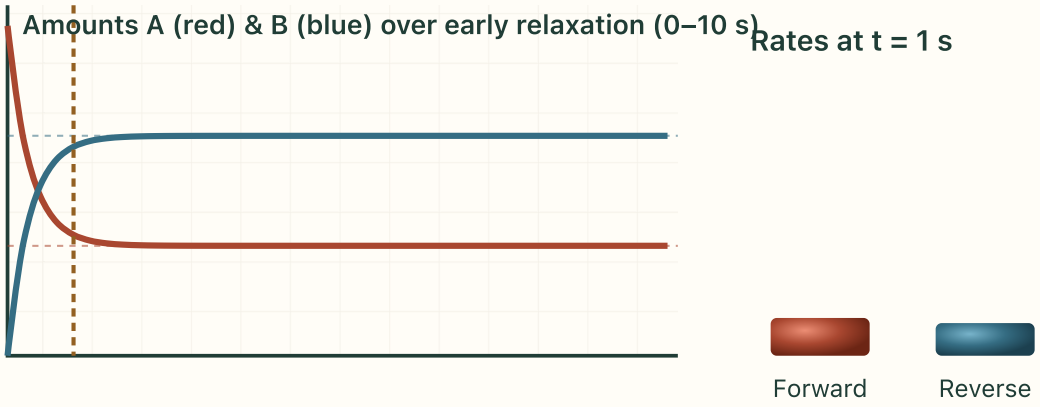
Why adding reactant or removing product drives a net shift through unequal forward and reverse rates

When a chemical system at dynamic equilibrium is disturbed by adding reactant or removing product, the forward and reverse rates become temporarily unequal (Q ≠ K), driving a net shift until a new equilibrium with the same ratio K = k_f / k_r is restored.

Bench Observation: Adding extra reactant A to an equilibrated mixture temporarily spikes the forward rate k_f[A], producing more B until k_f[A] = k_r[B] again.

Investigation Task: Set Time t = 6, k_f = 2, k_r = 1, and compare Perturbation 0 (None), Perturbation 1 (+4 units Reactant A), Perturbation 2 (Remove Product B), and Perturbation 3 (Catalyst).

Perturbing a dynamic equilibrium



Synthetic closed A ⇌ B relaxation. At dynamic equilibrium, forward and reverse rate bars are equal even when amounts A and B differ.

Stock versus flow rate

At equilibrium the opposing rates match. The inventories need not be equal. Added or removed material crosses the system boundary.

The bookkeeping idea: Like a stockroom with deliveries and dispatches: stock is how much is present; flow rate is how much changes per second. Equal opposing rates hold stocks steady without making the two stocks equal.

Current explanatory plate · intro-models-2

Amount A / model units	3.665247122452426 model units
Amount B / model units	6.334752877547574 model units
Forward rate / model units per s	7.330494244904852 model units/s
Reverse rate / model units per s	6.334752877547574 model units/s

Stock A + stock B = 3.66525 + 6.33475 = 10 model units. Forward rate 7.33049; reverse rate 6.33475 model units/s.

Model note: Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 6.3 Supplement 3–4. Equal rates and responses to changed conditions Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.6.2.3–7 (conditions: HT). Equilibrium and changing conditions Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 5.7: Deterministic particle & mathematical plate for *Perturbing a dynamic equilibrium*. First Principle: Perturbing an equilibrium mixture by adding reactant or removing product makes forward and reverse collision rates temporarily unequal until net reaction restores the equilibrium ratio K = k_f / k_r.

Historical & Experimental Evidence: Perturbing iron(III) thiocyanate or NO_2 / N_2O_4 mixtures at equilibrium

When a blood-red equilibrium mixture of $Fe^{3+}(aq) + SCN^-(aq) \rightleftharpoons FeSCN^{2+}(aq)$ is disturbed by adding extra colourless Fe^{3+} ions, the solution deepens to a darker red as more $FeSCN^{2+}$ forms. Why? The mixture does not have feelings or 'want' to oppose you! Instead, raising $[Fe^{3+}]$ immediately increases the frequency of forward $Fe^{3+} + SCN^-$ collisions (forward rate > reverse rate) until enough $FeSCN^{2+}$ accumulates for the reverse rate to catch up.

Key Terms

Reaction quotient vs. equilibrium (Q vs. K)
When a perturbation changes reactant or product amounts at fixed temperature, the current ratio Q no longer equals K, so forward and reverse rates become unequal until Q = K again.

Le Châtelier's principle (kinetic mechanism)
A perturbed equilibrium shifts in the direction where the reaction rate is temporarily higher until forward and reverse rates equalize again.

Worked Reasoning Trace

Held fixed: Time t = 6, $k_f = 2$, $k_r = 1$ (equilibrium ratio B/A = 2:1).

Changed: Switched Perturbation from 0 (None: A = 4, B = 8) to 1 (Add +4 units Reactant A at t = 2).

Model result: Injecting +4 units of A spikes the forward rate $k_f \times A$, converting most of the added A into B until the mixture settles near A = 5.33, B = 10.67 (preserving the 2:1 equilibrium ratio!).

Model boundary: Adding a catalyst (Perturbation 3: 2× both k_f and k_r) speeds up both rates equally and therefore causes zero shift in equilibrium amounts.

Common Misconception Trap · Perturbing a dynamic equilibrium

Intuitive Trap: "I think adding reactant a permanently increases the equilibrium constant k — because the equilibrium constant $K = k_f / k_r$ increases to accommodate the extra reactant."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Higher [A] immediately increases the forward reaction rate ($k_f[A] > k_r[B]$), so net forward conversion occurs until forward and reverse rates become equal again. More A particles per unit volume immediately increase the frequency of forward A → B transitions ($k_f[A] > k_r[B]$) while the rate constants k_f and k_r stay unchanged."

Why Claim A fails: The equilibrium constant K depends on temperature, not on the initial or perturbed amounts. · Why Claim B holds up: Exact mechanism: at fixed temperature, k_f and k_r stay fixed; adding A raises the forward rate above the reverse rate until net conversion restores $B/A = k_f/k_r$.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Unperturbed equilibrium (k_f = 2, k_r = 1, t = 10 s)	Forward rate equals reverse rate at a 2:1 B:A equilibrium ratio.	Synthetic closed A ⇌ B first-order model, initially 10 units A and 0 B. Equal rates can coexist with unequal amounts. Multiplying both constants by the same factor changes approach speed, not the equilibrium ratio.
Add +4 Reactant A → shifts forward to restore 2:1 ratio	Added A boosts forward rate until the 2:1 B:A ratio is restored at higher total amount.	Le Châtelier perturbation response: Adding +4 units of Reactant A raises forward collision rate (Q < K), shifting net conversion forward until a new equilibrium with the same B/A ratio (k_f/k_r) is restored.
Remove half of Product B → shifts forward to replenish B	Removing B drops the reverse rate below the forward rate, driving net A → B conversion.	Le Châtelier perturbation response: Removing half of Product B lowers reverse rate (Q < K), so net forward reaction replenishes B until B/A returns to k_f/k_r.

Check Question: At fixed temperature, when extra Reactant A is injected into a closed reversible reaction A ⇌ B already at equilibrium, why does the amount of Product B increase? · **Transfer Challenge:** Transfer challenge: Select Perturbation 3 (Catalyst: 2× both rate constants) at t = 6. Why do both the forward rate and reverse rate double while the equilibrium amounts of A and B remain completely unchanged?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 5 · The Stalled Industrial Reactor

A chemical plant runs a reversible exothermic synthesis A ⇌ B (ΔH = −90 kJ/mol, K = k_f/k_r = 2). The reactor reaches equilibrium at 67% B, 33% A. An engineer adds an expensive solid catalyst hoping to push the equilibrium yield of B from 67% up to 95%.

- Clue 1 (Pathway Peak): The catalyst lowers the transition-state peak height by 30 units without changing reactant or product energy levels.
- Clue 2 (Rate Constants): Lowering the peak doubles both k_f (from 2 to 4) and k_r (from 1 to 2).
- Clue 3 (Perturbation Test): Removing product B continuously or injecting extra reactant A changes the collision rate balance (Q ≠ K).

Resolution: Catalysts accelerate how fast equilibrium is reached by lowering both barriers equally, leaving K unchanged; shifting equilibrium composition requires perturbing amounts (adding A / removing B) or changing temperature.

What our upgraded model now explains: Explains exothermic vs. endothermic bond energy ledgers, activation energy barriers, catalysis, reaction rates, dynamic equilibrium, and Le Châtelier perturbations. · **Next boundary:** Knowing the energy profile and equilibrium ratio still leaves a

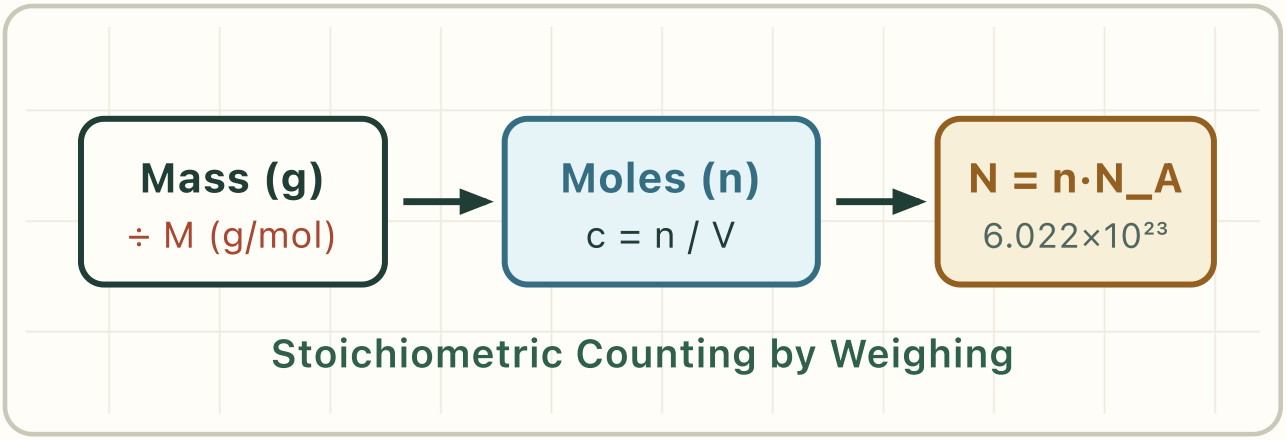
practical laboratory question: how many grams or litres of reactants should we weigh out? → motivating Chapter 6 (A Quantitative Language).

VOLUME II · CHANGE, QUANTITIES & DRIVING FORCES · CHAPTER 6

Chapter 6: A quantitative language

Build from ratios to moles, concentration and unit-factor chains, one unit at a time.

Chapter 6 Prelude · How Do Chemists Count 10^{23} Reacting Molecules with a Scale and a Flask?: When an airbag inflates in milliseconds or a pharmacist prepares an intravenous saline solution, being off by a factor of two can be catastrophic. We cannot count molecules with tweezers—we weigh powders in grams and measure liquids in litres. How do we translate fluently between grams, moles, limiting reactants, and molar concentration?



Macroscopic Bench: Analytical balance masses (g), volumetric flask volumes (L), and leftover unreacted reagents.

Submicroscopic Particle Model: Bundled stoichiometric reaction groups, excess intact particles, and solute crowding per litre.

Symbolic Notation: Dimensional unit chains: n = m / M, c = n / V, and limiting-reactant group stoichiometry.

INVESTIGATION 6.1 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: FORMULA

Ratios tell a story

Keeping track of whether a ratio counts atoms, molecules, or grams

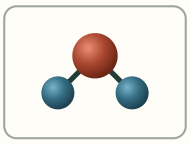
A ratio compares quantities. Keep track of what is being counted: molecules, atoms and formula units are not interchangeable.

Bench Observation: Water has two H atoms for each O atom. That is an atom ratio, not a 2:1 mass ratio.

Investigation Task: Compare one and five water molecules. Show what scales and which ratio stays fixed.

Ratios tell a story

Symbolic multiplier: 1 × H_2O



Solid boxes group the 1 discrete H_2O molecules; subscripts stay fixed inside each molecule.

Inventory per molecule or formula unit

Subscripts specify composition; a coefficient multiplies the whole inventory.

The bookkeeping idea: Like an itemized packing list: subscripts count atoms inside one molecule or formula unit; a coefficient multiplies the whole list. For an ionic formula, the signed charge entries must also reconcile.

Current explanatory plate · intro-models-2

H atoms 2

O atoms 1

Model note: 1 × H_2O. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.1 Core 1–4. Reading formulae and counting atoms Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.1.1–2. Atom inventories and relative formula mass Only the named concept is connected. Practical work and the

complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1 (pp. 11–12). Formulae and formula-unit masses Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 6.1: Deterministic particle & mathematical plate for *Ratios tell a story*. First Principle: Chemical formulas specify fixed ratios of particle counts (atoms per molecule), which remain invariant regardless of sample size.

Historical & Experimental Evidence: Electrolysis gives a 2:1 gas volume ratio, but an 1:8 mass ratio

When electric current decomposes liquid water into hydrogen and oxygen gases, the collection tubes show twice the volume of H_2 gas as O_2 gas (a 2:1 particle ratio). Yet if you weigh the collected gases, the oxygen weighs 8 times as much as the hydrogen (16 g of O for every 2 g of H)! Confusing an atom-count ratio with a mass ratio is one of the most common pitfalls in chemistry.

Key Terms

Atom ratio
The ratio of the numbers of atoms of each element in a substance (e.g., 2 H atoms : 1 O atom in H_2O).

Specified entity
The exact particle being counted—such as an individual H atom, a diatomic H_2 molecule, or an NaCl formula unit.

Worked Reasoning Trace

Held fixed: Formula H_2O (2 H atoms per 1 O atom).

Changed: Scaled molecule count from 1 × H_2O to 5 × H_2O.

Model result: Total atoms scale from (2 H, 1 O) to (10 H, 5 O), while the H : O atom ratio remains invariant at 2 : 1.

Model boundary: Because one oxygen atom is about 16 times as massive as one hydrogen atom, a 2:1 atom ratio in water corresponds to roughly a 2:16 (or 1:8) mass ratio.

Common Misconception Trap · Ratios tell a story

Intuitive Trap: "I think their masses — because hydrogen and oxygen atoms have the exact same atomic mass."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Numbers of atoms. Each H_2O molecule contains 2 hydrogen atoms per 1 oxygen atom, whereas each oxygen atom is ~16 times as massive as a hydrogen atom."

Why Claim A fails: An oxygen-16 atom is roughly 16 times as massive as a hydrogen-1 atom. · Why Claim B holds up: Exact mechanism: chemical subscripts count discrete atoms (2 H : 1 O), whereas mass weights each atom by its atomic mass (~2 u : 16 u = 1 : 8).

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
1 × H_2O (2 H : 1 O)	2 hydrogen atoms and 1 oxygen atom.	1 × H_2O. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.
5 × H_2O (10 H : 5 O)	10 hydrogen atoms and 5 oxygen atoms—still a 2:1 atom ratio.	5 × H_2O. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.
10 × H_2O (20 H : 10 O)	20 hydrogen atoms and 10 oxygen atoms—still 2:1.	10 × H_2O. Subscripts specify composition; the multiplier changes the amount. The formula does not show the three-dimensional molecular shape.

Check Question: The 2:1 H:O ratio in H_2O is a ratio of… · **Transfer Challenge:** Transfer challenge: Carbon dioxide has formula CO_2 (1 C atom : 2 O atoms). Given that carbon-12 has mass ~12 u and oxygen-16 has mass ~16 u, explain why the C : O atom ratio is 1:2 while the C : O mass ratio is roughly 12:32 (3:8).

INVESTIGATION 6.2 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: MOLE

Counting beyond what we can see

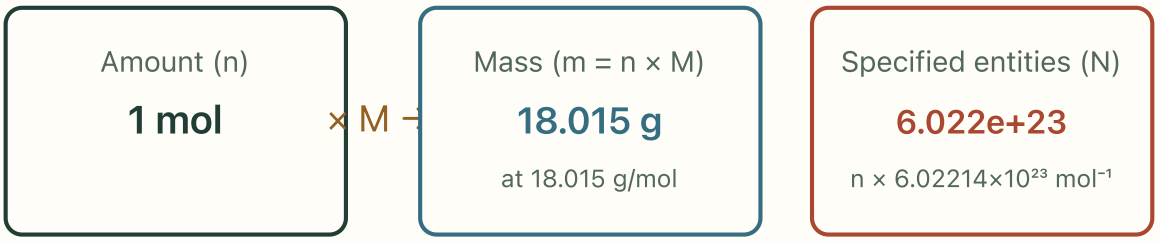
How Avogadro's constant bridges lab-scale amounts and individual particles

A mole is an amount containing exactly 6.02214076 × 10^{23} specified entities. It connects particle counting with measurable quantities.

Bench Observation: Equal mole amounts contain equal numbers of specified entities, but can have different masses.

Investigation Task: Keep amount at 1 mol while changing molar mass. Compare which result changes and which stays fixed.

Counting beyond what we can see



Unit bridge connecting molar amount (mol), laboratory mass (g) and number of specified entities (atoms, molecules or formula units).

Amount-to-mass account

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like converting a counted order to its mass: n tells how many moles, M gives grams per mole, and n × M gives grams. Counting units does not change the substance.

Current explanatory plate · intro-models-2

Mass / g	18.015 g
Number of specified entities	6.02214076e+23 entities

m = n × M = 1 mol × 18.015 g/mol = 18.015 g

Model note: m = nM; N = n × 6.02214076 × 10²³ mol⁻¹. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.

Scientific inputs · scientific-values-2

Avogadro constant: 6.02214076e+23 mol⁻¹ · defined. Counts specified entities per mole; state whether atoms, molecules or formula units. Exact SI defining constant. Source

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.2; 3.3 Supplement 2–3, 5. Relative masses, moles and reacting quantities Only the named concept is

connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.2.1–2 (HT). Moles, molar masses and reacting amounts Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 6.2: Deterministic particle & mathematical plate for *Counting beyond what we can see*. First Principle: Because single atoms are too tiny to weigh individually, the mole groups exactly 6.02214076 × 10^{23} specified entities so equal moles always contain equal particle counts.

Historical & Experimental Evidence: SI definition of the mole (6.02214076 × 10^{23} specified entities)

A single water molecule has a mass near 3 × 10^{-23} grams—far too tiny for any laboratory balance to weigh one by one. Just as a baker counts eggs by the dozen (12), chemists count atoms, molecules or ions by the mole: since the 2019 SI revision, 1 mole is defined as containing exactly 6.02214076 × 10^{23} specified elementary entities (Avogadro's constant N_A).

Key Terms

The mole (mol)
The SI base unit of amount of substance; 1 mol contains exactly 6.02214076 × 10^{23} specified elementary entities.

Avogadro's constant (N_A)
The scaling factor 6.02214076 × 10^{23} mol^{-1} linking amount in moles (n) to number of entities (N = n × N_A).

Worked Reasoning Trace

Held fixed: Amount n = 1.0 mol.

Changed: Changed Molar mass M from 12.011 g/mol (Carbon) to 18.015 g/mol (Water).

Model result: Mass changes from 12.011 g to 18.015 g, while the number of specified entities stays identical at 6.022141 × 10^{23}.

Model boundary: You must always name the entity: 1 mol of H_2O molecules contains 6.022 × 10^{23} molecules, which equals 3 × 6.022 × 10^{23} = 1.807 × 10^{24} total atoms!

Common Misconception Trap · Counting beyond what we can see

Intuitive Trap: "I think masses — because a mole of water contains more molecules than a mole of carbon contains atoms."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Numbers of specified entities. Both samples contain the exact same number of entities (6.022 × 10^{23}), but each H_2O molecule (~18 u) is heavier than each C atom (~12 u)."

Why Claim A fails: By definition, 1 mol of any specified entity contains the exact same number of entities. · Why Claim B holds up: Exact mechanism: 1 mol always counts 6.02214076 × 10^{23} specified entities, so sample mass scales with the mass of each individual entity.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
1.0 mol of Carbon (M = 12.011 g/mol)	Mass = 12.011 g; entities = 6.022141×10^{23} carbon atoms.	$m = nM$; $N = n \times 6.02214076 \times 10^{23}$ mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.
1.0 mol of Water (M = 18.015 g/mol)	Mass = 18.015 g; entities = 6.022141×10^{23} H_2O molecules.	$m = nM$; $N = n \times 6.02214076 \times 10^{23}$ mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.
0.5 mol of Water (M = 18.015 g/mol)	Half a mole has half the mass (9.008 g) and half the entities (3.011×10^{23}).	$m = nM$; $N = n \times 6.02214076 \times 10^{23}$ mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.

Check Question: One mole of carbon atoms and one mole of water molecules have equal… · **Transfer Challenge:** Transfer challenge: If you have 1.0 mol of oxygen gas molecules (O_2), how many O_2 molecules do you have, and how many individual oxygen atoms (O) does that sample contain?

INVESTIGATION 6.3 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: MOLE

From moles to mass

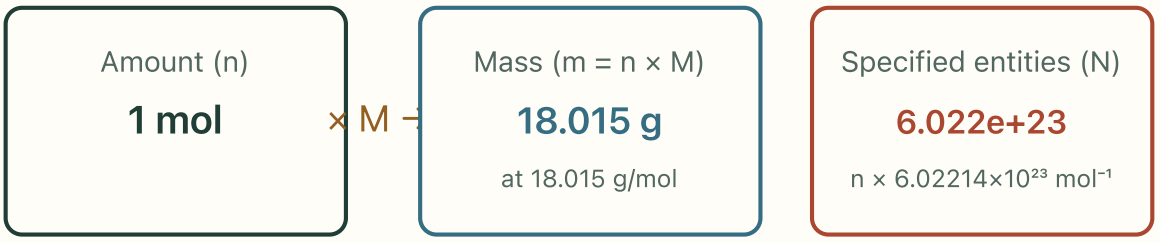
Converting between measurable mass in grams and chemical amount in moles

Molar mass is mass per mole. Multiply amount in mol by molar mass in g/mol to obtain mass in grams.

Bench Observation: A value near 18.015 g/mol is a commonly used molar mass for water with ordinary isotopic composition; it is not a mass number.

Investigation Task: Use 18.015 g/mol and compare 1 mol with 2 mol. Explain how the units cancel.

From moles to mass



Unit bridge connecting molar amount (mol), laboratory mass (g) and number of specified entities (atoms, molecules or formula units).

Amount-to-mass account

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like converting a counted order to its mass: n tells how many moles, M gives grams per mole, and n × M gives grams. Counting units does not change the substance.

Current explanatory plate · intro-models-2

Mass / g	18.015 g
Number of specified entities	6.02214076e+23 entities

m = n × M = 1 mol × 18.015 g/mol = 18.015 g

Model note: m = nM; N = n × 6.02214076 × 10²³ mol⁻¹. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.

Scientific inputs · scientific-values-2

Avogadro constant: 6.02214076e+23 mol⁻¹ · defined. Counts specified entities per mole; state whether atoms, molecules or formula units. Exact SI defining constant. Source

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.2; 3.3 Supplement 2–3, 5. Relative masses, moles and reacting quantities Only the named concept is

connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.2.1–2 (HT). Moles, molar masses and reacting amounts Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 6.3: Deterministic particle & mathematical plate for *From moles to mass*. First Principle: Molar mass (g/mol) translates between particle count (mol) and laboratory balance mass (g) so dimensional units cancel algebraically: n (mol) × M (g/mol) = m (g).

Historical & Experimental Evidence: Weighing stoichiometric amounts on an analytical balance

In a laboratory, you cannot set a dial to count out 3.01×10^{23} molecules directly. Instead, you use a substance's molar mass M (in grams per mole, g/mol) to convert the desired molar amount n (in mol) into a mass $m = n \times M$ (in grams) that you can weigh on a balance.

Key Terms

Molar mass (M)
The mass of one mole of a substance, usually expressed in grams per mole (g/mol or g mol^{-1}).

Unit cancellation (mol × g/mol = g)
Multiplying an amount in mol by molar mass in g/mol cancels 'mol' in numerator and denominator, leaving mass in grams (g).

Worked Reasoning Trace

Held fixed: Molar mass M = 18.015 g/mol (water).

Changed: Doubled amount n from 1.0 mol to 2.0 mol.

Model result: Mass m = n × M doubles from 18.015 g to 36.03 g.

Model boundary: 18.015 g/mol is a reference average molar mass for ordinary terrestrial water; it is not an integer mass number.

Common Misconception Trap · From moles to mass

Intuitive Trap: "I think 88 g — because adds mol and g/mol together into a combined unit."

Particle Mechanism Refutation: "Wait — look at the particle ledger: 22 g. Scales the number of moles by the mass of one mole, cancelling 'mol' in numerator and denominator (mol × g/mol = g)."

Why Claim A fails: Quantities with different dimensions cannot be added; they multiply or divide so units cancel. · Why Claim B holds up: Exact mechanism: molar mass M is the conversion factor (g per 1 mol), so n (mol) × M (g/mol) cancels mol and leaves mass in grams.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
1.0 mol Water (M = 18.015 g/mol)	1.0 mol × 18.015 g/mol = 18.015 g.	m = nM; N = n × 6.02214076 × 10^{23} mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.
2.0 mol Water (M = 18.015 g/mol)	2.0 mol × 18.015 g/mol = 36.03 g.	m = nM; N = n × 6.02214076 × 10^{23} mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.
0.5 mol CO_2 (M = 44.000 g/mol)	0.5 mol × 44 g/mol = 22 g.	m = nM; N = n × 6.02214076 × 10^{23} mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.

Check Question: At M = 44 g/mol, what is the mass of 0.5 mol? · **Transfer Challenge:** Transfer challenge: If a chemist weighs out 36.03 g of pure water (M = 18.015 g/mol), how many moles of water is that, and how do the units cancel when dividing grams by g/mol?

INVESTIGATION 6.4 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: LIMITING

Which reactant runs out first?

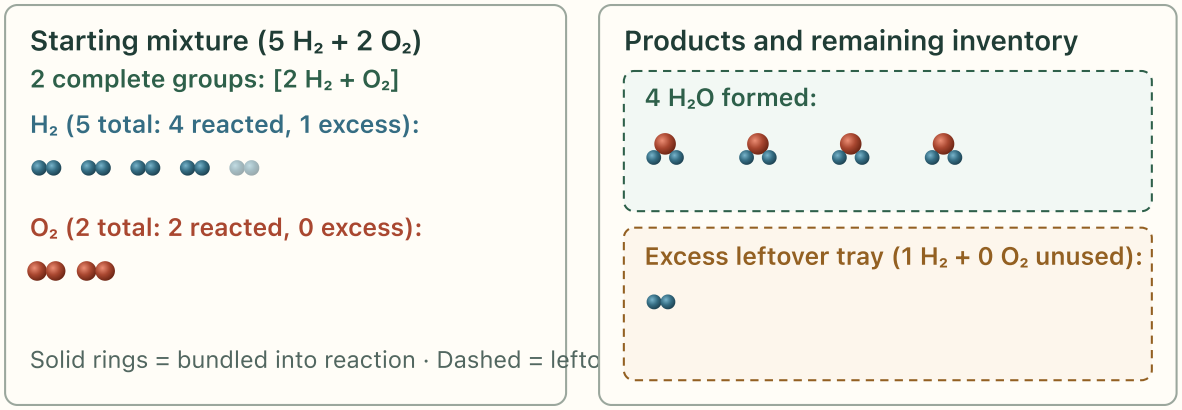
Why reaction yield stops when the first stoichiometric reactant is exhausted

Reactants are used in the ratios in a balanced equation. The reactant that permits the least product limits the amount that can form under the stated complete-reaction assumption.

Bench Observation: Leftover reactant can remain even when no more product can form from the available mixture.

Investigation Task: Start with 5 H_2 and 2 O_2 molecules. Count products and leftovers. Then add one O_2 and explain whether another full reaction group can occur.

Which reactant runs out first?



Whole-molecule stoichiometric grouping assuming complete reaction in a closed vessel. Up to ten starting units and eight product/leftover units are sketched per species; read the labels for full counts. Fe_2O_3 sketches count formula units in an extended solid.

Reacting bundles & stock remaining

Use complete reacting sets. Count products and excess separately, then reconcile each element.

The bookkeeping idea: Like packing orders from a stockroom: the first ingredient to run out limits complete reacting sets. Starting atoms = atoms in products + atoms in leftovers, for every element separately.

Current explanatory plate · intro-models-2	
H_2O molecules made	4 molecules
H_2 molecules left	1 molecules
O_2 molecules left	0 molecules

H: 10 starting = 10 in products + leftovers · O: 4 starting = 4 in products + leftovers

Model note: Whole-molecule accounting for 2 H_2 + O_2 → 2 H_2O, assuming complete reaction in a closed system. This is not a home experiment.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.3 Supplement 5. Reacting amounts and limiting reactants Only the named concept is connected.

Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.2.4 (HT). Limiting reactants Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 6.4: Deterministic particle & mathematical plate for *Which reactant runs out first?*. First Principle: Stoichiometric coefficients act like a strict assembly recipe—the reactant that can form the fewest complete reaction groups runs out first and caps product yield.

Historical & Experimental Evidence: Complete reaction groups and unreacted excess in closed mixtures

Suppose a bicycle frame requires 1 frame and 2 wheels. If a workshop has 5 frames and 6 wheels, they can build at most 3 complete bicycles—at which point the wheels run out (the limiting component) and 2 frames remain unused in excess. Chemical reactions obey the exact same stoichiometric grouping rule dictated by the balanced equation.

Key Terms

Limiting reactant
The reactant that is completely consumed first according to the balanced stoichiometric ratio, limiting the maximum amount of product that can form.

Excess reactant
Any reactant that remains partially unreacted after the limiting reactant has been completely used up.

Worked Reasoning Trace

Held fixed: Balanced reaction $2\ H_2 + O_2 \rightarrow 2\ H_2O$.

Changed: Started with 5 H_2 and 2 O_2, then increased O_2 from 2 to 3.

Model result: With 5 H_2 and 2 O_2, two reaction groups use 4 H_2 + 2 O_2 to make 4 H_2O, leaving 1 H_2 and 0 O_2 (O_2 limits). Adding a 3rd O_2 still makes only 4 H_2O because only 1 H_2 is left (less than the 2 H_2 needed for another group; now H_2 limits!).

Model boundary: Simply picking the reactant with the smaller starting number is a trap: with 5 H_2 and 3 O_2, O_2 has the smaller count (3 < 5), yet H_2 runs out first because H_2 is consumed twice as fast!

Common Misconception Trap · Which reactant runs out first?

Intuitive Trap: "I think 6 h_2o molecules — because all 6 H_2 molecules turn into 6 H_2O molecules by creating extra oxygen atoms."

Particle Mechanism Refutation: "Wait — look at the particle ledger: 4 H_2O molecules. Each O_2 molecule requires 2 H_2 partners to form 2 H_2O, so 2 O_2 consume 4 H_2 to make 4 H_2O and leave 2 H_2 molecules with no oxygen left to react with."

Why Claim A fails: Atoms are conserved; water cannot form once oxygen atoms are exhausted. · Why Claim B holds up: Exact mechanism: reaction stops after 2 complete reaction groups (4 H_2 + 2 O_2 → 4 H_2O) because 0 O_2 molecules remain.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
5 H_2 + 2 O_2 (O_2 runs out first)	Makes 4 H_2O molecules with 1 H_2 molecule left over and 0 O_2 left.	Whole-molecule accounting for 2 H_2 + O_2 → 2 H_2O, assuming complete reaction in a closed system. This is not a home experiment.
5 H_2 + 3 O_2 (H_2 limits despite 5 > 3!)	Still makes 4 H_2O molecules: 1 H_2 and 1 O_2 remain because a full group needs 2 H_2.	Whole-molecule accounting for 2 H_2 + O_2 → 2 H_2O, assuming complete reaction in a closed system. This is not a home experiment.
Methane: 3 CH_4 + 4 O_2 (O_2 limits)	4 O_2 can combust at most 2 CH_4 into 2 CO_2 + 4 H_2O, leaving 1 CH_4 unreacted.	Whole-molecule accounting for CH_4 + 2 O_2 → CO_2 + 2 H_2O (2 CO_2 and 4 H_2O made), assuming complete reaction in a closed system.
Ammonia: 2 N_2 + 5 H_2 (H_2 limits)	Each N_2 requires 3 H_2; 5 H_2 completes only 1 reaction group (2 NH_3), leaving 1 N_2 and 2 H_2.	Whole-molecule accounting for N_2 + 3 H_2 → 2 NH_3, assuming complete reaction in a closed system.

Check Question: For 2 H_2 + O_2 → 2 H_2O, 6 H_2 and 2 O_2 can form at most… · **Transfer Challenge:** Transfer challenge: Explain why a mixture of 5 H_2 molecules and 3 O_2 molecules has H_2 as the limiting reactant, even though there are more H_2 molecules (5) than O_2 molecules (3) at the start.

INVESTIGATION 6.5 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: SOLUTION

How much in each litre?

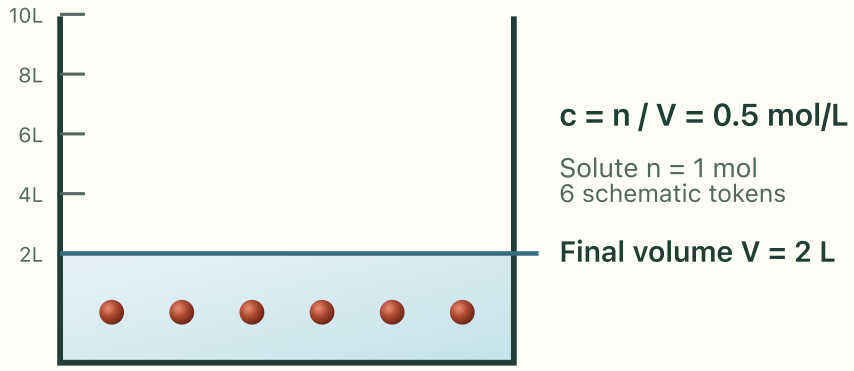
Separating how crowded a solution is (mol/L) from the total amount dissolved (mol)

Molar concentration is amount of dissolved substance divided by final solution volume. Diluting without reaction or solute loss preserves solute amount.

Bench Observation: A larger final volume with the same amount of solute has a lower concentration.

Investigation Task: Compare 1 mol in 2 L with 1 mol in 4 L, then find two different mixtures with the same concentration.

How much in each litre?



Schematic beaker showing final solution volume V and dispersed solute tokens (6 tokens per mole).

Same stock, more space

Adding solvent increases final solution volume. In this ideal dilution model, solute amount remains in the account.

The bookkeeping idea: Like spreading the same stock across more shelves: dilution changes the space available, not the amount owned. Count solute in mol, volume in L, and stock per litre in mol/L. Never add these unlike units together.

Current explanatory plate · intro-models-2

Solute amount	1 mol
Solution volume	2 L
Concentration	0.5 mol/L

c = n / V = 1 mol ÷ 2 L = 0.5 mol/L

Model note: Final solution volume is used, not solvent volume. No reaction, solute loss or nonideal volume mixing is simulated.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.3 Core 1; Supplement 5. Concentration and solution quantities Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.2.5; 4.3.4 (chemistry, HT). Mass and molar concentrations Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Mixtures and their Separation · C-1.2 (pp. 7–8). Concentration; molarity is enrichment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 6.5: Deterministic particle & mathematical plate for *How much in each litre?*. First Principle: Molarity (c = n/V) measures particle crowding per litre of final solution—doubling volume with fixed solute halves the particle density in every drop.

Historical & Experimental Evidence: Volumetric flasks and Beer–Lambert colour intensity

If you prepare a 0.25 mol/L aqueous copper(II) sulfate solution in a 2.0 L volumetric flask and pour out a 100 mL cup of it, the liquid in the cup has the exact same blue colour intensity—and the exact same 0.25 mol/L concentration—as the liquid in the 2.0 L flask, even though the cup contains only one-twentieth as many moles of solute.

Key Terms

Molarity / Molar concentration (mol/L)
An intensive ratio measuring moles of dissolved solute per litre of final solution (c = n / V).

Intensive vs. Extensive quantity
Concentration (mol/L) describes crowding per litre, whereas total dissolved amount (mol) scales with how much solution you take.

Worked Reasoning Trace

Held fixed: Target concentration c = 0.50 mol/L.

Changed: Compared (1.0 mol in 2.0 L) with (2.0 mol in 4.0 L).

Model result: Both mixtures have the exact same concentration c = 0.50 mol/L, even though the second mixture contains twice the total solute and twice the total volume.

Model boundary: Equal concentrations only mean equal amounts of solute when the solution volumes are also equal (n = c × V).

Common Misconception Trap · How much in each litre?

Intuitive Trap: "I think 1 mol/l — because the total mass of the beaker plus water."

Particle Mechanism Refutation: "Wait — look at the particle ledger: 0.25 mol/L. The crowding of dissolved solute particles per unit volume of final solution (moles per litre), so 0.5 mol spread across 2 L is 0.25 mol per 1 L."

Why Claim A fails: Molarity is an intensive ratio of solute amount to solution volume (mol/L). · Why Claim B holds up: Exact mechanism: dividing 0.5 mol of solute equally across 2 L of solution places 0.25 mol in each 1 L of solution.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
1.0 mol in 2.0 L (0.50 mol/L)	c = 1.0 mol / 2.0 L = 0.50 mol/L.	c = n/V. V is the final solution volume, not the volume of water poured in. Assumes all of the specified solute is dissolved and no reaction occurs.
1.0 mol in 4.0 L (0.25 mol/L)	Doubling final volume with the same 1.0 mol solute halves concentration to 0.25 mol/L.	c = n/V. V is the final solution volume, not the volume of water poured in. Assumes all of the specified solute is dissolved and no reaction occurs.
2.0 mol in 4.0 L (Same 0.50 mol/L)	Doubling both solute and final volume keeps concentration at 0.50 mol/L.	c = n/V. V is the final solution volume, not the volume of water poured in. Assumes all of the specified solute is dissolved and no reaction occurs.

Check Question: What is the concentration of 0.5 mol in 2 L of solution? · **Transfer Challenge:** Transfer challenge: Bottle A holds 1.0 L of 0.50 mol/L NaCl solution, and Bottle B holds 4.0 L of 0.25 mol/L NaCl solution. Which bottle tastes saltier per sip (higher concentration), and which bottle contains the larger total number of moles of NaCl?

INVESTIGATION 6.6 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: MOLE

From laboratory grams to product grams

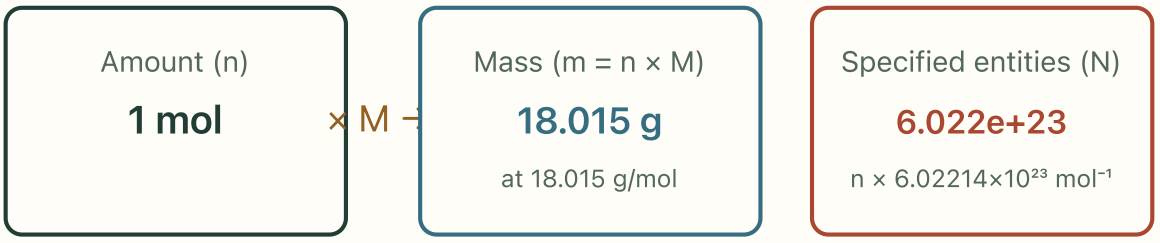
Why laboratory gram masses must always pass through the mole bridge before using equation coefficients

Balanced chemical equations relate numbers of moles, never grams directly. To predict reaction masses, chemists convert reactant mass (g) → reactant moles (mol) → product moles (mol) → product mass (g).

Bench Observation: Reacting 4.0 g of H_2 (about 2.0 mol) requires 32.0 g of O_2 (1.0 mol)—not 4.0 g of O_2—to form 36.0 g of H_2O (2.0 mol).

Investigation Task: Compare 2.0 mol of H_2O (M = 18 g/mol, mass = 36 g) with 1.0 mol of O_2 (M = 32 g/mol, mass = 32 g). Explain why a 2:1 mole ratio does not mean a 2:1 gram ratio.

From laboratory grams to product grams



Unit bridge connecting molar amount (mol), laboratory mass (g) and number of specified entities (atoms, molecules or formula units).

Amount-to-mass account

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like converting a counted order to its mass: n tells how many moles, M gives grams per mole, and n × M gives grams. Counting units does not change the substance.

Current explanatory plate · intro-models-2

Mass / g	18.015 g
Number of specified entities	6.02214076e+23 entities

m = n × M = 1 mol × 18.015 g/mol = 18.015 g

Model note: m = nM; N = n × 6.02214076 × 10²³ mol⁻¹. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.

Scientific inputs · scientific-values-2

Avogadro constant: 6.02214076e+23 mol⁻¹ · defined. Counts specified entities per mole; state whether atoms, molecules or formula units. Exact SI defining constant. Source

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 3.2; 3.3 Supplement 2–3, 5. Relative masses, moles and reacting quantities Only the named concept is

connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.3.2.1–2 (HT). Moles, molar masses and reacting amounts Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 6.6: Deterministic particle & mathematical plate for *From laboratory grams to product grams*. First Principle: Balanced equation coefficients count mole ratios, never gram ratios; molar mass M (g/mol) translates between laboratory grams and stoichiometric moles.

Historical & Experimental Evidence: Gravimetric analysis and Lavoisier–Proust quantitative synthesis

In a balanced equation like $2\ H_2 + O_2 \rightarrow 2\ H_2O$, the 2 : 1 : 2 coefficients count molecules or moles—never grams! Because each O_2 molecule (32.0 g/mol) is 16 times as massive as an H_2 molecule (2.0 g/mol), 2 moles of H_2 weigh 4.0 g while 1 mole of O_2 weighs 32.0 g, reacting completely to produce 36.0 g (2 moles) of H_2O. Notice how both moles (in the 2:1:2 ratio) and total grams (4.0 g + 32.0 g = 36.0 g) are consistent!

Key Terms

Mass–mole–mass stoichiometry chain
The three-step quantitative bridge: (1) divide reactant grams by molar mass M to get moles, (2) multiply by the balanced equation mole ratio, and (3) multiply by product molar mass M to get product grams.

Stoichiometric equivalence
Amounts of reactants that match the exact mole ratio of the balanced chemical equation so neither reactant is in excess.

Worked Reasoning Trace

Held fixed: Reaction $2\ H_2 + O_2 \rightarrow 2\ H_2O$ (requiring 2 mol H_2O per 1 mol O_2).

Changed: Compared 1.00 mol O_2 (M = 32 g/mol) with 2.00 mol H_2O (M = 18 g/mol).

Model result: 1.00 mol of O_2 has mass 32 g (6.022×10^{23} O_2 molecules); 2.00 mol of H_2O has mass 36 g (1.204×10^{24} H_2O molecules).

Model boundary: Calculated theoretical yield in grams assumes 100% reaction completion with no side reactions or mechanical transfer losses.

Common Misconception Trap · From laboratory grams to product grams

Intuitive Trap: "I think every molecule in a balanced equation has the exact same mass in grams — because laboratory balances measure moles directly rather than gravitational mass."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Coefficients count relative numbers of molecules or moles, and each substance has a different molar mass (g/mol). Coefficients count relative numbers of discrete molecules (or moles), and molecules of different substances have different masses per mole (g/mol)."

Why Claim A fails: Balances measure mass in grams; molar mass M (g/mol) is required to convert grams into moles. · Why Claim B holds up: Exact mechanism: 1 mol of O_2 (32 g) and 2 mol of H_2 (4 g) react in a 1:2 mole ratio, which corresponds to a 32:4 (8:1) mass ratio.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
1.00 mol O_2 reactant (M = 32 g/mol → 32 g)	1.00 mol of O_2 gas weighs 32.0 g.	m = nM; N = n × 6.02214076 × 10^{23} mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.
2.00 mol H_2O product (M = 18 g/mol → 36 g)	2.00 mol of H_2O product weighs 36.0 g (32.0 g from O_2 + 4.0 g from 2 mol H_2).	m = nM; N = n × 6.02214076 × 10^{23} mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.
0.50 mol CO_2 product (M = 44 g/mol → 22 g)	0.50 mol of CO_2 weighs 22.0 g.	m = nM; N = n × 6.02214076 × 10^{23} mol^{-1}. Specify whether entities are atoms, molecules or formula units. The molar mass is an input, not a provider lookup.

Check Question: Why can’t you read the coefficients in 2 H_2 + O_2 → 2 H_2O as '2 grams of H_2 react with 1 gram of O_2'? · **Transfer Challenge:** Transfer challenge: In CH_4 + 2 O_2 → CO_2 + 2 H_2O, burning 0.50 mol of CH_4 (M = 16 g/mol, so 8.0 g CH_4) produces 0.50 mol of CO_2 (M = 44 g/mol). What mass in grams of CO_2 is formed?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 6 · The Missing Yield in the Hydrogen Burner

A technician mixes 12.0 g of H_2 gas (M = 2.0 g/mol) with 64.0 g of O_2 gas (M = 32.0 g/mol) in a sealed reaction chamber and sparks the mixture (2 H_2 + O_2 → 2 H_2O, M_H2O = 18.0 g/mol). They expect 12.0 g + 64.0 g = 76.0 g of liquid water to form, but only 72.0 g of H_2O forms!

- Clue 1 (Mole Conversion): 12.0 g H_2 ÷ 2.0 g/mol = 6.0 mol H_2; 64.0 g O_2 ÷ 32.0 g/mol = 2.0 mol O_2.
- Clue 2 (Stoichiometric Groups): Each 1 mol O_2 requires 2 mol H_2 to form 2 mol H_2O.
- Clue 3 (Mass Conservation): After the reaction, 4.0 g of unreacted colourless gas remains in the chamber alongside 72.0 g of H_2O.

Resolution: Always convert grams to moles first: 2.0 mol O_2 can react with at most 4.0 mol H_2 to produce 4.0 mol H_2O (72.0 g), leaving 6.0 – 4.0 = 2.0 mol H_2 (4.0 g) in excess.

What our upgraded model now explains: Unites atom/mole ratios, Avogadro’s number, molar mass (g/mol), mass-to-mass reaction stoichiometry, limiting reactants with leftovers, and solution molarity (mol/L). · **Next boundary:** Counting total dissolved solute (mol/L) treats every dissolved substance alike—it cannot explain why 0.10 M HCl is strongly acidic, 0.10 M CH_3COOH is weakly acidic, and 0.10 M NaOH is basic → motivating Chapter 7 (Acids, Bases & Buffers).

FROM QUANTITATIVE ANALYST TO CHEMISTRY HERO (CHAPTERS 7–11)

Volume III · Chemical Systems & The Real World

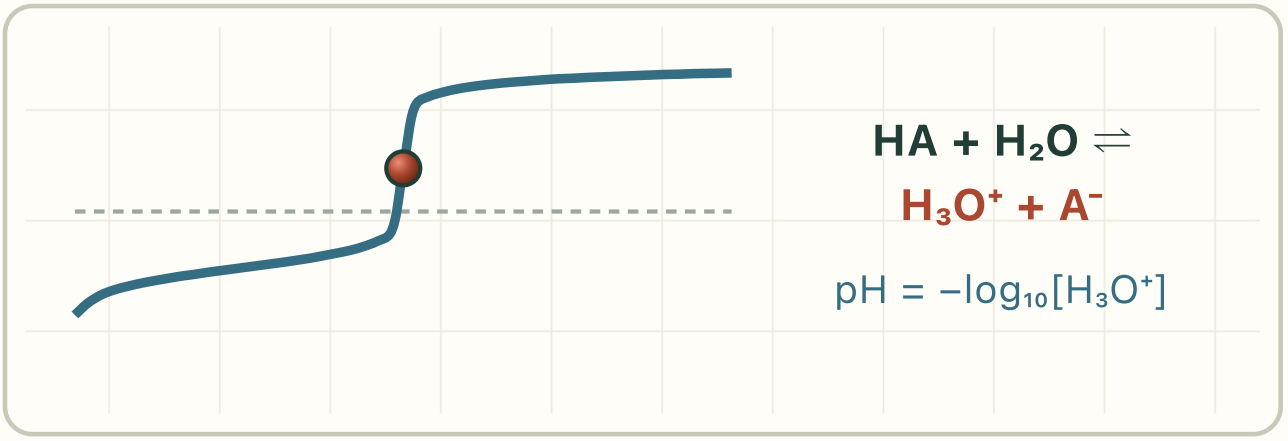
Synthesize proton transfer, titration buffers, electrochemistry, periodic trends, carbon functional groups, and planetary materials into real-world chemical problem solving.

VOLUME III · CHEMICAL SYSTEMS & THE REAL WORLD · CHAPTER 7

Chapter 7: Acids, bases and proton transfer

Follow proton (H^+) transfer in water, separate acid strength from concentration, and connect each pH unit to a tenfold change in $[H_3O^+]$.

Chapter 7 Prelude · Why Does Lemon Juice Taste Sour While Blood Stays Locked at pH 7.4?: Vinegar and battery acid can be prepared at the exact same 0.10 mol/L concentration, yet one goes on your salad and the other corrodes metal. Meanwhile, human blood stays within a razor-thin window of pH 7.35–7.45 even when your muscles produce lactic acid during a sprint. How does proton (H^+) transfer, logarithmic pH, and conjugate acid–base buffering control aqueous chemistry?



Macroscopic Bench: pH meter readings, indicator colour shifts, conductivity differences, and flat titration buffer plateaus.

Submicroscopic Particle Model: Proton (H^+) transfer forming H_3O^+ and A^-; 100% strong ionization vs. partial weak equilibrium; HA / A^- conjugate buffer reservoirs.

Symbolic Notation: $HA + H_2O \rightleftharpoons H_3O^+ + A^-$, K_a, $pH = -\log_{10}[H_3O^+]$, and Henderson–Hasselbalch ratio $[A^-]/[HA]$.

INVESTIGATION 7.1 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: ACIDBASE

What makes a solution acidic?

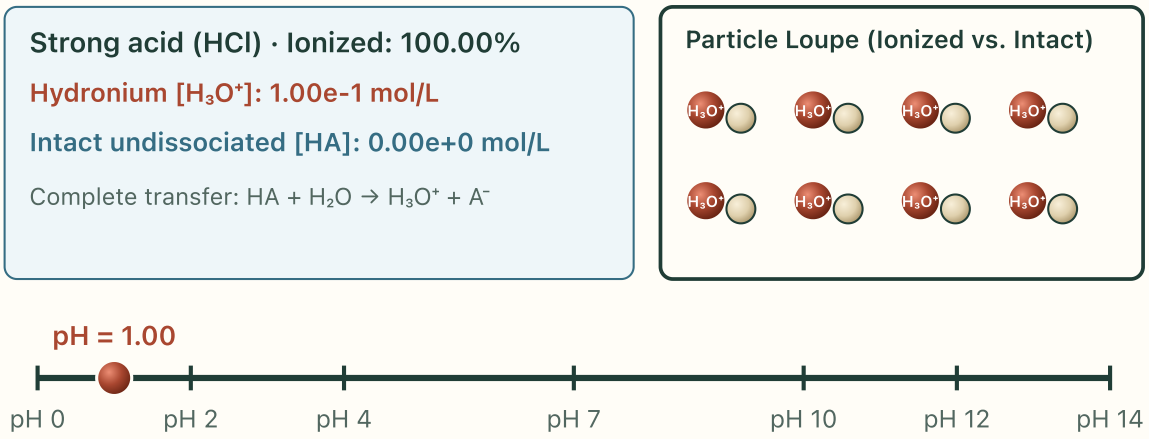
How Brønsted–Lowry acids and bases exchange a bare proton (H^+) in water

In the Brønsted–Lowry model, an acid transfers a proton (a bare hydrogen nucleus, H^+) to a base. In water, proton transfer from an acid HA forms hydronium ions (H_3O^+) and conjugate base anions (A^-).

Bench Observation: Acidic aqueous solutions conduct electricity and shift indicator colours because mobile H_3O^+ and anion particles are present.

Investigation Task: Compare pure water with a 0.10 mol/L strong acid (HCl) and a 0.10 mol/L strong base (NaOH). Explain which particle concentration rises in an acidic solution.

What makes a solution acidic?



Logarithmic pH scale ($-\log_{10}[H_3O^+]$) and submicroscopic particle loupe at 25 °C. Each 1-unit pH step represents a 10× change in $[H_3O^+]$.

Concentration and ionization account

Distinguish total solute from ionized species. This dilute, 25 °C concentration model approximates activities.

The bookkeeping idea: Keep separate accounts for total dissolved acid and the fraction ionized. Concentration and acid strength answer different questions. pH is logarithmic; pH numbers cannot be added like amounts.

Current explanatory plate · intro-models-2	
Total solute concentration / mol per L	0.1 mol/L
H_3O^+ concentration / mol per L	0.1 mol/L
Ionized or dissociated fraction / %	100 %
Calculated pH (25 °C ideal model)	1 dimensionless

$pH \approx -\log_{10}([H_3O^+]/(1\ mol/L)) = -\log_{10}(0.1) \approx 1$. The displayed concentration is rounded; pH is calculated before display rounding.

Model note: Strong acid (HCl) at 0.1 mol/L: essentially 100% proton transfer to water ($HCl + H_2O \rightarrow H_3O^+ + Cl^-$), giving pH = 1. Each 10× dilution raises pH by 1 unit.

Scientific inputs · scientific-values-2

Dilute-solution concentration approximation for K_w: 1e-14 $(mol/L)^2$ · reference. 25 °C; thermodynamic equilibrium constants are expressed through activities. Concentrations approximate activities in this model. Introductory rounded approximation. Source

Ethanoic-acid dissociation parameter: 0.000018 concentration-model parameter · reference. 25 °C, dilute aqueous model. Concentration approximation to activity-based equilibrium. Two significant figures. Source

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 7.1 Core; Supplement 9–10. Acid behaviour and strong versus weak acids Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.4.2.4; 4.4.2.6 (HT). pH and acid strength versus concentration Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Acids, Bases and Salts (pp. 4–5). Acids and pH; logarithms and K_a are enrichment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 7.1: Deterministic particle & mathematical plate for *What makes a solution acidic?*. First Principle: Brønsted–Lowry acid-base reactions transfer a bare proton (H^+, +1 charge) from an acid to a base, conserving both nuclear counts and net electric charge.

Historical & Experimental Evidence: Conductivity and spectroscopy of aqueous acids

Pure liquid hydrogen chloride (HCl) at low temperature is a molecular liquid that conducts almost no electricity. Yet the moment HCl gas dissolves in water, the solution conducts strongly and infrared spectroscopy reveals hydrated hydronium ions (H_3O^+). Johannes Brønsted and Thomas Lowry explained this in 1923: HCl transfers a proton (H^+) to a water molecule, forming H_3O^+ and Cl^-.

Key Terms

Brønsted–Lowry acid
A chemical species that donates a proton (H^+, a hydrogen nucleus with zero electrons) in a reaction.

Brønsted–Lowry base
A chemical species that accepts a proton (H^+) by sharing a pair of electrons to form a new covalent bond.

Hydronium ion (H_3O^+)
The hydrated proton formed when a water molecule accepts H^+ in aqueous solution.

Worked Reasoning Trace

Held fixed: Initial acid concentration c = 0.10 mol/L in water.

Changed: Compared Strong acid HCl (system 0) with Neutral water baseline (system 2).

Model result: In 0.10 mol/L HCl(aq), nearly 100% of HCl molecules transfer H^+ to H_2O, yielding $[H_3O^+]$ = 0.10 mol/L and [HA] = 0 mol/L.

Model boundary: A transferred 'proton' (H^+) is the hydrogen nucleus stripping away from its shared bonding electrons; it never changes the nuclear proton count of chlorine (Z = 17) or oxygen (Z = 8).

Common Misconception Trap · What makes a solution acidic?

Intuitive Trap: "I think oh^- only, with no positive ion — because a proton is ejected from the chlorine nucleus, turning chlorine into sulfur."

Particle Mechanism Refutation: "Wait — look at the particle ledger: H_3O^+ (hydronium ion). An H^+ nucleus (proton) leaves the H–Cl bond—leaving its bonding electron pair on Cl^-—and forms a coordinate covalent bond with a lone pair on H_2O to make H_3O^+."

Why Claim A fails: Acid-base proton transfer moves a hydrogen nucleus (H^+), never a nucleon from inside a chlorine or oxygen nucleus. · Why Claim B holds up: Exact mechanism: Brønsted–Lowry acid-base reactions transfer H^+ without its bonding electrons, creating Cl^- (–1) and H_3O^+ (+1) while conserving all nuclear atomic numbers.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Strong acid HCl at 0.10 mol/L	Complete proton transfer to water: $[H_3O^+]$ = 0.10 mol/L, pH = 1.00.	Strong acid (HCl) at 0.1 mol/L: essentially 100% proton transfer to water ($HCl + H_2O \rightarrow H_3O^+ + Cl^-$), giving pH = 1. Each 10× dilution raises pH by 1 unit.
Weak acid CH_3COOH at 0.10 mol/L	Reversible partial proton transfer: most CH_3COOH molecules remain intact.	Weak acid (CH_3COOH, $K_a = 1.8\times10^{-5}$) at 0.1 mol/L: only 1.33% of molecules transfer a proton at equilibrium ($CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$), leaving most molecules intact and giving pH = 2.875.
Pure water at 25 °C	Baseline autoionization: $[H_3O^+] = 1.0 \times 10^{-7}$ mol/L, pH = 7.00.	Pure water at 25 °C: autoionization $2\ H_2O \rightleftharpoons H_3O^+ + OH^-$ gives $[H_3O^+] = [OH^-] = 1.0 \times 10^{-7}$ mol/L and neutral pH = 7.00.

Check Question: When a Brønsted–Lowry acid HA transfers a proton to water (H_2O), which ion is formed? · **Transfer Challenge:** Transfer challenge: When ammonia (NH_3) dissolves in water, the reversible reaction $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$ occurs. Which molecule acts as the proton donor (acid) and which acts as the proton acceptor (base)?

INVESTIGATION 7.2 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: ACIDBASE

Strong or concentrated?

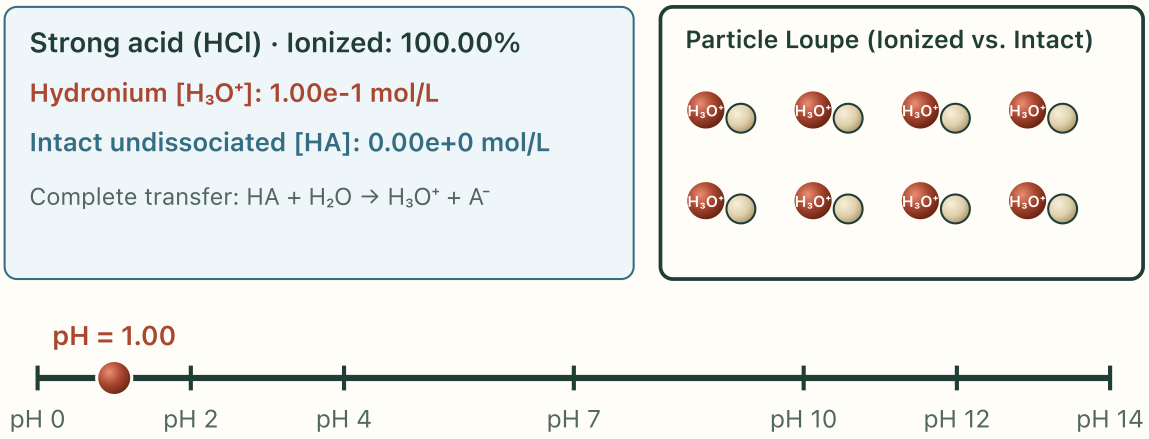
Separating the fraction of molecules that ionize (strength) from moles per litre (concentration)

Acid strength and solution concentration answer two completely different questions: concentration (mol/L) counts how much solute was dissolved per litre, whereas strength describes what fraction of those acid molecules ionize by transferring H^+ to water.

Bench Observation: At the same 0.10 mol/L concentration, hydrochloric acid (HCl) conducts electric current much more strongly and has a much lower pH than acetic acid (CH_3COOH).

Investigation Task: Compare 0.01 mol/L strong acid (HCl) with 1.00 mol/L weak acid (CH_3COOH). Save both examples and compare their ionized fractions and $[H_3O^+]$.

Strong or concentrated?



Logarithmic pH scale ($-\log_{10}[H_3O^+]$) and submicroscopic particle loupe at 25 °C. Each 1-unit pH step represents a 10× change in $[H_3O^+]$.

Concentration and ionization account

Distinguish total solute from ionized species. This dilute, 25 °C concentration model approximates activities.

The bookkeeping idea: Keep separate accounts for total dissolved acid and the fraction ionized. Concentration and acid strength answer different questions. pH is logarithmic; pH numbers cannot be added like amounts.

Current explanatory plate · intro-models-2	
Total solute concentration / mol per L	0.1 mol/L
H_3O^+ concentration / mol per L	0.1 mol/L
Ionized or dissociated fraction / %	100 %
Calculated pH (25 °C ideal model)	1 dimensionless

$pH \approx -\log_{10}([H_3O^+]/(1\ mol/L)) = -\log_{10}(0.1) \approx 1$. The displayed concentration is rounded; pH is calculated before display rounding.

Model note: Strong acid (HCl) at 0.1 mol/L: essentially 100% proton transfer to water ($HCl + H_2O \rightarrow H_3O^+ + Cl^-$), giving pH = 1. Each 10× dilution raises pH by 1 unit.

Scientific inputs · scientific-values-2

Dilute-solution concentration approximation for K_w: 1e-14 $(mol/L)^2$ · reference. 25 °C; thermodynamic equilibrium constants are expressed through activities. Concentrations approximate activities in this model. Introductory rounded approximation. Source

Ethanoic-acid dissociation parameter: 0.000018 concentration-model parameter · reference. 25 °C, dilute aqueous model. Concentration approximation to activity-based equilibrium. Two significant figures. Source

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 7.1 Core; Supplement 9–10. Acid behaviour and strong versus weak acids Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.4.2.4; 4.4.2.6 (HT). pH and acid strength versus concentration Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Acids, Bases and Salts (pp. 4–5). Acids and pH; logarithms and K_a are enrichment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 7.2: Deterministic particle & mathematical plate for *Strong or concentrated?*. First Principle: Concentration (mol/L) counts how many acid molecules were added per litre, whereas acid strength (K_a) governs the statistical fraction of those molecules that ionize into H_3O^+.

Historical & Experimental Evidence: Electrical conductivity of dilute HCl vs. concentrated acetic acid

If you test 0.010 mol/L hydrochloric acid (HCl) and 1.00 mol/L acetic acid (CH_3COOH) with a conductivity probe and pH meter, the dilute 0.010 mol/L HCl has a higher hydronium concentration ($[H_3O^+]$ = 0.010 mol/L, pH 2.00) than the 100-times more concentrated 1.00 mol/L acetic acid ($[H_3O^+] \approx 0.0042$ mol/L, pH 2.37)! Strength measures the equilibrium fraction ionized, whereas concentration measures total solute per litre.

Key Terms

Strong vs. Weak acid
A strong acid ionizes nearly 100% in water; a weak acid reaches dynamic equilibrium with only a small fraction ionized.

Concentrated vs. Dilute
Concentrated means a large total molar amount of solute per litre of solution; dilute means a small total molar amount per litre.

Worked Reasoning Trace

Held fixed: Aqueous solution at 25 °C.

Changed: Compared 0.010 mol/L HCl (system 0, c = 0.01) with 1.00 mol/L CH_3COOH (system 1, c = 1.00).

Model result: 0.010 mol/L HCl is 100% ionized ($[H_3O^+]$ = 0.010 mol/L, [HA] = 0). 1.00 mol/L CH_3COOH is only ~0.42% ionized ($[H_3O^+]$ = 0.0042 mol/L, intact [HA] = 0.9958 mol/L).

Model boundary: Adding water to a strong acid makes it dilute, never 'weak'; evaporating water from acetic acid makes it concentrated, never 'strong'.

Common Misconception Trap · Strong or concentrated?

Intuitive Trap: "I think a concentrated weak acid always has 100% ionization because more solute is present — because diluting any acid forces more water molecules to turn into HCl."

Particle Mechanism Refutation: "Wait — look at the particle ledger: HCl is nearly 100% ionized into H_3O^+ and Cl^-, whereas CH_3COOH remains mostly intact molecules. Nearly 100% of the 0.010 mol/L HCl molecules ionize into H_3O^+ and Cl^-, whereas over 99% of the 1.00 mol/L CH_3COOH molecules remain intact undissociated molecules at equilibrium."

Why Claim A fails: Water is the solvent and proton acceptor; it never turns into chlorine or HCl. · Why Claim B holds up: Exact mechanism: $[H_3O^+]$ depends on the product of initial concentration and fraction ionized; 100% of 0.010 M (0.010 M) beats 0.42% of 1.00 M (0.0042 M).

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Dilute strong acid: 0.010 mol/L HCl	100% ionized: $[H_3O^+]$ = 0.010 mol/L, pH = 2.00.	Strong acid (HCl) at 0.01 mol/L: essentially 100% proton transfer to water ($HCl + H_2O \rightarrow H_3O^+ + Cl^-$), giving pH = 2. Each 10× dilution raises pH by 1 unit.
Concentrated weak acid: 1.00 mol/L CH_3COOH	Only 0.42% ionized: $[H_3O^+]$ = 0.0042 mol/L, pH = 2.37, intact [HA] = 0.996 mol/L.	Weak acid (CH_3COOH, $K_a = 1.8\times10^{-5}$) at 1 mol/L: only 0.423% of molecules transfer a proton at equilibrium ($CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$), leaving most molecules intact and giving pH = 2.373.
Moderately concentrated strong acid: 0.50 mol/L HCl	Both strong (100% ionized) and higher in concentration ($[H_3O^+]$ = 0.50 mol/L).	Strong acid (HCl) at 0.5 mol/L: essentially 100% proton transfer to water ($HCl + H_2O \rightarrow H_3O^+ + Cl^-$), giving pH = 0.301. Each 10× dilution raises pH by 1 unit.

Check Question: Which statement accurately compares a dilute strong acid (0.01 mol/L HCl) with a concentrated weak acid (1.0 mol/L CH_3COOH)? · **Transfer Challenge:** Transfer challenge: Sketch or describe the particles inside two beakers: (A) a concentrated weak acid and (B) a dilute strong acid. How do you show both concentration and strength in the particle count?

INVESTIGATION 7.3 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: ACIDBASE

Every step on the pH scale is tenfold

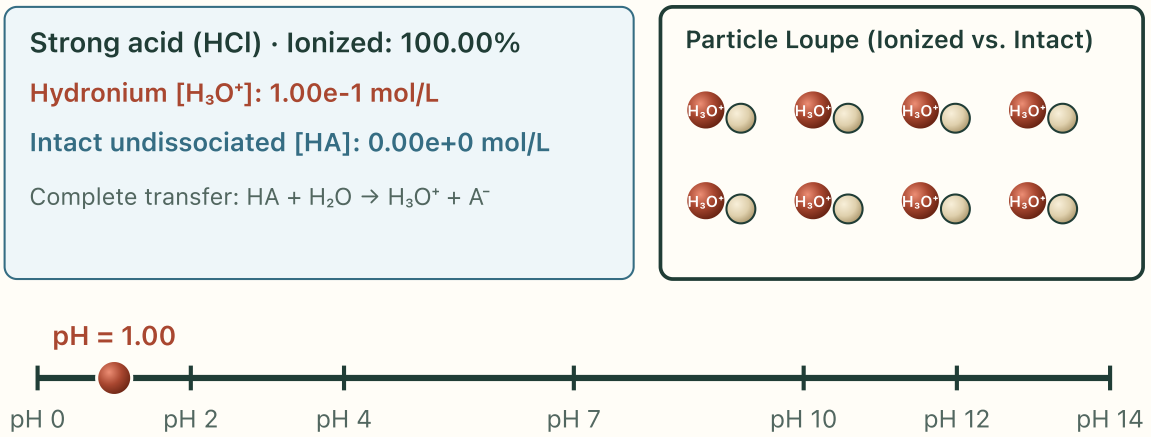
Why every 1-unit drop in pH means a tenfold increase in hydronium ion concentration

Because pH is defined logarithmically as $pH = -\log_{10}[H_3O^+]$ (in ideal dilute solution), a decrease of 1 pH unit means $[H_3O^+]$ is 10 times larger, and a 10-fold dilution of a strong acid raises pH by 1 unit.

Bench Observation: Diluting 0.10 mol/L HCl tenfold to 0.010 mol/L shifts its measured pH from 1.0 to 2.0; diluting a hundredfold to 0.0010 mol/L shifts pH to 3.0.

Investigation Task: Hold the system on Strong acid (HCl) at 0.10 mol/L and increase the tenfold dilution steps from 0 to 1, 2 and 3. Track both $[H_3O^+]$ and pH.

Every step on the pH scale is tenfold



Logarithmic pH scale ($-\log_{10}[H_3O^+]$) and submicroscopic particle loupe at 25 °C. Each 1-unit pH step represents a 10× change in $[H_3O^+]$.

Concentration and ionization account

Distinguish total solute from ionized species. This dilute, 25 °C concentration model approximates activities.

The bookkeeping idea: Keep separate accounts for total dissolved acid and the fraction ionized. Concentration and acid strength answer different questions. pH is logarithmic; pH numbers cannot be added like amounts.

Current explanatory plate · intro-models-2	
Total solute concentration / mol per L	0.1 mol/L
H_3O^+ concentration / mol per L	0.1 mol/L
Ionized or dissociated fraction / %	100 %
Calculated pH (25 °C ideal model)	1 dimensionless

$pH \approx -\log_{10}([H_3O^+]/(1\ mol/L)) = -\log_{10}(0.1) \approx 1$. The displayed concentration is rounded; pH is calculated before display rounding.

Model note: Strong acid (HCl) at 0.1 mol/L: essentially 100% proton transfer to water ($HCl + H_2O \rightarrow H_3O^+ + Cl^-$), giving pH = 1. Each 10× dilution raises pH by 1 unit.

Scientific inputs · scientific-values-2

Dilute-solution concentration approximation for K_w: 1e-14 $(mol/L)^2$ · reference. 25 °C; thermodynamic equilibrium constants are expressed through activities. Concentrations approximate activities in this model. Introductory rounded approximation. Source

Ethanoic-acid dissociation parameter: 0.000018 concentration-model parameter · reference. 25 °C, dilute aqueous model. Concentration approximation to activity-based equilibrium. Two significant figures. Source

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 7.1 Core; Supplement 9–10. Acid behaviour and strong versus weak acids Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.4.2.4; 4.4.2.6 (HT). pH and acid strength versus concentration Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Acids, Bases and Salts (pp. 4–5). Acids and pH; logarithms and K_a are enrichment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 7.3: Deterministic particle & mathematical plate for *Every step on the pH scale is tenfold*. First Principle: Because pH = $-\log_{10}[H_3O^+]$, every 1-unit drop in pH means a 10-fold multiplication in hydronium ion concentration—not a linear addition.

Historical & Experimental Evidence: Sørensen's logarithmic pH scale and glass electrode potentiometry

In 1909, Søren Peder Lauritz Sørensen introduced the pH scale at the Carlsberg Laboratory because aqueous hydronium concentrations span more than 14 powers of ten—from 1 mol/L in strong acid down to 10^{-14} mol/L in strong base. Defining pH = $-\log_{10}([H_3O^+])$ compresses each factor-of-10 change in $[H_3O^+]$ into a 1-unit step on the pH scale.

Key Terms

pH definition
For dilute aqueous solutions, pH ≈ $-\log_{10}([H_3O^+] / 1\ \text{mol}\ L^{-1})$, or equivalently $[H_3O^+] = 10^{-pH}$ mol/L.

Logarithmic decade step
Decreasing pH by 1 unit multiplies $[H_3O^+]$ by 10; decreasing pH by 2 units multiplies $[H_3O^+]$ by $10^2 = 100$.

Worked Reasoning Trace

Held fixed: Strong acid system (HCl, 100% ionized).

Changed: Increased concentration tenfold from 0.010 mol/L to 0.100 mol/L.

Model result: $[H_3O^+]$ increases tenfold from 0.010 mol/L to 0.100 mol/L, causing pH to decrease by exactly 1.00 unit from 2.00 to 1.00.

Model boundary: Because of the negative sign in $-\log_{10}$, a lower pH means a higher $[H_3O^+]$ concentration; and a 2-unit change is a factor of $10^2 = 100$, never 2×.

Common Misconception Trap · Every step on the pH scale is tenfold

Intuitive Trap: "I think 2 times greater in solution x — because each 1-unit decrease in pH adds 1 mol/L of H_3O^+ ions linearly."

Particle Mechanism Refutation: "Wait — look at the particle ledger: 100 times greater in Solution X. Each 1-unit decrease in pH corresponds to multiplying the number of H_3O^+ ions per litre by 10^1, so a 2-unit decrease multiplies $[H_3O^+]$ by $10^2 = 100$."

Why Claim A fails: The pH scale is logarithmic (powers of 10), not linear. · Why Claim B holds up: Exact mechanism: $10^{-1} / 10^{-3} = 10^2 = 100$ times as many hydronium ions per litre at pH 1 compared with pH 3.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
0.010 mol/L HCl (pH 2.00)	$[H_3O^+]$ = 1.0 × 10^{-2} mol/L gives pH = 2.00.	Strong acid (HCl) at 0.01 mol/L: essentially 100% proton transfer to water (HCl + H_2O → H_3O^+ + Cl^-), giving pH = 2. Each 10× dilution raises pH by 1 unit.
0.100 mol/L HCl (pH 1.00)	10× higher $[H_3O^+]$ (1.0 × 10^{-1} mol/L) lowers pH by 1 unit to 1.00.	Strong acid (HCl) at 0.1 mol/L: essentially 100% proton transfer to water (HCl + H_2O → H_3O^+ + Cl^-), giving pH = 1. Each 10× dilution raises pH by 1 unit.
0.0010 mol/L HCl (pH 3.00)	100× dilution from 0.10 mol/L raises pH by 2 units to 3.00.	Strong acid (HCl) at 0.001 mol/L: essentially 100% proton transfer to water (HCl + H_2O → H_3O^+ + Cl^-), giving pH = 3. Each 10× dilution raises pH by 1 unit.

Check Question: Solution X has pH = 2.0 and Solution Y has pH = 4.0. How does $[H_3O^+]$ in Solution X compare with Solution Y? · **Transfer Challenge:** Transfer challenge: How many litres of pure water would you need to add to 1.0 L of a pH 2.0 strong acid solution to raise its pH to 4.0?

INVESTIGATION 7.4 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: TITRATION

Titration curves and buffer resistance

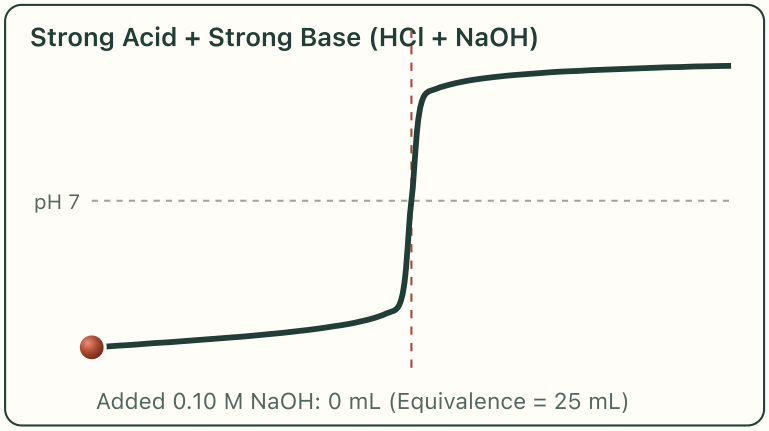
How weak-acid / conjugate-base buffers resist pH swings and why titration curves jump at equivalence

During an acid–base titration, added OH^- neutralizes HA into its conjugate base A^-. When comparable amounts of a weak acid (HA) and its conjugate base (A^-) coexist, the solution acts as a buffer that resists rapid pH swings.

Bench Observation: Adding 10 mL of 0.10 mol/L NaOH to an acetate buffer shifts pH by only ~0.37 units (4.74 → 5.11), whereas near the equivalence point (25 mL) a tiny drop causes a steep vertical pH jump.

Investigation Task: Step Added NaOH titrant from 0 mL to 12 mL, 24 mL, 26 mL, and 40 mL across System 0 (Strong acid), System 1 (Weak acid), and System 2 (Acetate buffer).

Titration curves and buffer resistance



Potentiometric pH titration curve (0–50 mL NaOH) and live millimole ledger of HA, conjugate base A^-, and excess OH^-.

Species, amounts & charge

Track the species present and distinguish amount, concentration and extent of ionization. Spectator ions also contribute to total charge.

The bookkeeping idea: Like matching an incoming delivery to the stock it consumes: calculate added hydroxide in mmol first, then reconcile neutralization. Concentrations require the new total volume; pH also requires an equilibrium calculation.

Current explanatory plate · intro-models-2

Solution pH (25 °C ideal model)	1
Unreacted acid H_3O^+ / HA remaining / mmol	2.5
Conjugate anion Cl^- / A^- in solution / mmol	0
Excess unreacted OH^- / mmol	0

Added hydroxide = 0.1 mol/L × 0 mL = 0 mmol. Calculated pH ≈ 1; neutralization and equilibrium approximations are separate steps.

Model note: Strong acid + strong base titration (0 mL NaOH added; equivalence at 25 mL, pH = 7.00): 2.5 mmol H_3O^+ remaining, 0 mmol excess OH^-, pH = 1.

Scientific inputs · scientific-values-2

Dilute-solution concentration approximation for K_w: 1e-14 $(mol/L)^2$ · reference. 25 °C; thermodynamic equilibrium constants are expressed through activities.

Concentrations approximate activities in this model. Introductory rounded approximation. Source

Ethanoic-acid dissociation parameter: 0.000018 concentration-model parameter · reference. 25 °C, dilute aqueous model. Concentration approximation to activity-based equilibrium. Two significant figures. Source

Initial acid solution volume: 25 mL · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Chosen acid and titrant concentration: 0.1 mol/L · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Initial buffer volume: 50 mL · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Initial amount of each buffer partner: 3 mmol · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Enrichment: No direct topic objective assigned. Model exploration beyond the mapped chemistry introduction Related prerequisite only: this investigation extends the cited outcome. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Enrichment: No direct topic objective assigned. Model exploration beyond the mapped chemistry introduction Related prerequisite only: this investigation extends the cited outcome. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 7.4: Deterministic particle & mathematical plate for *Titration curves and buffer resistance*. First Principle: A buffer containing comparable amounts of weak acid HA and conjugate base A^- absorbs added OH^- or H_3O^+ with minimal change in the $[A^-]/[HA]$ ratio and pH until buffer capacity is exhausted.

Historical & Experimental Evidence: Potentiometric pH titration curves and biological / acetate buffer plateaus

If you add 1.0 mmol of strong base (NaOH) to 1 L of pure water, the pH rockets from 7.0 to 11.0 (a 10,000-fold change in $[H_3O^+]$!). Yet if you add that same NaOH to a solution containing both weak acetic acid (CH_3COOH, HA) and its conjugate base acetate (CH_3COO^-, A^-), the pH barely nudges by a tenth of a unit. Why? The reserve of intact HA molecules neutralizes the added OH^- ($HA + OH^- \rightarrow A^- + H_2O$), keeping the $[A^-]/[HA]$ ratio—and therefore the pH—nearly constant until HA runs out at the equivalence point.

Key Terms

Buffer solution
A solution containing comparable amounts of a weak acid (HA) and its conjugate base (A^-) that resists large pH changes when small amounts of strong acid or strong base are added.

Equivalence point
The point in a titration where the molar amount of added titrant (e.g., OH^-) exactly equals the initial molar amount of analyte (e.g., HA).

Worked Reasoning Trace

Held fixed: System 1: Weak acid (2.5 mmol CH_3COOH) titrated with 0.10 mol/L NaOH.

Changed: Increased Added NaOH titrant from 12 mL (near half-equivalence buffer region) to 26 mL (just past the 25 mL equivalence point).

Model result: At 12 mL (1.20 mmol OH^- added), 1.30 mmol HA and 1.20 mmol A^- coexist in a buffer with pH = 4.71; at 26 mL (past equivalence), 0 mmol HA remains, 0.10 mmol excess OH^- accumulates, and pH jumps to 11.29!

Model boundary: Henderson–Hasselbalch buffer calculations lose accuracy at extreme dilutions or very high ionic strengths where activity coefficients deviate from 1.

Common Misconception Trap · Titration curves and buffer resistance

Intuitive Trap: "I think a buffer destroys added oh^- ions without conserving hydrogen or oxygen atoms — because the buffer prevents NaOH from dissolving in water."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Intact weak acid molecules (HA) react with added OH^- to form A^- and H_2O, preventing free OH^- from accumulating while keeping the $[A^-]/[HA]$ ratio nearly steady. Added OH^- ions are consumed by proton transfer from the reserve of intact HA molecules ($HA + OH^- \rightarrow A^- + H_2O$), so $[A^-]/[HA]$ and $[H_3O^+]$ change only slightly until HA is depleted."

Why Claim A fails: NaOH dissociates completely in aqueous buffer solutions. · Why Claim B holds up: Exact mechanism: as long as both HA and A^- are present in comparable millimole amounts, added OH^- converts HA → A^- rather than accumulating as free OH^-.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Weak acid half-equivalence buffer (12 mL NaOH, pH ≈ 4.71)	1.30 mmol HA and 1.20 mmol A^- coexist; pH sits right near pK_a (4.74) in the flat buffer region.	Weak acid (CH_3COOH) + strong base titration (12 mL NaOH added): Buffer regime where both HA (1.3 mmol) and A^- (1.2 mmol) coexist, resisting steep pH jumps (pH = 4.71).
Weak acid equivalence point (24–26 mL NaOH, steep pH jump)	All 2.50 mmol HA has been converted to A^-; excess OH^- drives a steep vertical jump in pH.	Weak acid (CH_3COOH) + strong base titration (26 mL NaOH added): pH = 11.29.
Acetate buffer + 10 mL NaOH (pH shifts only 4.74 → 5.11)	Starting with 2.50 mmol HA + 2.50 mmol A^- buffers the added base smoothly.	Conjugate acid–base buffer (CH_3COOH / CH_3COO^- with 1.0 mmol OH^- added): Intact HA neutralizes added OH^- ($HA + OH^- \rightarrow A^- + H_2O$), holding pH near pKa = 4.74 (current pH = 5.05).

Check Question: Why does a weak acid + conjugate base buffer (HA / A^-) resist large pH changes when a small amount of strong base (OH^-) is added? · **Transfer Challenge:** Transfer challenge: Compare System 0 (Strong acid HCl + NaOH) and System 1 (Weak acid CH_3COOH + NaOH) at 12 mL of added NaOH. Why is the pH of the weak acid titration higher and flatter in the middle of the titration, even though both require the exact same 25 mL of NaOH to reach equivalence?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 7 · The Two Bottles at pH 3.0

Two unlabelled flasks both read pH = 3.00 on a calibrated pH meter ($[H_3O^+] = 1.0 \times 10^{-3}$ mol/L). Flask A contains a strong acid (HCl), and Flask B contains a weak acid (CH_3COOH). You titrate 25 mL of each flask with 0.10 mol/L NaOH.

- Clue 1 (Initial Concentration): Because HCl is 100% ionized, pH 3.0 requires only 0.0010 mol/L HCl; because CH_3COOH is < 2% ionized, reaching pH 3.0 requires ~0.06 mol/L total CH_3COOH.
- Clue 2 (Titration Addition): Adding 5 mL of 0.10 M NaOH immediately shoots one flask to pH > 11, while the other flask stays near pH 4.7 in a flat buffer plateau.
- Clue 3 (Buffer Reserve): Intact weak acid HA reacts with added OH^- to form a conjugate HA / A^- buffer.

Resolution: Equal pH means equal free $[H_3O^+]$, NOT equal total acid concentration! Weak acid Flask B holds ~60× more total acid molecules (mostly intact HA), creating an HA/A^- buffer plateau during titration.

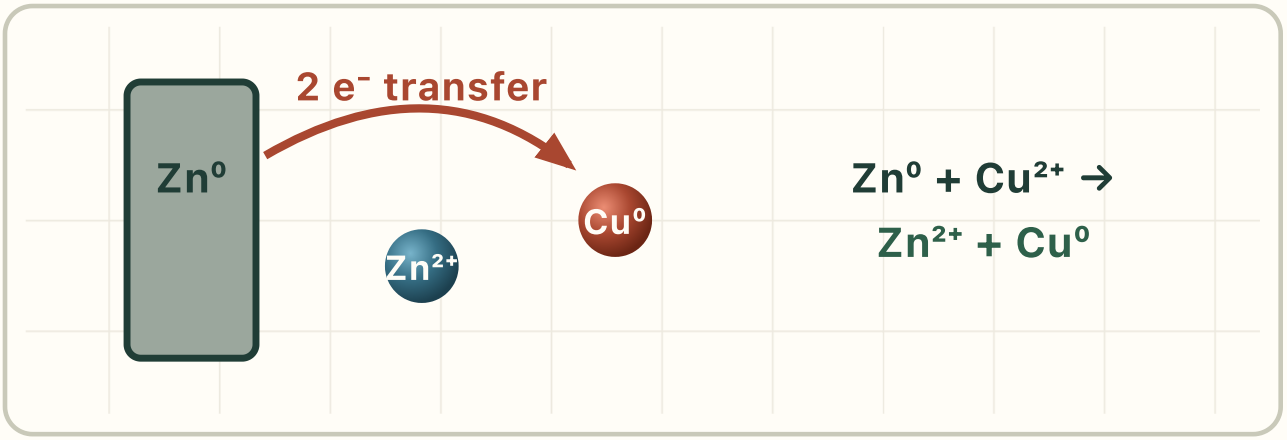
What our upgraded model now explains: Explains Brønsted–Lowry proton transfer, strong vs. weak acid ionization, concentrated vs. dilute solutions, the base-10 logarithmic pH scale, titration curves, and buffer capacity. · **Next boundary:** Acids and bases transfer bare protons (H^+); what happens when reacting species transfer ELECTRONS instead, plating metals and generating electrical voltage? → motivating Chapter 8 (Redox & Electrochemistry).

VOLUME III · CHEMICAL SYSTEMS & THE REAL WORLD · CHAPTER 8

Chapter 8: Electron transfer and electrochemistry

Track electron transfer alongside atom and charge conservation when metals and ions react.

Chapter 8 Prelude · How Does a Chemical Reaction Push Electrons Through a Wire?: When you dip a zinc strip into blue copper(II) sulfate, copper metal plates onto the zinc while the solution warms up. If you separate those exact same reactants into two half-cells connected by a wire and a salt bridge, the electron transfer powers a lightbulb or a clock! How do electron conservation, half-reaction stoichiometry, and thermodynamic driving force govern redox chemistry?



Macroscopic Bench: Metal plating, solution colour changes, voltmeter readings (+E_cell), and electrolytic charging.

Submicroscopic Particle Model: Valence electron loss (oxidation at anode) and gain (reduction at cathode) coupled with ion migration across a salt bridge.

Symbolic Notation: Half-reactions (Zn → Zn^{2+} + 2 e^-, Cu^{2+} + 2 e^- → Cu), oxidation states, and charge-balanced net ionic equations.

INVESTIGATION 8.1 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: REDOX

When metals and ions trade electrons

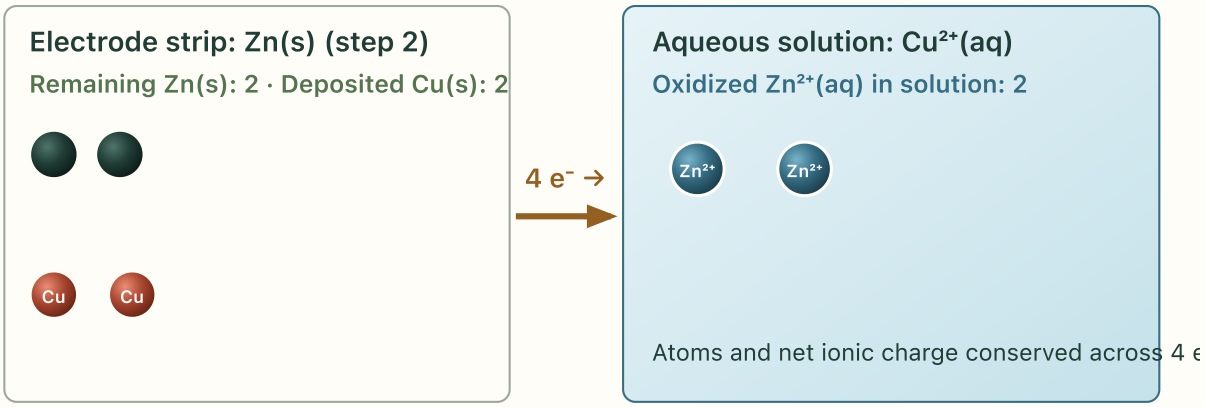
Connecting macroscopic metal displacement to submicroscopic electron transfer

In an oxidation-reduction (redox) reaction, electrons transfer from one species to another. Oxidation is the loss of electrons; reduction is the gain of electrons.

Bench Observation: When a grey zinc strip is placed in blue aqueous copper(II) sulfate, reddish-brown solid copper plates onto the strip while the blue Cu^{2+} colour fades and Zn^{2+} enters solution.

Investigation Task: Step the Zn(s) + Cu^{2+}(aq) reaction progress from 0 to 4 groups. Track which species loses electrons and which gains electrons.

When metals and ions trade electrons



Submicroscopic electron-transfer accounting at the metal–solution interface.

Matched electron transfers

Electrons lost must equal electrons gained in a complete redox account. Follow the half-reactions and ionic charge separately.

The bookkeeping idea: Like a matched transfer between two accounts: electrons leaving the donor must equal electrons arriving at the acceptor. An electron is negatively charged, so losing it raises the donor's charge. Count atoms separately.

Current explanatory plate · intro-models-2

Starting metal atoms on strip	2 atoms
Oxidized metal cations in solution	2 ions
Reduced metal atoms plated out	2 atoms
Electrons transferred	4 electrons

Electrons lost = electrons gained = 4 e^-; remaining metal = 2 atoms

Model note: Zn(s) + Cu^{2+}(aq) → Zn^{2+}(aq) + Cu(s): 2 Zn atom(s) lost 4 e^- (oxidized to 2 Zn^{2+}) while 2 Cu^{2+} ion(s) gained 4 e^- (reduced to 2 solid Cu).

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 6.4 Supplement 6–8. Oxidation, reduction and electron transfer Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.4.1.4 (HT). Electron accounting in redox Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Chemical Reactions and Equations (p. 4). Redox; electron accounting extends the treatment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 8.1: Deterministic particle & mathematical plate for *When metals and ions trade electrons*. First Principle: Redox reactions transfer valence electrons from a less electronegative species (oxidized) to a more electron-attracting species (reduced) while nuclei remain unchanged.

Historical & Experimental Evidence: Zinc in copper(II) sulfate: fading blue colour and copper plating

When a silvery strip of zinc metal is immersed in blue aqueous copper(II) sulfate, reddish-brown solid copper coats the strip while the blue colour of hydrated $Cu^{2+}(aq)$ gradually fades. Chemical analysis shows colourless $Zn^{2+}(aq)$ ions entering the solution: each neutral zinc atom transfers two electrons to a Cu^{2+} ion.

Key Terms

Oxidation
Loss of electrons by a chemical species (for example, $Zn(s) \rightarrow Zn^{2+}(aq) + 2\ e^-$).

Reduction
Gain of electrons by a chemical species (for example, $Cu^{2+}(aq) + 2\ e^- \rightarrow Cu(s)$).

Redox (reduction–oxidation) reaction
A chemical reaction in which electrons are transferred from one reactant to another; oxidation and reduction always occur together.

Worked Reasoning Trace

Held fixed: System 0: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$.

Changed: Increased reaction progress from 1 group to 3 groups.

Model result: 3 Zn atoms transfer 6 electrons in total (2 e^- per Zn) to reduce 3 Cu^{2+} ions into 3 neutral Cu(s) atoms.

Model boundary: Free electrons do not swim around loose in water; in direct immersion, electrons transfer directly at the metal-solution interface (or through a wire in a galvanic cell).

Common Misconception Trap · When metals and ions trade electrons

Intuitive Trap: "I think it gains 2 protons in its nucleus to become copper — because zinc atoms transfer 2 protons from their nuclei to copper nuclei."

Particle Mechanism Refutation: "Wait — look at the particle ledger: It loses 2 electrons and becomes a dissolved Zn^{2+} ion. Each neutral Zn atom (30p, 30e) loses 2 valence electrons to become $Zn^{2+}(aq)$ (30p, 28e), while each Cu^{2+} ion (29p, 27e) gains those 2 electrons to become neutral Cu(s) (29p, 29e)."

Why Claim A fails: Nuclear proton counts never change in chemical redox reactions. · Why Claim B holds up: Exact mechanism: redox reactions change oxidation states by transferring electrons between species while nuclear proton counts (Z = 30 and Z = 29) remain fixed.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
1 group reacted (2 e^- transferred)	1 Zn atom oxidizes to Zn^{2+}, transferring 2 e^- to reduce 1 Cu^{2+} to Cu(s).	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$: 1 Zn atom(s) lost 2 e^- (oxidized to 1 Zn^{2+}) while 1 Cu^{2+} ion(s) gained 2 e^- (reduced to 1 solid Cu).
3 groups reacted (6 e^- transferred)	3 Zn and 3 Cu^{2+} exchange 6 electrons.	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$: 3 Zn atom(s) lost 6 e^- (oxidized to 3 Zn^{2+}) while 3 Cu^{2+} ion(s) gained 6 e^- (reduced to 3 solid Cu).
4 groups reacted (8 e^- transferred)	All 4 starting Zn atoms on the strip have oxidized to $Zn^{2+}(aq)$.	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$: 4 Zn atom(s) lost 8 e^- (oxidized to 4 Zn^{2+}) while 4 Cu^{2+} ion(s) gained 8 e^- (reduced to 4 solid Cu).

Check Question: In $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$, what happens to each Zn atom that reacts? · **Transfer Challenge:** Transfer challenge: In the reaction $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$, sulfate ions ($SO_4^{2-}$) are also present in the beaker. Why are sulfate ions called 'spectator ions' in this electron transfer?

INVESTIGATION 8.2 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: REDOX

Balancing electrons as well as atoms

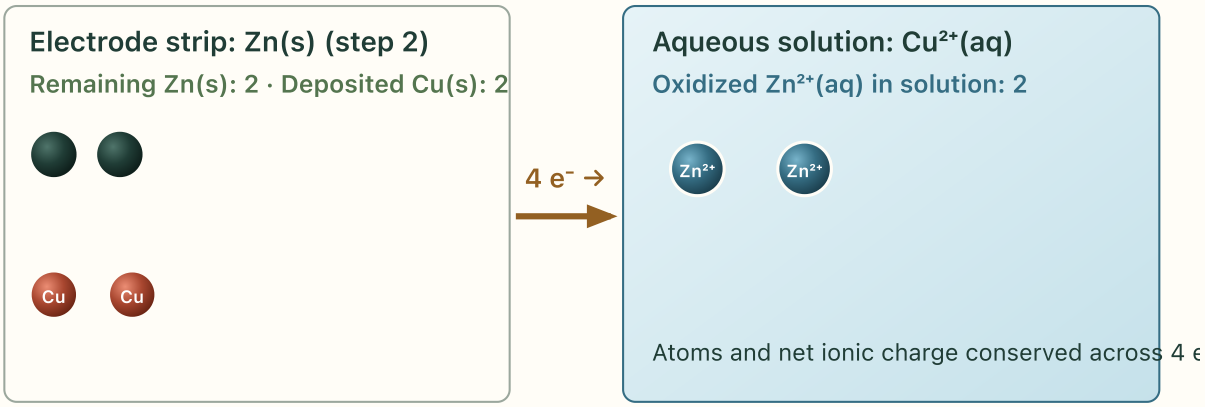
Why redox equations must conserve both elemental atoms and total electric charge

A balanced redox equation must conserve both every element's atom count and net electric charge—meaning the total number of electrons lost in oxidation must equal the total number of electrons gained in reduction.

Bench Observation: When solid magnesium reacts with aqueous silver ions (Ag^+), each Mg atom loses 2 electrons to form Mg^{2+}, whereas each Ag^+ ion can accept only 1 electron to form Ag(s).

Investigation Task: Select Mg(s) + 2 Ag^+(aq) → Mg^{2+}(aq) + 2 Ag(s) and step the reaction progress. Compare the number of Mg^{2+} ions formed with the number of Ag atoms plated.

Balancing electrons as well as atoms



Submicroscopic electron-transfer accounting at the metal–solution interface.

Matched electron transfers

Electrons lost must equal electrons gained in a complete redox account. Follow the half-reactions and ionic charge separately.

The bookkeeping idea: Like a matched transfer between two accounts: electrons leaving the donor must equal electrons arriving at the acceptor. An electron is negatively charged, so losing it raises the donor's charge. Count atoms separately.

Current explanatory plate · intro-models-2

Starting metal atoms on strip	2 atoms
Oxidized metal cations in solution	2 ions
Reduced metal atoms plated out	2 atoms
Electrons transferred	4 electrons

Electrons lost = electrons gained = 4 e^-; remaining metal = 2 atoms

Model note: Zn(s) + Cu^{2+}(aq) → Zn^{2+}(aq) + Cu(s): 2 Zn atom(s) lost 4 e^- (oxidized to 2 Zn^{2+}) while 2 Cu^{2+} ion(s) gained 4 e^- (reduced to 2 solid Cu).

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 6.4 Supplement 6–8. Oxidation, reduction and electron transfer Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.4.1.4 (HT). Electron accounting in redox Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Chemical Reactions and Equations (p. 4). Redox; electron accounting extends the treatment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 8.2: Deterministic particle & mathematical plate for *Balancing electrons as well as atoms*. First Principle: Electrons cannot accumulate freely in solution, so oxidation and reduction half-reactions must scale until electrons lost exactly equal electrons gained.

Historical & Experimental Evidence: Stoichiometry of silver crystal trees grown on magnesium or copper

When magnesium metal reacts with aqueous silver nitrate, shimmering needles of pure silver metal (Ag) grow while Mg^{2+} enters solution. Weighing the dissolved magnesium and deposited silver shows that exactly 2 moles of Ag(s) form for every 1 mole of Mg(s) consumed—because each Mg atom gives up 2 electrons while each Ag^{+} ion can accept only 1 electron!

Key Terms

Conservation of charge
In any balanced ionic equation, the sum of electric charges on the reactant side must equal the sum of electric charges on the product side.

Half-reaction electron balance
The total number of electrons lost in oxidation must equal the total number of electrons gained in reduction.

Worked Reasoning Trace

Held fixed: System 1: Mg(s) + 2 Ag^+(aq) → Mg^{2+}(aq) + 2 Ag(s).

Changed: Stepped reaction progress from 1 group to 2 groups.

Model result: 2 Mg atoms lose 4 electrons to form 2 Mg^{2+} (+4 total charge), while 4 Ag^+ (+4 total charge) gain 4 electrons to form 4 Ag(s).

Model boundary: Writing 'Mg + Ag^+ → Mg^{2+} + Ag' balances Mg and Ag atom counts (1 = 1), but violates conservation of charge (+1 ≠ +2) because 1 electron is unaccounted for!

Common Misconception Trap · Balancing electrons as well as atoms

Intuitive Trap: "I think one mg atom splits into two ag atoms during the reaction — because silver atoms are half the mass of magnesium atoms."

Particle Mechanism Refutation: "Wait — look at the particle ledger: One Mg atom loses 2 electrons, which reduces two Ag^+ ions (1 electron each) so electrons and net charge balance. Oxidation of 1 Mg → Mg^{2+} releases 2 electrons, and each Ag^+ → Ag reduction can accept only 1 electron, so 2 Ag^+ are required so no free electrons are created or left over."

Why Claim A fails: Silver (M ≈ 107.9 g/mol) is actually more than four times as massive as magnesium (M ≈ 24.3 g/mol); stoichiometry counts electrons and atoms, not mass. · Why Claim B holds up: Exact mechanism: electrons are conserved in redox reactions—the 2 electrons lost by 1 Mg atom must be accepted by 2 Ag^+ ions.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Mg + 2 Ag^+ (1 group: 2 e^-, 2 Ag plated)	1 Mg(s) + 2 Ag^+(aq) → 1 Mg^{2+}(aq) + 2 Ag(s); net charge is conserved.	Mg(s) + 2 Ag^+(aq) → Mg^{2+}(aq) + 2 Ag(s): 1 Mg atom(s) transferred 2 e^- to 2 Ag^+ ion(s), forming 1 Mg^{2+}(aq) and plating 2 Ag(s). Both atom counts and net ionic charge (+8 e) are conserved.
Mg + 2 Ag^+ (2 groups: 4 e^-, 4 Ag plated)	4 Ag(s) atoms plate out as 2 Mg(s) atoms oxidize.	Mg(s) + 2 Ag^+(aq) → Mg^{2+}(aq) + 2 Ag(s): 2 Mg atom(s) transferred 4 e^- to 4 Ag^+ ion(s), forming 2 Mg^{2+}(aq) and plating 4 Ag(s). Both atom counts and net ionic charge (+8 e) are conserved.
2 Al + 3 Cu^{2+} (1 group: 6 e^-, 3 Cu plated)	2 Al atoms (3 e^- each) reduce 3 Cu^{2+} ions (2 e^- each) across 6 electrons.	2 Al(s) + 3 Cu^{2+}(aq) → 2 Al^{3+}(aq) + 3 Cu(s): 2 Al atom(s) transferred 6 e^- to 3 Cu^{2+} ion(s), forming 2 Al^{3+}(aq) and plating 3 Cu(s).

Check Question: Why does 1 Mg(s) atom react with 2 Ag^+(aq) ions rather than 1 Ag^+(aq) ion? · **Transfer Challenge:** Transfer challenge: Solid aluminium (Al, which forms Al^{3+}) reacts with Cu^{2+}(aq) to form Al^{3+}(aq) and Cu(s). What is the smallest whole-number ratio of Al atoms to Cu^{2+} ions so that electrons lost equal electrons gained?

INVESTIGATION 8.3 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: REDOX

Why doesn’t copper dissolve in zinc sulfate?

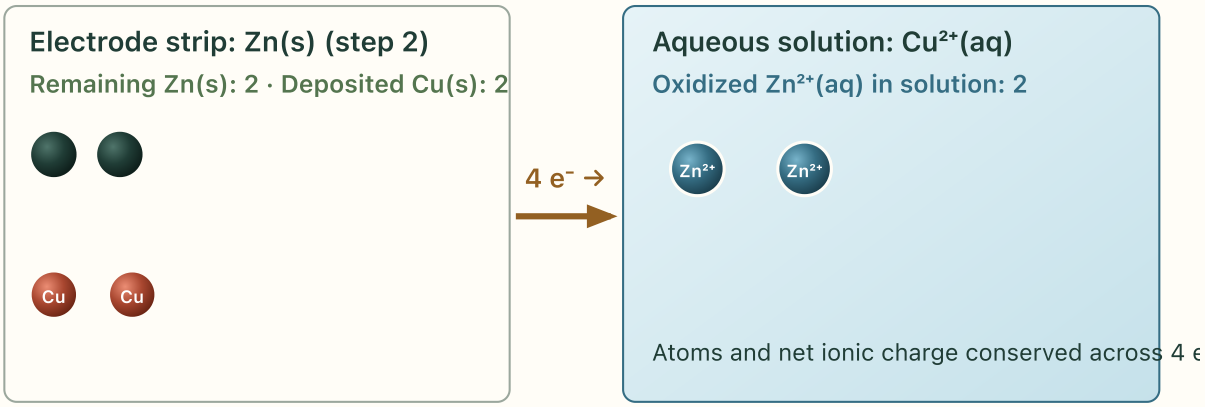
Why electron transfer has a preferred thermodynamic direction under given conditions

Electron transfer between a metal and an aqueous cation has a spontaneous thermodynamic direction under given conditions. Zinc spontaneously transfers electrons to $Cu^{2+}(aq)$, whereas solid copper placed in $Zn^{2+}(aq)$ shows no spontaneous reaction.

Bench Observation: A copper wire immersed in colourless aqueous zinc sulfate remains unchanged, even though we could write a balanced equation $Cu + Zn^{2+} \rightarrow Cu^{2+} + Zn$ on paper.

Investigation Task: Compare System 0 (Zn in Cu^{2+}) with System 3 (Cu in Zn^{2+}). Explain how this reinforces the rule that a balanced equation alone does not prove a reaction occurs.

Why doesn’t copper dissolve in zinc sulfate?



Submicroscopic electron-transfer accounting at the metal–solution interface.

Matched electron transfers

Electrons lost must equal electrons gained in a complete redox account. Follow the half-reactions and ionic charge separately.

The bookkeeping idea: Like a matched transfer between two accounts: electrons leaving the donor must equal electrons arriving at the acceptor. An electron is negatively charged, so losing it raises the donor’s charge. Count atoms separately.

Current explanatory plate · intro-models-2

Starting metal atoms on strip	2 atoms
Oxidized metal cations in solution	2 ions
Reduced metal atoms plated out	2 atoms
Electrons transferred	4 electrons

Electrons lost = electrons gained = 4 e^-; remaining metal = 2 atoms

Model note: Zn(s) + Cu^{2+}(aq) → Zn^{2+}(aq) + Cu(s): 2 Zn atom(s) lost 4 e^- (oxidized to 2 Zn^{2+}) while 2 Cu^{2+} ion(s) gained 4 e^- (reduced to 2 solid Cu).

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Enrichment: 6.4 Supplement 6–8. Oxidation, reduction and electron transfer Related prerequisite only: this investigation extends the cited outcome. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Enrichment: 4.4.1.4 (HT). Electron accounting in redox Related prerequisite only: this investigation extends the cited outcome. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: X · Unit I · Chemical Reactions and Equations (p. 4). Redox; electron accounting extends the treatment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 8.3: Deterministic particle & mathematical plate for *Why doesn't copper dissolve in zinc sulfate?*. First Principle: Spontaneous electron transfer flows toward the species that holds electrons at lower potential energy (higher reduction potential); reversing that direction requires external work.

Historical & Experimental Evidence: Asymmetry of metal displacement and galvanic cell voltage

Placing zinc metal into $Cu^{2+}(aq)$ spontaneously plates copper and releases thermal energy (or produces +1.10 V in a Daniell electrochemical cell). Yet if you place a strip of pure copper metal into colourless aqueous zinc sulfate (Zn^{2+}), nothing happens for hours or days! The reverse reaction requires an external electrical power source (electrolysis) to force electrons uphill.

Key Terms

Spontaneous redox direction
The direction of electron transfer that lowers the system's Gibbs free energy under the given conditions without external electrical work.

Galvanic vs. Electrolytic process
A galvanic cell harnesses a spontaneous redox reaction to do electrical work; electrolysis uses an external voltage to drive a non-spontaneous redox reaction.

Worked Reasoning Trace

Held fixed: Reaction progress set to 2 groups at 25 °C.

Changed: Switched from System 0 (Zn + Cu^{2+}) to System 3 (Cu + Zn^{2+} reverse test).

Model result: In System 0, 4 e^- transfer and 2 Cu(s) form. In System 3 (Cu + Zn^{2+}), 0 e^- transfer and 0 Zn(s) form spontaneously.

Model boundary: Just like any balanced equation, 'Cu(s) + Zn^{2+}(aq) → Cu^{2+}(aq) + Zn(s)' is balanced on paper for atoms and charge, yet it does not proceed spontaneously in a beaker.

Common Misconception Trap · Why doesn't copper dissolve in zinc sulfate?

Intuitive Trap: "I think copper atoms have no electrons at all — because the equation Cu + Zn^{2+} → Cu^{2+} + Zn cannot be balanced for atoms."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Electron transfer from Cu(s) to Zn^{2+}(aq) is thermodynamically uphill under these conditions. Copper holds its valence electrons more tightly than zinc (higher reduction potential), so transferring electrons from Cu(s) to Zn^{2+}(aq) is thermodynamically uphill without an external voltage source."

Why Claim A fails: Cu + Zn^{2+} → Cu^{2+} + Zn is completely balanced for both atoms and charge (+2 = +2), illustrating that balance alone does not imply spontaneity. · Why Claim B holds up: Exact mechanism: redox spontaneity depends on the relative free energy / reduction potentials of the two half-reactions, not on whether the equation can be written balanced on paper.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Forward spontaneous: Zn(s) + Cu^{2+}(aq)	Proceeds spontaneously: 4 e^- transferred, 2 Cu(s) formed.	Zn(s) + Cu^{2+}(aq) → Zn^{2+}(aq) + Cu(s): 2 Zn atom(s) lost 4 e^- (oxidized to 2 Zn^{2+}) while 2 Cu^{2+} ion(s) gained 4 e^- (reduced to 2 solid Cu).
Reverse test: Cu(s) + Zn^{2+}(aq)	No spontaneous reaction: 0 e^- transferred, all 4 Cu(s) remain unreacted.	Cu(s) placed in Zn^{2+}(aq): 0 electrons transferred. Copper metal does not spontaneously reduce Zn^{2+}(aq) under standard conditions; electron transfer has a preferred thermodynamic direction.

Check Question: When solid copper is placed in aqueous Zn^{2+} solution at standard conditions, no reaction occurs because... · **Transfer Challenge:** Transfer challenge: Connect this investigation back to Chapter 4 ('What a balanced equation does and does not say'). How does testing Cu(s) + Zn^{2+}(aq) reinforce the rule that a balanced equation does not guarantee a reaction occurs?

INVESTIGATION 8.4 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: REDOX

Galvanic cells vs. electrolytic charging

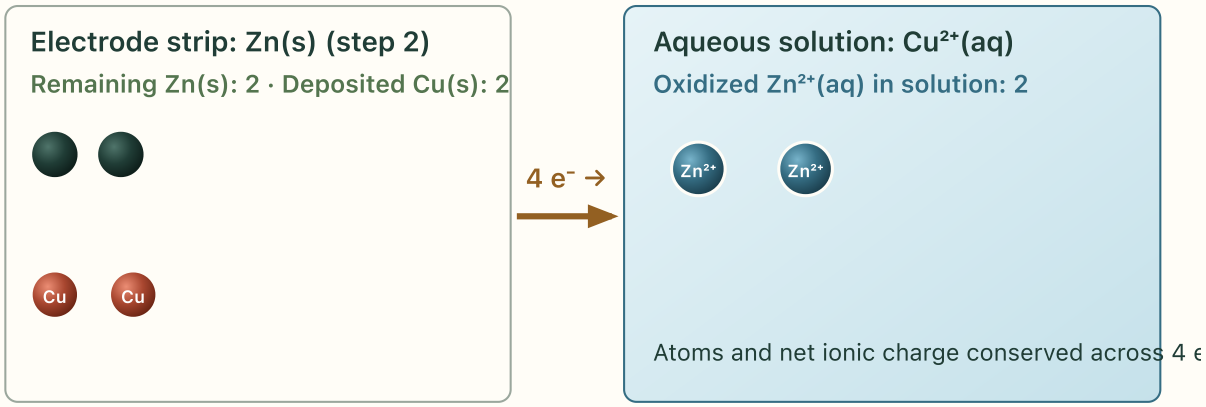
Harnessing spontaneous electron transfer through a wire while ions balance charge across a salt bridge

When oxidation and reduction half-reactions are separated into two half-cells connected by a wire and an ion-conducting salt bridge, a spontaneous redox reaction (E°_cell > 0) drives electric current through the external circuit (a galvanic cell). Driving the reverse uphill reaction requires an external power source.

Bench Observation: A Zn(s) | Zn^{2+}(aq) || Cu^{2+}(aq) | Cu(s) Daniell cell generates +1.10 V as electrons flow from the zinc anode to the copper cathode and ions migrate across the salt bridge to maintain charge neutrality.

Investigation Task: Compare System 0 (Zn + Cu^{2+}, spontaneous galvanic discharge) with System 3 (Cu + Zn^{2+}, uphill electrolytic direction). Explain the distinct roles of the wire and the salt bridge.

Galvanic cells vs. electrolytic charging



Submicroscopic electron-transfer accounting at the metal–solution interface.

Matched electron transfers

Electrons lost must equal electrons gained in a complete redox account. Follow the half-reactions and ionic charge separately.

The bookkeeping idea: Like a matched transfer between two accounts: electrons leaving the donor must equal electrons arriving at the acceptor. An electron is negatively charged, so losing it raises the donor's charge. Count atoms separately.

Current explanatory plate · intro-models-2

Starting metal atoms on strip	2 atoms
Oxidized metal cations in solution	2 ions
Reduced metal atoms plated out	2 atoms
Electrons transferred	4 electrons

Electrons lost = electrons gained = 4 e^-; remaining metal = 2 atoms

Model note: Zn(s) + Cu^{2+}(aq) → Zn^{2+}(aq) + Cu(s): 2 Zn atom(s) lost 4 e^- (oxidized to 2 Zn^{2+}) while 2 Cu^{2+} ion(s) gained 4 e^- (reduced to 2 solid Cu).

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Enrichment: 6.4 Supplement 6–8. Oxidation, reduction and electron transfer Related prerequisite only: this investigation extends the cited outcome. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Enrichment: 4.4.1.4 (HT). Electron accounting in redox Related prerequisite only: this investigation extends the cited outcome. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: X · Unit I · Chemical Reactions and Equations (p. 4). Redox; electron accounting extends the treatment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 8.4: Deterministic particle & mathematical plate for *Galvanic cells vs. electrolytic charging*. First Principle: Separating oxidation and reduction into half-cells channels electrons through an external wire while salt-bridge ion migration maintains electrical neutrality in both solutions.

Historical & Experimental Evidence: The Daniell cell (Zn | Zn^{2+} || Cu^{2+} | Cu) and salt-bridge open-circuit tests

If Zn(s) is placed directly into Cu^{2+}(aq), electrons hop straight from zinc to copper ions at the metal surface and the energy is released mostly as heat. In 1836, John Frederic Daniell separated the two half-reactions into two beakers connected by a metal wire and an ion-conducting porous barrier (salt bridge). Electrons now flow from the Zn anode through the external wire to the Cu cathode—performing electrical work (+1.10 V)—while spectator ions migrate through the salt bridge to prevent charge buildup. Removing the salt bridge immediately drops current to zero!

Key Terms

Anode and Cathode
Oxidation (loss of electrons) always occurs at the anode; reduction (gain of electrons) always occurs at the cathode.

Salt bridge
A pathway containing mobile electrolyte ions that migrate between half-cells to maintain electrical neutrality in each solution while electrons travel through the external wire.

Worked Reasoning Trace

Held fixed: System 0: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$.

Changed: Increased reaction progress from 0 to 3 electron-transfer groups.

Model result: 3 Zn atoms oxidize at the anode (releasing 6 e^- into the wire and 3 Zn^{2+} into the anode beaker) while 3 Cu^{2+} ions reduce at the cathode (consuming 6 e^- from the wire); anions migrate across the salt bridge to balance the +6 charge shift.

Model boundary: Electrons travel only through the metal electrodes and external wire; in the aqueous solution and salt bridge, electric current is carried by moving ions (cations and anions).

Common Misconception Trap · Galvanic cells vs. electrolytic charging

Intuitive Trap: "I think free electrons swim directly through the aqueous salt bridge solution — because electrons need to swim through the liquid inside the salt bridge to complete the loop."

Particle Mechanism Refutation: "Wait — look at the particle ledger: It allows mobile ions to migrate between half-cells to balance electric charge as Zn^{2+} enters one solution and Cu^{2+} plates out of the other. Without mobile counter-ions migrating between half-cells, oxidation builds up net positive charge in the anode solution and reduction leaves net negative charge in the cathode solution, electrostatically opposing further electron flow."

Why Claim A fails: Electrons travel exclusively through the external metal wire and electrodes, never as free solvated electrons in water. · Why Claim B holds up: Exact mechanism: electrons flow through the metal wire while ions migrate through the salt bridge to keep both half-cell solutions electrically neutral.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Galvanic discharge: Zn | Zn^{2+} || Cu^{2+} | Cu (2 groups, 4 e^-)	Spontaneous electron flow from Zn anode to Cu cathode transfers 4 e^-.	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$: 2 Zn atom(s) lost 4 e^- (oxidized to 2 Zn^{2+}) while 2 Cu^{2+} ion(s) gained 4 e^- (reduced to 2 solid Cu).
Galvanic discharge: 2 Al + 3 Cu^{2+} (1 group, 6 e^-)	2 Al atoms release 6 e^- through the wire to plate 3 Cu atoms.	$2\ Al(s) + 3\ Cu^{2+}(aq) \rightarrow 2\ Al^{3+}(aq) + 3\ Cu(s)$: 2 Al atom(s) transferred 6 e^- to 3 Cu^{2+} ion(s), forming 2 $Al^{3+}(aq)$ and plating 3 Cu(s).
Uphill reverse direction: Cu + Zn^{2+} (0 e^- without external voltage)	Requires an external power supply (electrolysis) to drive electrons from Cu to Zn^{2+}.	Cu(s) placed in $Zn^{2+}(aq)$: 0 electrons transferred. Copper metal does not spontaneously reduce $Zn^{2+}(aq)$ under standard conditions; electron transfer has a preferred thermodynamic direction.

Check Question: In a galvanic cell based on $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$, why is a salt bridge or porous separator needed between the two half-cells? · **Transfer Challenge:** Transfer challenge: As the Zn | Zn^{2+} || Cu^{2+} | Cu galvanic cell discharges, which electrode gains mass and which electrode loses mass?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 8 · The Dead Battery with Fresh Electrodes

A student builds a galvanic cell with a fresh magnesium anode in $MgSO_4(aq)$ and a silver cathode in $AgNO_3(aq)$, connected by a copper wire and a voltmeter. To keep the solutions pure, they omit the salt bridge between the two beakers. The voltmeter flashes a reading for a fraction of a millisecond and drops to 0.00 V!

- Clue 1 (Electron Stoichiometry): Each $Mg \rightarrow Mg^{2+} + 2\ e^-$ oxidation releases 2 electrons into the wire to reduce $2\ Ag^+ + 2\ e^- \rightarrow 2\ Ag(s)$.
- Clue 2 (Charge Accumulation): Every Mg^{2+} entering the left beaker adds +2 charge, and every 2 Ag^+ leaving the right beaker leaves behind 2 unbalanced NO_3^- (–2 charge).
- Clue 3 (Circuit Completion): When a KNO_3 salt bridge is inserted between the beakers, steady current flows and 2 mol Ag plate per 1 mol Mg dissolved.

Resolution: Electrons travel through the wire, while mobile ions migrate through the salt bridge to keep both half-cell solutions electrically neutral.

What our upgraded model now explains: Tracks electron transfer, balances both atoms and electric charge across unequal electron stoichiometries (Mg + 2 Ag^+, 2 Al + 3 Cu^{2+}), and separates spontaneous galvanic cells from electrolytic processes. · **Next boundary:** Why does zinc give up

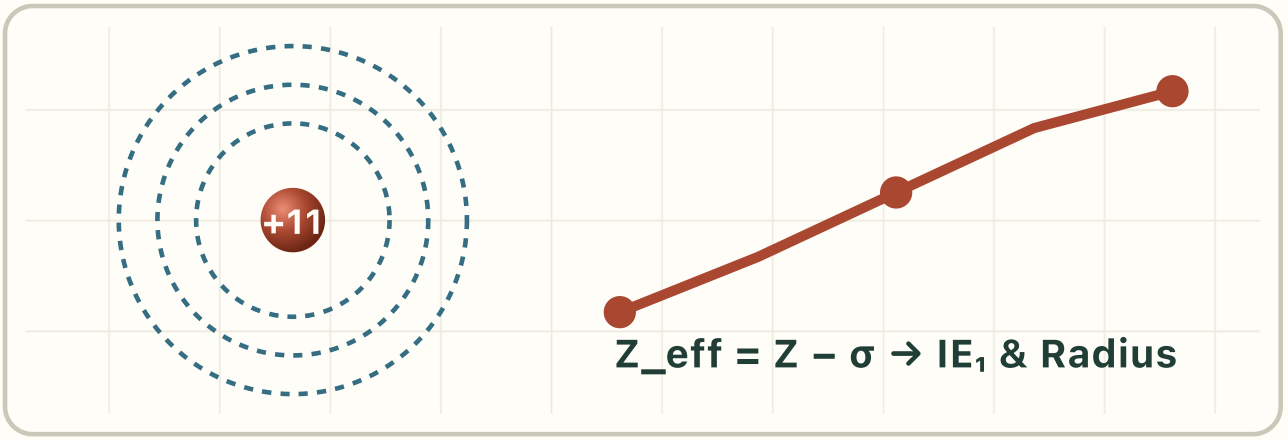
electrons more easily than copper, and why do atomic size, ionization energy, and electronegativity follow systematic patterns across the periodic table? → motivating Chapter 9 (Periodic Trends & Orbitals).

VOLUME III · CHEMICAL SYSTEMS & THE REAL WORLD · CHAPTER 9

Chapter 9: Periodic trends and quantum orbitals

Explain why atomic radius, ionization energy, and electronegativity shift systematically across the periodic table through nuclear charge and shielding.

Chapter 9 Prelude · Why Does the Periodic Table Have Its Shape?: Dmitri Mendeleev organized the elements by recurring chemical properties decades before anyone knew about protons or electron shells. Today we can explain the entire periodic table from first principles: why do atoms shrink as you move from left to right across a row even though you are adding more protons, neutrons, and electrons? And why does removing an electron ALWAYS cost energy?



Macroscopic Bench: Periodic trends in atomic density, reactivity of alkali metals vs. noble gases, and photoelectron ionization spectra.

Symbolic Notation: Shell/orbital configurations, Z_eff, covalent radius r (pm), and first ionization energy IE_1 (kJ/mol).

Submicroscopic Particle Model: Effective nuclear charge (Z_eff = Z – S) pulling valence shells inward against core-electron shielding and shell distance r.

INVESTIGATION 9.1 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: FORCES

Core shielding and effective nuclear charge

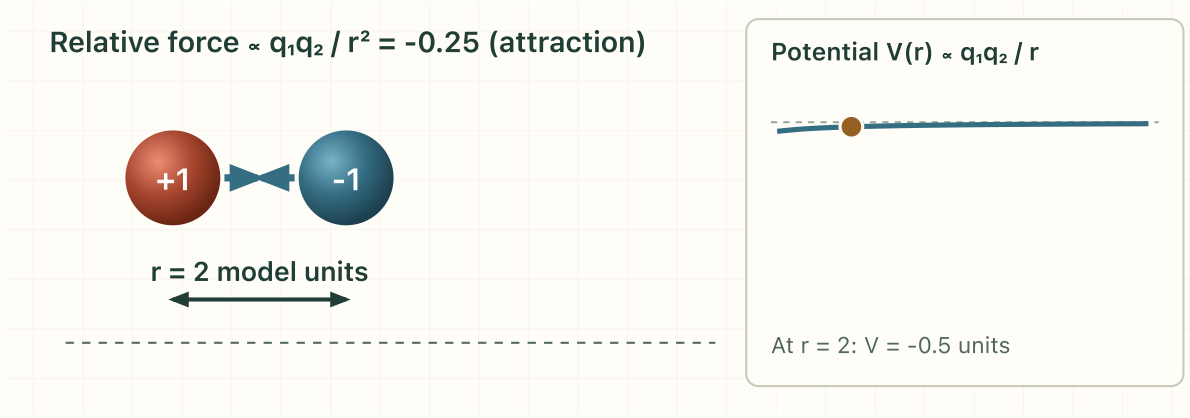
Why atoms shrink across a period as nuclear charge (+Z) outpaces core electron shielding

Across a period from left to right, each added proton increases nuclear charge (+Z) while added valence electrons enter the same principal shell and shield one another poorly. The rising effective nuclear charge (Z_eff) pulls the valence shell closer through stronger electrostatic attraction.

Bench Observation: Moving across Period 2 from lithium (Z = 3) to fluorine (Z = 9), covalent atomic radius shrinks markedly even though more electrons are present.

Investigation Task: Use the electrostatic force model to increase positive charge q_1 (representing Z_eff) from +1 to +3 while holding negative valence electron charge $q_2 = -1$, and then decrease separation r.

Core shielding and effective nuclear charge



Idealized point-charge schematic with live Coulomb potential curve V(r). Vector arrow lengths scale with $|q_1q_2|/r^2$.

Signed interaction account

The charge product determines the sign of this ideal electrostatic contribution; separation controls its magnitude.

The bookkeeping idea: Use a signed calculation with a declared reference: charge signs set attraction or repulsion, and separation changes the magnitude. Relative model units are not newtons, and the result does not establish a stable bond.

Current explanatory plate · intro-models-2

Signed relative electrostatic force -0.25 relative model units

$F_{rel} = q_1q_2/r^2 = (1 \times -1) / 2^2 = -0.25$ model units (not newtons)

Model note: Opposite charges attract. Point-charge model proportional to q_1q_2/r^2; negative denotes attraction. It omits quantum structure, screening and other interactions, so cannot alone predict a chemical bond.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Enrichment: No direct topic objective assigned. Model exploration beyond the mapped chemistry introduction Related prerequisite only: this investigation extends the cited outcome. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Enrichment: No direct topic objective assigned. Model exploration beyond the mapped chemistry introduction Related prerequisite

only: this investigation extends the cited outcome. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 9.1: Deterministic particle & mathematical plate for *Core shielding and effective nuclear charge*. First Principle: Across a period, increasing nuclear proton count Z outpaces core electron shielding, raising effective nuclear charge (Z_eff) and pulling valence electrons closer by Coulomb attraction.

Historical & Experimental Evidence: X-ray crystallographic covalent radii and photoelectron spectroscopy across Period 2

Beginners often guess that adding more protons and electrons across a row of the periodic table—from lithium (Z = 3, 3 e^-) to fluorine (Z = 9, 9 e^-)—should make the atom bigger. Experimental covalent radii reveal the exact opposite: lithium's radius is ~128 pm, carbon's is ~76 pm, and fluorine's is just ~57 pm! Why? All added electrons across Period 2 enter the same n = 2 valence shell, while the 2 inner core electrons ($1s^2$) shield them while nuclear charge climbs from +3 to +9. The rising effective nuclear charge ($Z_{eff} \approx Z - S$) pulls the valence shell inward by Coulomb's law.

Key Terms

Effective nuclear charge (Z_eff)
The net positive nuclear charge experienced by a valence electron after accounting for repulsive shielding by inner core electrons (approximated as Z_eff ≈ Z – S).

Core electron shielding
Inner-shell electrons repel outer valence electrons, partially cancelling the attraction of the positively charged nucleus.

Worked Reasoning Trace

Held fixed: Valence electron charge q_2 = –1 and shell distance r = 2.

Changed: Increased effective nuclear charge q_1 from +1 (alkali metal, Li) to +2 (next element across the period, Be).

Model result: Attractive electrostatic force magnitude doubles from |F| = 0.25 to |F| = 0.50 units, pulling the valence shell closer and shrinking atomic radius.

Model boundary: A two-particle point-charge calculation illustrates the directional Coulomb trend of Z_eff, whereas real multi-electron orbitals require quantum mechanical wavefunctions.

Common Misconception Trap · Core shielding and effective nuclear charge

Intuitive Trap: "I think more electrons always take up more space, so atoms grow larger from left to right — because electrons in the same shell completely block 100% of each added proton's positive charge."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Increasing effective nuclear charge (Z_eff) pulls electrons in the same valence shell more strongly toward the nucleus. Each step adds +1 proton to the nucleus while adding 1 electron to the SAME principal valence shell (which shields other valence electrons poorly), increasing net inward Coulomb attraction (Z_eff)."

Why Claim A fails: Same-shell electrons shield poorly (~0.35 per electron), so Z_eff increases steadily across a period. · Why Claim B holds up: Exact mechanism: core electrons shield effectively, but electrons in the same valence shell shield one another poorly, so Z_eff rises across a period and pulls the electron cloud closer.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Low Z_eff (+1 attracting –1 at r = 2 → F = –0.25)	Models a weakly held alkali-metal valence electron (e.g., Li, Z_eff ≈ +1).	Opposite charges attract. Point-charge model proportional to q_1q_2/r^2; negative denotes attraction. It omits quantum structure, screening and other interactions, so cannot alone predict a chemical bond.
Higher Z_eff (+2 attracting –1 at r = 2 → F = –0.50)	Moving right across a period increases Z_eff, doubling the inward electrostatic pull.	Opposite charges attract. Point-charge model proportional to q_1q_2/r^2; negative denotes attraction. It omits quantum structure, screening and other interactions, so cannot alone predict a chemical bond.
Higher Z_eff + contracted radius (+2 and –1 at r = 1 → F = –2.00)	Stronger Z_eff pulls the valence shell to a smaller radius r, further strengthening attraction.	Opposite charges attract. Point-charge model proportional to q_1q_2/r^2; negative denotes attraction. It omits quantum structure, screening and other interactions, so cannot alone predict a chemical bond.

Check Question: Why do atoms generally become smaller as you move from left to right across a row (period) of the periodic table? · **Transfer Challenge:** Transfer challenge: When you move DOWN a group (for example, from Li in n = 2 to Na in n = 3 to K in n = 4), why does atomic radius increase even though nuclear proton count Z is much larger?

INVESTIGATION 9.2 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: ENERGY

Why removing an electron costs energy

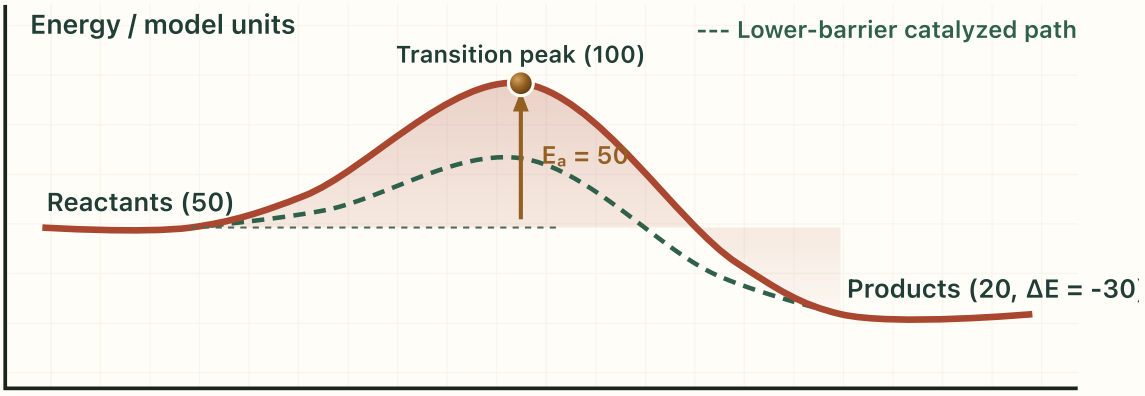
Why removing an electron always requires an energy input and tracks effective nuclear charge

First ionization energy (IE_1) is the energy required to remove the highest-energy electron from a gaseous atom: $X(g) \rightarrow X^+(g) + e^-$. Because electrostatic attraction holds the electron to the nucleus, removing an electron is always endothermic (requires energy input).

Bench Observation: First ionization energy generally increases across a period (e.g., Li = 520 kJ/mol vs. Ne = 2081 kJ/mol) and decreases down a group as valence electrons sit farther from the nucleus.

Investigation Task: Model ionization as an uphill energy transition (Products above Reactants) and compare a shallow energy cost (alkali metal, low Z_eff, large r) with a steep energy cost (noble gas, high Z_eff, small r).

Why removing an electron costs energy



Smooth reaction-coordinate energy profile showing activation barrier E_a, net energy change ΔE, and a lower-barrier catalyzed pathway.

Energy levels & barrier account

Compare levels using one reference. An activation barrier is not the net energy change.

The bookkeeping idea: Like comparing opening and closing balances in the same currency: net change = final level – initial level. The temporary amount needed to cross the barrier is a separate requirement, not the net change.

Current explanatory plate · intro-models-2	
Product minus reactant energy	-30 model energy units
Forward barrier	50 model energy units
Reverse barrier	80 model energy units

ΔE = products – reactants = 20 – 50 = -30 model units; E_a = peak – reactants = 100 – 50 = 50 model units

Model note: Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Enrichment: 5.1 Core 1–3; Supplement 4–6. Energy changes, profiles and activation barriers Related prerequisite only: this

investigation extends the cited outcome. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Enrichment: 4.5.1.1–2. Energy changes and reaction profiles Related prerequisite only: this investigation extends the cited outcome. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: X · Unit I · Chemical Reactions and Equations (p. 4). Exothermic/endothermic changes; barriers are enrichment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 9.2: Deterministic particle & mathematical plate for *Why removing an electron costs energy*. First Principle: Removing an electron from an atom or ion always requires an endothermic energy input to overcome electrostatic attraction to the positively charged nucleus.

Historical & Experimental Evidence: First ionization energies (IE_1) and photoelectron spectroscopy across the periodic table

How tightly does an atom hold its outermost electron? Measuring the minimum energy required to knock an electron out of a gas-phase atom—$X(g) \rightarrow X^+(g) + e^-$—shows that first ionization energy is ALWAYS positive (endothermic): +520 kJ/mol for lithium, +1086 kJ/mol for carbon, and +2081 kJ/mol for neon! Even for an alkali metal like lithium or sodium, removing an electron requires an energy input to overcome electrostatic attraction to the nucleus.

Key Terms

First ionization energy (IE_1)
The minimum energy required to remove the most loosely held electron from 1 mole of neutral gaseous atoms: $X(g) \rightarrow X^+(g) + e^-$ (always endothermic, $\Delta E > 0$).

Periodic trend in IE_1
IE_1 generally increases left-to-right across a period (due to rising Z_eff and smaller radius) and decreases down a group (due to larger principal shell distance r).

Worked Reasoning Trace

Held fixed: Reactant energy = 20 units (neutral atom ground state).

Changed: Increased Product energy (ionized $X^+ + e^-$ state) from 45 units (low IE_1, alkali metal) to 95 units (high IE_1, noble gas).

Model result: Net energy change ΔE rises from +25 units to +75 units—both are positive (energy absorbed from surroundings) because pulling a negative electron away from a positive cation always climbs an electrostatic hill.

Model boundary: Gas-phase ionization energy measures an isolated atom $X(g) \rightarrow X^+(g) + e^-$; in aqueous or crystal environments, subsequent hydration or lattice formation can release energy.

Common Misconception Trap · Why removing an electron costs energy

Intuitive Trap: "I think removing an electron from lithium releases energy because lithium wants to lose its electron — because alkali metals release energy when ionized because they want to have a full octet."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Neon's valence electrons experience a much larger effective nuclear charge at a smaller average distance, requiring a greater energy input to overcome electrostatic attraction. Work must be done (energy supplied) to pull a negatively charged electron away from the electrostatic attraction of the positively charged nucleus."

Why Claim A fails: Even lithium and sodium require +520 and +496 kJ/mol to ionize in the gas phase; the 'atoms want to lose electrons' story is a teleological misconception. · Why Claim B holds up: Exact mechanism: opposite charges attract, so separating an electron (–1) from the resulting cation (+1) always climbs an electrostatic potential energy barrier.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Alkali metal ionization (Low uphill cost: ΔE = +20 units)	Low Z_eff and larger radius make Li/Na easier to ionize, though still endothermic (+ΔE > 0).	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.
Mid-period atom ionization (Moderate uphill cost: ΔE = +40 units)	Higher Z_eff requires a larger energy input to remove a valence electron.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.
Noble gas ionization (Steep uphill cost: ΔE = +60 units)	High Z_eff and small radius in Ne hold valence electrons in a deep electrostatic well.	Schematic energy profile in model units, not measured enthalpy. Lower products indicate energy release in this model; a barrier can remain. Changing the peak alone leaves the overall difference unchanged.

Check Question: Why is the first ionization energy of neon (Z = 10) much larger than that of lithium (Z = 3)? · **Transfer Challenge:** Transfer challenge: Why is the SECOND ionization energy of lithium (removing a core 1s electron from Li^+, 7298 kJ/mol) more than fourteen times larger than its first ionization energy (removing the 2s valence electron, 520 kJ/mol)?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 9 · The Ionization Energy Cliff

Spectroscopists measure the successive ionization energies of an unknown Period 2 element X: removing the 1st electron (IE_1) costs 520 kJ/mol, but removing the 2nd electron (IE_2) suddenly skyrockets to 7,298 kJ/mol—more than 14 times higher!

- Clue 1 (Endothermic Sign): Both IE_1 (+520 kJ/mol) and IE_2 (+7,298 kJ/mol) are positive because removing an electron always requires energy input.
- Clue 2 (Shell Jump): A 14-fold jump in ionization energy indicates that the 1st electron came from an outer valence shell (n = 2, heavily shielded), while the 2nd electron was pulled from an inner core shell (n = 1, close to the nucleus with minimal shielding).
- Clue 3 (Period 2 Candidates): Lithium (Z = 3, 2·1), Beryllium (Z = 4, 2·2), Carbon (Z = 6, 2·4), Neon (Z = 10, 2·8).

Resolution: Lithium (Z = 3) has 1 valence electron in n = 2 (IE_1 = 520 kJ/mol) and 2 core electrons in n = 1; once the single n = 2 electron is removed, the next electron must be pulled from the inner n = 1 core shell.

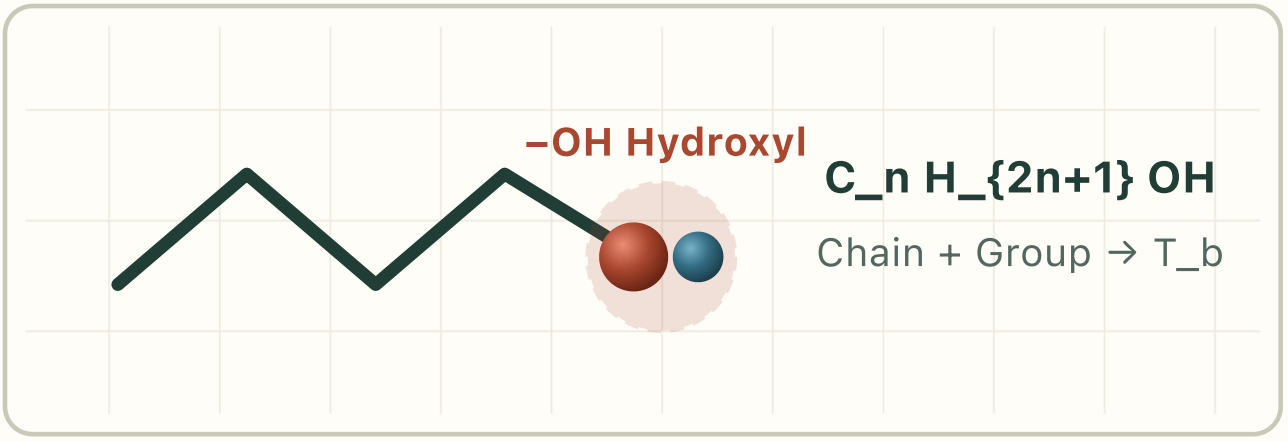
What our upgraded model now explains: Connects Coulomb's law ($F \propto Z_{eff} / r^2$) and core shielding to atomic radius shrinkage across periods, expansion down groups, and endothermic ionization energy trends. · **Next boundary:** Sitting right in the middle of Period 2 (Z = 6, 4 valence electrons, intermediate Z_{eff} and electronegativity) is carbon—whose unique ability to bond with itself launches Chapter 10 (Organic Chemistry & Life).

VOLUME III · CHEMICAL SYSTEMS & THE REAL WORLD · CHAPTER 10

Chapter 10: Carbon chains and the chemistry of life

Discover why carbon's four valence electrons build stable chains, rings, and functional groups that determine molecular polarity and biological function.

Chapter 10 Prelude · Why Is Life Built on Carbon and Functional Groups?: Over 95% of all known chemical compounds contain carbon! From methane gas and plastic polymers to DNA, glucose, and cell membranes, carbon's four valence electrons snap together into chains, branched skeletons, and rings. Yet a pure carbon–hydrogen chain is greasy and insoluble in water—how does swapping a single hydrogen atom for an –OH or $–NH_2$ group transform a nonpolar gas into a water-soluble molecule of life?



Macroscopic Bench: Volatility of hydrocarbon fuels vs. water miscibility and high boiling points of alcohols, sugars, and amino acids.

Symbolic Notation: Structural formulas, isomer connectivity, and functional group notation (R–OH, $R–NH_2$).

Submicroscopic Particle Model: Tetrahedral (109.5°) sp^3 carbon backbones decorated with polar functional groups (–OH, $–NH_2$, –COOH) that form hydrogen bonds.

INVESTIGATION 10.1 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: COMPOSITION

Carbon chains, rings and 3D networks

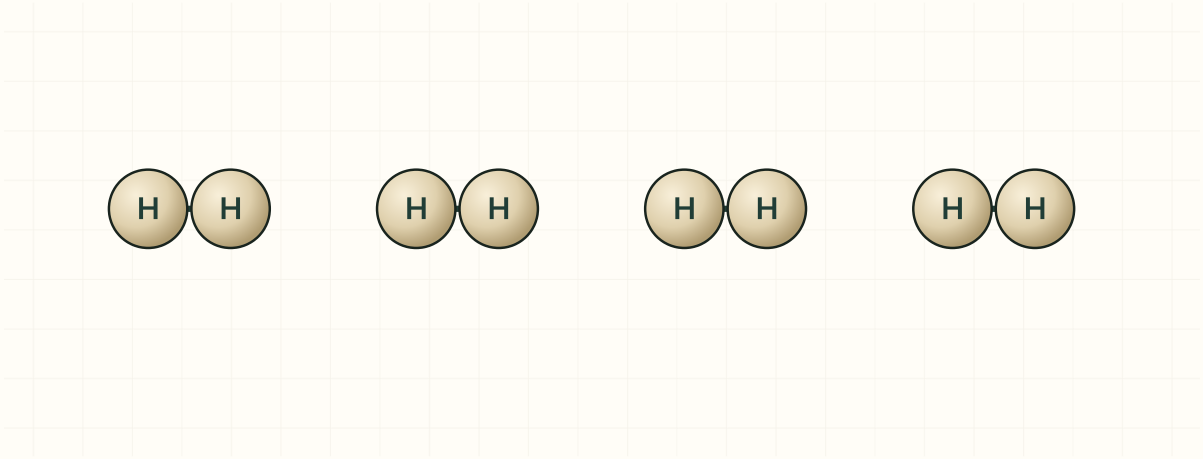
Why carbon's four valence electrons build tetrahedral chains, rings, and contrasting allotropes

With four valence electrons, carbon forms four strong covalent bonds with itself and with hydrogen, oxygen, and nitrogen—building tetrahedral (sp^3) chains like methane and diamond as well as planar layered sheets like graphite.

Bench Observation: Methane (CH_4) is a discrete tetrahedral gas molecule, diamond is a rigid 3D tetrahedral carbon network, and graphite consists of stacked 2D carbon sheets that conduct electricity along the layers.

Investigation Task: Compare Diamond (C) with Pure H_2O and Pure O_2 in the composition model, and inspect the 3D diamond and graphite structures.

Carbon chains, rings and 3D networks



Grouped atom symbols show molecular composition. Lines mark connectivity, not exact bond order, electron positions or geometry.

Account for the model

Keep quantities, units and assumptions together. A computed output is not experimental evidence.

The bookkeeping idea: Like sorting stock by identity: count element identities, chemical substances and particles in separate columns. Mixing two stocks is not evidence that they reacted.

Current explanatory plate · intro-models-2

Substances represented 1

Model note: H_2 is an element: each molecule contains only hydrogen atoms. A molecule need not be a compound.

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 2.1; 2.4–2.7. Substances, bonding and structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.2.1–2. Structure–property explanations Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: IX · Atoms and Molecules · C-1.1, C-7.1 (pp. 11–12). Compounds, bonding and fixed proportions Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 10.1: Deterministic particle & mathematical plate for *Carbon chains, rings and 3D networks*. First Principle: Carbon's four valence electrons form up to four strong covalent bonds in tetrahedral chains, rings, or planar sheets, allowing one element to build vast structural diversity.

Historical & Experimental Evidence: Tetrahedral bond angles (van 't Hoff–Le Bel) and X-ray structures of diamond vs. graphite

Why is terrestrial biochemistry built on carbon rather than silicon or boron? With atomic number Z = 6 (2 core electrons and 4 valence electrons), carbon forms four strong covalent bonds with itself ('catenation') and with H, O, and N. Because four bonding pairs around a carbon atom repel one another symmetrically in 3D space, saturated carbon centres adopt a 109.5° tetrahedral geometry—Seen both in discrete methane (CH_4) molecules and in the continuous 3D covalent network of diamond!

Key Terms

Catenation
The ability of an element (especially carbon) to form strong covalent bonds with other atoms of the same element in long chains, branched skeletons, and rings.

Tetrahedral carbon centre (109.5°)
A carbon atom bonded to four neighbours arranged toward the corners of a regular tetrahedron so electron-pair repulsions are minimized in 3D space.

Worked Reasoning Trace

Held fixed: Pure substance covalent bonding comparison.

Changed: Switched from Sample 3 (Pure H_2O discrete molecules) to Sample 5 (Diamond extended carbon network).

Model result: Both use directional covalent bonding, but H_2O forms discrete 3-atom molecules whereas diamond links every carbon atom to 4 neighbours in a continuous 3D crystal.

Model boundary: Chemical formulas like CH_4 or C(diamond) state composition; you must inspect 3D connectivity to distinguish isomers and allotropes.

Common Misconception Trap · Carbon chains, rings and 3D networks

Intuitive Trap: "I think carbon atoms only form ionic lattices like nacl in living cells — because carbon atoms transfer 4 electrons completely to form C^{4+} and C^{4-} ionic crystals in water."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Each carbon atom can form four strong, stable covalent bonds with other carbon atoms and heteroatoms in chains, branches, and rings. Its small atomic radius and four valence electrons allow strong directional C–C and C–H covalent bonds with high bond dissociation enthalpies."

Why Claim A fails: Forming a bare C^{4+} ion requires far too much ionization energy; carbon bonds covalently. · Why Claim B holds up: Exact mechanism: strong overlap of carbon's n = 2 valence orbitals creates robust C–C (~348 kJ/mol) and C–H (~413 kJ/mol) covalent frameworks.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Diamond network (Pure carbon, 3D tetrahedral covalent network)	Each carbon atom is covalently bonded to 4 neighbours at 109.5° in an extended 3D lattice.	Diamond is an element with an extended covalent structure of carbon. The formula C alone does not specify structure or hardness.
Discrete covalent molecule (H_2O, 1 compound)	Finite molecules with strong internal covalent bonds and weaker attractions between molecules.	H_2 and O_2 together form a mixture. Their proportions can vary. Placing them together in a diagram does not show a reaction.
Extended ionic crystal (NaCl, 1:1 ion lattice)	Contrast: NaCl is an extended electrostatic ion lattice, whereas diamond is an extended covalent network.	Solid NaCl is an ionic compound with an extended arrangement of Na^+ and Cl^- ions. NaCl gives a 1:1 ratio, not a discrete molecule in the solid.

Check Question: What structural feature allows carbon to serve as the backbone for millions of organic and biological molecules? · **Transfer Challenge:** Transfer challenge: Butane and isobutane both have the molecular formula C_4H_{10}, yet one is a straight 4-carbon chain and the other is a branched 3-carbon chain with a central carbon bonded to three carbons. Why do they have different boiling points despite sharing the formula C_4H_{10}?

INVESTIGATION 10.2 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: POLARITY

How functional groups transform properties

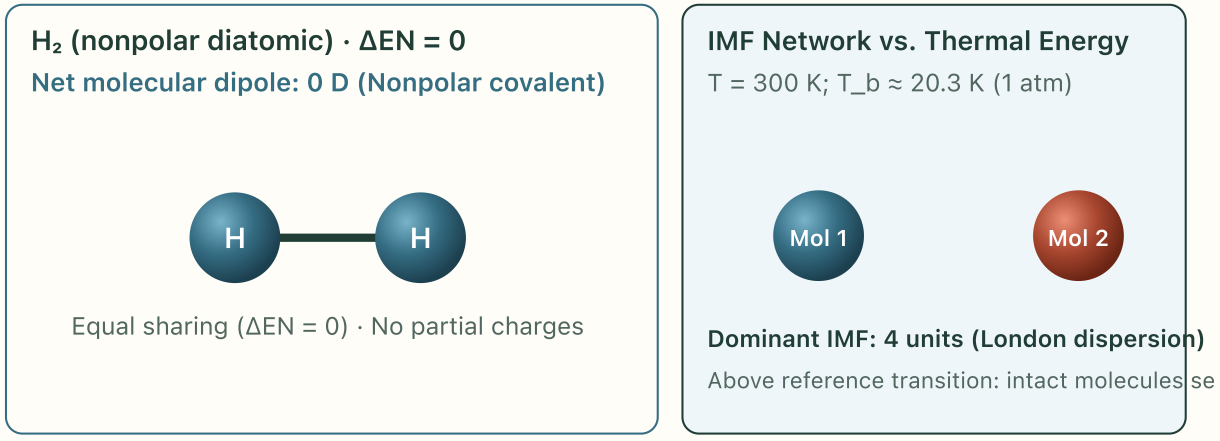
How attaching polar –OH or –NH_2 functional groups transforms molecular polarity and boiling point

Replacing a nonpolar C–H bond on a hydrocarbon skeleton with a polar functional group—such as a hydroxyl (–OH) or amino (–NH_2) group—introduces a permanent dipole and hydrogen-bonding sites that dramatically increase water solubility and boiling point.

Bench Observation: Methane (CH_4) is a nonpolar gas (boiling point 112 K) that barely dissolves in water, whereas adding polar O–H or N–H groups creates hydrogen-bonding networks with high boiling points and water miscibility.

Investigation Task: Compare nonpolar CH_4 (0 D) with polar H–Cl (1.08 D), NH_3 (1.47 D), and H_2O (1.85 D) across temperatures.

How functional groups transform properties



Left: 3D bond dipole addition ($\delta^+ \rightarrow \delta^-$). Right: Intermolecular attraction dashed bridge vs. thermal kinetic motion at 300 K.

Vector contributions

Bond dipoles add as vectors in space. Equal scalar magnitudes can cancel when their directions oppose.

The bookkeeping idea: This account needs arrows, not just totals: dipole contributions add with direction. Opposing vectors can cancel even though every bond contribution is nonzero.

Current explanatory plate · intro-models-2	
Bond electronegativity difference ΔEN	0 dimensionless
Net molecular dipole moment / D	0 D
Relative intermolecular attraction / model units	4 relative model units
Reference 1 atm boiling/sublimation point / K	20.3 K

Model note: Hydrogen (H_2) at 300 K (Above reference transition at 1 atm via weak London dispersion only): Two identical H nuclei share electrons equally (ΔEN = 0, dipole = 0 D). Weak dispersion attractions are easily overcome by thermal motion above 20.3 K.

Scientific inputs · scientific-values-2

Hydrogen (H_2): deltaEN: 0 Pauling scale · reference. Difference of rounded Pauling electronegativities; not a measured bond charge. Introductory approximation. Checked 2026-10-28. Source

Hydrogen (H_2): dipole: 0 D · reference. Rounded gas-phase dipole; zero values reflect the geometry shown. Introductory approximation. Checked 2026-10-28. Source

Hydrogen (H_2): boiling: 20.3 K · reference. Approximate pure-substance normal boiling point at 1 atm. Introductory approximation. Checked 2026-10-28. Source

Hydrogen (H_2): attraction: 4 model units · illustrative. Author-selected comparison score, not an energy. Introductory approximation. Checked 2026-10-28.

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Enrichment: 2.5 Supplement 5. Qualitative intermolecular attractions in small molecules Related prerequisite only: this investigation extends the cited outcome. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Enrichment: 4.2.2.4. Molecular attractions; dipoles are enrichment Related prerequisite only: this investigation extends the cited outcome. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 10.2: Deterministic particle & mathematical plate for *How functional groups transform properties*. First Principle: Polar functional groups (like –OH and $–NH_2$) attach strong bond dipoles and hydrogen-bonding sites onto carbon skeletons, transforming intermolecular attractions and solubility.

Historical & Experimental Evidence: Comparing hydrocarbons (CH_4, C_2H_6) with alcohols (CH_3OH, C_2H_5OH) and amines

Ethane (C_2H_6, 30 g/mol) is a colourless nonpolar gas that boils at −89 °C and refuses to dissolve in water. Replace just one hydrogen atom with a hydroxyl functional group (–OH) to make methanol (CH_3OH, 32 g/mol)—nearly the exact same molar mass!—and it transforms into a liquid boiling at +65 °C that mixes with water in all proportions. Functional groups act as the chemical control knobs of organic molecules by introducing local bond dipoles (δ^+/δ^-) and hydrogen-bonding sites.

Key Terms

Functional group
A specific atom or group of bonded atoms (such as hydroxyl –OH, amino $–NH_2$, or carboxyl –COOH) that imparts characteristic polarity, intermolecular forces, and chemical reactivity to an organic carbon skeleton.

Hydrophilic vs. Hydrophobic regions
Polar, hydrogen-bonding functional groups interact favourably with water (hydrophilic), whereas nonpolar C–H hydrocarbon chains experience only dispersion forces (hydrophobic).

Worked Reasoning Trace

Held fixed: Thermal energy T = 300 K.

Changed: Switched from nonpolar CH_4 (System 4: ΔEN = 0.35, 0.00 D) to polar NH_3 (System 5: ΔEN = 0.84, 1.47 D) and H_2O (System 2: ΔEN = 1.24, 1.85 D).

Model result: Replacing symmetric C–H bonds with polar N–H or O–H groups + lone pairs creates a strong net dipole and hydrogen-bonding network, raising boiling point from 112 K to 240 K and 373 K.

Model boundary: As a nonpolar hydrocarbon chain grows very long (e.g., octanol $C_8H_{17}OH$), cumulative dispersion forces along the carbon tail begin to dominate over a single –OH group for water solubility.

Common Misconception Trap · How functional groups transform properties

Intuitive Trap: "I think adding –oh removes all covalent bonds and turns the molecule into a noble gas — because the –OH group doubles the molar mass of every carbon atom in the chain."

Particle Mechanism Refutation: "Wait — look at the particle ledger: The polar –OH group enables strong intermolecular hydrogen bonding, raising the boiling point and water solubility compared with nonpolar CH_4. The large electronegativity difference in the O–H bond creates δ^+ H and δ^- O lone pairs that form strong intermolecular hydrogen bonds with neighbouring molecules and with H_2O."

Why Claim A fails: Each carbon atom retains its atomic mass of ~12 u. · Why Claim B holds up: Exact mechanism: polar hydroxyl groups introduce directional hydrogen-bonding attractions that require much higher thermal energy to separate and mix favourably with water's H-bond network.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Nonpolar hydrocarbon baseline: CH_4 (0.00 D, T_b = 112 K)	Symmetric C–H bonds cancel to zero net dipole; only weak dispersion forces act between molecules.	Methane (CH_4) at 300 K (Above reference transition at 1 atm via weak London dispersion (symmetric tetrahedron)): Four weakly polar C–H bonds arranged symmetrically in a 109.5° tetrahedron cancel to net dipole = 0 D, leaving only weak dispersion attractions (boiling point 111.7 K).
Polar amino-type group: NH_3 (1.47 D, T_b = 240 K)	Polar N–H bonds and a nitrogen lone pair enable intermolecular hydrogen bonding.	Ammonia (NH_3) at 300 K (Above reference transition at 1 atm via hydrogen bonding + dipole–dipole + dispersion): Trigonal pyramidal geometry gives a net dipole of 1.47 D and N–H···N hydrogen bonding, raising its boiling point (239.8 K) well above nonpolar CH_4 (111.7 K).
Polar hydroxyl-type network: H_2O (1.85 D, T_b = 373 K)	Strong O–H bond dipoles (ΔEN = 1.24) form an extensive hydrogen-bonded liquid at 300 K.	Water (H_2O) at 300 K (Below reference transition; solid/liquid not distinguished via strong hydrogen bonding + dipole + dispersion): Bent 104.5° geometry prevents the two polar O–H bond dipoles from cancelling (net dipole = 1.85 D), enabling directional O–H···O hydrogen bonding and an unusually high boiling point (373.15 K).

Check Question: How does replacing a hydrogen atom on methane (CH_4) with a polar hydroxyl group (–OH, forming methanol CH_3OH) change its intermolecular interactions? · **Transfer Challenge:** Transfer challenge: Soap and cell-membrane phospholipid molecules have a long nonpolar hydrocarbon tail and a charged/polar head group. How does having both regions in one molecule allow soap to bridge grease droplets and water?

INVESTIGATION 10.3 · PRINCIPLE 3: THERMAL MOTION VS. ATTRACTION · MODEL: ORGANIC

Hydrocarbon chain length and dispersion forces

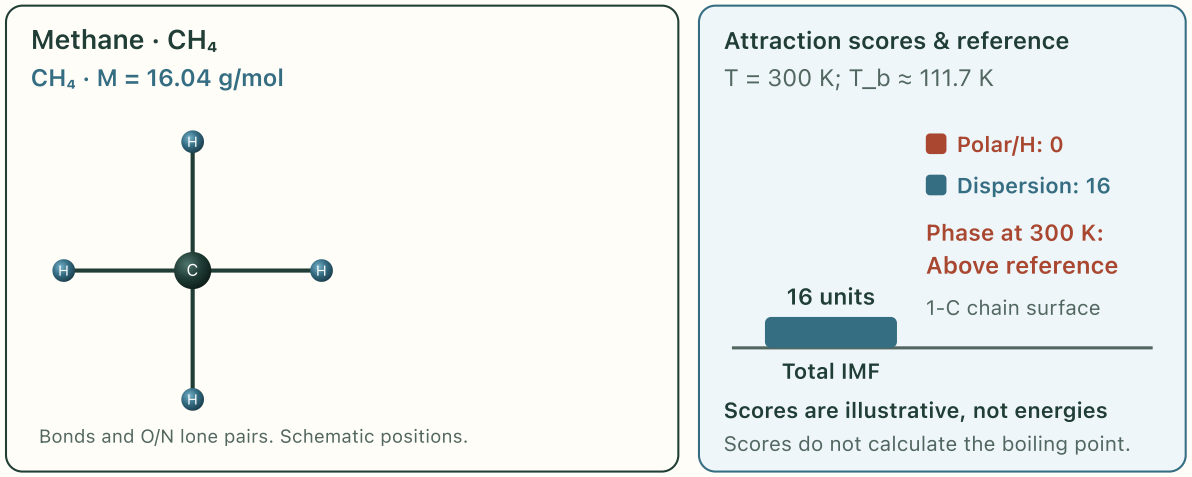
Why adding –CH_2– units turns nonpolar hydrocarbon gases into liquids and waxy solids

Alkanes (C_nH_{2n+2}) contain only nonpolar C–C and C–H covalent bonds (net dipole = 0 D), yet methane (CH_4) is a cryogenic gas (T_b = 112 K), butane (C_4H_{10}) liquefies in a pocket lighter (T_b = 273 K), octane (C_8H_{18}) is liquid petrol, and C_{20}+ alkanes are solid paraffin wax.

Bench Observation: As each –CH_2– unit is added along the homologous series (CH_4 → C_2H_6 → C_3H_8 → n-C_4H_{10}), net dipole stays 0.00 D while molar mass, contact surface area, and boiling point rise steadily.

Investigation Task: Step Carbon skeleton from C_1 Methane through C_4 n-Butane with Functional group set to Hydrocarbon only (–H), and inspect how London dispersion attraction and boiling point scale.

Hydrocarbon chain length and dispersion forces



Left: 2D all-atom Lewis connectivity of Methane (CH_4). Right: Stacked London dispersion (16) + functional-group polar/H-bond (0) scores; approximate normal boiling point 111.7 K. Alkane (saturated C_1 hydrocarbon). Geometry guide: 109.5° tetrahedral (sp^3); positions are schematic.

Composition versus connectivity

Equal molecular inventories can have different connections and properties. A formula is not a complete structural account.

The bookkeeping idea: Like the same parts assembled in different ways: an inventory alone cannot describe connections. Structural isomers can share a formula while having different connectivity and properties.

Current explanatory plate · intro-models-2

Molar mass / g per mol	16.04 g/mol
Net molecular dipole moment / D	0 D
Total intermolecular attraction / model units	16 relative model units
Approximate normal boiling point / K	111.7 K

Model note: Methane (CH_4, CH_4; Alkane (saturated C_1 hydrocarbon)): Gas at 300 K (T ≥ T_b = 111.7 K / -161.4 °C). Geometry: 109.5° tetrahedral (sp^3) (Symmetric 3D tetrahedron). IMFs: Weak London dispersion only (small surface area) (Dispersion = 16 + Polar/H-bond = 0 → 16 units). Nonpolar hydrophobic gas; C–H bonds do not ionize in water.

Scientific inputs · scientific-values-2

Methane: normal boiling point approximation: 111.7 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish

solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties. Reference entry: 111.7 K. Checked 2026-10-28. Source

Methane: dipole approximation: 0 D · reference. Methane: zero by tetrahedral symmetry. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution. Reference entry: 0 D. Checked 2026-10-28. Source

Methane: dispersion score: 16 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: polar-H-bond score: 0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: formula molar mass: 16.04 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places. Source Working: 1 × 12.011 + 4 × 1.008 + 0 × 15.999 + 0 × 14.007

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 11.1 Core 1–3; Supplement 7–8. Displayed structures, functional groups and isomerism Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.7.1.1–3; 4.7.2.1, 3–4. Hydrocarbons, boiling trends, alcohols and carboxylic acids Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Carbon and its Compounds (p. 5). Carbon bonding and functional groups Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application

discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 10.3: Deterministic particle & mathematical plate for *Hydrocarbon chain length and dispersion forces*. First Principle: Even in nonpolar hydrocarbons (μ = 0 D), lengthening the carbon chain increases electron count and molecular contact area, strengthening cumulative London dispersion attractions and raising the boiling point.

Historical & Experimental Evidence: Fractional distillation of crude oil and boiling points of the C1–C4 homologous alkane series

Every straight-chain alkane (C_nH_{2n+2}) has a net dipole moment of 0.00 D because its C–C and C–H bonds are nearly nonpolar and arranged symmetrically. Yet at room temperature (298 K), methane (CH_4, 112 K) and ethane (C_2H_6, 185 K) are gases, n-butane (C_4H_{10}, 273 K) liquefies under slight pressure in a camp stove canister, octane (C_8H_{18}, 399 K) is liquid petrol, and octadecane ($C_{18}H_{38}$) is solid candle wax! Adding each $–CH_2–$ group adds 14 g/mol, 8 electrons, and extra molecular surface area, steadily amplifying temporary London dispersion attractions.

Key Terms

Homologous series (Alkanes C_nH_{2n+2})
A family of organic compounds that share the same functional-group structure and general formula, differing from each successive member by a single $–CH_2–$ (methylene) unit.

London dispersion forces & polarizability
Attractive intermolecular forces caused by temporary fluctuations in electron clouds (instantaneous dipoles); strength increases with total electron count and molecular contact surface area.

Worked Reasoning Trace

Held fixed: Functional group held fixed at Hydrocarbon only (–H, 0.00 D) and T = 300 K.

Changed: Stepped Carbon skeleton from C_1 Methane (16.04 g/mol) → C_2 Ethane (30.07 g/mol) → C_3 Propane (44.10 g/mol) → C_4 n-Butane (58.12 g/mol).

Model result: Net dipole remains 0.00 D across all four alkanes, while dispersion attraction climbs from 16 → 28 → 40 → 52 units and 1 atm boiling point climbs from 111.7 K → 184.6 K → 231.1 K → 272.7 K.

Model boundary: Boiling an alkane overcomes intermolecular dispersion attractions between intact molecules; it never breaks internal C–C (~348 kJ/mol) or C–H (~413 kJ/mol) covalent bonds.

Common Misconception Trap · Hydrocarbon chain length and dispersion forces

Intuitive Trap: "I think longer alkanes form strong ionic bonds between carbon nuclei that must be broken during boiling — because each –CH_2– group adds a permanent +1 ionic charge to the carbon chain."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Longer carbon chains have more electrons and a larger molecular surface area, strengthening temporary London dispersion attractions between neighbouring molecules. Larger electron clouds and longer molecular chains create stronger temporary instantaneous-dipole (London dispersion) attractions across a wider contact surface area, requiring higher thermal kinetic energy (k_B T) to separate intact molecules."

Why Claim A fails: Alkanes are electrically neutral covalent molecules with zero net charge and 0.00 D dipole moment. · Why Claim B holds up: Exact mechanism: dispersion forces scale with total polarizable electrons and molecular contact area; boiling separates intact molecules when k_B T overcomes those dispersion attractions.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
C_1 Methane (CH_4, 16.04 g/mol, T_b = 111.7 K)	Smallest alkane: compact 10-electron cloud gives weak dispersion forces (16 units).	Methane (CH_4, CH_4; Alkane (saturated C_1 hydrocarbon)): Gas at 300 K (T ≥ T_b = 111.7 K / -161.4 °C). Geometry: 109.5° tetrahedral (sp^3) (Symmetric 3D tetrahedron). IMFs: Weak London dispersion only (small surface area) (Dispersion = 16 + Polar/H-bond = 0 → 16 units). Nonpolar hydrophobic gas; C–H bonds do not ionize in water.
C_2 Ethane (C_2H_6, 30.07 g/mol, T_b = 184.6 K)	Adding one $–CH_2–$ unit increases surface contact and raises T_b by +72.9 K.	Ethane (C_2H_6, $CH_3–CH_3$; Alkane (saturated C_2 chain)): Gas at 300 K (T ≥ T_b = 184.6 K / -88.5 °C). Geometry: 109.5° tetrahedral centres (sp^3) (Free rotation around C–C single (σ) bond). IMFs: London dispersion (larger surface area than CH_4 raises T_b by +73 K) (Dispersion = 28 + Polar/H-bond = 0 → 28 units). Nonpolar hydrophobic gas; adding a $–CH_2–$ unit increases dispersion attraction.
C_4 n-Butane (C_4H_{10}, 58.12 g/mol, T_b = 272.7 K)	Four-carbon straight chain: dispersion reaches 52 units, condensing into a liquid below 272.7 K (–0.5 °C).	n-Butane (C_4H_{10}, $CH_3–CH_2–CH_2–CH_3$; Straight-chain alkane (unbranched C_4H_{10} isomer)): Below boiling reference at 250 K (solid/liquid not distinguished). Geometry: 109.5° extended tetrahedral zig-zag (Free C–C single-bond rotation; extended chain maximizes intermolecular contact). IMFs: Stronger London dispersion than branched isobutane due to elongated contact area (Dispersion = 52 + Polar/H-bond = 0 → 52 units). Constitutional isomer of isobutane (both C_4H_{10}, 58.12 g/mol); boils 11.3 K higher than isobutane.

Check Question: Why does the boiling point of straight-chain alkanes increase steadily from methane (CH_4, 112 K) to n-butane (C_4H_{10}, 273 K) even though every alkane has a net dipole moment of 0 D? · **Transfer Challenge:** Transfer challenge: Why can a gecko walk up a vertical glass wall using millions of microscopic hair-like setae on its toes, even though neither the gecko's keratin hairs nor the glass rely on permanent ionic bonds?

INVESTIGATION 10.4 · PRINCIPLE 1: CONSERVATION OF COUNT & CHARGE · MODEL: ORGANIC

Same formula, different molecule: Constitutional isomers

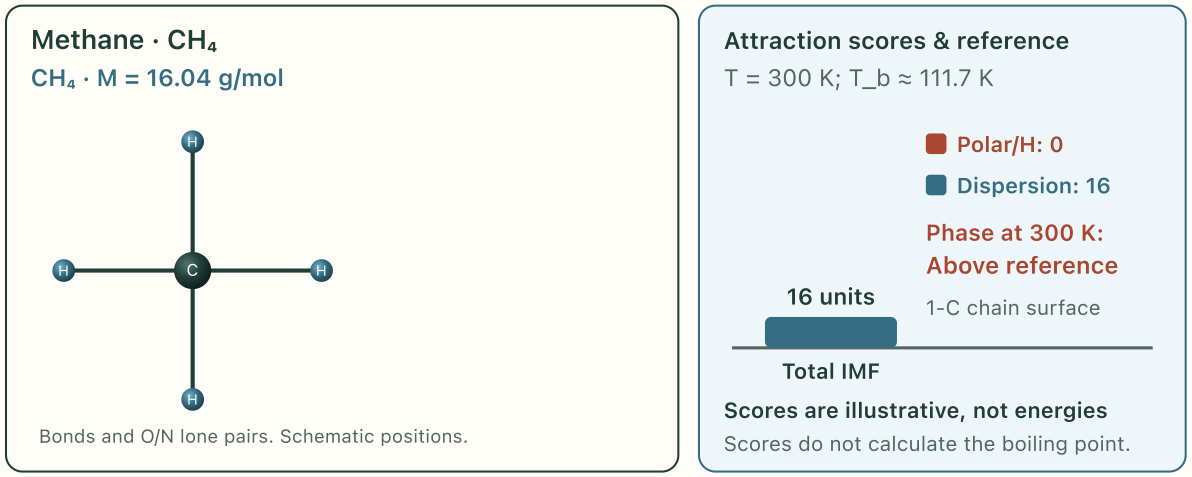
Why C_4H_{10} represents two different hydrocarbons and C_2H_6O represents both a liquid alcohol and a gaseous ether

In organic chemistry, a molecular formula like C_4H_{10} or C_2H_6O is never enough to identify a substance. Constitutional (structural) isomers contain the exact same atom inventory and molar mass, but wire those atoms together in a different bonding order.

Bench Observation: n-Butane and isobutane both have formula C_4H_{10} (58.12 g/mol), yet branched isobutane boils 11.3 K lower; ethanol (CH_3CH_2OH) and dimethyl ether (CH_3OCH_3) both have formula C_2H_6O (46.07 g/mol), yet ethanol boils 102.5 K higher!

Investigation Task: Compare C_4 n-Butane vs. C_4 Isobutane (both 58.12 g/mol), and then compare C_2 Ethanol (–OH) vs. C_2 Dimethyl ether (C–O–C bridge, both 46.07 g/mol) at 300 K.

Same formula, different molecule: Constitutional isomers



Left: 2D all-atom Lewis connectivity of Methane (CH_4). Right: Stacked London dispersion (16) + functional-group polar/H-bond (0) scores; approximate normal boiling point 111.7 K. Alkane (saturated C_1 hydrocarbon). Geometry guide: 109.5° tetrahedral (sp^3); positions are schematic.

Composition versus connectivity

Equal molecular inventories can have different connections and properties. A formula is not a complete structural account.

The bookkeeping idea: Like the same parts assembled in different ways: an inventory alone cannot describe connections. Structural isomers can share a formula while having different connectivity and properties.

Current explanatory plate · intro-models-2

Molar mass / g per mol	16.04 g/mol
Net molecular dipole moment / D	0 D
Total intermolecular attraction / model units	16 relative model units
Approximate normal boiling point / K	111.7 K

Model note: Methane (CH_4, CH_4; Alkane (saturated C_1 hydrocarbon)): Gas at 300 K (T ≥ T_b = 111.7 K / -161.4 °C). Geometry: 109.5° tetrahedral (sp^3) (Symmetric 3D tetrahedron). IMFs: Weak London dispersion only (small surface area) (Dispersion = 16 + Polar/H-bond = 0 → 16 units). Nonpolar hydrophobic gas; C–H bonds do not ionize in water.

Scientific inputs · scientific-values-2

Methane: normal boiling point approximation: 111.7 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish

solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties. Reference entry: 111.7 K. Checked 2026-10-28. Source

Methane: dipole approximation: 0 D · reference. Methane: zero by tetrahedral symmetry. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution. Reference entry: 0 D. Checked 2026-10-28. Source

Methane: dispersion score: 16 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: polar-H-bond score: 0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: formula molar mass: 16.04 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places. Source Working: 1 × 12.011 + 4 × 1.008 + 0 × 15.999 + 0 × 14.007

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 11.1 Core 1–3; Supplement 7–8. Displayed structures, functional groups and isomerism Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.7.1.1–3; 4.7.2.1, 3–4. Hydrocarbons, boiling trends, alcohols and carboxylic acids Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Carbon and its Compounds (p. 5). Carbon bonding and functional groups Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application

discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 10.4: Deterministic particle & mathematical plate for *Same formula, different molecule: Constitutional isomers*. First Principle: Constitutional isomers conserve the exact same atom counts and molar mass, yet wiring those atoms in a different connectivity changes 3D contact area or hydrogen-bonding capacity.

Historical & Experimental Evidence: Wöhler, Liebig & Berzelius on isomerism (1828–1830): identical elemental composition, distinct substances

In the 1820s, chemists believed that every chemical formula corresponded to one unique substance. Then Friedrich Wöhler and Jöns Jacob Berzelius discovered 'isomers' (from Greek isos + meros, 'equal parts'): compounds with the exact same molecular formula and molar mass that behave like strangers! Butane (C_4H_{10}, 58.12 g/mol) exists as straight-chain n-butane (T_b = −0.5 °C) and branched isobutane (T_b = −11.7 °C). Even more dramatically, C_2H_6O (46.07 g/mol) can be wired as ethanol (CH_3CH_2OH, boiling at +78.3 °C) or as dimethyl ether (CH_3OCH_3, boiling at −24.2 °C)!

Key Terms

Constitutional (structural) isomers
Compounds that have the exact same molecular formula (identical atom counts and molar mass) but different atom-to-atom bonding connectivity.

Chain branching vs. contact area
Branching makes an alkane more spherical and compact, reducing the surface area available for intermolecular London dispersion contact and lowering its boiling point.

Worked Reasoning Trace

Held fixed: Molecular formula held fixed at C_4H_{10} (58.12 g/mol, Functional group = Hydrocarbon only) or C_2H_6O (46.07 g/mol, C_2 skeleton).

Changed: Switched connectivity from straight C_4 n-Butane [3, 0, 265] to branched C_4 Isobutane [4, 0, 265], and from Ethanol [1, 1, 300] to Dimethyl ether [1, 2, 300].

Model result: For C_4H_{10} (58.12 g/mol), branching drops dispersion from 52 to 46 units and T_b from 272.7 K to 261.4 K. For C_2H_6O (46.07 g/mol), moving O from –OH to C–O–C eliminates O–H hydrogen bonding and drops T_b from 351.5 K to 249.0 K.

Model boundary: A molecular formula (like C_4H_{10} or C_2H_6O) counts atoms per molecule but hides connectivity; structural or skeletal formulas are required to identify an organic compound.

Common Misconception Trap · Same formula, different molecule: Constitutional isomers

Intuitive Trap: "I think n-butane has a larger molar mass and more protons than isobutane — because straight-chain isomers contain heavier carbon isotopes than branched isomers."

Particle Mechanism Refutation: "Wait — look at the particle ledger: The extended straight chain of n-butane provides a larger surface area for intermolecular contact than compact, spherical isobutane, resulting in stronger London dispersion attractions. In C_4H_{10}, straight chains have greater surface contact for dispersion than compact branched spheres; in C_2H_6O, ethanol's O–H bond enables intermolecular hydrogen bonding whereas dimethyl ether's C–O–C bridge has no H bonded to O."

Why Claim A fails: Isomers have the exact same isotopic composition and molar mass. · Why Claim B holds up: Exact mechanism: constitutional isomers have identical atom and electron counts, so their property differences come strictly from 3D contact geometry and functional-group bond dipoles/H-bonding.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Straight-chain isomer: n-Butane (C_4H_{10}, 58.12 g/mol, T_b = 272.7 K)	Unbranched 4-carbon chain maximizes surface contact (52 dispersion units; liquid at 265 K).	n-Butane (C_4H_{10}, $CH_3–CH_2–CH_2–CH_3$; Straight-chain alkane (unbranched C_4H_{10} isomer)): Below boiling reference at 265 K (solid/liquid not distinguished). Geometry: 109.5° extended tetrahedral zig-zag (Free C–C single-bond rotation; extended chain maximizes intermolecular contact). IMFs: Stronger London dispersion than branched isobutane due to elongated contact area (Dispersion = 52 + Polar/H-bond = 0 → 52 units). Constitutional isomer of isobutane (both C_4H_{10}, 58.12 g/mol); boils 11.3 K higher than isobutane.
Branched isomer: Isobutane (C_4H_{10}, 58.12 g/mol, T_b = 261.4 K)	Same C_4H_{10} formula & mass, but compact branched shape lowers dispersion (46 units; gas at 265 K).	Isobutane (2-methylpropane) (C_4H_{10}, $CH_3–CH(CH_3)–CH_3$; Branched alkane (branched C_4H_{10} constitutional isomer)): Gas at 265 K (T ≥ T_b = 261.4 K / -11.8 °C). Geometry: 109.5° branched central carbon bonded to 3 CH_3 groups (Compact spherical 3D shape reduces surface-to-surface contact between neighbours). IMFs: Lower London dispersion (46 vs. 52 units) than straight-chain n-butane despite identical 58.12 g/mol mass (Dispersion = 46 + Polar/H-bond = 0 → 46 units). Constitutional isomer of n-butane: same C_4H_{10} formula, different connectivity and lower boiling point.
Functional-group isomer pair: Dimethyl ether vs. Ethanol (C_2H_6O, 46.07 g/mol)	$CH_3–O–CH_3$ (T_b = 249.0 K) has no O–H bond; switch Functional group to Hydroxyl (–OH) to compare Ethanol (T_b = 351.5 K)!	Dimethyl ether (C_2H_6O isomer) (C_2H_6O, $CH_3–O–CH_3$; Ether (C_2H_6O constitutional isomer of ethanol)): Gas at 300 K (T ≥ T_b = 249 K / -24.1 °C). Geometry: Bent 111.7° C–O–C oxygen bridge; all 6 H atoms are bonded to C, none to O (Free rotation around C–O single bonds). IMFs: Dipole–dipole (14) + dispersion (29) = 43 units (vs. 82 units for ethanol CH_3CH_2OH) (Dispersion = 29 + Polar/H-bond = 14 → 43 units). Exact constitutional isomer of ethanol (46.07 g/mol): proves connectivity—not formula alone—determines hydrogen bonding and boiling point.

Check Question: Why does straight-chain n-butane ($CH_3CH_2CH_2CH_3$, T_b = 272.7 K) have a higher boiling point than its branched constitutional isomer isobutane ($CH_3CH(CH_3)CH_3$, T_b = 261.4 K) despite having the exact same formula C_4H_{10} and molar mass? · **Transfer Challenge:** Transfer challenge: Pentane (C_5H_{12}, 72.15 g/mol) has three constitutional isomers: straight-chain n-pentane, singly branched isopentane (2-methylbutane), and doubly branched neopentane (2,2-

dimethylpropane, shaped almost like a sphere). Predict the order of their boiling points from highest to lowest and explain why.

INVESTIGATION 10.5 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: ORGANIC

Double bonds, unsaturation and locked 3D geometry

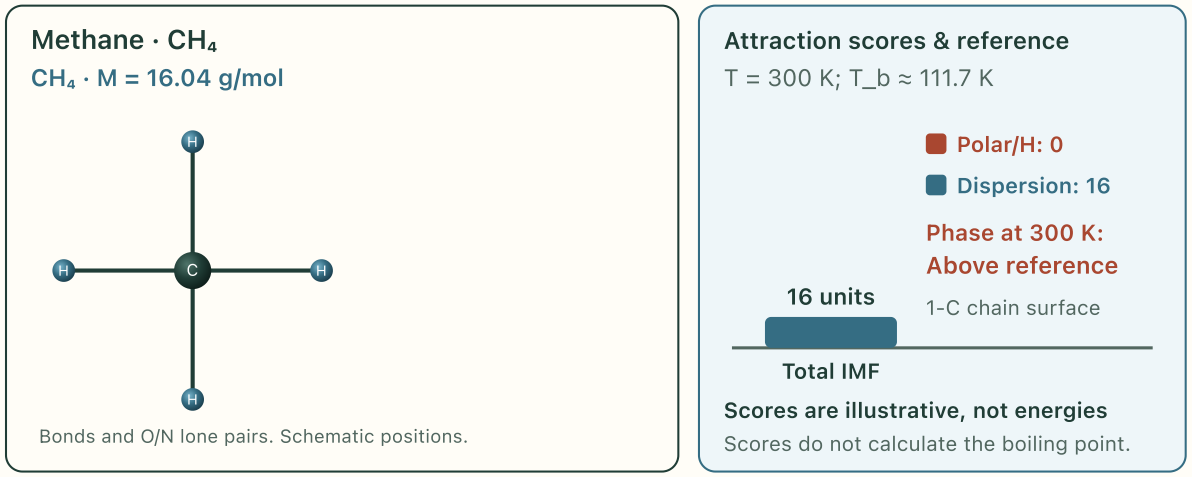
Why C–C single bonds rotate freely at 109.5° while C=C double bonds lock molecules into a flat 120° plane

Saturated alkanes like ethane (C_2H_6) have tetrahedral 109.5° carbon centres that rotate freely around their C–C single (σ) bond. Introducing a C=C double bond in ethene ($H_2C=CH_2$) removes two H atoms (one degree of unsaturation), flattens bond angles to 120° trigonal planar, and locks the bond against rotation.

Bench Observation: Replacing the C–C single bond of ethane (C_2H_6, 30.07 g/mol, 109.5° rotatable) with the C=C double bond of ethene (C_2H_4, 28.05 g/mol, 120° planar) locks all six atoms into a single rigid plane.

Investigation Task: Switch Carbon skeleton between C_2 Ethane (C–C single bond) and C_2 Ethene (C=C double bond) in both 2D and 3D views to compare bond angles, hydrogen counts, and rotational freedom.

Double bonds, unsaturation and locked 3D geometry



Left: 2D all-atom Lewis connectivity of Methane (CH_4). Right: Stacked London dispersion (16) + functional-group polar/H-bond (0) scores; approximate normal boiling point 111.7 K. Alkane (saturated C_1 hydrocarbon). Geometry guide: 109.5° tetrahedral (sp^3); positions are schematic.

Composition versus connectivity

Equal molecular inventories can have different connections and properties. A formula is not a complete structural account.

The bookkeeping idea: Like the same parts assembled in different ways: an inventory alone cannot describe connections. Structural isomers can share a formula while having different connectivity and properties.

Current explanatory plate · intro-models-2

Molar mass / g per mol	16.04 g/mol
Net molecular dipole moment / D	0 D
Total intermolecular attraction / model units	16 relative model units
Approximate normal boiling point / K	111.7 K

Model note: Methane (CH_4, CH_4; Alkane (saturated C_1 hydrocarbon)): Gas at 300 K (T ≥ T_b = 111.7 K / -161.4 °C). Geometry: 109.5° tetrahedral (sp^3) (Symmetric 3D tetrahedron). IMFs: Weak London dispersion only (small surface area) (Dispersion = 16 + Polar/H-bond = 0 → 16 units). Nonpolar hydrophobic gas; C–H bonds do not ionize in water.

Scientific inputs · scientific-values-2

Methane: normal boiling point approximation: 111.7 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish

solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties. Reference entry: 111.7 K. Checked 2026-10-28. Source

Methane: dipole approximation: 0 D · reference. Methane: zero by tetrahedral symmetry. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution. Reference entry: 0 D. Checked 2026-10-28. Source

Methane: dispersion score: 16 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: polar-H-bond score: 0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: formula molar mass: 16.04 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places. Source Working: 1 × 12.011 + 4 × 1.008 + 0 × 15.999 + 0 × 14.007

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 11.1 Core 1–3; Supplement 7–8. Displayed structures, functional groups and isomerism Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.7.1.1–3; 4.7.2.1, 3–4. Hydrocarbons, boiling trends, alcohols and carboxylic acids Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Carbon and its Compounds (p. 5). Carbon bonding and functional groups Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application

discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 10.5: Deterministic particle & mathematical plate for *Double bonds, unsaturation and locked 3D geometry*. First Principle: Replacing a rotatable tetrahedral C–C single bond (109.5°) with a C=C double bond removes two H atoms, flattens geometry to 120° trigonal planar, and locks the bond against rotation.

Historical & Experimental Evidence: Ethane vs. ethene bond lengths, infrared spectroscopy, and cis/trans geometric isomerism

In ethane ($H_3C–CH_3$), each carbon is bonded to four neighbours at 109.5° tetrahedral angles, and the two methyl groups spin past each other billions of times per second at room temperature because rotating a head-on σ bond does not break orbital overlap. In ethene ($H_2C=CH_2$), the two carbons share four electrons in a shorter, stronger C=C double bond (134 pm vs. 154 pm). Because the second bond (π bond) forms by side-by-side overlap above and below the bond axis, all six atoms of ethene are locked into a flat 120° trigonal planar sheet—twisting one end by 90° would cost ~260 kJ/mol!

Key Terms

Saturated vs. Unsaturated hydrocarbons
Saturated alkanes contain only C–C single bonds and carry the maximum hydrogen count (C_nH_{2n+2}); each C=C double bond or ring removes 2 H atoms (one degree of unsaturation, C_nH_{2n}).

Trigonal planar C=C geometry (120°) & locked rotation
Each carbon in a C=C double bond is surrounded by three electron domains arranged 120° apart in a flat plane; the π bond prevents free rotation around the C=C axis.

Worked Reasoning Trace

Held fixed: Two-carbon hydrocarbon skeleton (Functional group = Hydrocarbon only, T = 300 K).

Changed: Switched Carbon skeleton from C_2 Ethane (C–C single bond, [1, 0, 300]) to C_2 Ethene (C=C double bond, [5, 0, 300]).

Model result: Formula drops by 2 H atoms from C_2H_6 (30.07 g/mol) to C_2H_4 (28.05 g/mol), bond geometry shifts from 109.5° 3D tetrahedral (free rotation) to 120° flat trigonal planar (locked against rotation).

Model boundary: Ethene itself ($H_2C=CH_2$) has identical H atoms on each end; cis/trans geometric isomers appear when each carbon of the locked C=C bond carries two different groups (such as but-2-ene).

Common Misconception Trap · Double bonds, unsaturation and locked 3D geometry

Intuitive Trap: "I think it adds two extra hydrogen atoms (c_2h_6 → c_2h_8) and allows the double bond to spin faster than a single bond — because double bonds contain four protons inside the bond axis that repel hydrogen atoms into a square."

Particle Mechanism Refutation: "Wait — look at the particle ledger: It reduces the hydrogen count by two (C_2H_6 → C_2H_4), changes the geometry around each carbon from 109.5° tetrahedral to 120° trigonal planar, and locks the C=C bond against free rotation. Three electron domains around each sp^2 carbon repel to 120° trigonal planar angles, and rotating one end by 90° would break the side-by-side π electron overlap between the two carbons."

Why Claim A fails: Protons remain inside the carbon nuclei; covalent bonds consist of shared valence electron density. · Why Claim B holds up: Exact mechanism: the σ + π double bond locks the two carbons and their four attached atoms into a flat 120° plane because twisting destroys the parallel π overlap.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Saturated C_2H_6: Ethane (C–C single bond, 109.5° tetrahedral, rotatable)	Maximum H count (2n + 2 = 6 H); tetrahedral carbons rotate freely around the C–C axis.	Ethane (C_2H_6, CH_3–CH_3; Alkane (saturated C_2 chain)): Gas at 300 K (T ≥ T_b = 184.6 K / -88.5 °C). Geometry: 109.5° tetrahedral centres (sp^3) (Free rotation around C–C single (σ) bond). IMFs: London dispersion (larger surface area than CH_4 raises T_b by +73 K) (Dispersion = 28 + Polar/H-bond = 0 → 28 units). Nonpolar hydrophobic gas; adding a –CH_2– unit increases dispersion attraction.
Unsaturated C_2H_4: Ethene (C=C double bond, 120° planar, locked)	Two fewer H atoms (2n = 4 H); all 6 atoms lie in a rigid 120° plane that cannot rotate at 300 K.	Ethene (ethylene) (C_2H_4, $H_2C=CH_2$; Alkene (unsaturated C=C double bond)): Gas at 300 K (T ≥ T_b = 169.4 K / -103.7 °C). Geometry: 120° trigonal planar (sp^2) — all 6 atoms lie in one flat plane (Locked against rotation! Twisting a C=C double bond would break the side-by-side π overlap). IMFs: London dispersion only (2 fewer H atoms than ethane C_2H_6 → 1 degree of unsaturation) (Dispersion = 25 + Polar/H-bond = 0 → 25 units). Rigid C=C double bond enables cis/trans stereoisomers in longer alkenes and addition polymerizations.
Unsaturated conjugated aldehyde: Propenal ($H_2C=CH–CHO$)	Planar C=C double bond paired with a planar C=O carbonyl group (3.12 D dipole).	Propenal (acrolein) (C_3H_4O, $H_2C=CH–CH=O$; Conjugated enal (C=C double bond + C=O carbonyl)): Below boiling reference at 300 K (solid/liquid not distinguished). Geometry: Fully planar 120° conjugated π system ($H_2C=CH–CH=O$) (Restricted rotation across conjugated double bonds). IMFs: High dipole moment (3.12 D) + planar dispersion (Dispersion = 34 + Polar/H-bond = 29 → 63 units). Demonstrates how adjacent C=C and C=O double bonds create a flat, conjugated electron system.

Check Question: How does replacing a C–C single bond with a C=C double bond change the molecular formula and 3D geometry of a two-carbon hydrocarbon? · **Transfer Challenge:** Transfer challenge: Saturated fatty acids (in butter, all C–C single bonds) pack tightly into a solid at room temperature, whereas cis-unsaturated fatty acids (in olive oil, with a rigid 120° cis C=C kink in the chain) are liquids at room temperature. Use 3D chain packing and London dispersion forces to explain why!

INVESTIGATION 10.6 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: ORGANIC

Carboxylic acids, amines and condensation links of life

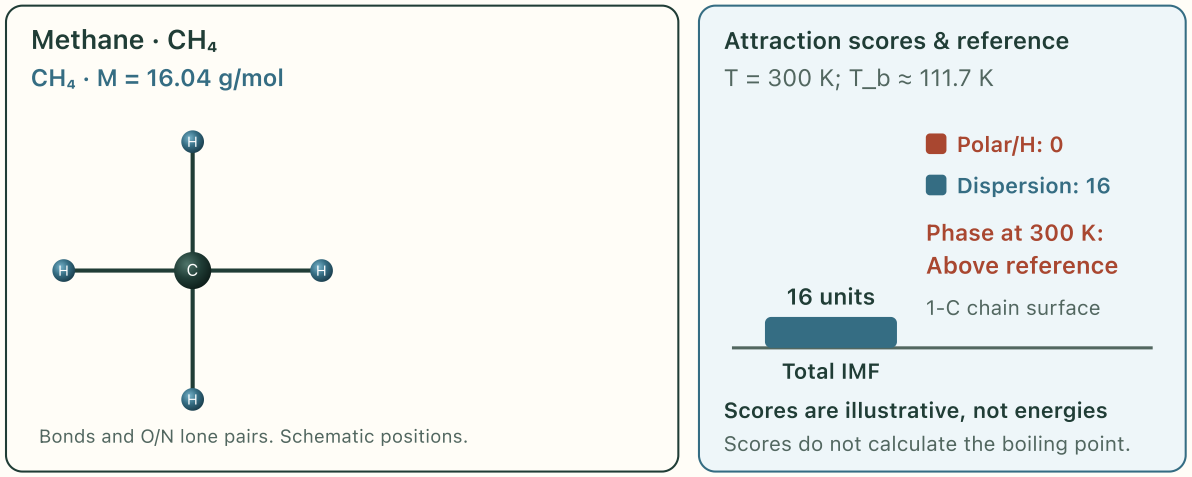
How –COOH acids, –NH_2 bases, and –OH alcohols react by proton transfer and water-splitting condensation

Living cells and materials rely on functional groups that do more than attract water—they transfer protons and link monomers into polymers! Carboxyl groups (–COOH) act as weak Brønsted–Lowry acids, amino groups (–NH_2) act as weak bases, and condensing –COOH with –OH (or –NH_2) splits out H_2O to forge ester (lipids/polyesters) and amide (peptide/protein) linkages.

Bench Observation: On a C_2 backbone, ethanoic acid (CH_3COOH, 60.05 g/mol, T_b = 391.1 K) donates H^+ in water and forms strong hydrogen-bonded dimers, whereas its ester isomer methyl methanoate ($HCOOCH_3$, 60.05 g/mol, T_b = 304.9 K) lacks an O–H bond and boils 86 K lower.

Investigation Task: Compare Carboxyl (–COOH), Amino (–NH_2), and Condensation ester (–$COOCH_3$ + H_2O) across the C_1–C_4 skeletons to trace acid–base behaviour and atom conservation during condensation.

Carboxylic acids, amines and condensation links of life



Left: 2D all-atom Lewis connectivity of Methane (CH_4). Right: Stacked London dispersion (16) + functional-group polar/H-bond (0) scores; approximate normal boiling point 111.7 K. Alkane (saturated C_1 hydrocarbon). Geometry guide: 109.5° tetrahedral (sp^3); positions are schematic.

Composition versus connectivity

Equal molecular inventories can have different connections and properties. A formula is not a complete structural account.

The bookkeeping idea: Like the same parts assembled in different ways: an inventory alone cannot describe connections. Structural isomers can share a formula while having different connectivity and properties.

Current explanatory plate · intro-models-2

Molar mass / g per mol	16.04 g/mol
Net molecular dipole moment / D	0 D
Total intermolecular attraction / model units	16 relative model units
Approximate normal boiling point / K	111.7 K

Model note: Methane (CH_4, CH_4; Alkane (saturated C_1 hydrocarbon)): Gas at 300 K (T ≥ T_b = 111.7 K / -161.4 °C). Geometry: 109.5° tetrahedral (sp^3) (Symmetric 3D tetrahedron). IMFs: Weak London dispersion only (small surface area) (Dispersion = 16 + Polar/H-bond = 0 → 16 units). Nonpolar hydrophobic gas; C–H bonds do not ionize in water.

Scientific inputs · scientific-values-2

Methane: normal boiling point approximation: 111.7 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish

solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties. Reference entry: 111.7 K. Checked 2026-10-28. Source

Methane: dipole approximation: 0 D · reference. Methane: zero by tetrahedral symmetry. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution. Reference entry: 0 D. Checked 2026-10-28. Source

Methane: dispersion score: 16 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: polar-H-bond score: 0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: formula molar mass: 16.04 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places. Source Working: 1 × 12.011 + 4 × 1.008 + 0 × 15.999 + 0 × 14.007

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Partial connection: 11.1 Core 1–3; Supplement 7–8. Displayed structures, functional groups and isomerism Only the named concept is connected. Practical work and the complete specified examples remain separate. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Partial connection: 4.7.1.1–3; 4.7.2.1, 3–4. Hydrocarbons, boiling trends, alcohols and carboxylic acids Only the named concept is connected. Practical work and the complete specified examples remain separate. Current web specification

CBSE Science 086 · Classes IX–X · Partial connection: X · Unit I · Carbon and its Compounds (p. 5). Carbon bonding and functional groups Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application

discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 10.6: Deterministic particle & mathematical plate for *Carboxylic acids, amines and condensation links of life*. First Principle: Carboxyl groups (–COOH) donate H^+ as weak acids, amino groups ($–NH_2$) accept H^+ as weak bases, and condensing functional groups joins monomers into esters or amides while conserving atoms by releasing H_2O.

Historical & Experimental Evidence: Fischer esterification, carboxylic acid pK_a measurements, and peptide/lipid condensation

Why is ethanol (CH_3CH_2OH) neutral in water while ethanoic acid (vinegar, CH_3COOH) turns litmus red and reacts with baking soda? In –COOH, the electronegative carbonyl oxygen (C=O) pulls electron density away from the O–H bond and spreads the negative charge of the resulting carboxylate ion (CH_3COO^-) equally over both oxygens! Conversely, an amino group ($–NH_2$) uses its nitrogen lone pair to accept H^+ as a weak base. And when a carboxylic acid meets an alcohol (or amine), they link covalently into an ester (–COO–) or amide (–CONH–) while splitting out one molecule of H_2O—the exact reaction that builds fats, polyesters, and proteins!

Key Terms

Carboxyl (–COOH) vs. Amino (–NH_2) groups
A carboxyl group combines C=O and –OH on the same carbon to act as a weak Brønsted–Lowry acid (R–COOH ⇌ R–COO^- + H^+); an amino group uses its N lone pair to act as a weak base (R–NH_2 + H^+ ⇌ R–NH_3^+).

Condensation reaction (Esterification & Amide link)
A reaction in which two molecules covalently join together by eliminating a small molecule such as water: R–COOH + HO–R′ ⇌ R–COO–R′ (ester) + H_2O.

Worked Reasoning Trace

Held fixed: Molecular formula $C_3H_6O_2$ (74.08 g/mol) at T = 300 K.

Changed: Compared Propanoic acid (C_3 carboxyl [2, 4, 300], CH_3CH_2COOH) with its ester isomer Methyl ethanoate (C_2 condensation ester [1, 6, 300], CH_3COOCH_3 formed from CH_3COOH + CH_3OH).

Model result: Both share molar mass 74.08 g/mol, but propanoic acid has an acidic O–H donor (104 IMF units, T_b = 414.3 K) whereas methyl ethanoate replaces –OH with –OCH_3 (63 IMF units, T_b = 330.1 K), dropping T_b by 84.2 K!

Model boundary: Esterification (RCOOH + R′OH ⇌ RCOOR′ + H_2O) is a reversible equilibrium (Chapter 5): removing H_2O or adding excess alcohol drives ester yield forward by Le Châtelier's principle.

Common Misconception Trap · Carboxylic acids, amines and condensation links of life

Intuitive Trap: "I think carbon atoms are destroyed so that the ester has a lower molar mass than the starting acid — because esters ionize 100% into free C^{4+} ions in water, which lowers their boiling point."

Particle Mechanism Refutation: "Wait — look at the particle ledger: They join covalently to form an ester linkage (R–COO–R′) while splitting out one water molecule (H_2O), conserving every C, H, and O atom and replacing the O–H hydrogen-bond donor. In R–COOH, the electronegative C=O group stabilizes the R–COO^- conjugate base (enabling H^+ transfer) and forms strong O–H···O=C hydrogen bonds, whereas condensing with HO–R′ replaces the acidic –OH donor with an –OR′ group and releases H_2O."

Why Claim A fails: Esters are neutral polar covalent molecules that do not ionize in water. · Why Claim B holds up: Exact mechanism: delocalizing negative charge over both oxygens makes –COOH a weak acid, and replacing the –OH hydrogen-bond donor with –OR′ during esterification eliminates H-bond donation between ester molecules.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Weak organic acid: Ethanoic acid (CH_3COOH, 60.05 g/mol, T_b = 391.1 K)	Carboxyl –COOH group forms doubly hydrogen-bonded dimers and partially donates H^+ in water (pK_a = 4.74).	Ethanoic acid (acetic acid) ($C_2H_4O_2$, CH_3–COOH; Carboxylic acid (–COOH weak acid, Ka = 1.8×10^{-5})): Below boiling reference at 300 K (solid/liquid not distinguished). Geometry: Tetrahedral CH_3 + 120° trigonal planar –COOH carboxyl group (Pairs into doubly hydrogen-bonded dimers across C=O and H–O). IMFs: Strongest IMFs of any C_2 functional group (98 units, T_b = 391.1 K / 118 °C) due to dual C=O···H–O hydrogen bonding (Dispersion = 34 + Polar/H-bond = 64 → 98 units). Weak Brønsted–Lowry acid (pK_a = 4.74, Chapter 7): unlike ethanol (R–OH), the electron-withdrawing C=O stabilizes the CH_3COO^- conjugate base so H^+ can transfer partially to water.
Weak organic base: Ethanamine ($CH_3CH_2NH_2$, 45.08 g/mol, T_b = 289.8 K)	Amino –NH_2 lone pair accepts H^+ from water or acids; N–H hydrogen bonds are weaker than O–H.	Ethanamine (ethylamine) (C_2H_7N, CH_3–CH_2–NH_2; Primary amine (C_2 chain + basic –NH_2 group)): Gas at 300 K ($T \geq T_b$ = 289.8 K / 16.7 °C). Geometry: Tetrahedral C_2 backbone + pyramidal –NH_2 group (Rotatable C–C and C–N single bonds). IMFs: N–H···N hydrogen bonding (35) + dispersion (31) = 66 units (T_b = 289.8 K sits between ethane 184.6 K and ethanol 351.5 K!) (Dispersion = 31 + Polar/H-bond = 35 → 66 units). Weak base (accepts H^+ at nitrogen lone pair to form $CH_3CH_2NH_3^+$); when paired with –COOH on the same carbon, forms an amino acid!

Preset	Physical Context	Computed Model Output
Condensed ester + H_2O: Methyl ethanoate (CH_3COOCH_3, 74.08 g/mol, T_b = 330.1 K)	CH_3COOH + $HOCH_3 \rightleftharpoons CH_3COOCH_3$ + H_2O: O–H donors are consumed to release H_2O, yielding a volatile fruity ester!	Methyl ethanoate (methyl acetate) ($C_3H_6O_2$, CH_3–COO–CH_3 (+ H_2O condensed); Ester (CH_3COOH + $HOCH_3 \rightleftharpoons CH_3COOCH_3$ + H_2O)): Below boiling reference at 300 K (solid/liquid not distinguished). Geometry: Tetrahedral CH_3 + planar –COO– ester link + tetrahedral –CH_3 (Ester linkage formed by splitting out 1 H_2O molecule (atom conservation!)). IMFs: Dipole–dipole (21) + dispersion (42) = 63 units (boils 84.2 K lower than its isomer propanoic acid!) (Dispersion = 42 + Polar/H-bond = 21 → 63 units). Condensation reaction: CH_3COOH (60.05) + CH_3OH (32.04) $\rightleftharpoons$ CH_3COOCH_3 (74.08) + H_2O (18.02). Volatile fruity aroma because O–H hydrogen-bond donors were consumed to make H_2O.

Check Question: What happens at the particle level when a carboxylic acid (R–COOH) and an alcohol (HO–R′) undergo an esterification condensation reaction? · **Transfer Challenge:** Transfer challenge: Every amino acid in your body contains BOTH a carboxyl group (–COOH) and an amino group (–NH_2) attached to the same central α-carbon. What happens when two amino acids undergo a condensation reaction, and why can the resulting dipeptide still grow into a 1,000-unit protein chain?

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 10 · The Isomer & Functional Group Swap

Two organic compounds share the exact same molecular formula C_2H_6O (M = 46 g/mol). Compound A (ethanol, CH_3–CH_2–OH) is a liquid at room temperature (T_b = 351 K / 78 °C) that mixes completely with water. Compound B (dimethyl ether, CH_3–O–CH_3) is a gas at room temperature (T_b = 249 K / –24 °C).

- Clue 1 (Atom Inventory): Both molecules contain 2 C, 6 H, and 1 O atom—so molar mass and total electron count are identical.
- Clue 2 (Bond Connectivity): Compound A contains a polar O–H hydroxyl bond (H directly bonded to O), whereas Compound B has only C–H and C–O bonds (no H bonded to O).
- Clue 3 (IMF Consequence): Hydrogen bonding between identical molecules requires both a δ^+ H (on N, O, or F) and a lone pair acceptor.

Resolution: Constitutional isomers share the same formula (C_2H_6O) and molar mass (46 g/mol), but ethanol's –OH group forms strong intermolecular hydrogen bonds while dimethyl ether (CH_3–O–CH_3) has only weaker dipole–dipole and dispersion forces.

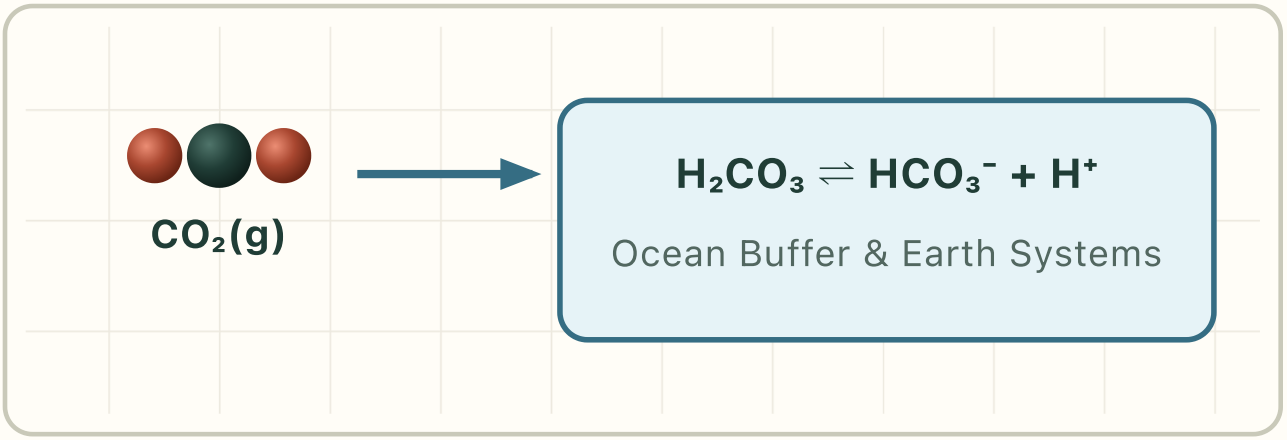
What our upgraded model now explains: Explains carbon catenation, 3D tetrahedral chains vs. planar sheets, structural isomerism, and how polar functional groups control dipole moments, hydrogen bonding, and solubility. · **Next boundary:** How do carbon fuels, atmospheric CO_2, ocean buffers, and electrochemical batteries interact across our entire planet? → motivating our capstone Chapter 11 (Earth, Materials & Energy Systems).

VOLUME III · CHEMICAL SYSTEMS & THE REAL WORLD · CHAPTER 11

Chapter 11: Earth, materials and energy systems

Unite stoichiometry, energetics, acid–base equilibria, and electrochemistry to analyze atmospheric carbon, ocean buffering, and rechargeable batteries.

Chapter 11 Prelude · Capstone: Uniting All Four First Principles on Planet Earth: You have travelled from counting protons in a single atom to analyzing bonds, gases, reaction energy ledgers, dynamic equilibria, moles, buffers, and electrochemical cells. In the real world, these topics never live in separate chapters! When humanity transitions from burning hydrocarbon fuels (releasing CO_2 that perturbs the ocean's carbonate buffer) to storing solar electricity in rechargeable lithium batteries, every single principle you have mastered works together simultaneously.



Macroscopic Bench: Global atmospheric CO_2 & ocean pH monitoring, fuel calorimetry, and lithium-ion battery energy density.

Submicroscopic Particle Model: Coupled combustion bond ledgers, seawater H_2CO_3 / HCO_3^- / CO_3^{2-} buffer shifts, and reversible Li^+ / e^- intercalation.

Symbolic Notation: Integrated stoichiometric, thermochemical (ΔH), buffer (pH / pK_a), and electrochemical (E_cell) system equations.

INVESTIGATION 11.1 · PRINCIPLE 4: DYNAMIC & STATISTICAL RATIOS · MODEL: TITRATION

Ocean acidification and carbonate buffering

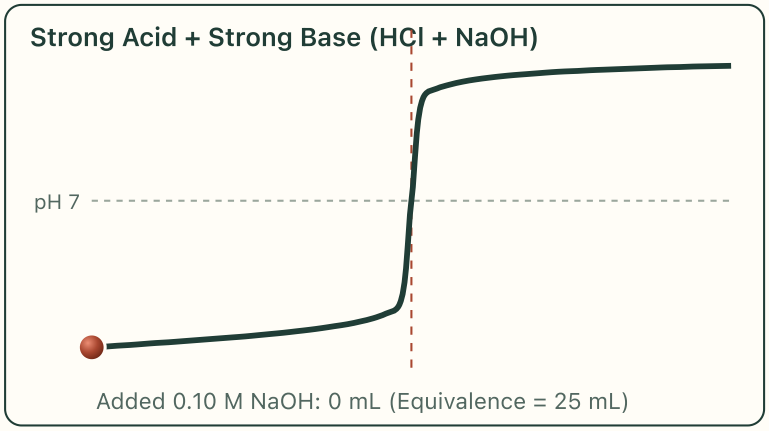
Synthesizing Le Châtelier's principle, logarithmic pH, and conjugate acid–base buffers in Earth's oceans

Earth's oceans act as a massive aqueous buffer system where dissolved carbon dioxide reacts with water to form carbonic acid (H_2CO_3), bicarbonate (HCO_3^-), and carbonate (CO_3^{2-}). Understanding ocean pH requires uniting gas solubility, Le Châtelier's principle, and conjugate acid–base buffers.

Bench Observation: Rising atmospheric CO_2 shifts the ocean carbonate equilibrium toward more H_3O^+ and HCO_3^-, lowering seawater pH from ~8.2 to ~8.1 (a ~26% increase in $[H_3O^+]$) and consuming CO_3^{2-} needed by shell-building organisms.

Investigation Task: Use the weak-acid buffer titration model to explore how a conjugate HA / A^- buffer resists pH swings within its buffer capacity but shifts sharply once one component is depleted.

Ocean acidification and carbonate buffering



Potentiometric pH titration curve (0–50 mL NaOH) and live millimole ledger of HA, conjugate base A^-, and excess OH^-.

Species, amounts & charge

Track the species present and distinguish amount, concentration and extent of ionization. Spectator ions also contribute to total charge.

The bookkeeping idea: Like matching an incoming delivery to the stock it consumes: calculate added hydroxide in mmol first, then reconcile neutralization. Concentrations require the new total volume; pH also requires an equilibrium calculation.

Current explanatory plate · intro-models-2

Solution pH (25 °C ideal model)	1
Unreacted acid H_3O^+ / HA remaining / mmol	2.5
Conjugate anion Cl^- / A^- in solution / mmol	0
Excess unreacted OH^- / mmol	0

Added hydroxide = 0.1 mol/L × 0 mL = 0 mmol. Calculated pH ≈ 1; neutralization and equilibrium approximations are separate steps.

Model note: Strong acid + strong base titration (0 mL NaOH added; equivalence at 25 mL, pH = 7.00): 2.5 mmol H_3O^+ remaining, 0 mmol excess OH^-, pH = 1.

Scientific inputs · scientific-values-2

Dilute-solution concentration approximation for K_w: 1e-14 $(mol/L)^2$ · reference. 25 °C; thermodynamic equilibrium constants are expressed through activities.

Concentrations approximate activities in this model. Introductory rounded approximation. Source

Ethanoic-acid dissociation parameter: 0.000018 concentration-model parameter · reference. 25 °C, dilute aqueous model. Concentration approximation to activity-based equilibrium. Two significant figures. Source

Initial acid solution volume: 25 mL · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Chosen acid and titrant concentration: 0.1 mol/L · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Initial buffer volume: 50 mL · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Initial amount of each buffer partner: 3 mmol · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Enrichment: No direct topic objective assigned. Model exploration beyond the mapped chemistry introduction Related prerequisite only: this investigation extends the cited outcome. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Enrichment: No direct topic objective assigned. Model exploration beyond the mapped chemistry introduction Related prerequisite only: this investigation extends the cited outcome. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: No direct IX–X topic objective assigned. Quantitative or mechanistic enrichment beyond the mapped IX–X outcomes Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 11.1: Deterministic particle & mathematical plate for *Ocean acidification and carbonate buffering*. First Principle: Dissolved CO_2 shifts the ocean's carbonic acid / bicarbonate / carbonate equilibrium, where carbonate ions buffer pH swings by converting into HCO_3^-.

Historical & Experimental Evidence: Keeling curve atmospheric CO_2 and Mauna Loa / ALOHA seawater pH and carbonate records

Since the industrial revolution, Earth's oceans have absorbed roughly a quarter of human-emitted CO_2. When CO_2 dissolves in seawater, it reacts with water to form carbonic acid (H_2CO_3), which transfers H^+ to carbonate ions: $H_2CO_3 + CO_3^{2-} \rightleftharpoons 2\ HCO_3^-$. Because seawater contains a bicarbonate/carbonate (HCO_3^- / CO_3^{2-}) buffer, ocean pH has shifted gradually from ~8.2 to ~8.1 rather than plunging to acidic pH 4—yet because pH is logarithmic, a 0.1-unit drop still represents a ~26% increase in $[H_3O^+]$ and consumes carbonate ions needed by corals and pteropods to build $CaCO_3$ shells!

Key Terms

Ocean carbonate buffer system
The coupled aqueous equilibria $CO_2(aq) + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons H^+ + HCO_3^- \rightleftharpoons 2\ H^+ + CO_3^{2-}$ that regulate seawater pH and carbonate mineral saturation.

Buffer capacity exhaustion
When enough acid or base is added to consume most of one member of the conjugate HA / A^- pair, pH begins to change rapidly per unit of added reactant.

Worked Reasoning Trace

Held fixed: System 2: Conjugate buffer (2.50 mmol HA + 2.50 mmol A^-).

Changed: Compared 0 mL to 14 mL of added reactant (within buffer plateau) versus 26 mL (buffer capacity exhausted).

Model result: Across the first 14 mL, pH shifts by only ~0.55 units because the conjugate pair absorbs the perturbation; past 25 mL, when the buffering partner is depleted, pH jumps by more than 6 units!

Model boundary: Real seawater buffering also couples to atmospheric gas exchange, temperature-dependent Henry's law solubility, and deep-ocean $CaCO_3$ sediment dissolution.

Common Misconception Trap · Ocean acidification and carbonate buffering

Intuitive Trap: "I think a decrease of 0.1 ph unit means hydronium ion concentration [h₃o⁺] decreased to zero — because cO₂ reacts with NaCl to precipitate solid sodium metal on the ocean floor."

Particle Mechanism Refutation: "Wait — look at the particle ledger: Carbonate ions (CO_3^{2-}) accept H^+ to form bicarbonate (HCO_3^-), moderating the drop in pH while reducing the concentration of free CO_3^{2-} available for $CaCO_3$ shells. Dissolved CO_2 forms carbonic acid (H_2CO_3), which transfers H^+ to water (raising $[H_3O^+]$) and to CO_3^{2-} buffer ions (converting CO_3^{2-} into HCO_3^-)."

Why Claim A fails: Sodium ions (Na^+) are spectator ions in seawater and cannot be reduced to Na(s) by CO_2. · Why Claim B holds up: Exact mechanism: $H_2CO_3 + CO_3^{2-} \rightleftharpoons 2\ HCO_3^-$ consumes carbonate ions to buffer the added H^+, moderating the pH drop while reducing CO_3^{2-} availability for $CaCO_3$ shells.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Balanced conjugate buffer (2.50 mmol HA / 2.50 mmol A^-, pH 4.74)	Equal molar amounts of weak acid and conjugate base provide maximum buffering in both directions.	Conjugate acid–base buffer (CH_3COOH / CH_3COO^- with 0.0 mmol OH^- added): Intact HA neutralizes added OH^- (HA + OH^- → A^- + H_2O), holding pH near pKa = 4.74 (current pH = 4.74).
Perturbed buffer within capacity (10 mL added, pH 5.11)	Conjugate buffer absorbs the perturbation with only a small pH shift while converting HA ⇔ A^-.	Conjugate acid–base buffer (CH_3COOH / CH_3COO^- with 1.0 mmol OH^- added): Intact HA neutralizes added OH^- (HA + OH^- → A^- + H_2O), holding pH near pKa = 4.74 (current pH = 5.05).
Buffer capacity exceeded (28 mL added, pH 11.40)	Once the reactive buffer partner is depleted (0 mmol remaining), pH shifts dramatically.	Conjugate acid–base buffer (CH_3COOH / CH_3COO^- with 2.8 mmol OH^- added): Intact HA neutralizes added OH^- (HA + OH^- → A^- + H_2O), holding pH near pKa = 4.74 (current pH = 6.21).

Check Question: When excess atmospheric CO_2 dissolves in seawater and forms carbonic acid, how does the ocean's carbonate/bicarbonate buffer respond? · **Transfer Challenge:** Transfer challenge: Using pH = $-\log_{10}[H_3O^+]$, verify by calculation that shifting from pH 8.20 ($[H_3O^+] = 10^{-8.20}$ M) to pH 8.10 ($[H_3O^+] = 10^{-8.10}$ M) increases hydronium ion concentration by ~26%.

INVESTIGATION 11.2 · PRINCIPLE 2: ELECTROSTATIC BALANCE · MODEL: REDOX

Energy storage in rechargeable cells

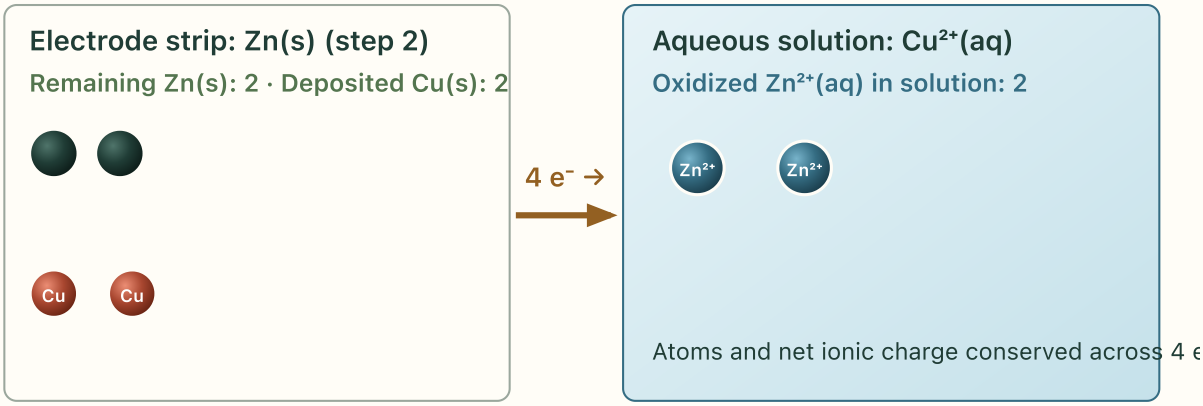
Uniting stoichiometry, molar mass, and reversible electrochemistry in modern energy storage

Modern clean-energy storage unites stoichiometry (molar mass and charge capacity), thermodynamics (spontaneous galvanic discharge vs. externally driven electrolytic recharge), and redox electron transfer.

Bench Observation: During battery discharge, oxidation at the anode releases electrons through the external load while cations migrate through the electrolyte; applying an external voltage reverses the electron transfer to recharge the cell.

Investigation Task: Step Reaction progress (electron transfer groups) across spontaneous discharge (System 0–2) and compare with uphill electrolytic recharge (System 3).

Energy storage in rechargeable cells



Submicroscopic electron-transfer accounting at the metal–solution interface.

Matched electron transfers

Electrons lost must equal electrons gained in a complete redox account. Follow the half-reactions and ionic charge separately.

The bookkeeping idea: Like a matched transfer between two accounts: electrons leaving the donor must equal electrons arriving at the acceptor. An electron is negatively charged, so losing it raises the donor's charge. Count atoms separately.

Current explanatory plate · intro-models-2

Starting metal atoms on strip	2 atoms
Oxidized metal cations in solution	2 ions
Reduced metal atoms plated out	2 atoms
Electrons transferred	4 electrons

Electrons lost = electrons gained = 4 e^-; remaining metal = 2 atoms

Model note: Zn(s) + Cu^{2+}(aq) → Zn^{2+}(aq) + Cu(s): 2 Zn atom(s) lost 4 e^- (oxidized to 2 Zn^{2+}) while 2 Cu^{2+} ion(s) gained 4 e^- (reduced to 2 solid Cu).

Scientific inputs · scientific-values-2

▼ Course connections · partial, not full coverage

Cambridge IGCSE Chemistry 0620 · Enrichment: 6.4 Supplement 6–8. Oxidation, reduction and electron transfer Related prerequisite only: this investigation extends the cited outcome. 2026–2028 · Version 2

AQA GCSE Chemistry 8462 · Enrichment: 4.4.1.4 (HT). Electron accounting in redox Related prerequisite only: this investigation extends the cited outcome. Current web specification

CBSE Science 086 · Classes IX–X · Enrichment: X · Unit I · Chemical Reactions and Equations (p. 4). Redox; electron accounting extends the treatment Consult the current NCERT text and school assessment plan. This model covers a concept, not the entire unit. 2026–27

IB MYP Sciences · Framework connection: Criteria A, B and C · inquiry framework. Explain a model (A), plan controlled comparisons (B), interpret calculated output and evaluate limitations (C). School-designed content and year-specific descriptors apply. Simulations do not fulfil practical investigation requirements. Application discussions may support D; no IB achievement level is assigned. Public subject brief · programme from 2014, ©2015

Figure 11.2: Deterministic particle & mathematical plate for *Energy storage in rechargeable cells*. First Principle: Rechargeable batteries unite reversible redox electron transfer, ion migration, and stoichiometry—releasing electrical work during spontaneous discharge and storing energy when external voltage reverses the reaction.

Historical & Experimental Evidence: Galvanic discharge vs. electrolytic recharge in lithium-ion and metal-ion cells

Why was lithium (Z = 3, M = 6.94 g/mol) chosen to power everything from smartphones and laptops to electric vehicles and grid storage? Look at the Periodic Table and our Four First Principles together: (1) Lithium is the lightest metal (only 6.94 g per mole of electrons!), maximizing energy stored per kilogram; (2) It has a strongly negative reduction potential, giving high cell voltage (> 3.6 V); and (3) During discharge, oxidation at the anode sends electrons through your device while cations intercalate into the cathode—and connecting an external charger reverses both the electron flow and ion migration without destroying a single atom!

Key Terms

Reversible electrochemical cycle (Discharge ⇔ Charge)
During discharge, the cell operates as a spontaneous galvanic cell delivering electrical work; during charging, an external power supply drives the non-spontaneous electrolytic reverse reaction.

Specific charge capacity (Ah / kg)
The amount of electric charge (moles of electrons × Faraday's constant) stored per unit mass of electrode material—inversely proportional to molar mass M per electron transferred.

Worked Reasoning Trace

Held fixed: Reversible metal/cation redox couple.

Changed: Stepped Reaction progress from 1 group (2 e^- transferred) to 3 groups (6 e^- transferred), then compared with the uphill reverse reaction (System 3).

Model result: Discharging 3 groups transfers 6 e^- through the external circuit while conserving all 8 metal atoms and net +8 ionic charge; recharging reverses the electron transfer back to the high-energy reactant state.

Model boundary: Real rechargeable batteries gradually lose capacity over hundreds of cycles due to solid-electrolyte interphase (SEI) side reactions and electrode structural fatigue—not because electrons or atoms vanish.

Common Misconception Trap · Energy storage in rechargeable cells

Intuitive Trap: "I think charging creates brand-new metal nuclei out of pure electrical energy — because pumps brand-new electrons out of the wall socket to permanently increase the total mass of the battery."

Particle Mechanism Refutation: "Wait — look at the particle ledger: An external voltage drives electrons in the reverse direction, reducing the oxidized anode species back toward its charged reactant state while conserving every atom and net charge. Applies an external voltage that drives electrons and mobile cations in the reverse direction, restoring the high-potential-energy anode and cathode reactants while conserving every atom and electron."

Why Claim A fails: Every electron pushed into the negative electrode during charging is pulled out of the positive electrode; total electron and atom counts inside the sealed cell stay constant. · Why Claim B holds up: Exact mechanism: recharging is an electrolytic process that uses electrical work to reverse the spontaneous galvanic discharge reaction in a closed mass-conserving system.

Guided Model Presets & Computed Outputs

Preset	Physical Context	Computed Model Output
Partially discharged cell (1 group, 2 e^- delivered)	1 oxidation event sends 2 e^- through the load; 3 unreacted anode units remain.	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$: 1 Zn atom(s) lost 2 e^- (oxidized to 1 Zn^{2+}) while 1 Cu^{2+} ion(s) gained 2 e^- (reduced to 1 solid Cu).
Deeply discharged cell (3 groups, 6 e^- delivered)	6 e^- have flowed from anode to cathode; total atom count and net charge remain strictly conserved.	$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$: 3 Zn atom(s) lost 6 e^- (oxidized to 3 Zn^{2+}) while 3 Cu^{2+} ion(s) gained 6 e^- (reduced to 3 solid Cu).
Uphill recharge direction (requires external voltage)	The reverse reaction is thermodynamically uphill (0 spontaneous progress) and requires an external charger.	Cu(s) placed in $Zn^{2+}(aq)$: 0 electrons transferred. Copper metal does not spontaneously reduce $Zn^{2+}(aq)$ under standard conditions; electron transfer has a preferred thermodynamic direction.

Check Question: What happens at the particle level when a rechargeable electrochemical cell is switched from discharging (powering a device) to charging (connected to an external power supply)? · **Transfer Challenge:** Capstone Hero Synthesis: Explain why lithium ($M \approx 6.94$ g/mol, 1 e^- per $Li \rightarrow Li^+$) can store far more moles of electrons per kilogram than lead ($M \approx 207.2$ g/mol, 2 e^- per $Pb \rightarrow Pb^{2+}$) in a traditional lead-acid car battery.

End-of-Chapter Synthesis & Forensic Mystery: Forensic Mystery 11 · The Grand Zero-to-Hero Synthesis Audit

An environmental engineering team compares two ways of delivering energy: (Path 1) Combusting 1.00 mol of methane ($CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$, $\Delta H = -802$ kJ) and absorbing the resulting CO_2 in seawater; (Path 2) Storing electrical energy in a rechargeable lithium cell ($M_{Li} = 6.94$ g/mol).

- Clue 1 (Stoichiometry & Bond Ledger): Burning 1.00 mol CH_4 (16.0 g) breaks 4 C–H and 2 O=O bonds (+2648 kJ input) and forms 2 C=O and 4 O–H bonds (−3450 kJ payback), producing 1.00 mol (44.0 g) of CO_2.
- Clue 2 (Ocean Buffer Perturbation): When that CO_2 dissolves in seawater, H_2CO_3 transfers H^+ to $CO_3^{2-} \rightarrow 2\ HCO_3^-$, buffering the pH drop while lowering free $[CO_3^{2-}]$.
- Clue 3 (Electrochemical Storage): In a rechargeable cell, spontaneous galvanic discharge sends electrons from the low-molar-mass anode through the load, and an external voltage reverses the redox reaction during charging while conserving all atoms.

Resolution: This capstone synthesis connects Principle 1 (16 g $CH_4 \rightarrow$ 44 g CO_2 mass stoichiometry & atom conservation in batteries), Principle 2 (reversible redox electron/ion transfer), Principle 3 (bond-making payback −3450 kJ vs. bond-breaking cost +2648 kJ), and Principle 4 (carbonate/bicarbonate buffer equilibrium).

What our upgraded model now explains: Synthesizes all Four First Principles—Conservation of Count & Charge, Electrostatic Balance, Thermal Motion vs. Barriers, and Dynamic Statistical Ratios—across Earth and clean-energy systems. · **Next boundary:** You have completed the Zero-to-Hero foundational chemistry series! Your next horizon: quantum molecular orbitals, multi-step organic synthesis, and statistical thermodynamics.

Scientific value register

264 entries · scientific-values-2. A reference approximation is distinct from a measured sample value; illustrative settings are author-selected. Derived outputs are local calculations. Source review: 28 October 2026.

Avogadro constant

6.02214076e+23 mol^{-1} · defined. Counts specified entities per mole; state whether atoms, molecules or formula units. Exact SI defining constant.

Read source

Molar gas constant R, rounded for the local model

8.314462618 kPa·L·mol^{-1}·K^{-1} · reference. Absolute temperature in kelvin. The ideal gas model ignores particle size and attractions. Rounded from the exact SI-derived gas constant.

Read source

Light particle in synthetic isotope mixture

12 u · illustrative. Defined teaching mass; not a measured isotope mass. Exact within the synthetic example.

Heavy particle in synthetic isotope mixture

13 u · illustrative. Defined teaching mass; not a measured isotope mass. Exact within the synthetic example.

Abridged standard atomic weight: C

12.011 relative atomic weight · reference. Normal terrestrial material; natural isotope composition varies. Used numerically in g/mol for introductory molar masses. Abridged value, not the exact mass of any individual isotope.

Read source

Abridged standard atomic weight: H

1.008 relative atomic weight · reference. Normal terrestrial material; natural isotope composition varies. Used numerically in g/mol for introductory molar masses. Abridged value, not the exact mass of any individual isotope.

Read source

Abridged standard atomic weight: O

15.999 relative atomic weight · reference. Normal terrestrial material; natural isotope composition varies. Used numerically in g/mol for introductory molar masses. Abridged value, not the exact mass of any individual isotope.

Read source

Abridged standard atomic weight: N

14.007 relative atomic weight · reference. Normal terrestrial material; natural isotope composition varies. Used numerically in g/mol for introductory molar masses. Abridged value, not the exact mass of any individual isotope.

Read source

Dilute-solution concentration approximation for K_w

1e-14 (mol/L)² · reference. 25 °C; thermodynamic equilibrium constants are expressed through activities. Concentrations approximate activities in this model. Introductory rounded approximation.

Read source

Ethanoic-acid dissociation parameter

0.000018 concentration-model parameter · reference. 25 °C, dilute aqueous model. Concentration approximation to activity-based equilibrium. Two significant figures.

Read source

Initial acid solution volume

25 mL · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Chosen acid and titrant concentration

0.1 mol/L · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Initial buffer volume

50 mL · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Initial amount of each buffer partner

3 mmol · illustrative. Chosen teaching preparation, not a measured experimental sample. Exact within the preparation model.

Synthetic 12 u / 13 u pair: light teaching mass

12 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture.

Read source

Synthetic 12 u / 13 u pair: heavy teaching mass

13 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture.

Read source

Chlorine-35 / Chlorine-37: light teaching mass

35 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture.

Read source

Chlorine-35 / Chlorine-37: heavy teaching mass

37 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture.

Read source

Bromine-79 / Bromine-81: light teaching mass

79 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture.

Read source

Bromine-79 / Bromine-81: heavy teaching mass

81 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture.

Read source

Boron-10 / Boron-11: light teaching mass

10 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture.

Read source

Boron-10 / Boron-11: heavy teaching mass

11 u · illustrative. Integer-mass approximation for weighted-average practice, not a measured isotope mass. Use the explicit reference lookup for measured isotope properties. Mass-number approximation. Natural abundance is not assumed by the adjustable mixture.

Read source

Methane: normal boiling point approximation

111.7 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 111.7 K

Read source

Methane: dipole approximation

0 D · reference. Methane: zero by tetrahedral symmetry. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 0 D

Read source

Methane: dispersion score

16 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: polar-H-bond score

0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methane: formula molar mass

16.04 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 1 × 12.011 + 4 × 1.008 + 0 × 15.999 + 0 × 14.007

Read source

Ethane: normal boiling point approximation

184.6 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 184.6 K

Read source

Ethane: dipole approximation

0 D · reference. Ethane: zero for the symmetry shown. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 0 D

Read source

Ethane: dispersion score

28 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethane: polar-H-bond score

0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethane: formula molar mass

30.07 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 6 × 1.008 + 0 × 15.999 + 0 × 14.007

Read source

Propane: normal boiling point approximation

231.1 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 231.1 K

Read source

Propane: schematic dipole parameter

0 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Propane: dispersion score

40 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propane: polar-H-bond score

0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propane: formula molar mass

44.1 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 8 × 1.008 + 0 × 15.999 + 0 × 14.007

Read source

n-Butane: normal boiling point approximation

272.7 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 272.7 K

Read source

n-Butane: dipole approximation

0 D · reference. Butane: anti conformer; other conformers can have small dipoles. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 0 D

Read source

n-Butane: dispersion score

52 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

n-Butane: polar-H-bond score

0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

n-Butane: formula molar mass

58.12 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 10 × 1.008 + 0 × 15.999 + 0 × 14.007

Read source

Isobutane (2-methylpropane): normal boiling point approximation

261.4 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 261.4 K

Read source

Isobutane (2-methylpropane): schematic dipole parameter

0.13 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Isobutane (2-methylpropane): dispersion score

46 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Isobutane (2-methylpropane): polar-H-bond score

0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Isobutane (2-methylpropane): formula molar mass

58.12 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 10 × 1.008 + 0 × 15.999 + 0 × 14.007

Read source

Ethene (ethylene): normal boiling point approximation

169.4 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 169.35 K

Read source

Ethene (ethylene): dipole approximation

0 D · reference. Ethylene: zero by symmetry. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 0 D

Read source

Ethene (ethylene): dispersion score

25 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethene (ethylene): polar-H-bond score

0 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethene (ethylene): formula molar mass

28.05 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 4 × 1.008 + 0 × 15.999 + 0 × 14.007

Read source

Methanol: normal boiling point approximation

337.8 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 337.8 K

Read source

Methanol: dipole approximation

1.7 D · reference. Methyl alcohol, gas-phase microwave value; model rounds to 1.7 D. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 1.672 D

Read source

Methanol: dispersion score

19 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methanol: polar-H-bond score

58 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methanol: formula molar mass

32.04 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 1 × 12.011 + 4 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Ethanol: normal boiling point approximation

351.5 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 351.44–351.56 K

Read source

Ethanol: schematic dipole parameter

1.69 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Ethanol: dispersion score

30 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethanol: polar-H-bond score

52 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethanol: formula molar mass

46.07 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 6 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Propan-1-ol: normal boiling point approximation

370.4 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 370.3 ± 0.5 K

Read source

Propan-1-ol: schematic dipole parameter

1.68 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Propan-1-ol: dispersion score

41 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propan-1-ol: polar-H-bond score

48 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propan-1-ol: formula molar mass

60.1 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 8 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Butan-1-ol: normal boiling point approximation

390.9 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 390.6 ± 0.8 K

Read source

Butan-1-ol: schematic dipole parameter

1.66 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Butan-1-ol: dispersion score

52 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Butan-1-ol: polar-H-bond score

44 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Butan-1-ol: formula molar mass

74.12 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 10 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

tert-Butanol (2-methylpropan-2-ol): normal boiling point approximation

355.6 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 355.5 ± 0.7 K

Read source

tert-Butanol (2-methylpropan-2-ol): schematic dipole parameter

1.66 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

tert-Butanol (2-methylpropan-2-ol): dispersion score

44 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

tert-Butanol (2-methylpropan-2-ol): polar-H-bond score

40 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

tert-Butanol (2-methylpropan-2-ol): formula molar mass

74.12 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 10 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Ethenol (vinyl alcohol tautomer): illustrative transition parameter

325 K · illustrative. Vinyl alcohol is a tautomeric teaching structure. A normal boiling point has not been established here; this value must not be treated as a measured phase boundary. Teaching approximation; inspect the source data and its uncertainties.

Ethenol (vinyl alcohol tautomer): schematic dipole parameter

1.42 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Ethenol (vinyl alcohol tautomer): dispersion score

27 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethenol (vinyl alcohol tautomer): polar-H-bond score

46 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethenol (vinyl alcohol tautomer): formula molar mass

44.05 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 4 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Dimethyl ether: normal boiling point approximation

249 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 248.2–249.5 K

Read source

Dimethyl ether: schematic dipole parameter

1.3 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Dimethyl ether: dispersion score

29 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Dimethyl ether: polar-H-bond score

14 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Dimethyl ether: formula molar mass

46.07 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 6 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Dimethyl ether (C_2H_6O isomer): normal boiling point approximation

249 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 248.2–249.5 K

Read source

Dimethyl ether (C_2H_6O isomer): schematic dipole parameter

1.3 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Dimethyl ether (C_2H_6O isomer): dispersion score

29 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Dimethyl ether (C_2H_6O isomer): polar-H-bond score

14 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Dimethyl ether (C_2H_6O isomer): formula molar mass

46.07 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 6 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Methoxyethane (ethyl methyl ether): normal boiling point approximation

280.5 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 279.75 K; source compilation contains differing determinations

Read source

Methoxyethane (ethyl methyl ether): schematic dipole parameter

1.17 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Methoxyethane (ethyl methyl ether): dispersion score

39 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methoxyethane (ethyl methyl ether): polar-H-bond score

12 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methoxyethane (ethyl methyl ether): formula molar mass

60.1 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 8 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Diethyl ether (ethoxyethane): normal boiling point approximation

307.8 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 307.7 ± 0.4 K

Read source

Diethyl ether (ethoxyethane): schematic dipole parameter

1.15 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Diethyl ether (ethoxyethane): dispersion score

49 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Diethyl ether (ethoxyethane): polar-H-bond score

10 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Diethyl ether (ethoxyethane): formula molar mass

74.12 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 10 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

2-Methoxypropane (methyl isopropyl ether): normal boiling point approximation

303.9 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 303.9 K

Read source

2-Methoxypropane (methyl isopropyl ether): schematic dipole parameter

1.15 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

2-Methoxypropane (methyl isopropyl ether): dispersion score

45 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

2-Methoxypropane (methyl isopropyl ether): polar-H-bond score

10 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

2-Methoxypropane (methyl isopropyl ether): formula molar mass

74.12 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 10 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Methoxyethene (methyl vinyl ether): normal boiling point approximation

278.7 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 278.7 K

Read source

Methoxyethene (methyl vinyl ether): schematic dipole parameter

1.1 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Methoxyethene (methyl vinyl ether): dispersion score

36 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methoxyethene (methyl vinyl ether): polar-H-bond score

12 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methoxyethene (methyl vinyl ether): formula molar mass

58.08 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 6 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Methanal (formaldehyde): normal boiling point approximation

254.1 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 253.85 K

Read source

Methanal (formaldehyde): schematic dipole parameter

2.33 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Methanal (formaldehyde): dispersion score

18 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methanal (formaldehyde): polar-H-bond score

26 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methanal (formaldehyde): formula molar mass

30.03 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 1 × 12.011 + 2 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Ethanal (acetaldehyde): normal boiling point approximation

293.4 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 294.0 ± 0.8 K

Read source

Ethanal (acetaldehyde): dipole approximation

2.75 D · reference. Acetaldehyde, gas-phase microwave value. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 2.75 D

Read source

Ethanal (acetaldehyde): dispersion score

28 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethanal (acetaldehyde): polar-H-bond score

28 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethanal (acetaldehyde): formula molar mass

44.05 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 4 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Propanone (acetone): normal boiling point approximation

329.2 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 329.2–329.45 K

Read source

Propanone (acetone): dipole approximation

2.88 D · reference. Acetone, gas-phase microwave value. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 2.88 D

Read source

Propanone (acetone): dispersion score

38 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propanone (acetone): polar-H-bond score

30 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propanone (acetone): formula molar mass

58.08 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 6 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Butanal: normal boiling point approximation

347.9 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 348 ± 2 K

Read source

Butanal: schematic dipole parameter

2.72 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Butanal: dispersion score

48 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Butanal: polar-H-bond score

26 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Butanal: formula molar mass

72.11 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 8 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

2-Methylpropanal (isobutyraldehyde): normal boiling point approximation

337.2 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 337.2 K

Read source

2-Methylpropanal (isobutyraldehyde): schematic dipole parameter

2.69 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

2-Methylpropanal (isobutyraldehyde): dispersion score

44 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

2-Methylpropanal (isobutyraldehyde): polar-H-bond score

25 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

2-Methylpropanal (isobutyraldehyde): formula molar mass

72.11 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 8 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Propenal (acrolein): normal boiling point approximation

325.8 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 325.42–325.7 K

Read source

Propenal (acrolein): schematic dipole parameter

3.12 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Propenal (acrolein): dispersion score

34 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propenal (acrolein): polar-H-bond score

29 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propenal (acrolein): formula molar mass

56.06 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 4 × 1.008 + 1 × 15.999 + 0 × 14.007

Read source

Methanoic acid (formic acid): normal boiling point approximation

373.9 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 373.9 ± 0.5 K

Read source

Methanoic acid (formic acid): schematic dipole parameter

1.41 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Methanoic acid (formic acid): dispersion score

24 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methanoic acid (formic acid): polar-H-bond score

66 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methanoic acid (formic acid): formula molar mass

46.02 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 1 × 12.011 + 2 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

Ethanoic acid (acetic acid): normal boiling point approximation

391.1 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 391.05–391.20 K

Read source

Ethanoic acid (acetic acid): schematic dipole parameter

1.74 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Ethanoic acid (acetic acid): dispersion score

34 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethanoic acid (acetic acid): polar-H-bond score

64 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethanoic acid (acetic acid): formula molar mass

60.05 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 4 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

Propanoic acid: normal boiling point approximation

414.3 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 414 ± 1 K

Read source

Propanoic acid: dipole approximation

1.75 D · reference. Propanoic acid, selected experimental value; phase/conformation matter. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 1.75 D

Read source

Propanoic acid: dispersion score

44 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propanoic acid: polar-H-bond score

60 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propanoic acid: formula molar mass

74.08 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 6 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

Butanoic acid (butyric acid): normal boiling point approximation

436.9 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 436 ± 2 K

Read source

Butanoic acid (butyric acid): schematic dipole parameter

1.75 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Butanoic acid (butyric acid): dispersion score

54 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Butanoic acid (butyric acid): polar-H-bond score

56 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Butanoic acid (butyric acid): formula molar mass

88.11 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 8 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

2-Methylpropanoic acid (isobutyric acid): normal boiling point approximation

427.6 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 427 ± 2 K

Read source

2-Methylpropanoic acid (isobutyric acid): schematic dipole parameter

1.75 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

2-Methylpropanoic acid (isobutyric acid): dispersion score

49 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

2-Methylpropanoic acid (isobutyric acid): polar-H-bond score

55 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

2-Methylpropanoic acid (isobutyric acid): formula molar mass

88.11 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 8 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

Propenoic acid (acrylic acid): normal boiling point approximation

414.2 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 412.2–414.8 K

Read source

Propenoic acid (acrylic acid): schematic dipole parameter

1.98 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Propenoic acid (acrylic acid): dispersion score

40 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propenoic acid (acrylic acid): polar-H-bond score

62 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propenoic acid (acrylic acid): formula molar mass

72.06 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 4 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

Methanamine (methylamine): normal boiling point approximation

266.8 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 266.8 K

Read source

Methanamine (methylamine): dipole approximation

1.31 D · reference. Methylamine, microwave value. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 1.31 D

Read source

Methanamine (methylamine): dispersion score

20 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methanamine (methylamine): polar-H-bond score

38 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methanamine (methylamine): formula molar mass

31.06 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 1 × 12.011 + 5 × 1.008 + 0 × 15.999 + 1 × 14.007

Read source

Ethanamine (ethylamine): normal boiling point approximation

289.8 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 289.8 K

Read source

Ethanamine (ethylamine): dipole approximation

1.22 D · reference. Ethylamine, selected experimental value. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 1.22 D

Read source

Ethanamine (ethylamine): dispersion score

31 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethanamine (ethylamine): polar-H-bond score

35 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Ethanamine (ethylamine): formula molar mass

45.09 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 7 × 1.008 + 0 × 15.999 + 1 × 14.007

Read source

Propan-1-amine: normal boiling point approximation

321 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 322 ± 2 K

Read source

Propan-1-amine: dipole approximation

1.17 D · reference. 1-Propanamine, selected experimental value. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 1.17 D

Read source

Propan-1-amine: dispersion score

42 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propan-1-amine: polar-H-bond score

33 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Propan-1-amine: formula molar mass

59.11 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 9 × 1.008 + 0 × 15.999 + 1 × 14.007

Read source

Butan-1-amine: normal boiling point approximation

350.9 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 351 ± 1 K

Read source

Butan-1-amine: schematic dipole parameter

1.37 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Butan-1-amine: dispersion score

53 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Butan-1-amine: polar-H-bond score

32 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Butan-1-amine: formula molar mass

73.14 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 11 × 1.008 + 0 × 15.999 + 1 × 14.007

Read source

tert-Butylamine (2-methylpropan-2-amine): normal boiling point approximation

317.6 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 317.6 K

Read source

tert-Butylamine (2-methylpropan-2-amine): schematic dipole parameter

1.29 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

tert-Butylamine (2-methylpropan-2-amine): dispersion score

45 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

tert-Butylamine (2-methylpropan-2-amine): polar-H-bond score

28 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

tert-Butylamine (2-methylpropan-2-amine): formula molar mass

73.14 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 11 × 1.008 + 0 × 15.999 + 1 × 14.007

Read source

Prop-2-en-1-amine (allylamine): normal boiling point approximation

326.5 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 326.2 K

Read source

Prop-2-en-1-amine (allylamine): schematic dipole parameter

1.2 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Prop-2-en-1-amine (allylamine): dispersion score

38 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Prop-2-en-1-amine (allylamine): polar-H-bond score

33 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Prop-2-en-1-amine (allylamine): formula molar mass

57.1 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 7 × 1.008 + 0 × 15.999 + 1 × 14.007

Read source

Methyl methanoate (methyl formate): normal boiling point approximation

304.9 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 305 ± 1 K

Read source

Methyl methanoate (methyl formate): schematic dipole parameter

1.77 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Methyl methanoate (methyl formate): dispersion score

32 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl methanoate (methyl formate): polar-H-bond score

22 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl methanoate (methyl formate): formula molar mass

60.05 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 2 × 12.011 + 4 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

Methyl ethanoate (methyl acetate): normal boiling point approximation

330.1 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 330.02–330.05 K

Read source

Methyl ethanoate (methyl acetate): dipole approximation

1.72 D · reference. Methyl acetate, selected experimental value. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Reference entry: 1.72 D

Read source

Methyl ethanoate (methyl acetate): dispersion score

42 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl ethanoate (methyl acetate): polar-H-bond score

21 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl ethanoate (methyl acetate): formula molar mass

74.08 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 3 × 12.011 + 6 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

Methyl propanoate: normal boiling point approximation

352.9 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 352.9 K

Read source

Methyl propanoate: schematic dipole parameter

1.7 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Methyl propanoate: dispersion score

52 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl propanoate: polar-H-bond score

20 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl propanoate: formula molar mass

88.11 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 8 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

Methyl butanoate (apple/pineapple ester): normal boiling point approximation

375.9 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 375.9 K

Read source

Methyl butanoate (apple/pineapple ester): schematic dipole parameter

1.71 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Methyl butanoate (apple/pineapple ester): dispersion score

62 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl butanoate (apple/pineapple ester): polar-H-bond score

19 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl butanoate (apple/pineapple ester): formula molar mass

102.13 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 5 × 12.011 + 10 × 1.008 + 2 × 15.999 + 0 × 14.007

Read source

Methyl 2-methylpropanoate: normal boiling point approximation

365.4 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 365 ± 1 K

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Methyl 2-methylpropanoate: schematic dipole parameter

1.7 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Methyl 2-methylpropanoate: dispersion score

57 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl 2-methylpropanoate: polar-H-bond score

19 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl 2-methylpropanoate: formula molar mass

102.13 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 5 × 12.011 + 10 × 1.008 + 2 × 15.999 + 0 × 14.007

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Methyl propenoate (methyl acrylate): normal boiling point approximation

353.3 K · reference. Pure substance, nominal normal boiling at 101.325 kPa. This single threshold does not distinguish solid from liquid below boiling; source compilations contain differing determinations. Teaching approximation; inspect the source data and its uncertainties.

Reference entry: 353–354.1 K

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Methyl propenoate (methyl acrylate): schematic dipole parameter

1.75 D · illustrative. Author-selected parameter for qualitative comparison; no species-specific measured value is claimed. Dipoles depend on geometry/conformation and phase; the drawing does not solve an electron distribution.

Methyl propenoate (methyl acrylate): dispersion score

48 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl propenoate (methyl acrylate): polar-H-bond score

21 model units · illustrative. Author-selected qualitative score, not an energy in joules or a computed intermolecular potential. Only comparable within this teaching scale.

Methyl propenoate (methyl acrylate): formula molar mass

86.09 g/mol · derived. Calculated from abridged standard atomic weights, not an isotope-specific mass. Rounded to two decimal places.

Working: 4 × 12.011 + 6 × 1.008 + 2 × 15.999 + 0 × 14.007

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Hydrogen (H_2): deltaEN

0 Pauling scale · reference. Difference of rounded Pauling electronegativities; not a measured bond charge. Introductory approximation.

Read source

Hydrogen (H_2): dipole

0 D · reference. Rounded gas-phase dipole; zero values reflect the geometry shown. Introductory approximation.

Read source

Hydrogen (H_2): boiling

20.3 K · reference. Approximate pure-substance normal boiling point at 1 atm. Introductory approximation.

Read source

Hydrogen (H_2): attraction

4 model units · illustrative. Author-selected comparison score, not an energy. Introductory approximation.

Hydrogen chloride (HCl): deltaEN

0.96 Pauling scale · reference. Difference of rounded Pauling electronegativities; not a measured bond charge. Introductory approximation.

Read source

Hydrogen chloride (HCl): dipole

1.08 D · reference. Rounded gas-phase dipole; zero values reflect the geometry shown. Introductory approximation.

Read source

Hydrogen chloride (HCl): boiling

188.1 K · reference. Approximate pure-substance normal boiling point at 1 atm. Introductory approximation.

Read source

Hydrogen chloride (HCl): attraction

38 model units · illustrative. Author-selected comparison score, not an energy. Introductory approximation.

Water (H_2O): deltaEN

1.24 Pauling scale · reference. Difference of rounded Pauling electronegativities; not a measured bond charge. Introductory approximation.

Read source

Water (H_2O): dipole

1.85 D · reference. Rounded gas-phase dipole; zero values reflect the geometry shown. Introductory approximation.

Read source

Water (H_2O): boiling

373.15 K · reference. Approximate pure-substance normal boiling point at 1 atm. Introductory approximation.

Read source

Water (H_2O): attraction

92 model units · illustrative. Author-selected comparison score, not an energy. Introductory approximation.

Carbon dioxide (CO_2): deltaEN

0.89 Pauling scale · reference. Difference of rounded Pauling electronegativities; not a measured bond charge. Introductory approximation.

Read source

Carbon dioxide (CO_2): dipole

0 D · reference. Rounded gas-phase dipole; zero values reflect the geometry shown. Introductory approximation.

Read source

Carbon dioxide (CO_2): boiling

194.7 K · reference. CO_2: sublimation at 1 atm, not a normal liquid boiling point. Introductory approximation.

Read source

Carbon dioxide (CO_2): attraction

28 model units · illustrative. Author-selected comparison score, not an energy. Introductory approximation.

Methane (CH_4): deltaEN

0.35 Pauling scale · reference. Difference of rounded Pauling electronegativities; not a measured bond charge. Introductory approximation.

Read source

Methane (CH_4): dipole

0 D · reference. Rounded gas-phase dipole; zero values reflect the geometry shown. Introductory approximation.

Read source

Methane (CH_4): boiling

111.7 K · reference. Approximate pure-substance normal boiling point at 1 atm. Introductory approximation.

Read source

Methane (CH_4): attraction

16 model units · illustrative. Author-selected comparison score, not an energy. Introductory approximation.

Ammonia (NH_3): deltaEN

0.84 Pauling scale · reference. Difference of rounded Pauling electronegativities; not a measured bond charge. Introductory approximation.

Read source

Ammonia (NH_3): dipole

1.47 D · reference. Rounded gas-phase dipole; zero values reflect the geometry shown. Introductory approximation.

Read source

Ammonia (NH_3): boiling

239.8 K · reference. Approximate pure-substance normal boiling point at 1 atm. Introductory approximation.

Read source

Ammonia (NH_3): attraction

68 model units · illustrative. Author-selected comparison score, not an energy. Introductory approximation.

$2\ H_2 + O_2 \rightarrow 2\ H_2O$: bond-break teaching estimate

1370 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input.

Working: Breaking 2 H–H (2×436) + 1 O=O (498) absorbs +1,370 kJ

Read source

$2\ H_2 + O_2 \rightarrow 2\ H_2O$: bond-make teaching estimate

1854 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input.

Working: Forming 4 O–H bonds (4×463.5) releases –1,854 kJ

Read source

$N_2 + 3\ H_2 \rightarrow 2\ NH_3$: bond-break teaching estimate

2253 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input.

Working: Breaking 1 N≡N (945) + 3 H–H (3×436) absorbs +2,253 kJ

Read source

$N_2 + 3\ H_2 \rightarrow 2\ NH_3$: bond-make teaching estimate

2345 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input.

Working: Forming 6 N–H bonds (6×390.8) releases −2,345 kJ

Read source

$N_2 + O_2 \rightarrow 2\ NO$: bond-break teaching estimate

1443 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input.

Working: Breaking 1 N≡N (945) + 1 O=O (498) absorbs +1,443 kJ

Read source

$N_2 + O_2 \rightarrow 2\ NO$: bond-make teaching estimate

1263 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input.

Working: Forming 2 N=O bonds (2×631.5) releases −1,263 kJ

Read source

$CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$: bond-break teaching estimate

2648 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input.

Working: Breaking 4 C–H (4×413) + 2 O=O (2×498) absorbs +2,648 kJ

Read source

$CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$: bond-make teaching estimate

3450 kJ per reaction amount as written · illustrative. Author-selected aggregate bond-energy estimate. It is not a measured reaction enthalpy; chemical environment and product phase affect real values. Rounded aggregate teaching input.

Working: Forming 2 C=O (2×799) + 4 O–H (4×463) releases –3,450 kJ

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